Tame and wild theories

An introduction to the logic of dependence and independence

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Introduction

This book is a study of three dividing lines of Shelah's classification theory: stability, the negation of the independence property (NIP), and simplicity. Each of them is drawn by a configuration that a single formula can encode. In stable and simple theories, where the configuration is absent, an abstract relation of independence takes its place, and we follow these two lines from the configuration that each forbids to the independence that each produces. The lines are drawn for the formulas of first-order logic and, in their real-valued forms, for bounded functions on arbitrary sets, where they meet the analysis of Grothendieck and Rosenthal.

Dependence and independence are fundamental notions throughout mathematics. Random variables are independent when their joint distribution is the product of their individual distributions. Vectors are linearly independent when no nontrivial linear combination of them vanishes, and elements of a field extension are algebraically independent over a subfield when they satisfy no nonzero polynomial equation with coefficients in the subfield. Model theory considers variables in idealized form, as *types*. The types of a theory T form a topological space, whose interest lies less in its individual points than in the configurations, combinatorial and topological, that the points form among themselves. These configurations record how idealized variables can stand to one another in all the models of T, and a single formula consistent with T can encode one of them: a tree every branch of which is consistent with T, for instance. The dividing lines of classification theory are drawn by configurations of this kind. A theory is called *tame* when the configurations that exhibit independence, that is, freeness, among types are forbidden, and *wild* when they are realizable. Dependence, in Shelah's sense, is the impossibility of these configurations, and for this reason the NIP theories are also called *dependent*.

Shelah introduced stability in his paper *Stable theories* of 1969 [**She69**]. In 1971 he passed from theories to single formulas, and defined there both the order property and the independence property [**She71**]. In the same year Vapnik and Chervonenkis introduced shattering, the combinatorial core of the independence property, in probability theory [**VC71**], and in 1972 Sauer bounded, in combinatorics, the number of traces of a family that shatters no large set [**Sau72**]. Laskowski [**Las92**] observed that a formula is NIP exactly when the family of sets that it defines has finite Vapnik–Chervonenkis dimension. In 1980 Shelah singled out the simple theories [**She80**], those in which no formula has the tree property.

The three configurations are the following. In a stable theory no formula encodes binary trees of arbitrarily large finite height, or, equivalently, an infinite ladder. In an NIP theory, for each formula $\varphi(x, y)$ there is a finite bound on the size of the sets of parameters that φ shatters. Here a set B is *shattered* by φ if for every subset C of B some element a satisfies $\varphi(a, b)$ exactly for the parameters b in C. In a simple theory no formula has the tree property. A formula has this property when it encodes a tree in which each node has infinitely many

successors, every branch is consistent, and, for some fixed integer $k \geq 2$, any k distinct successors of a node are inconsistent.

Stability and simplicity are also characterized by abstract relations of independence. In every simple theory, Shelah's forking independence is a relation between types that has finite character and local character, and, by Kim's theorem [**Kim98**], it is symmetric and transitive. The relation also allows every type to be extended to any larger set of parameters without forking, and a theory is simple exactly when forking independence has local character. The *independence theorem* amalgamates independent extensions of one type over a model to sets that are independent of each other (see Chapter 7). By the theorem of Kim and Pillay [**KP97**], a theory is simple exactly when it carries some relation with the properties listed above that also satisfies the independence theorem. A theory is stable exactly when, in addition, every type over a model has a unique nonforking extension to each larger set (see Chapter 6 and Section 7.5).

In first-order logic a formula is true or false of each pair of elements, and the forbidden configurations are patterns of truth values. Read as a $\{0, 1\}$-valued function on a product of two sets, a formula becomes a special case of a bounded real-valued function, and stability and NIP extend to all such functions. Stability becomes the exchange of two iterated limits, the condition by which Grothendieck characterized in 1952 the relatively weakly compact bounded sets of continuous functions on a compact Hausdorff space [**Gro52**]. Definability of types becomes uniform approximation by polynomials in the instances of the function. NIP becomes the existence of pointwise convergent subsequences, one side of Rosenthal's dichotomy [**Ros74**]. The implication from stability to simplicity becomes the theorem that a stable function has no binary tree of infinite height with a gap, proved by counting approximations up to a tolerance.

The book begins, in Chapter 1, with the types of a formula over a model, definable types and the order property. There we state that a formula is stable, that is, forbids arbitrarily long ladders, exactly when all its types are definable, and we give the real-valued form of stability as the equality of two iterated limits. The proof of this definability theorem occupies Chapter 2, in which we show that the absence of arbitrarily long ladders, the absence of binary trees of arbitrarily large finite height, and the definability of types coincide. Over countable models in a countable language, the countability of type spaces is equivalent to these conditions as well. In Chapter 3 we make these arguments quantitative: we bound the height of a binary tree that forces a ladder of given length, and we obtain the definitions of types from a rank that measures the trees above a formula. In Chapter 4 we turn to the independence property. Stable formulas are NIP, the order of the rationals is NIP and unstable, and a bounded real-valued function is NIP exactly when every sequence of its instances has a subsequence that converges pointwise on the space of types. Dividing and the tree property enter in Chapter 5, where we prove that the stable theories are the simple NIP theories. In the same chapter we show that the random graph and the Fraïssé limits in a finite relational language with disjoint 2- and 3-amalgamation are simple. In the last two chapters we pass from the forbidden configurations to the independence relations that their absence produces. In Chapter 6 we show that, in a stable theory, independence between types over models, obtained from their definitions, satisfies all the usual axioms, uniqueness of independent extensions included, and that a theory that carries such a relation is stable. In Chapter 7 we do the same for simplicity: in a simple theory, nonforking independence is symmetric and transitive and satisfies the independence theorem, simplicity together with uniqueness is stability, and we conclude with the Kim–Pillay theorem. In the appendices

we prove Grothendieck's criterion and develop the set-theoretic tools for counting types and finding common bases.

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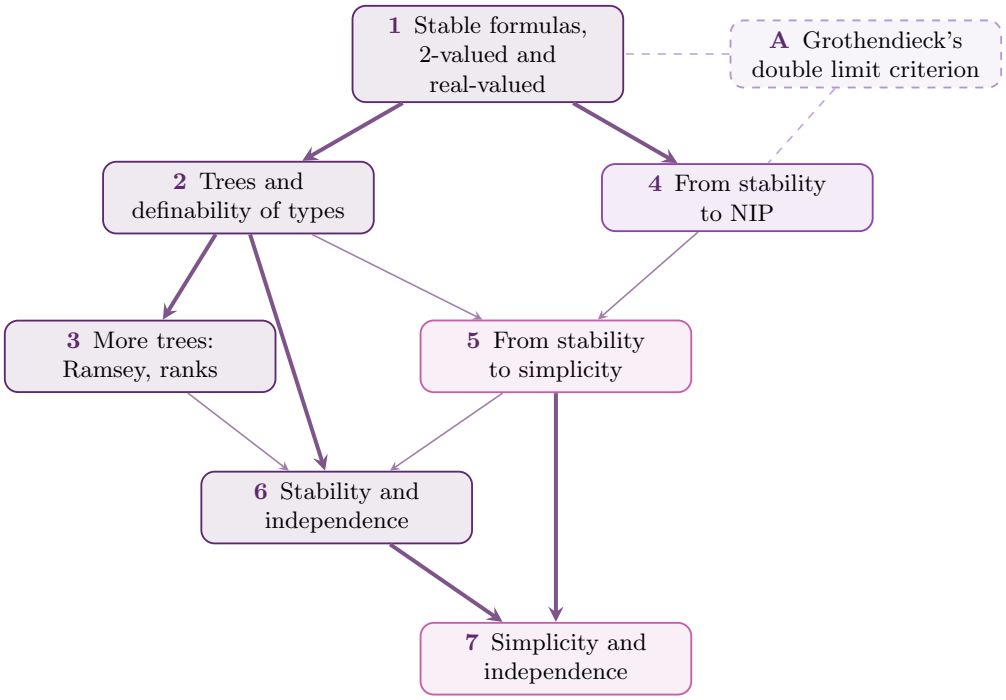
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Leitfaden

The chapters do not form a single line. Their trunk is Chapter 1, in which we introduce stable formulas, in the 2-valued and in the real-valued form. In that chapter we also state the definability of types, which we prove in Chapter 2. From there the book branches three ways: toward Ramsey bounds and ranks (Chapter 3), toward NIP (Chapter 4), and toward simplicity and independence (Chapters 5–7). In the first five chapters we study the dividing lines from inside the formulas: definability and trees for stability, shattering for NIP, dividing and the tree property for simplicity. The three dividing lines meet in the theorem that the stable theories are the simple NIP theories (see Theorem 5.3.4). In the last two we describe stability and simplicity globally, by the independence relations that stable and simple theories carry: uniqueness of independent extensions for stability, amalgamation for simplicity. A reader interested in simple theories may read Chapters 1, 2, 5, 6 and 7, with Sections 4.1 and 4.2; a reader interested in NIP may read Chapters 1 and 4, with Appendix A. In the diagram below, an arrow from one chapter to another means that the second uses results of the first. A heavy arrow marks heavy use, and arrows that follow from others are omitted.



In Appendix A (dashed) we collect the analytic background shared by Chapters 1 and 4. The set-theoretic tools for counting types and for capturing small sets in chains of models are

developed in Appendix B; they may be read as the arguments call for them. In Chapter 6 we use from Chapter 5 only the conventions on monster models (see Section 5.1), the lemma on λ-stability (see Lemma 5.1.2) and the axioms for independence (see Definition 5.4.7). The real source of Chapter 6 is the definability of types of Chapters 1 and 2.

Chapter

1

Stable formulas, both 2-valued and real-valued

Roadmap

Stability is the first of the three dividing lines. Shelah localized it in 1971 to a single formula, and in this chapter we introduce it in this local form, for a formula in two variables. In Section 1.1 we define the types of the formula over a model and the compact space that they form (see Definition 1.1.1). We also define there the definable types, whose instances are decided by a single formula with parameters from the model (see Definition 1.1.3). Finally, we define there the order property, the ladder configuration that stability forbids (see Definition 1.1.4). The central result is the definability theorem: a formula has no arbitrarily long ladders in any model exactly when all its types over all models are definable (see Theorem 1.1.6); its proof occupies Chapter 2. In Section 1.2 the formula becomes a bounded real-valued function and its types become ultrafilter limits of realized types. In that section, the absence of arbitrarily long ladders becomes the equality of the two iterated limits (see Proposition 1.2.4 and Definition 1.2.5). This equality is equivalent to the separate continuity of the extension of the function to pairs of types (see Proposition 1.2.8). It is also equivalent to a factorization through the pairing between the unit balls of a reflexive Banach space and of its dual (see Theorem 1.2.9). A third equivalent condition is the uniform approximation, in each variable, of that extension by polynomials in the instances of the function. This approximation is the real-valued form of definability (see Theorem 1.2.11), and from it we obtain a second, analytic proof that stable formulas have definable types (see Corollary 1.2.13). Chapters 2 and 3 rest only on Section 1.1.

1.1 The 2-valued case: φ-types and type definability

The stability of a formula is read off from its types, the ways in which the formula sees an element of an elementary extension against the parameters of a model. In this section we organize these φ-types into a compact space. We then single out the types whose behavior is captured by one formula of the model, the definable types, and we isolate the ladder configuration that, by the main theorem of the chapter, is the only obstruction to definability.

Throughout this chapter and Chapters 2 and 3, we fix a first-order theory T in a language L and an L-formula $\varphi(x,y)$ consistent with T, where x and y are finite tuples of variables. In Section 1.2 the letter φ also stands for an arbitrary function of two variables, and at each point we say which is meant. In the present section, the letter M always denotes a model of T. Tuples of elements are written $a, b, \ldots$ and are always of the appropriate length. We write $b \in M$ when b is a tuple of elements of M, and, by the same abuse, we call a set of such tuples a subset of M. Finally, we write $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$.

Definition 1.1.1. Let M be a model of T.

(a) A *φ-formula over M* is a formula $\varphi(x,b)^u$ with $b \in M$ and $u \in \{0,1\}$.

(b) A *complete φ-type over M* (in the variables x) is a set p of φ-formulas over M such that

(i) the set p is consistent with the elementary diagram of M, and

(ii) for each $b \in M$, exactly one of $\varphi(x,b)$ and $\neg\varphi(x,b)$ belongs to p.

The set of complete φ-types over M is denoted $S_\varphi(M)$.

If $N \succeq M$ and $c \in N$, the set

$$\mathrm{tp}_\varphi(c/M) = \{\varphi(x,b)^u : b \in M,\ u \in \{0,1\},\ N \models \varphi(c,b)^u\}$$

is a complete φ-type over M. Conversely, every complete φ-type p over M is consistent with the elementary diagram of M and is therefore realized by some element c of some $N \succeq M$. For such an element we have $p = \mathrm{tp}_\varphi(c/M)$, because $p \subseteq \mathrm{tp}_\varphi(c/M)$ and both sets are complete φ-types.

For $p \in S_\varphi(M)$ we write $D_p = \{b \in M : \varphi(x,b) \in p\}$. By condition (ii), the type p equals $\{\varphi(x,b) : b \in D_p\} \cup \{\neg\varphi(x,b) : b \in M \setminus D_p\}$ and is therefore determined by D_p. Each formula $\varphi(x,b)$ is, as Frege would say, unsaturated: it has an empty place, and it becomes true or false only when an object is placed there. The set D_p thus records, for every parameter b, whether the objects that realize p fall under $\varphi(x,b)$.

A type over M need not be realized in M, but each finite part of it is realized there.

Lemma 1.1.2 (Finite satisfiability). *Let $p \in S_\varphi(M)$ be given. For every finite subset $\{\varphi(x,b_l)^{u_l} : l < k\}$ of p, there is $a \in M$ with $M \models \bigwedge_{l<k} \varphi(a,b_l)^{u_l}$.*

Proof. The $L(M)$-sentence $\exists x \bigwedge_{l<k} \varphi(x,b_l)^{u_l}$ is consistent with the elementary diagram of M. The elementary diagram is a complete $L(M)$-theory, and therefore the sentence belongs to it, that is, holds in M. ☺

Finite satisfiability gives φ-types a natural topological home. The sets $\varphi(M,b) = \{a \in M^{|x|} : M \models \varphi(a,b)\}$, for $b \in M^{|y|}$, generate a Boolean algebra $\mathcal{B}_\varphi(M)$ of subsets of $M^{|x|}$. For $p \in S_\varphi(M)$, the sets $\varphi(M,b)$ with $\varphi(x,b) \in p$ and the complements $M^{|x|} \setminus \varphi(M,b)$ with $\neg\varphi(x,b) \in p$ have the finite intersection property by Lemma 1.1.2. This family contains, for every generator of $\mathcal{B}_\varphi(M)$, either the generator or its complement, so it generates an ultrafilter on $\mathcal{B}_\varphi(M)$. Conversely, an ultrafilter on $\mathcal{B}_\varphi(M)$ determines a set of φ-formulas over M that is finitely satisfiable in M, hence consistent with the elementary diagram of M, and so a complete φ-type. A complete φ-type over M is therefore the same thing as an ultrafilter on $\mathcal{B}_\varphi(M)$, and we regard $S_\varphi(M)$ as the Stone space of $\mathcal{B}_\varphi(M)$. As such, the space $S_\varphi(M)$ is compact, Hausdorff and zero-dimensional, and its basic clopen sets consist of the types containing a given finite set of φ-formulas over M.

In general, the set D_p of a type $p \in S_\varphi(M)$ is an arbitrary subset of M, and nothing makes it describable inside M. Indeed, when M is infinite and $|L| \leq |M|$, there are $2^{|M|}$ subsets of M but only $|M|$ formulas over M. A definable type, in the sense of the next

definition, is one whose set D_p is the extension of a single formula. Realized types are the basic example, since for $c \in M$ the set D_p of the type $p = \mathrm{tp}_\varphi(c/M)$ is the extension of the formula $\varphi(c, y)$. A definable type thus behaves like the type of an element of M, even when the type is realized only in an elementary extension.

Definition 1.1.3. Let $p \in S_\varphi(M)$ be given.

(a) The type p is *definable* if there is an $L(M)$-formula $\psi(y)$, a *φ-definition* of p, such that for all $b \in M$, we have

$$\varphi(x, b) \in p \quad \text{if and only if} \quad M \models \psi(b).$$

Equivalently, the set $D_p = \{b \in M : \varphi(x, b) \in p\}$ is the definable subset $\psi(M)$ of M.

(b) A *positive Boolean combination of instances of φ over M* is a formula of the form

$$\bigvee_{i<n} \bigwedge_{a \in W_i} \varphi(a, y),$$

where $n < \omega$ and $W_0, \ldots, W_{n-1}$ are finite subsets of M. By convention the empty disjunction is $\bot$ and the empty conjunction is $\top$.

Definability also has a practical side. A φ-definition $\psi(y)$ of a type $p \in S_\varphi(M)$ makes sense in every elementary extension $N \succeq M$. The definition therefore tells us how to extend p to parameters in N: include $\varphi(x, b)$ in the extension exactly when $N \models \psi(b)$. The result is a complete φ-type over N. Indeed, if we write $\psi^1 = \psi$ and $\psi^0 = \neg\psi$, then for every positive integer k and every $u = (u_1, \ldots, u_k) \in \{0, 1\}^k$ the $L(M)$-sentence $\forall y_1 \ldots \forall y_k \big(\bigwedge_{1 \le l \le k} \psi(y_l)^{u_l} \to \exists x \bigwedge_{1 \le l \le k} \varphi(x, y_l)^{u_l}\big)$ holds in M by Lemma 1.1.2, and hence in N. This way of extending a definable type is the source of the canonical extensions used to define independence in Chapter 6.

The simplest type that is not definable comes from an order. In $(\mathbb{Q}, <)$, let $\varphi(x, y)$ be $x < y$ and let p be the type of $\sqrt{2}$ over $\mathbb{Q}$. The set $D_p = \{b \in \mathbb{Q} : b > \sqrt{2}\}$ is a cut with an irrational edge. The subsets of $\mathbb{Q}$ definable in $(\mathbb{Q}, <)$ with parameters, on the other hand, are finite unions of intervals with rational or infinite endpoints; hence the type p is not definable. The type of an element just above a rational q, by contrast, is defined by $y > q$ (see Figure 1.1).

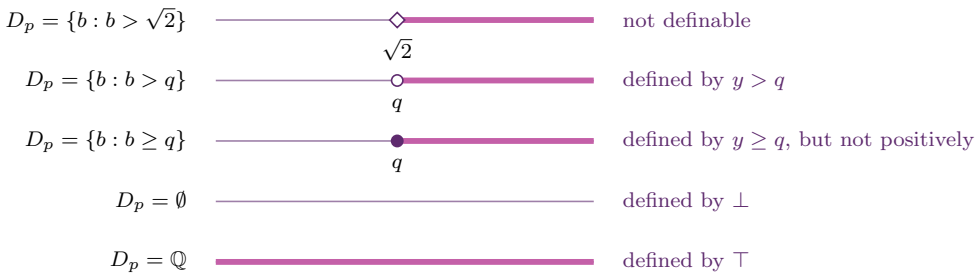


FIGURE 1.1. Five φ-types over $\mathbb{Q}$, where $\varphi(x, y)$ is the formula $x < y$ in $(\mathbb{Q}, <)$, each type p represented by its set $D_p = \{b \in \mathbb{Q} : \varphi(x, b) \in p\}$, a final segment of the rational line.

In $(\mathbb{Q}, <)$ the cut at $\sqrt{2}$ escapes every formula because the formula $x < y$ orders the rationals. The next definition isolates this behavior in finite pieces.

Definition 1.1.4. Let M be a model of T, and let n be a positive integer. An *n-ladder for φ in M* consists of tuples $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ of elements of M such that, for all $1 \leq i, j \leq n$, we have

$$M \models \varphi(a_i, b_j) \quad \text{if and only if} \quad i \leq j.$$

We say that φ has the *order property* if for every positive integer n some model of T contains an n-ladder for φ, and that φ is *stable* otherwise.

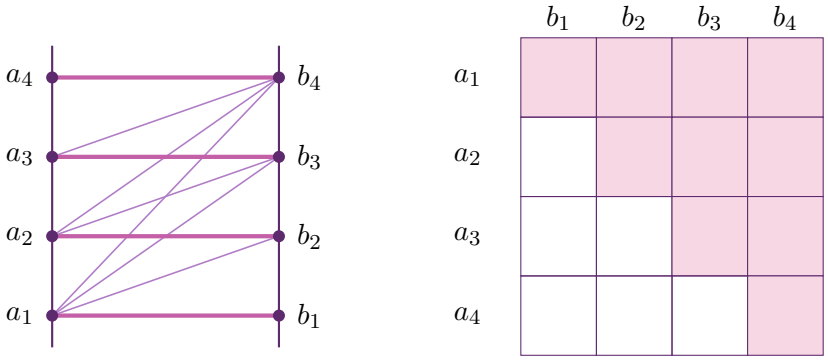


FIGURE 1.2. A 4-ladder for φ (see Definition 1.1.4), drawn as a graph joining a_i to b_j when $\varphi(a_i, b_j)$ holds (left) and as the truth table of $\varphi(a_i, b_j)$, with pink cells where φ holds (right).

The name *ladder* comes from the picture in Figure 1.2. Place the tuples a_i of an n-ladder on one rail and its tuples b_j on the other, both in increasing order, and join a_i to b_j whenever $\varphi(a_i, b_j)$ holds. The "horizontal" pairs (a_i, b_i) are then the rungs of a ladder. What matters is the asymmetry between the two directions: each a_i is related to every b_j with $j \geq i$ and to none with $j < i$. In this sense the formula φ linearly orders the rungs.

Remark 1.1.5. For every positive integer n, an $(n+1)$-ladder contains an n-ladder. More generally, any k rungs of a ladder, taken in increasing order, form a k-ladder. Hence, by compactness, the formula φ has the order property if and only if some model of T contains an infinite sequence $(a_i, b_i)_{i<\omega}$ with $\varphi(a_i, b_j) \iff i \leq j$ for all $i, j < \omega$. The choice of "$\leq$" is immaterial. Suppose that $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ are tuples in a model of T such that $\varphi(a_i, b_j) \iff i > j$ for $1 \leq i, j \leq n$. Reversing the indices by letting $a'_i = a_{n+1-i}$ and $b'_j = b_{n+1-j}$ gives $\varphi(a'_i, b'_j) \iff i < j$. Shifting by letting $b''_j = b'_{j+1}$ then produces $\varphi(a'_i, b''_j) \iff i \leq j$ for $1 \leq i, j \leq n-1$. Thus a "$>$"-pattern of length n yields an $(n-1)$-ladder.

By the main theorem of this chapter, the order property is the only obstruction to definability. When the order property is absent, the definitions can moreover be chosen of the simplest possible kind, namely positive Boolean combinations of instances of φ (see Definition 1.1.3(b)). By distributivity, these are, up to equivalence, all the formulas built from instances $\varphi(a, y)$ with $a \in M$ using only $\wedge$ and $\vee$ (together with $\bot$ and $\top$). If φ is quantifier-free, so is every such combination.

Theorem 1.1.6 (Definability of types). *The following are equivalent:*

(1) *The formula φ does not have the order property.*

(2) *For every model M of T, every $p \in S_\varphi(M)$ has a φ-definition that is a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$.*

(3) *For every model M of T, every $p \in S_\varphi(M)$ is definable.*

The implication (2) $\Rightarrow$ (3) is trivial, and we prove (1) $\Rightarrow$ (2) in Section 2.2. The remaining implication (3) $\Rightarrow$ (1) takes a detour through binary trees. In Sections 2.3–2.4 we show that if φ has the order property, then the formula φ encodes binary trees of arbitrarily large finite height. We show there, too, that such trees produce a φ-type that is not definable.

1.2 The real-valued case

Once a formula is identified with a $\{0,1\}$-valued function, stability becomes a property that makes sense for every bounded real-valued function: the exchangeability of iterated limits. Seen this way, stability meets the reflexivity of Banach spaces, through a characterization of reflexivity due to James [**Jam64**] whose combinatorial form closely resembles the order property. In the subsections that follow we characterize real-valued stability by continuous extension to types, by reflexive Banach spaces and by polynomial approximation; the treatment follows Iovino [**Iov99**].

1.2.1 Stability and iterated limit exchange

We first extend the order property from formulas to arbitrary 2-valued functions of two variables. We then show that the stable functions, those of which neither the function nor its negation has the order property, are exactly those whose two iterated limits can be exchanged.

Let M be an L-structure and $\varphi(x, y)$ an L-formula. We identify φ with the function $\varphi\colon M^{|x|} \times M^{|y|} \to \{0,1\}$ that takes the value 1 at (a, b) if and only if $M \models \varphi(a, b)$. More generally, a function $\varphi\colon A \times B \to \{0,1\}$ on arbitrary sets A, B has the *order property* if there are sequences $(a_m)_{m\in\mathbb{N}}$ in A and $(b_n)_{n\in\mathbb{N}}$ in B such that

$$\varphi(a_m, b_n) = 1 \quad \text{if and only if} \quad m \leq n.$$

Definition 1.2.1. A function $\varphi\colon A \times B \to \{0,1\}$ is *unstable* if either φ or $\neg\varphi = 1 - \varphi$ has the order property, and *stable* otherwise. For an L-structure M, an L-formula $\varphi(x, y)$ is *stable on M* if the corresponding function on $M^{|x|} \times M^{|y|}$ is stable.

Remark 1.2.2. A formula $\varphi(x, y)$ is stable in the sense of Definition 1.1.4 if and only if φ is stable on every model of T. Indeed, if φ has the order property, then by compactness (see Remark 1.1.5) some model contains an infinite ladder, so the formula φ is unstable on that model. Conversely, if φ or $\neg\varphi$ has the order property on some model M of T, then the model M contains n-ladders for φ for every positive integer n. In the first case the order property provides these ladders directly; in the second case they come from Remark 1.1.5, since sequences $(a_m)_{m\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ with $\neg\varphi(a_m, b_n) \iff m \leq n$ satisfy $\varphi(a_m, b_n) \iff m > n$.

To state the analytic form of stability, we use limits along ultrafilters. For a family $(a_i)_{i\in I}$ in a topological space X, a point $a \in X$, and an ultrafilter $\mathcal{U}$ on I, we write

$$\lim_{i,\mathcal{U}} a_i = a \tag{1.1}$$

if, for every neighborhood O of a, the set $\{i \in I : a_i \in O\}$ belongs to $\mathcal{U}$. We use freely the standard facts about such limits: limits along ultrafilters are unique in Hausdorff spaces, always exist in compact spaces, describe closures, and are preserved by continuous maps.

A pair of sequences that witnesses the order property of a function $\varphi\colon A\times B \to \{0,1\}$ is an *infinite ladder* for φ. On an infinite ladder, the two iterated limits of φ along a nonprincipal ultrafilter on $\mathbb{N}$ already differ. For fixed m the values $\varphi(a_m, b_n)$ are eventually 1, and for fixed n the values $\varphi(a_m, b_n)$ are eventually 0, so one iterated limit is 1 and the other is 0. This disagreement of the two iterated limits is the phenomenon that the analytic form of stability forbids. Ladders, however, are indexed by $\mathbb{N}$, whereas the limits met in practice (for instance in the spaces of types of the next subsection) are taken along ultrafilters on arbitrary index sets. The next proposition, which we use several times, bridges the gap: from two iterated limits taken along arbitrary families, it extracts sequences on which both limits are already visible. The first limit appears above the diagonal and the second below it.

Proposition 1.2.3. *Let $\varphi\colon A \times B \to [0,1]$ be a function, let $(a_i)_{i\in I}$ be a family in A and $(b_j)_{j\in J}$ a family in B, and let $\mathcal{U}$, $\mathcal{V}$ be ultrafilters on I, J with*

$$\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} \varphi(a_i, b_j) = \alpha, \qquad \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} \varphi(a_i, b_j) = \beta.$$

Then there are $i_0, i_1, \ldots \in I$ and $j_0, j_1, \ldots \in J$ such that, for all $m, n \in \mathbb{N}$, we have

$$|\varphi(a_{i_m}, b_{j_n}) - \alpha| \le 2^{1-m} \quad \text{if } m < n, \qquad |\varphi(a_{i_m}, b_{j_n}) - \beta| \le 2^{1-n} \quad \text{if } n < m.$$

Proof. The indices are chosen alternately, each so deep in its ultrafilter that it approximates both the outer limit and the inner limits at the indices already chosen. By definition of the limits, for every $\varepsilon > 0$ there are $U_\varepsilon \in \mathcal{U}$ and $V_\varepsilon \in \mathcal{V}$ such that

$$i \in U_\varepsilon \implies \big|\lim_{j,\mathcal{V}} \varphi(a_i, b_j) - \alpha\big| \le \varepsilon,$$
$$j \in V_\varepsilon \implies \big|\lim_{i,\mathcal{U}} \varphi(a_i, b_j) - \beta\big| \le \varepsilon.$$

Moreover, for all $\hat{\imath} \in I$, $\hat{\jmath} \in J$ and $\varepsilon > 0$ there are $V_\varepsilon^{\hat{\imath}} \in \mathcal{V}$ and $U_\varepsilon^{\hat{\jmath}} \in \mathcal{U}$ such that

$$j \in V_\varepsilon^{\hat{\imath}} \implies \big|\varphi(a_{\hat{\imath}}, b_j) - \lim_{j,\mathcal{V}} \varphi(a_{\hat{\imath}}, b_j)\big| \le \varepsilon,$$
$$i \in U_\varepsilon^{\hat{\jmath}} \implies \big|\varphi(a_i, b_{\hat{\jmath}}) - \lim_{i,\mathcal{U}} \varphi(a_i, b_{\hat{\jmath}})\big| \le \varepsilon.$$

We choose recursively

$$i_N \in U_{2^{-N}} \cap \bigcap_{k<N} U_{2^{-N}}^{j_k}, \qquad j_N \in V_{2^{-N}} \cap \bigcap_{k\le N} V_{2^{-N}}^{i_k}.$$

These sets are finite intersections of members of an ultrafilter, hence nonempty. If $m < n$, then the indices satisfy $j_n \in V_{2^{-n}}^{i_m}$ and $i_m \in U_{2^{-m}}$, and therefore $|\varphi(a_{i_m}, b_{j_n}) - \alpha| \le 2^{-n} + 2^{-m} \le 2^{1-m}$. If $n < m$, then instead $i_m \in U_{2^{-m}}^{j_n}$ and $j_n \in V_{2^{-n}}$, whence $|\varphi(a_{i_m}, b_{j_n}) - \beta| \le 2^{-m} + 2^{-n} \le 2^{1-n}$. ☺

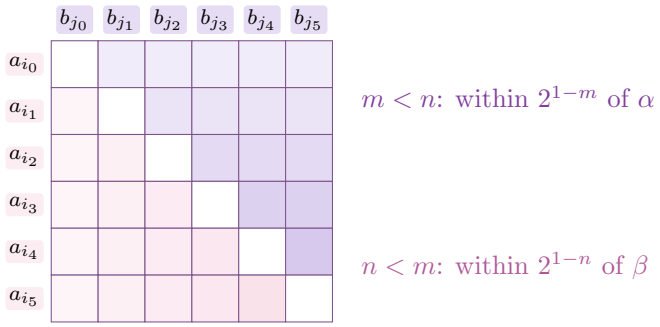


FIGURE 1.3. The array $\varphi(a_{i_m}, b_{j_n})$, for $m, n \in \mathbb{N}$, that Proposition 1.2.3 extracts from two families whose iterated limits are α and β.

The array produced by Proposition 1.2.3 is drawn in Figure 1.3.

With Proposition 1.2.3 in hand, we can show that a $\{0,1\}$-valued function is stable, that is, neither the function nor its negation has the order property, exactly when its two iterated limits can always be exchanged.

Proposition 1.2.4. *Let $\varphi\colon A \times B \to \{0,1\}$ be a function. The following are equivalent:*

(1) *The function φ is stable.*

(2) *For all sequences $(a_m)_{m\in\mathbb{N}}$ in A and $(b_n)_{n\in\mathbb{N}}$ in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have*

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} \varphi(a_m, b_n) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} \varphi(a_m, b_n).$$

Proof. Both implications are proved by contraposition: a ladder separates the two iterated limits, and, conversely, two distinct iterated limits yield through Proposition 1.2.3 an array of values of φ that thins out to a ladder for φ or for $\neg\varphi$.

(2) $\Rightarrow$ (1). Suppose that φ is unstable. If φ has the order property, there are sequences (a_m) in A and (b_n) in B with $\varphi(a_m, b_n) = 1 \iff m \leq n$. Let $\mathcal{U} = \mathcal{V}$ be a nonprincipal ultrafilter on $\mathbb{N}$. For fixed m we have $\varphi(a_m, b_n) = 1$ for all $n \geq m$, so the left-hand side of (2) is 1. For fixed n, the values $\varphi(a_m, b_n)$ vanish for all $m > n$, whence the right-hand side is 0. If instead $\neg\varphi$ has the order property, the two sides are 0 and 1. In either case (2) fails.

(1) $\Rightarrow$ (2). Suppose that (2) fails: for some sequences (a_m), (b_n) and ultrafilters $\mathcal{U}$, $\mathcal{V}$, the two iterated limits are $\alpha \neq \beta$. Both limits lie in $\{0,1\}$, because the values of φ do. By Proposition 1.2.3 there are sequences (a'_m) in A and (b'_n) in B with $|\varphi(a'_m, b'_n) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|\varphi(a'_m, b'_n) - \beta| \leq 2^{1-n}$ for $n < m$. For $m \geq 2$ the tolerance 2^{1-m} is at most $1/2$, and a $\{0,1\}$-valued function within $1/2$ of 0 or 1 equals it; hence $\varphi(a'_m, b'_n) = \alpha$ for $2 \leq m < n$ and $\varphi(a'_m, b'_n) = \beta$ for $2 \leq n < m$. We pass to the subsequences $c_m = a'_{2m+2}$ and $d_n = b'_{2n+3}$. Since $2m + 2 < 2n + 3 \iff m \leq n$, we get $\varphi(c_m, d_n) = \alpha$ if $m \leq n$ and $\varphi(c_m, d_n) = \beta$ if $m > n$. Thus the function φ has the order property if $\alpha = 1$, and the function $\neg\varphi$ has it if $\alpha = 0$; in either case φ is unstable. ☺

Condition (2) of Proposition 1.2.4 makes sense for every bounded real-valued function, and we take this condition as the definition of stability in the real-valued setting.

Definition 1.2.5. A bounded function $\varphi\colon A \times B \to \mathbb{R}$ is *stable* if, for all sequences $(a_m)_{m\in\mathbb{N}}$ in A and $(b_n)_{n\in\mathbb{N}}$ in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} \varphi(a_m, b_n) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} \varphi(a_m, b_n).$$

By Proposition 1.2.4, the new definition agrees with Definition 1.2.1 for $\{0,1\}$-valued functions. For a bounded set of continuous functions on a compact space, the same condition is the hypothesis of Grothendieck's double limit criterion (see Appendix A). In view of Proposition 1.2.3, the sequences in Definition 1.2.5 may be replaced by arbitrary families, as the next corollary records. We state it for functions with values in $[0,1]$, to which every bounded function reduces by an affine change of scale that preserves stability.

Corollary 1.2.6. *If $\varphi\colon A \times B \to [0,1]$ is stable, then we have $\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} \varphi(a_i, b_j) = \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} \varphi(a_i, b_j)$ for all families $(a_i)_{i\in I}$ in A and $(b_j)_{j\in J}$ in B and all ultrafilters $\mathcal{U}$ on I and $\mathcal{V}$ on J.*

Proof. If the two limits are $\alpha \neq \beta$, we take (i_m), (j_n) as in Proposition 1.2.3 and a nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$. For fixed m, the limit $\lim_{n,\mathcal{W}} \varphi(a_{i_m}, b_{j_n})$ is within 2^{1-m} of α, so we have $\lim_{m,\mathcal{W}} \lim_{n,\mathcal{W}} \varphi(a_{i_m}, b_{j_n}) = \alpha$. Symmetrically, the other iterated limit is β, which contradicts the stability of φ. ☺

The proof of Corollary 1.2.6 used a single nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$, in both coordinates, and this is no accident. For ultrafilters $\mathcal{U}$ on I and $\mathcal{V}$ on J, let the tensor product $\mathcal{U} \otimes \mathcal{V}$ be the ultrafilter on $I \times J$ consisting of the sets X with $\{i : \{j : (i,j) \in X\} \in \mathcal{V}\} \in \mathcal{U}$. For every bounded $f\colon I \times J \to \mathbb{R}$ we have $\lim_{(i,j),\mathcal{U}\otimes\mathcal{V}} f(i,j) = \lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} f(i,j)$. Consequently, the two iterated limits of Definition 1.2.5 are the limits of $\varphi(a_m, b_n)$ along $\mathcal{U} \otimes \mathcal{V}$ and along the transpose of $\mathcal{V} \otimes \mathcal{U}$, that is, its image under the map $(n,m) \mapsto (m,n)$. These two ultrafilters can differ, because the tensor product of ultrafilters is not commutative. The simplest witness already lives on $\mathbb{N}\times\mathbb{N}$: if $\mathcal{W}$ is a nonprincipal ultrafilter on $\mathbb{N}$, then the upper triangle $\{(m,n) : m < n\}$ belongs to $\mathcal{W} \otimes \mathcal{W}$, because each of its rows is cofinite. The lower triangle $\{(m,n) : m > n\}$, by contrast, does not belong to $\mathcal{W} \otimes \mathcal{W}$, because each of its rows is finite. A ladder is exactly a pair of sequences along which φ is the indicator function of the closed upper triangle $\{(m,n) : m \leq n\}$ (see Figure 1.4). Thus the order property says that φ can tell $\mathcal{W} \otimes \mathcal{W}$ from its own transpose. The proofs of Proposition 1.2.4 and Corollary 1.2.6 yield a stronger statement, which reads as follows. A bounded function $\varphi\colon A \times B \to \mathbb{R}$ is stable exactly when, for one (equivalently, for every) nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$ and all sequences $(a_m)_{m\in\mathbb{N}}$ in A and $(b_n)_{n\in\mathbb{N}}$ in B, we have

$$\lim_{m,\mathcal{W}} \lim_{n,\mathcal{W}} \varphi(a_m, b_n) = \lim_{n,\mathcal{W}} \lim_{m,\mathcal{W}} \varphi(a_m, b_n).$$

Stable functions are blind to the noncommutativity of the tensor square of a single ultrafilter.

1.2.2 Stability and continuous extendability

Exchanging iterated limits is a statement about continuity in disguise. In Proposition 1.2.8 below we show that a function of two variables with values in $[0,1]$ is stable exactly when it extends to a separately continuous function on pairs of types. The types in question live in compact spaces built from the two domains of the function.

Definition 1.2.7. Let $\varphi\colon A \times B \to [0,1]$ be a function. For $a \in A$, the *left φ-type* of a is the function $\mathrm{ltp}_\varphi(a)\colon y \mapsto \varphi(a,y)$ on B. For $b \in B$, the *right φ-type* of b is the function $\mathrm{rtp}_\varphi(b)\colon x \mapsto \varphi(x,b)$ on A. The *space of left φ-types* $\mathrm{LS}(\varphi)$ is the closure of $\{\mathrm{ltp}_\varphi(a) : a \in A\}$ in $[0,1]^B$ with the product topology, and the *space of right φ-types* $\mathrm{RS}(\varphi)$ is the closure of $\{\mathrm{rtp}_\varphi(b) : b \in B\}$ in $[0,1]^A$.

By Tychonov's theorem, the spaces $[0,1]^A$ and $[0,1]^B$ are compact, and so are their closed subspaces $\mathrm{RS}(\varphi)$ and $\mathrm{LS}(\varphi)$. Both spaces are quotients of Stone–Čech compactifications. The map $a \mapsto \mathrm{ltp}_\varphi(a)$ from the discrete space A into the compact space $[0,1]^B$ extends to

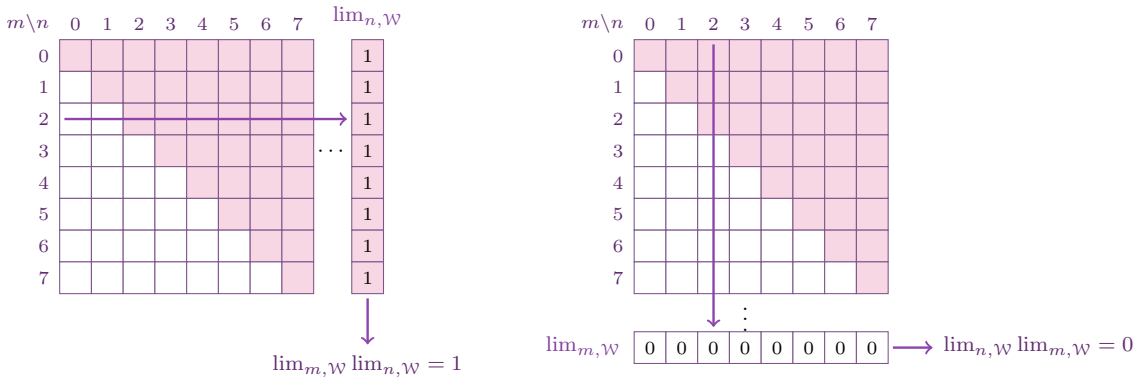


FIGURE 1.4. An infinite ladder, given by sequences $(a_m)_{m\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ with $\varphi(a_m, b_n) = 1$ if and only if $m \leq n$, shown for $0 \leq m, n \leq 7$ (pink cells are 1) with its iterated limits along a nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$, rows first (left) and columns first (right).

a continuous map $\beta A \to [0,1]^B$, namely $\mathcal{U} \mapsto \lim_{a,\mathcal{U}} \mathrm{ltp}_\varphi(a)$. The image of this extension is compact and contains the realized types. Since the image consists of limits of realized types, the image is also contained in their closure, and therefore equals $\mathrm{LS}(\varphi)$. Thus every left φ-type is the limit of the realized types along an ultrafilter on A, and the principal ultrafilters give the realized types. Hence the space $\mathrm{LS}(\varphi)$ is a continuous image of βA, and likewise the space $\mathrm{RS}(\varphi)$ is a continuous image of βB.

Proposition 1.2.8. *For $\varphi\colon A \times B \to [0,1]$, the following are equivalent:*

(1) *The function φ is stable.*

(2) *There is a separately continuous function $\hat{\varphi}\colon \mathrm{LS}(\varphi) \times \mathrm{RS}(\varphi) \to [0,1]$ such that*

$$\hat{\varphi}\big(\mathrm{ltp}_\varphi(a), \mathrm{rtp}_\varphi(b)\big) = \varphi(a,b) \qquad \text{for all } (a,b) \in A \times B.$$

Proof. (1) $\Rightarrow$ (2). The extension is forced: by the computation in the proof of (2) $\Rightarrow$ (1) below, a separately continuous extension takes at a pair of types the iterated limit of φ along families of realized types that converge to them along ultrafilters. We define $\hat{\varphi}$ in this way. Stability, through Corollary 1.2.6, makes the two orders of the limits agree, and it remains to be shown that the common value does not depend on the families chosen, that $\hat{\varphi}$ extends φ, and that $\hat{\varphi}$ is separately continuous. Let $p \in \mathrm{LS}(\varphi)$ and $q \in \mathrm{RS}(\varphi)$ be given. Since every type is a limit of realized types along an ultrafilter, there are families $(a_i)_{i\in I}$ in A, $(b_j)_{j\in J}$ in B and ultrafilters $\mathcal{U}$ on I, $\mathcal{V}$ on J with

$$p = \lim_{i,\mathcal{U}} \mathrm{ltp}_\varphi(a_i), \qquad q = \lim_{j,\mathcal{V}} \mathrm{rtp}_\varphi(b_j).$$

By Corollary 1.2.6, we have $\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} \varphi(a_i, b_j) = \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} \varphi(a_i, b_j)$, and this common value is by definition $\hat{\varphi}(p,q)$. Since limits in the product topology are computed coordinate-wise, the value $\hat{\varphi}(p,q)$ also satisfies

$$\hat{\varphi}(p,q) = \lim_{i,\mathcal{U}} q(a_i) = \lim_{j,\mathcal{V}} p(b_j). \tag{1.2}$$

The map $\hat{\varphi}$ is well defined. Suppose also that $p = \lim_{i,\mathcal{U}'} \mathrm{ltp}_\varphi(a'_i)$ and $q = \lim_{j,\mathcal{V}'} \mathrm{rtp}_\varphi(b'_j)$. Then we have $\lim_{j,\mathcal{V}} \varphi(x, b_j) = q(x) = \lim_{j,\mathcal{V}'} \varphi(x, b'_j)$ for every $x \in A$, and $\lim_{i,\mathcal{U}} \varphi(a_i, y) =$

$p(y) = \lim_{i,\mathcal{U}'} \varphi(a'_i, y)$ for every $y \in B$. Hence, using Corollary 1.2.6 for the middle step, we obtain

$$\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} \varphi(a_i, b_j) = \lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}'} \varphi(a_i, b'_j) = \lim_{j,\mathcal{V}'} \lim_{i,\mathcal{U}} \varphi(a_i, b'_j) = \lim_{j,\mathcal{V}'} \lim_{i,\mathcal{U}'} \varphi(a'_i, b'_j).$$

The value $\hat{\varphi}(p, q)$ therefore does not depend on the families and ultrafilters chosen.

The map $\hat{\varphi}$ extends φ. Given $(a, b) \in A \times B$, we use the constant families $a_i = a$ and $b_j = b$ in (1.2), which gives $\hat{\varphi}\big(\mathrm{ltp}_\varphi(a), \mathrm{rtp}_\varphi(b)\big) = \mathrm{rtp}_\varphi(b)(a) = \varphi(a, b)$.

The map $\hat{\varphi}$ is separately continuous. Fix p, let $(q_k)_{k\in K}$ be a family in $\mathrm{RS}(\varphi)$, and let $\mathcal{W}$ be an ultrafilter on K with $q = \lim_{k,\mathcal{W}} q_k$. Each q_k has the form $\lim_{j,\mathcal{V}_k} \mathrm{rtp}_\varphi(b_{k,j})$ for a family $(b_{k,j})_{j\in J_k}$ in B and an ultrafilter $\mathcal{V}_k$ on J_k. On $J' = \{(k, j) : k \in K,\ j \in J_k\}$ let $\mathcal{V}'$ be the ultrafilter of all $X \subseteq J'$ with $\{k : \{j : (k, j) \in X\} \in \mathcal{V}_k\} \in \mathcal{W}$. For every family $(z_{k,j})$ in a compact Hausdorff space we have $\lim_{(k,j),\mathcal{V}'} z_{k,j} = \lim_{k,\mathcal{W}} \lim_{j,\mathcal{V}_k} z_{k,j}$. It follows that $q = \lim_{(k,j),\mathcal{V}'} \mathrm{rtp}_\varphi(b_{k,j})$, and equation (1.2) gives

$$\hat{\varphi}(p, q) = \lim_{(k,j),\mathcal{V}'} p(b_{k,j}) = \lim_{k,\mathcal{W}} \lim_{j,\mathcal{V}_k} p(b_{k,j}) = \lim_{k,\mathcal{W}} \hat{\varphi}(p, q_k).$$

Thus the function $\hat{\varphi}(p, \cdot)$ is continuous, and the continuity of $\hat{\varphi}(\cdot, q)$ is proved symmetrically.

(2) $\Rightarrow$ (1). Given sequences (a_m) in A and (b_n) in B and ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, let p and q be the limit types $\lim_{m,\mathcal{U}} \mathrm{ltp}_\varphi(a_m)$ and $\lim_{n,\mathcal{V}} \mathrm{rtp}_\varphi(b_n)$. Then the pair (p, q) lies in $\mathrm{LS}(\varphi) \times \mathrm{RS}(\varphi)$, and by separate continuity

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} \varphi(a_m, b_n) = \lim_{m,\mathcal{U}} \hat{\varphi}\big(\mathrm{ltp}_\varphi(a_m), q\big) = \hat{\varphi}(p, q) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} \varphi(a_m, b_n). \qquad \text{☺}$$

1.2.3 Stability and reflexivity

By Proposition 1.2.8, the stable functions are those that extend to separately continuous functions on products of compact spaces, and functional analysis supplies a canonical example of such a function. The unit balls of a reflexive Banach space and of its dual are compact in their weak topologies, and the pairing between them is separately continuous. In Theorem 1.2.9 below we state Iovino's theorem that this example is universal: the stable functions are exactly those obtained by composing such a pairing with maps into the two balls.

For a Banach space E with dual E^*, we write B_E for the closed unit ball of E and $\langle x, y\rangle$ for $y(x)$, where $x \in E$ and $y \in E^*$. Recall that E is *reflexive* if the canonical embedding of E into E^{**} is surjective.

Theorem 1.2.9 (Iovino). *For $\varphi\colon A \times B \to [0, 1]$, the following are equivalent:*

(1) *The function φ is stable.*

(2) *There are a reflexive Banach space E and maps $u\colon A \to B_E$ and $v\colon B \to B_{E^*}$ such that*

$$\varphi(x, y) = \langle u(x), v(y)\rangle \qquad \text{for all } (x, y) \in A \times B.$$

Proof. We only sketch the argument; the details are in Theorem 3.3 of Iovino's paper [**Iov99**]. For (1) $\Rightarrow$ (2), stability is used to show that the operator $R\colon \ell^\infty(A)^* \to \ell^\infty(B)$ given by $R\mu(y) = \int \varphi(x, y)\, d\mu(x)$ is weakly compact; this step goes back to Krivine and Maurey [**KM81**] and Raynaud [**Ray83**]. By the Davis–Figiel–Johnson–Pełczyński factorization theorem [**DFJP74**], the operator R then factors as $R = j \circ i$ through a reflexive space E. If δ_x and δ_y are the evaluation functionals at x and y, the maps $u(x) = i(\delta_x)$ and $v(y) = j^*(\delta_y)$ satisfy $\varphi(x, y) = \langle u(x), v(y)\rangle$. They take values in the unit balls once E is given the equivalent norm $\|e\|' = \max\big(\|e\|/c,\ \sup_{y\in B} |\langle e, v(y)\rangle|\big)$, where $c = \max(1, \|i\|)$, and the space E remains reflexive under it. The implication (2) $\Rightarrow$ (1) follows from the weak compactness of

B_E and the weak* compactness of B_{E^*}, since the pairing is separately continuous for these topologies (see Exercise 10). ☺

Translated back into model theory, the theorem gives every stable formula a Banach-space picture. By Remark 1.2.2, a formula $\varphi(x,y)$ is stable in the sense of Definition 1.1.4 if and only if it is stable on every model M of T. In addition, by Proposition 1.2.4, stability on M in the sense of Definition 1.2.1 agrees with stability in the sense of Definition 1.2.5. Hence the formula $\varphi(x,y)$ is stable exactly when, for every model M of T, some reflexive Banach space E and some maps $u\colon M^{|x|} \to B_E$ and $v\colon M^{|y|} \to B_{E^*}$ satisfy $\varphi(a,b) = \langle u(a), v(b)\rangle$ for all $a \in M^{|x|}$ and $b \in M^{|y|}$. Here the space E depends on M, and the formula φ is identified with its $\{0,1\}$-valued function on M.

1.2.4 Stability and polynomial approximation

The last characterization of stability returns to definability, the theme of Section 1.1. By Theorem 1.1.6, if φ does not have the order property, then every φ-type over a model of T is defined by a positive Boolean combination of instances $\varphi(a,y)$ with a in the model. On $\{0,1\}$ the Boolean connectives are polynomials: negation is subtraction from 1, and conjunction is multiplication. The real-valued analogue of a definition is therefore a *uniform limit of polynomials in instances of* φ. In Theorem 1.2.11 below we use the Stone–Weierstrass theorem to show that stability is exactly what is needed for such approximations to exist. Continuous logic treats definability in the same way (see [**BYU10**]). Ben Yaacov [**BY14**] derived the definability of types in stable theories from Grothendieck's double limit criterion (see the historical remarks at the end of this chapter, and Appendix A). We use the Stone–Weierstrass theorem in the following form.

Fact 1.2.10 (Stone–Weierstrass). *Let K be a compact Hausdorff space and let $\mathcal{F} \subseteq C(K)$ be a family of continuous real-valued functions that separates the points of K. Then every $f \in C(K)$ is a uniform limit on K of functions of the form $P(g_1,\dots,g_k)$, where $g_1,\dots,g_k \in \mathcal{F}$ and P is a real polynomial in k variables.*

To apply the Stone–Weierstrass theorem to a function $\varphi\colon A \times B \to [0,1]$, which we now fix, we need continuous functions on its spaces of left and right types that separate points. The instances of φ supply such functions. For $a \in A$, let $e_a\colon \mathrm{RS}(\varphi) \to [0,1]$ be the evaluation map $e_a(q) = q(a)$, and for $b \in B$, let $e_b\colon \mathrm{LS}(\varphi) \to [0,1]$ be the evaluation map $e_b(p) = p(b)$. These maps are continuous, because the spaces $\mathrm{RS}(\varphi)$ and $\mathrm{LS}(\varphi)$ carry the product topology. On realized types they are instances of φ: the identities $e_a(\mathrm{rtp}_\varphi(b)) = \varphi(a,b) = e_b(\mathrm{ltp}_\varphi(a))$ hold for all $a \in A$ and $b \in B$. Finally, the family $\{e_a : a \in A\}$ separates the points of $\mathrm{RS}(\varphi)$, since two distinct functions $q \neq q'$ in $[0,1]^A$ differ at some $a \in A$. Likewise, the family $\{e_b : b \in B\}$ separates the points of $\mathrm{LS}(\varphi)$.

Theorem 1.2.11. *For $\varphi\colon A \times B \to [0,1]$, the following are equivalent:*

(1) *The function φ is stable.*

(2) *There is a function $\hat{\varphi}\colon \mathrm{LS}(\varphi) \times \mathrm{RS}(\varphi) \to [0,1]$ with $\hat{\varphi}(\mathrm{ltp}_\varphi(a), \mathrm{rtp}_\varphi(b)) = \varphi(a,b)$ for all $(a,b) \in A \times B$, such that for every $\varepsilon > 0$:*

   (a) *for every $p \in \mathrm{LS}(\varphi)$ there are $a_1,\dots,a_k \in A$ and a real polynomial P with*

$$\big|\hat{\varphi}(p,q) - P\big(q(a_1),\dots,q(a_k)\big)\big| \leq \varepsilon \qquad \text{for all } q \in \mathrm{RS}(\varphi);$$

   (b) *for every $q \in \mathrm{RS}(\varphi)$ there are $b_1,\dots,b_l \in B$ and a real polynomial Q with*

$$\big|\hat{\varphi}(p,q) - Q\big(p(b_1),\dots,p(b_l)\big)\big| \leq \varepsilon \qquad \text{for all } p \in \mathrm{LS}(\varphi).$$

Proof. (1) ⇒ (2). Let $\hat{\varphi}$ be the separately continuous function given by Proposition 1.2.8. For fixed p, the function $\hat{\varphi}(p,\cdot)$ is continuous on the compact Hausdorff space $\mathrm{RS}(\varphi)$. Applying Fact 1.2.10 with $\mathcal{F} = \{e_a : a \in A\}$ gives (a). Clause (b) follows in the same way, using the continuity of $\hat{\varphi}(\cdot, q)$ on $\mathrm{LS}(\varphi)$ and $\mathcal{F} = \{e_b : b \in B\}$.

(2) ⇒ (1). Each function $q \mapsto P(q(a_1), \ldots, q(a_k))$ is continuous on $\mathrm{RS}(\varphi)$. By (a), every function $\hat{\varphi}(p,\cdot)$ is a uniform limit of such functions, hence continuous; clause (b) gives the continuity of every $\hat{\varphi}(\cdot, q)$ in the same way. Thus the extension $\hat{\varphi}$ is separately continuous, and Proposition 1.2.8 shows that φ is stable. ☺

Clause (a) has a concrete meaning at realized right types: there it says that every left type is approximated by polynomials in instances. To see this, suppose that φ is stable, and let $\hat{\varphi}$ be the separately continuous extension given by Proposition 1.2.8, as in the proof of (1) ⇒ (2). Then we have $\hat{\varphi}(p, \mathrm{rtp}_\varphi(b)) = p(b)$ for $p \in \mathrm{LS}(\varphi)$ and $b \in B$: if $p = \lim_{i,\mathcal{U}} \mathrm{ltp}_\varphi(a_i)$ for a family $(a_i)_{i\in I}$ in A and an ultrafilter $\mathcal{U}$ on I, continuity in the first variable gives $\hat{\varphi}(p, \mathrm{rtp}_\varphi(b)) = \lim_{i,\mathcal{U}} \varphi(a_i, b) = p(b)$. Taking $q = \mathrm{rtp}_\varphi(b)$ in (a) therefore yields

$$\big|p(b) - P\big(\varphi(a_1, b), \ldots, \varphi(a_k, b)\big)\big| \leq \varepsilon \qquad \text{for all } b \in B.$$

Thus every left φ-type is a uniform limit on B of polynomials in instances $\varphi(a, y)$ with $a \in A$, and symmetrically for right types.

In Theorem 1.2.11 the approximation is required on all of $\mathrm{RS}(\varphi)$ (respectively, $\mathrm{LS}(\varphi)$), not only at realized types, and the requirement is essential. In the next example, every type coincides at realized types with an instance or a constant, and yet the function has the order property.

Example 1.2.12. Let $A = B = \mathbb{N}$, and define $\varphi(m, n) = 1$ if $m \leq n$ and $\varphi(m, n) = 0$ otherwise. Then the function φ has the order property and is therefore unstable. Here the left type $\mathrm{ltp}_\varphi(m)$ is the indicator function of $[m, \infty)$, and these functions converge pointwise to the constant function $\mathbf{0}$ as $m \to \infty$. Hence the space $\mathrm{LS}(\varphi)$ consists of the functions $\mathrm{ltp}_\varphi(m)$ together with $\mathbf{0}$. Likewise, the right type $\mathrm{rtp}_\varphi(n)$ is the indicator function of $[0, n]$, and the space $\mathrm{RS}(\varphi)$ consists of these functions together with the constant function $\mathbf{1}$. Thus every $p \in \mathrm{LS}(\varphi)$, viewed as a function on B, is either an instance $\varphi(m,\cdot)$ or the constant 0, which is trivially a polynomial in instances. Symmetrically, every $q \in \mathrm{RS}(\varphi)$ is an instance or the constant 1. The obstruction sits at the unrealized pair $(\mathbf{0}, \mathbf{1})$ (see Figure 1.5). If $\hat{\varphi}$ were separately continuous and extended φ as in Theorem 1.2.11, then we would have $\hat{\varphi}(\mathbf{0}, \mathrm{rtp}_\varphi(n)) = \mathbf{0}(n) = 0$ for all n, and since $\mathrm{rtp}_\varphi(n) \to \mathbf{1}$, we would get $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 0$. On the other hand, we would also have $\hat{\varphi}(\mathrm{ltp}_\varphi(m), \mathbf{1}) = \mathbf{1}(m) = 1$ for all m, and since $\mathrm{ltp}_\varphi(m) \to \mathbf{0}$, we would get $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 1$. Hence no separately continuous extension exists, and since a function satisfying condition (2) of Theorem 1.2.11 is separately continuous (see the proof of (2) ⇒ (1)), no function $\hat{\varphi}$ satisfies that condition.

We now return to the formula $\varphi(x, y)$ fixed in Section 1.1, whose left types are its φ-types in another guise. Let M be a model of T, and identify each complete φ-type $p \in S_\varphi(M)$ with the characteristic function on $M^{|y|}$ of the set $D_p = \{b \in M : \varphi(x, b) \in p\}$. With this identification we have $\mathrm{LS}(\varphi) = S_\varphi(M)$. Indeed, for $a \in M$ the left type $\mathrm{ltp}_\varphi(a)$ is $\mathrm{tp}_\varphi(a/M)$. By Lemma 1.1.2, every $p \in S_\varphi(M)$ agrees on each finite set of parameters with some realized type, so the type p is a pointwise limit of realized types. Conversely, a pointwise limit of

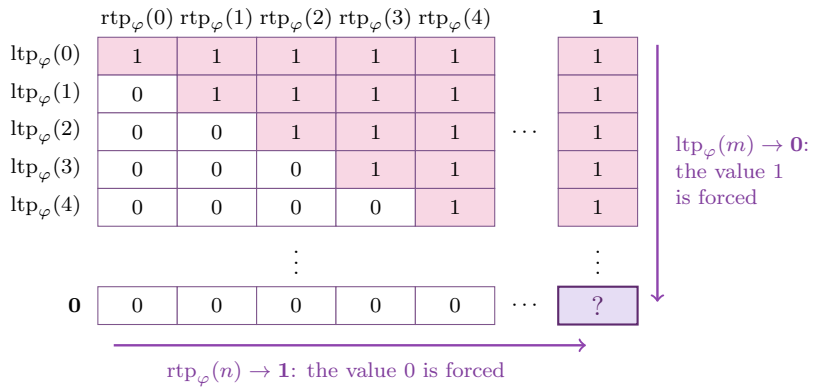


FIGURE 1.5. The ladder of Example 1.2.12, seen on pairs of types, with the extra row indexed by $\mathbf{0}$, the extra column indexed by $\mathbf{1}$, and the corner $(\mathbf{0}, \mathbf{1})$ at which no separately continuous extension $\hat{\varphi}$ can take a value.

realized types determines a set of φ-formulas over M that is finitely satisfiable in M. This set is therefore consistent with the elementary diagram of M, and so it is a complete φ-type.

In this two-valued setting, the quotient map $\beta M^{|x|} \to S_\varphi(M)$, described for $\mathrm{LS}(\varphi)$ after Definition 1.2.7, is transparent. Recall from Section 1.1 that $S_\varphi(M)$ is the Stone space of the Boolean algebra $\mathcal{B}_\varphi(M)$ generated by the sets $\varphi(M, b)$ with $b \in M^{|y|}$. An ultrafilter $\mathcal{U}$ on $M^{|x|}$ goes to its trace on $\mathcal{B}_\varphi(M)$, that is, to the type containing $\varphi(x, b)$ exactly when $\varphi(M, b) \in \mathcal{U}$. The realized types are the traces of principal ultrafilters. Every type is the trace of some ultrafilter, since by Lemma 1.1.2 it is an ultrafilter on $\mathcal{B}_\varphi(M)$, and an ultrafilter on $\mathcal{B}_\varphi(M)$ extends to one on $M^{|x|}$.

For $\{0,1\}$-valued types, an approximation within $\frac{1}{3}$ can be rounded, and polynomials in instances become Boolean combinations of instances.

Corollary 1.2.13. *Let M be an L-structure and let $\varphi(x, y)$ be a formula that is stable on M. Then every $p \in \mathrm{LS}(\varphi)$ is defined by a Boolean combination of instances: there are $a_1, \ldots, a_k \in M^{|x|}$ and a Boolean combination $\psi(y)$ of $\varphi(a_1, y), \ldots, \varphi(a_k, y)$ such that, for all $b \in M^{|y|}$, we have*

$$p(b) = 1 \quad \textit{if and only if} \quad M \models \psi(b).$$

Proof. Every $p \in \mathrm{LS}(\varphi)$ is a pointwise limit of $\{0,1\}$-valued functions, and is therefore $\{0,1\}$-valued. Applying Theorem 1.2.11 with $\varepsilon = \frac{1}{3}$ and evaluating clause (a) at realized right types as above, we find $a_1, \ldots, a_k$ and P with $|p(b) - P(t_b)| \leq \frac{1}{3}$ for all $b \in M^{|y|}$, where $t_b = (\varphi(a_1, b), \ldots, \varphi(a_k, b))$. Since $p(b)$ and the entries of t_b lie in $\{0,1\}$, we have $p(b) = 1$ if and only if the tuple t_b belongs to $S = \{t \in \{0,1\}^k : P(t) > \frac{1}{2}\}$. Let $\psi(y)$ be the formula $\bigvee_{t \in S} \bigwedge_{1 \leq i \leq k} \varphi(a_i, y)^{t_i}$ (recall that $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$). The formula $\psi(b)$ then holds in M exactly when $t_b \in S$, that is, exactly when $p(b) = 1$. ☺

A formula without the order property is stable on every model of T (see Remark 1.2.2). Through the identification $\mathrm{LS}(\varphi) = S_\varphi(M)$, valid for every model M of T, Corollary 1.2.13 therefore already gives (1) $\Rightarrow$ (3) of Theorem 1.1.6, by analytic means. The definitions obtained in this way are Boolean combinations of instances, not necessarily positive, and their size depends on the type being defined. The positive definitions of Theorem 1.1.6(2) come from the combinatorial proof in Section 2.2.

Remark 1.2.14. Polynomials are not the only operations that serve in Theorem 1.2.11.

(1) *Lattice version.* The lattice form of the Stone–Weierstrass theorem replaces polynomials by the operations max, min and affine maps. Continuous logic uses this form to show that a small set of connectives generates all continuous connectives up to uniform approximation (see [**BYU10**, Corollaries 1.4–1.6]).

(2) *Convex combinations.* Stability gives more than a polynomial approximation. In Proposition A.1.7 below we show, using Grothendieck's criterion, that for a stable function $\varphi\colon A \times B \to [0,1]$ the set $\{e_a : a \in A\}$ is relatively weakly compact in $C(\mathrm{RS}(\varphi))$. Let $\hat{\varphi}$ be the separately continuous extension of φ given by Proposition 1.2.8, and let $p \in \mathrm{LS}(\varphi)$ be the limit $\lim_{i,\mathcal{U}} \mathrm{ltp}_\varphi(a_i)$ of a family $(a_i)_{i\in I}$ in A along an ultrafilter $\mathcal{U}$ on I. Then equation (1.2) shows that the function $\hat{\varphi}(p,\cdot)$ is the pointwise limit of the functions e_{a_i} along $\mathcal{U}$, and relative weak compactness makes it their weak limit. By Mazur's lemma [**Maz33**], the function $\hat{\varphi}(p,\cdot)$ is then a uniform limit of convex combinations of the functions e_{a_i}. Thus the left type p is a uniform limit on B of convex combinations of instances $\varphi(a,y)$. Ben Yaacov [**BY14**] uses this convex approximation to recover the familiar forms of the defining formulas.

1.3 Exercises

The exercises follow the order of the sections, and the mark $(*)$ singles out those that ask for more than a direct application of the text. Exercises 9, 10, 13 and 14 share a single test case, the *damped ladder*: the function $\theta\colon \mathbb{N}\times\mathbb{N} \to [0,1]$ given by $\theta(m,n) = 2^{-m-1}$ if $m \le n$ and $\theta(m,n) = 0$ if $m > n$ (see Figure 1.6).

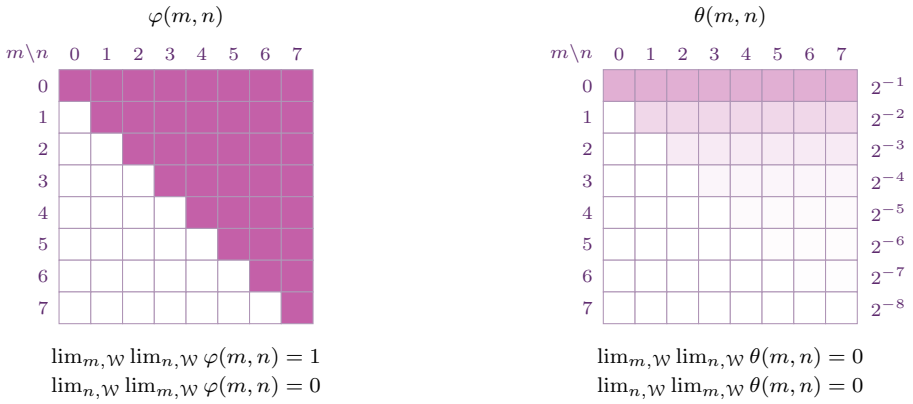


Figure 1.6. The ladder φ of Example 1.2.12 (left) and the damped ladder θ (right) on $\{0,\dots,7\}^2$, drawn as heat maps in which the intensity of each cell is proportional to its value; below each panel, its two iterated limits along a nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$.

1. *(Section 1.1.)* Let M be an infinite model of a theory T, and let $\varphi(x,y)$ be the formula $x = y$, where x and y are single variables. Recall that $S_\varphi(M)$ is the space of complete φ-types over M, and that $D_p = \{b \in M : \varphi(x,b) \in p\}$ for $p \in S_\varphi(M)$. Show that, for every $p \in S_\varphi(M)$, the set D_p is either a singleton $\{a\}$ with $a \in M$ or empty, and that

each of these sets is D_p for exactly one $p \in S_\varphi(M)$. For each type, find a φ-definition that is a positive Boolean combination of instances of φ (see Definition 1.1.3). Let p_∞ be the type with $D_{p_\infty} = \emptyset$. Show that p_∞ is the only point of $S_\varphi(M)$ that is not isolated, and conclude that the space $S_\varphi(M)$ is the one-point compactification of the discrete space M. What changes when M is finite? Next, let T_E be the theory of an equivalence relation E with infinitely many classes, all of them infinite. Let M be a model of T_E, and let $\varphi(x, y)$ be the formula xEy. Show that, for every $p \in S_\varphi(M)$, the set D_p is either an E-class of M or empty, and that every E-class of M and the empty set occur in this way. Show also that the space $S_\varphi(M)$ is the one-point compactification of the discrete set of E-classes of M. (Hint: if b and b' belong to D_p, then the formula $xEb \wedge xEb'$ is consistent with the elementary diagram of M, which contains the transitivity of E. Hence the elements b and b' lie in the same E-class. A set of φ-formulas over M that is finitely satisfiable in M is consistent with the elementary diagram of M, by compactness.)

2. *(Section 1.1.)* Let K be a finite field, and let T be the theory of infinite vector spaces over K. The language of T consists of $+$, $-$, 0 and a unary function symbol for multiplication by each $\lambda \in K$. Let n be a positive integer, and let $\varphi(x_1, \ldots, x_n; y)$ be the formula

$$\bigvee_{(\lambda_1,\ldots,\lambda_n)\in K^n} y = \lambda_1 x_1 + \cdots + \lambda_n x_n,$$

which says that y lies in the span of $x_1, \ldots, x_n$. The object variables are $x = (x_1, \ldots, x_n)$, and the parameter variable is y. Let M be a model of T. Show that, for every $p \in S_\varphi(M)$, the set D_p is a subspace of M of dimension at most n. Conversely, show that every subspace of M of dimension at most n is D_p for some type p realized in M. Show also that every $p \in S_\varphi(M)$ is defined by a single instance $\varphi(d, y)$ with $d \in M^n$. Conclude that every complete φ-type over M is realized in M. Let $N \succeq M$ be given, and let $c \in N^n$ be a tuple whose span meets M only in 0. Show that the tuples c and $(0, \ldots, 0)$ have the same φ-type over M, and that they have different complete types over M when $c \neq (0, \ldots, 0)$. (Hint: if $c \in N^n$ realizes p, then the set D_p equals $M \cap \mathrm{span}(c)$.)

3. *(Section 1.1.)* Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$.

(a) Show that, for every $p \in S_\varphi(\mathbb{Q})$, the set D_p is a final segment of $\mathbb{Q}$: every element of $\mathbb{Q}$ above an element of D_p belongs to D_p. Show that every final segment of $\mathbb{Q}$ is D_p for exactly one $p \in S_\varphi(\mathbb{Q})$.

(b) Show that the final segments of $\mathbb{Q}$ are $\emptyset$, $\mathbb{Q}$, the sets $\{b : b \geq q\}$ and $\{b : b > q\}$ with $q \in \mathbb{Q}$, and the sets $\{b : b > r\}$ with r irrational. Conclude that the set $S_\varphi(\mathbb{Q})$ has cardinality $2^{\aleph_0}$.

(c) Let $A \subseteq \mathbb{Q}$ be finite. Show that every subset of $\mathbb{Q}$ definable in $(\mathbb{Q}, <)$ with parameters from A is a union of singletons $\{a\}$ with $a \in A$ and of open intervals whose endpoints are consecutive elements of $A \cup \{-\infty, +\infty\}$. (Hint: such a set is invariant under the automorphisms of $(\mathbb{Q}, <)$ that fix A pointwise. If c and d lie in the same such interval, a piecewise linear automorphism fixing A pointwise maps c to d.)

(d) Deduce that the types p with D_p equal to $\emptyset$, $\mathbb{Q}$, $\{b : b \geq q\}$ or $\{b : b > q\}$ are definable, and give their definitions. Deduce also that the types with $D_p = \{b : b > r\}$ and r irrational are not definable.

(e) Let $q \in \mathbb{Q}$ be given. Show that the type p with $D_p = \{b : b \geq q\}$ has no φ-definition that is a positive Boolean combination of instances $\varphi(a, y)$ with $a \in \mathbb{Q}$. (Hint: compute the set defined by $\bigvee_{i<n} \bigwedge_{a \in W_i} a < y$, where $n < \omega$ and $W_0, \ldots, W_{n-1}$ are finite subsets of $\mathbb{Q}$.)

(f) Show directly that φ has the order property, with $a_i = i$ and $b_j = j + \frac{1}{2}$ for positive integers i and j. Explain why (d), (e) and (f) agree with Theorem 1.1.6.

4. $(*)$ *(Section 1.1.)* Show that no structure contains a 2-ladder (see Definition 1.1.4) for the formula $x = y$, and that no model of T_E (see Exercise 1) contains a 2-ladder for the formula xEy. Compare with Exercise 1 and Theorem 1.1.6. Next, in the setting of Exercise 2, show that every model of T contains an $(n + 1)$-ladder for φ and that no model of T contains an $(n + 2)$-ladder for φ. (Hint: given a positive integer m and an m-ladder $a_1, \ldots, a_m$, $b_1, \ldots, b_m$, let V_i be the span of the entries of a_i, and write $W_j = V_1 \cap \cdots \cap V_j$. Show that W_{j+1} is a proper subspace of W_j for $1 \leq j < m$. For the $(n + 1)$-ladder, let $e_1, \ldots, e_n$ be linearly independent elements of a model of T, and let $a_i = (e_i, \ldots, e_n, 0, \ldots, 0)$ for $i \leq n$. Complete the ladder with $a_{n+1} = (0, \ldots, 0)$, $b_j = e_j$ for $j \leq n$ and $b_{n+1} = 0$.)

5. *(Section 1.1.)* Let φ^{op} be the formula $\varphi(x, y)$ read with y as the tuple of object variables and x as the tuple of parameter variables. Thus, for a positive integer n, an n-ladder for φ^{op} consists of tuples $c_1, \ldots, c_n$ of the length of y and $d_1, \ldots, d_n$ of the length of x. These tuples are subject to the condition that $\varphi(d_j, c_i)$ holds if and only if $i \leq j$. Show that the following are equivalent: the formula φ has the order property; the formula $\neg\varphi$ has the order property; the formula φ^{op} has the order property. (Hint: for $\neg\varphi$, use Remark 1.1.5; for φ^{op}, reverse the indices of an n-ladder for φ.) Conclude from Exercise 3 that the formula $x \leq y$ has the order property in $(\mathbb{Q}, <)$.

6. *(Section 1.1.)* Assume that φ is stable, and let M be a model of T. Show that $|S_\varphi(M)| \leq |M| + \aleph_0$. (Hint: by Theorem 1.1.6, each $p \in S_\varphi(M)$ has a definition that is a positive Boolean combination of instances of φ over M. Such a combination is determined by a finite set of finite subsets of $M^{|x|}$, and the set D_p determines p.) Deduce from Exercise 3 a second proof that the formula $x < y$ has the order property in the theory of $(\mathbb{Q}, <)$. Compare with Exercise 1, where the φ-types over M correspond to the E-classes of M, with one extra type for the empty set.

7. *(Subsection 1.2.1.)* Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$.

(a) Let $\varphi\colon \mathbb{N} \times \mathbb{N} \to \{0, 1\}$ be given by $\varphi(m, n) = 1$ if $m < n$ and $\varphi(m, n) = 0$ otherwise. Compute $\lim_{m,\mathcal{U}} \lim_{n,\mathcal{U}} \varphi(m, n)$ and $\lim_{n,\mathcal{U}} \lim_{m,\mathcal{U}} \varphi(m, n)$. Find sequences $(a_m)_{m\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ in $\mathbb{N}$ with $\varphi(a_m, b_n) = 1$ if and only if $m \leq n$.

(b) Let $\varphi\colon A \times B \to \mathbb{R}$ be a bounded function, let $(a_m)_{m\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be sequences in A and B, and let $\mathcal{V}$ and $\mathcal{W}$ be ultrafilters on $\mathbb{N}$. Show that, if $\mathcal{V}$ or $\mathcal{W}$ is principal, then the two iterated limits $\lim_{m,\mathcal{V}} \lim_{n,\mathcal{W}} \varphi(a_m, b_n)$ and $\lim_{n,\mathcal{W}} \lim_{m,\mathcal{V}} \varphi(a_m, b_n)$ are equal. Thus only pairs of nonprincipal ultrafilters can witness instability.

(c) Show that the function $(x, y) \mapsto xy$ on $[0, 1] \times [0, 1]$ is stable. (Hint: given sequences $(a_m)_{m\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ in $[0, 1]$ and ultrafilters $\mathcal{V}$ and $\mathcal{W}$ on $\mathbb{N}$, compute both iterated limits in terms of $\lim_{m,\mathcal{V}} a_m$ and $\lim_{n,\mathcal{W}} b_n$.)

(d) Let $\delta\colon [0,1]\times[0,1] \to \{0,1\}$ be given by $\delta(x,y) = 1$ if $x = y$ and $\delta(x,y) = 0$ otherwise. Show that δ is stable, although no function $\delta(\cdot, y)$ is continuous on $[0,1]$. (Hint: regard $[0,1]$ as a model of the theory of infinite sets in the empty language, and use Exercise 4 and Remark 1.2.2.)

8. *(Subsection 1.2.1.)*

(a) Let $\varphi, \psi\colon A \times B \to \mathbb{R}$ be bounded stable functions, and let $c \in \mathbb{R}$ be given. Show that the functions $\varphi + \psi$, $c\varphi$ and $\varphi\psi$ are stable, and that every constant function is stable. (Hint: limits along an ultrafilter commute with sums and products of bounded real sequences.)

(b) Let $(\varphi_k)_{k\in\mathbb{N}}$ be a sequence of stable functions that converges uniformly on $A \times B$ to a function φ. Show that φ is stable.

(c) Let φ be a stable function with values in an interval $[c,d]$, and let $g\colon [c,d] \to \mathbb{R}$ be continuous. Show that $g \circ \varphi$ is stable. (Hint: apply Fact 1.2.10 with $K = [c,d]$ and $\mathcal{F}$ consisting of the identity.)

(d) Let $\varphi(x,y)$ and $\psi(x,y)$ be stable formulas. Show that $\neg\varphi$, $\varphi \wedge \psi$ and $\varphi \vee \psi$ are stable, and hence that every Boolean combination of φ and ψ is stable. (Hint: on a model of T, the function of $\varphi \wedge \psi$ is the product of the functions of φ and ψ, and the function of $\neg\varphi$ is $1 - \varphi$. Use Proposition 1.2.4 and Remark 1.2.2.) Explain how to treat a formula $\psi(x,z)$ whose parameter variables differ from those of φ.

9. $(*)$ *(Subsection 1.2.1.)* Let $\varphi\colon A \times B \to \mathbb{R}$ be a bounded function. Show that φ is unstable if and only if there are sequences $(a_m)_{m\in\mathbb{N}}$ in A and $(b_n)_{n\in\mathbb{N}}$ in B and real numbers $r \neq s$ with the following property: if $\varepsilon = |r-s|/3$, then $|\varphi(a_m, b_n) - r| \leq \varepsilon$ whenever $m \leq n$, and $|\varphi(a_m, b_n) - s| \leq \varepsilon$ whenever $m > n$. (Hint: for the forward direction, rescale φ to take values in $[0,1]$, apply Proposition 1.2.3, and pass to subsequences as in the proof of Proposition 1.2.4.) Check that, for $\{0,1\}$-valued functions, the condition says that φ or $\neg\varphi$ has the order property. Next, show that the damped ladder satisfies $\theta(m,n) > 0$ if and only if $m \leq n$, and that both iterated limits of $\theta(m,n)$ along a nonprincipal ultrafilter on $\mathbb{N}$ are 0. Thus, in the real-valued case, a ladder witnesses instability only when its values are separated by a margin. (The function θ is in fact stable; see Exercises 10 and 13.)

10. *(Subsection 1.2.2.)* Let X and Y be compact Hausdorff spaces, and let $f\colon X \times Y \to \mathbb{R}$ be a bounded function that is separately continuous. Show that f is stable, as a function on the sets X and Y. (Hint: given sequences $(a_m)_{m\in\mathbb{N}}$ in X and $(b_n)_{n\in\mathbb{N}}$ in Y and ultrafilters $\mathcal{U}$ and $\mathcal{V}$ on $\mathbb{N}$, let $a = \lim_{m,\mathcal{U}} a_m$ and $b = \lim_{n,\mathcal{V}} b_n$, and show that both iterated limits equal $f(a,b)$.) Apply this result to the following functions.

(a) The function f on $[0,1]\times[0,1]$ given by $f(x,y) = 2xy/(x^2+y^2)$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$. Show that f is stable and takes values in $[0,1]$, but is not continuous at $(0,0)$.

(b) The damped ladder θ. Let $\mathbb{N} \cup \{\infty\}$ be the one-point compactification of $\mathbb{N}$, and define $\theta(m,\infty) = 2^{-m-1}$, $\theta(\infty,n) = 0$ and $\theta(\infty,\infty) = 0$. Show that the extended function is separately continuous, and conclude that θ is stable.

The function δ of Exercise 7(d) is stable, but it is not separately continuous on $[0,1]\times[0,1]$ with the usual topology. Explain why this does not contradict Proposition 1.2.8.

11. *(Subsection 1.2.2.)* Let A be an infinite set, and let $\varphi\colon A \times A \to \{0,1\}$ be given by $\varphi(a,b) = 1$ if $a = b$ and $\varphi(a,b) = 0$ otherwise. This function is stable by the argument of Exercise 7(d). For $a \in A$, let $\mathbf{1}_a$ be the indicator function of $\{a\}$.

(a) Show that the space $\mathrm{LS}(\varphi)$ of left φ-types consists of the functions $\mathbf{1}_a$ with $a \in A$ together with the constant function $\mathbf{0}$. Show also that each point $\mathbf{1}_a$ is isolated in $\mathrm{LS}(\varphi)$, and that every neighborhood of $\mathbf{0}$ contains all but finitely many of the points $\mathbf{1}_a$. Prove that these three statements hold as well for the space $\mathrm{RS}(\varphi)$ of right φ-types. Compare $\mathrm{LS}(\varphi)$ with the space $S_\varphi(M)$ of Exercise 1.

(b) Let $\hat{\varphi}$ be the function given by Proposition 1.2.8. Using (1.2), show that $\hat{\varphi}(p,q) = 1$ if $p = q = \mathbf{1}_a$ for some $a \in A$, and $\hat{\varphi}(p,q) = 0$ otherwise.

(c) Show that the function $\hat{\varphi}$ is not continuous at $(\mathbf{0},\mathbf{0})$ as a function on $\mathrm{LS}(\varphi) \times \mathrm{RS}(\varphi)$. (Hint: take the limit of the pairs $(\mathbf{1}_a, \mathbf{1}_a)$ along a nonprincipal ultrafilter on A.) Thus the extension in Proposition 1.2.8 is separately continuous, and in general no more.

12. *(Subsection 1.2.3.)* Here Theorem 1.2.9 meets the classical sequence spaces. The spaces in this exercise are real, and for each $k \in \mathbb{N}$ we write e_k for the k-th unit vector.

(a) Show that the function $(x,y) \mapsto \langle x,y\rangle$ on $B_{\ell^2} \times B_{\ell^2}$ is stable. (Hint: the ball B_{ℓ^2} is compact and Hausdorff in the weak topology, and the inner product is separately continuous for it; use Exercise 10.)

(b) Let $y_n = (n+1)^{-1/2} \sum_{k \le n} e_k$. Show that $y_n \in B_{\ell^2}$, that $\langle e_m, y_n\rangle > 0$ if and only if $m \le n$, and that both iterated limits of $\langle e_m, y_n\rangle$ along a nonprincipal ultrafilter on $\mathbb{N}$ are 0. Compare with Exercise 9.

(c) In ℓ^1, whose dual is ℓ^∞, let $f_n = \sum_{k \le n} e_k \in B_{\ell^\infty}$. Let $A = \{e_m : m \in \mathbb{N}\}$ and $B = \{f_n : n \in \mathbb{N}\}$. Show that the pairing restricted to $A \times B$ takes values in $\{0,1\}$ and has the order property. Deduce from Theorem 1.2.9 that ℓ^1 is not reflexive.

(d) In c_0, whose dual is ℓ^1, let $s_m = \sum_{k \le m} e_k \in B_{c_0}$. Compute both iterated limits of $\langle s_m, e_n\rangle$ along a nonprincipal ultrafilter on $\mathbb{N}$, and deduce that c_0 is not reflexive.

13. *(Subsection 1.2.3.)* For each of the following functions $\varphi\colon A \times B \to [0,1]$, find a reflexive Banach space X and maps $u\colon A \to B_X$ and $v\colon B \to B_{X^*}$ with $\varphi(x,y) = \langle u(x), v(y)\rangle$. The space X and the maps u and v then witness condition (2) of Theorem 1.2.9, with X in place of E.

(a) The function of Exercise 11 on an infinite set A. (Hint: take $X = \ell^2(A)$.)

(b) The function of the formula xEy on a model M of T_E (see Exercise 1). (Hint: let C be the set of E-classes of M, take $X = \ell^2(C)$, and send each element of M to the unit vector of its class.)

(c) The damped ladder θ. (Hint: take $X = \ell^2$, $u(m) = e_m$ and $v(n) = \sum_{k \le n} 2^{-k-1} e_k$, where e_k is the k-th unit vector of ℓ^2 as in Exercise 12, and check that $\|v(n)\| \le 1$.) This provides a second proof that θ is stable.

Finally, let $\lambda\colon \mathbb{N} \times \mathbb{N} \to \{0,1\}$ be the ladder of Example 1.2.12, given by $\lambda(m,n) = 1$ if $m \le n$ and $\lambda(m,n) = 0$ otherwise. Show that the choice $u(m) = e_m$ and $v(n) = \sum_{k \le n} e_k$ in ℓ^2 gives $\lambda(m,n) = \langle u(m), v(n)\rangle$, but that the norms of the vectors $v(n)$ are unbounded. Use Theorem 1.2.9 to explain why no reflexive space X admits maps u and v into

B_X and B_{X^*} (or, after rescaling, into any bounded subsets of X and X^*) such that $\lambda(m,n) = \langle u(m), v(n)\rangle$. Compare this conclusion with Exercise 12(c).

14. *(Subsection 1.2.4.)* The damped ladder θ is stable by Exercise 10. Its types show that the approximate definitions of Theorem 1.2.11 cannot, in general, be replaced by exact definitions.

(a) Show that $\mathrm{LS}(\theta)$ consists of the functions $\mathrm{ltp}_\theta(m)$ with $m \in \mathbb{N}$ together with the constant function $\mathbf{0}$. Show that $\mathrm{RS}(\theta)$ consists of the functions $\mathrm{rtp}_\theta(n)$ with $n \in \mathbb{N}$ together with the function q_∞ given by $q_\infty(m) = 2^{-m-1}$, and that q_∞ is not a realized type.

(b) Let $\hat\theta$ be the function given by Proposition 1.2.8. Using (1.2), show that $\hat\theta(\mathrm{ltp}_\theta(m), q) = q(m)$, $\hat\theta(\mathbf{0}, q) = 0$ and $\hat\theta(p, \mathrm{rtp}_\theta(n)) = p(n)$ for all $m, n \in \mathbb{N}$, $p \in \mathrm{LS}(\theta)$ and $q \in \mathrm{RS}(\theta)$.

(c) Let $k \in \mathbb{N}$ be given. Show that $|\hat\theta(p, q_\infty) - p(k)| \leq 2^{-k-2}$ for all $p \in \mathrm{LS}(\theta)$. This is clause (b) of Theorem 1.2.11 for q_∞, with $l = 1$, $b_1 = k$ and $Q(t) = t$. Evaluated at realized types, it says that $|q_\infty(m) - \theta(m,k)| \leq 2^{-k-2}$ for all m: the type q_∞ is the uniform limit of the instances $\theta(x,k)$ as $k \to \infty$. Compare q_∞ with the norm limit of the vectors $v(n)$ of Exercise 13(c).

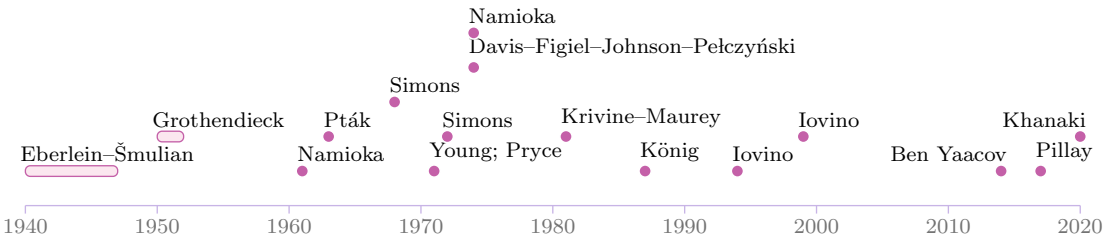
(d) Show that no exact definition exists: there are no $b_1, \ldots, b_l \in \mathbb{N}$ and no polynomial Q with $\hat\theta(p, q_\infty) = Q(p(b_1), \ldots, p(b_l))$ for all $p \in \mathrm{LS}(\theta)$. (Hint: evaluate at $p = \mathrm{ltp}_\theta(m)$ with $m > \max_i b_i$.) Contrast this with Corollary 1.2.13, where rounding turns approximate definitions of $\{0,1\}$-valued types into exact ones.

1.4 Historical remarks

Section 1.1. Counting types begins with Morley's proof of Łoś's conjecture [Mor65], that a countable theory categorical in one uncountable cardinality is categorical in all of them. Morley introduced the totally transcendental theories, defined by a condition on the topology of their type spaces. He showed that for countable theories the condition amounts to a count: over a countable set there are only countably many types. This is the condition now called ω-stability. The general notion was not long in coming. Setting out to extend Morley's categoricity theorem to uncountable theories, Rowbottom generalized ω-stability to λ-*stability*, for an infinite cardinal λ. In Rowbottom's definition, a theory is λ-stable if there are at most λ types over sets of size λ [Row64]. Shelah then called a theory *stable* when it is λ-stable for some λ, and began the study of stable theories in general in his paper *Stable theories* [She69]. In a paper of 1971 [She71] he moved from theories to single formulas. There he defined a formula $\varphi(x,y)$ to be *stable in* λ when it has at most λ φ-types over sets of size λ [She71, Definition 2.1]. He also defined the *order property* [She71, Definition 2.2] and proved that stability is equivalent to the absence of the order property. The ladder formulation of Definition 1.1.4 is a finitary restatement of Shelah's condition. Definable types were found independently by Shelah and by Gaifman, who met them in his study of end extensions of models of Peano arithmetic [Gai76]. The theorem that φ-types are definable when φ is stable, Theorem 3.1 of Shelah's 1971 paper [She71] (see also [She90, Chapter II]), is often called the fundamental theorem of stability theory. Four decades later, Ben Yaacov [BY14] observed that the theorem is an immediate consequence

of Grothendieck's work on weak compactness in function spaces [Gro52], published two decades before Shelah's (see Appendix A). This analytic viewpoint, in which the definability of φ-types is derived from the topology of spaces of functions, is that of Section 1.2. There, however, definability is obtained not from Grothendieck's criterion but from the separately continuous extension of φ to pairs of types and from the Stone–Weierstrass theorem (see Proposition 1.2.8 and Corollary 1.2.13).

Section 1.2.1. Interchanging iterated limits is an old theme. For double sequences of reals, the classical sufficient condition is uniform convergence in one variable (the Moore–Osgood theorem). The condition of Definition 1.2.5 requires the iterated limits to agree along *all* pairs of sequences. This condition comes from functional analysis, and it reached model theory by way of the geometry of Banach spaces, along the road traced in the timeline below.



1940–1947 **Eberlein–Šmulian.** Šmulian [Š40] (1940) and Eberlein [Ebe47] (1947) prove that in a Banach space weak compactness and weak sequential compactness coincide. Weak compactness can therefore be tested along sequences.

1950–1952 **Grothendieck.** In a note of 1950 [Gro50], Grothendieck announces general criteria of compactness in locally convex spaces, extending results of Eberlein and Krein. The *double limit criterion* itself appears in his paper of 1952 [Gro52], received at the end of 1950. For a compact Hausdorff space K, the criterion reads as follows. A bounded set $H \subseteq C(K)$ is relatively weakly compact if and only if $\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$ for all sequences $(f_n)_{n\in\mathbb{N}}$ in H and $(x_m)_{m\in\mathbb{N}}$ in K for which both iterated limits exist. For a countably compact space, the same condition decides when pointwise limits of functions in H are continuous. See Appendix A.

1961 **Namioka.** Namioka proves a combinatorial "substitute for Lebesgue's bounded convergence theorem" [Nam61]. From this result he obtains, without measure theory, Krein's theorem: in a Banach space, the closed convex hull of a weakly compact set is weakly compact.

1963 **Pták.** Pták [Pt́a63a] finds a purely combinatorial lemma about averages, from which both the double limit criterion and Krein's theorem follow. In a second paper [Pt́a63b] he considers bounded functions on a product $X \times Y$ that are continuous in each variable separately. By an interchange of iterated limits he decides which of these functions extend to bilinear forms with the same kind of continuity.

1968 **Simons.** Simons [Sim68] introduces the *Fubini characterization* of the double limit condition. According to this characterization, a bounded function

on a product satisfies the condition exactly when the two iterated integrals of the function agree, for every pair of finitely additive probability measures on the factors.

1971 **Young; Pryce.** N. J. Young [You71] gives topological proofs of Pták's double limit theorems, based on Grothendieck's work rather than combinatorics, and generalizes Pták's bilinear extension theorem. Pryce [Pry71] uses a trick of Whitley's to simplify proofs about pointwise compact sets of continuous functions.

1972 **Simons.** In three consecutive papers, Simons finds an elementary proof of Pták's lemma [Sim72c]. He also proves an inequality about suprema and upper limits, now known as *Simons's inequality* [Sim72a]. In the third paper he relates minimax theorems, which say when $\max\min = \min\max$, to James's characterization of reflexivity [Sim72b].

1974 **Davis–Figiel–Johnson–Pełczyński.** Davis, Figiel, Johnson and Pełczyński [DFJP74] prove that every weakly compact operator factors through a reflexive Banach space. This is the key step in the proof of Theorem 1.2.9.

1974 **Namioka.** Namioka [Nam74] studies when separately continuous functions on products are jointly continuous at many points. The question arises for the extension $\hat{\varphi}$ of Proposition 1.2.8, which is in general only separately continuous (see Exercise 11). His theorem applies to $\hat{\varphi}$: since $\mathrm{LS}(\varphi)$ and $\mathrm{RS}(\varphi)$ are compact, there is a dense G_δ set $D \subseteq \mathrm{LS}(\varphi)$ such that $\hat{\varphi}$ is jointly continuous at every point of $D \times \mathrm{RS}(\varphi)$.

1981 **Krivine–Maurey.** Krivine and Maurey [KM81] apply the interchange of limits to the norm. A Banach space is *stable* in their sense if $\lim_m \lim_n \|x_m + y_n\| = \lim_n \lim_m \|x_m + y_n\|$ for all bounded sequences $(x_m)_{m\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ in the space for which both limits exist. They prove that every stable Banach space contains, for some p with $1 \le p < \infty$, copies of ℓ^p that are as close to isometric as desired. This result generalizes a theorem of Aldous about subspaces of L^1.

1987 **König.** König [Kön87] studies the double limit condition in angelic spaces, where compactness can be tested by sequences.

1994 **Iovino.** In his Ph.D. thesis [Iov94], Iovino introduces stability for real-valued logic.

1999 **Iovino.** Iovino [Iov99] observes that Shelah's order property is the 2-valued case of the same condition, and proves Theorem 1.2.9. Both Proposition 1.2.3 and Proposition 1.2.4 are taken from his paper.

2014 **Ben Yaacov.** Ben Yaacov [BY14] derives the definability of types in stable theories directly from Grothendieck's criterion.

2017 **Pillay.** Pillay [Pil17] comments on the model-theoretic reading of Grothendieck's criterion, and derives from the criterion the positive form of the definitions.

2020 **Khanaki.** Khanaki [Kha20] studies stability, NIP (see Chapter 4), and related properties systematically, through topological properties of function spaces.

Section 1.2.2. Spaces of left and right φ-types (see Definition 1.2.7) are the analytic counterpart of Shelah's spaces of φ-types. The use of *types* in Banach space theory, as limits of the functions $y \mapsto \|x_n + y\|$ for a bounded sequence $(x_n)_{n\in\mathbb{N}}$ in a Banach space, is due to Krivine and Maurey [KM81]. The characterization of stability by the existence of a separately continuous extension to pairs of types (see Proposition 1.2.8) is Proposition 2.6 of Iovino [Iov99]. Separately continuous functions on products of compact spaces, and their relation to iterated limits, are the subject of Grothendieck's paper [Gro52].

Section 1.2.3. James [Jam64] characterized the reflexivity of a Banach space E in terms of its unit ball B_E and the unit ball B_{E^*} of its dual. The space E is reflexive if and only if there are no sequences $(a_m)_{m\in\mathbb{N}}$ in B_E and $(f_n)_{n\in\mathbb{N}}$ in B_{E^*} whose pairing takes one value for $m < n$ and another for $m \geq n$. This configuration of sequences is a Banach-space form of the order property. In the early 1980s, Krivine and Maurey [KM81] and Raynaud [Ray83] showed that stable functions give rise to weakly compact operators. The Davis–Figiel–Johnson–Pełczyński factorization theorem [DFJP74] (1974) says that every weakly compact operator factors through a reflexive space. Iovino [Iov99] combined these results to prove Theorem 1.2.9: stable functions are exactly the functions that factor through the pairing between the unit balls of a reflexive space and of its dual.

Section 1.2.4. Weierstrass proved in 1885, in his seventieth year, that every continuous function on a compact interval is a uniform limit of polynomials [Wei85]. Half a century later, Stone replaced the interval by an arbitrary compact Hausdorff space and the coordinate function by any family of functions that separates points [Sto37, Sto48]. Ben Yaacov and Usvyatsov [BYU10] used the Stone–Weierstrass theorem in the foundations of continuous first-order logic. In that setting the theorem shows that finitely many connectives suffice up to uniform approximation, and that the definable predicates are exactly the continuous functions on type spaces. Ben Yaacov [BY14] combined Grothendieck's criterion with Mazur's lemma [Maz33], by which a weakly convergent sequence has convex combinations converging in norm to its limit. He thus obtained the defining formulas of stable types as uniform limits of averages of instances (see Remark 1.2.14).

Chapter
2

Stability, trees, and definability of types

Roadmap

The definability theorem stated in Chapter 1 is proved here, and with it the fact that the configurations forbidden by stability are one. The equivalence, stated in Section 2.1, joins four conditions on a formula. They are instability, the order property, the encoding of binary trees of arbitrarily large finite height, and the existence of a type over some model that is not definable (see Theorem 2.1.3). In Section 2.2 we show that, in the absence of arbitrarily long ladders, every type is defined by a positive Boolean combination of instances with parameters from a finite set (see Theorem 2.2.1). In Section 2.3 a ladder is folded into a binary tree (see Proposition 2.3.1). In Section 2.4 a tree of infinite height over a small model produces more types than there are formulas to define them (see Proposition 2.4.2). In Section 2.5, for a countable language, the same division between stable and unstable formulas appears in the type spaces over countable models. These spaces are countable, and scattered, exactly when the formula is stable; here a space is scattered if every nonempty subset has a point isolated within that subset. When the formula is unstable, one of these spaces contains a copy of the Cantor space (see Theorem 2.5.1). In Chapter 3 we make two of these arguments quantitative and explicit: we bound the height of a tree that forces a ladder of given length, and we obtain definitions from a rank.

As in Section 1.1, we fix a first-order theory T in a language L and an L-formula $\varphi(x, y)$ consistent with T.

2.1 The equivalence proved in this chapter

In Theorem 2.1.3 below we show that four conditions on φ coincide: instability, the order property, the presence of tall binary trees, and the existence of a non-definable φ-type. The notions of φ-type, definable type, ladder, order property and stability come from Section 1.1. The newcomer, a binary tree for φ, needs a little notation for finite 0–1 sequences.

Notation 2.1.1. Recall that $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$. For an ordinal μ, we write 2^μ for the set of 0–1 sequences of length μ and $2^{<\mu}$ for the set of those of length $<\mu$. For $n < \omega$ we also write $2^{\leq n} = 2^{<n} \cup 2^n$. We write $\sigma \sqsubseteq \tau$ if σ is an initial segment of τ, and $\sigma \sqsubset \tau$ if moreover $\sigma \neq \tau$. The length of σ is denoted $|\sigma|$, and the extension of σ by one entry u is denoted $\sigma^\frown\langle u\rangle$. Finally, for a sequence η of length $>k$, the initial segment of length k of η is denoted $\eta|k$.

Definition 2.1.2. For $n < \omega$, a *tree of height n for φ in a model M of T* consists of families $(b_\sigma : \sigma \in 2^{<n})$ and $(a_\eta : \eta \in 2^n)$ of tuples of M such that, for all $\sigma \in 2^{<n}$, $\eta \in 2^n$ and $u \in \{0, 1\}$, we have

$$\sigma^\frown\langle u\rangle \sqsubseteq \eta \implies M \models \varphi(a_\eta, b_\sigma)^u.$$

In other words, for every σ with $\sigma \sqsubset \eta$, the leaf a_η satisfies $\varphi(x, b_\sigma)$ exactly when the branch η turns right at the node σ, that is, when $\sigma^\frown\langle 1\rangle \sqsubseteq \eta$. We say that φ *encodes binary trees of arbitrarily large finite height* if for every $n < \omega$ some model of T contains a tree of height n for φ.

A tree of height 3 is drawn in Figure 2.1. Each node b_σ splits the leaves above it into two halves: the leaves on the right satisfy $\varphi(x, b_\sigma)$, and the leaves on the left satisfy $\neg\varphi(x, b_\sigma)$. The branch leading to a leaf a_η is thus a record, node by node, of which of the formulas $\varphi(x, b_\sigma)$ the leaf satisfies. A tree of height n realizes all 2^n such records.

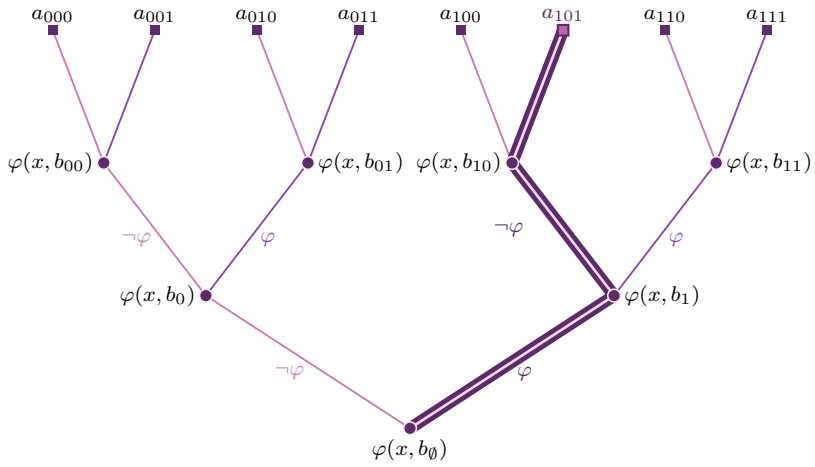


FIGURE 2.1. A tree of height 3 for φ in a model M of T, with the edges for φ in purple, those for $\neg\varphi$ in pink, and the branch to the leaf a_{101} drawn thick.

Like ladders (see Remark 1.1.5), trees can be shortened. A tree of height $n + 1$ restricts to one of height n: keep the nodes b_σ with $\sigma \in 2^{<n}$, and for each $\eta \in 2^n$ keep the leaf $a_{\eta^\frown\langle 0\rangle}$. Hence a formula encodes binary trees of arbitrarily large finite height as soon as, for infinitely many n, some model of T contains a tree of height n for it. We can now state the equivalence that gives the chapter its title.

Theorem 2.1.3. *Let $\varphi(x, y)$ be an L-formula. The following are equivalent:*

(1) *The formula φ is unstable (see Definition 1.1.4).*

(2) *The formula φ has the order property (see Definition 1.1.4).*

(3) *The formula φ encodes binary trees of arbitrarily large finite height.*

(4) *Some φ-type over some model of T is not definable (see Definitions 1.1.1 and 1.1.3).*

The proof goes around a cycle of implications. The equivalence (1) $\Leftrightarrow$ (2) is the definition of stability (see Definition 1.1.4). Long ladders are turned into tall trees in Section 2.3, which gives (2) $\Rightarrow$ (3) (see Proposition 2.3.1). In Section 2.4 we use tall trees to produce more φ-types than there are formulas to define them; this is (3) $\Rightarrow$ (4) (see Proposition 2.4.2). The cycle closes with the substantial half of the definability theorem, which we prove first, in Section 2.2: if φ does not have the order property, then every φ-type over a model is definable. The contrapositive of this half is (4) $\Rightarrow$ (2) (see Theorem 2.2.1 and Corollary 2.2.2).

In Chapter 3 two of these implications receive alternative proofs, each of which yields further information. In this chapter the way back from tall trees to the order property passes through non-definable types, and hence through a new model obtained by compactness and a count of types. In Section 3.1 we prove (3) $\Rightarrow$ (2) directly, by finite combinatorics based on Ramsey's theorem (see Proposition 3.1.5 and Corollary 3.1.6). This direct proof stays inside a single model and is quantitative: for every $n \geq 1$, a tree of height $2\binom{2n}{n} - 1$ for φ in a model M of T already forces an n-ladder in M. In Section 3.2 the classical rank argument proves (4) $\Rightarrow$ (3), again in contrapositive form: by Proposition 3.2.3, the absence of tall trees alone makes every φ-type definable, without a detour through the order property. The φ-definitions produced in this way are given explicitly in terms of a rank, whereas those of Section 2.2 are positive Boolean combinations of instances of φ.

2.2 From the absence of the order property to type definability

We begin with the implication of the definability theorem that carries the weight, namely (1) $\Rightarrow$ (2) of Theorem 1.1.6. This implication asserts that, if φ does not have the order property, then every φ-type over a model has a definition built from instances of φ using only $\wedge$ and $\vee$. The candidate definitions are easy to write down, and the work lies in choosing the finitely many parameters that they use.

Let M be a model of T, let $p \in S_\varphi(M)$ be given, and recall that $D_p = \{b \in M : \varphi(x, b) \in p\}$. For a finite set $F \subseteq M$, we shall write

$$\mathcal{W}_F = \big\{\{a \in F : M \models \varphi(a, b)\} : \ b \in D_p\big\} \quad \text{and} \quad d_F\varphi(y) = \bigvee_{W \in \mathcal{W}_F} \bigwedge_{a \in W} \varphi(a, y).$$

The family $\mathcal{W}_F$ consists of subsets of the finite set F, so the family is finite and the formula $d_F\varphi$ is a positive Boolean combination of instances of φ over M. In words, the formula $d_F\varphi(y)$ says that, for some $b' \in D_p$, every $a \in F$ with $M \models \varphi(a, b')$ also satisfies $\varphi(a, y)$. Every parameter $b \in D_p$ satisfies $d_F\varphi$, whatever F is (take $b' = b$). The task is to choose F in such a way that no parameter outside D_p satisfies $d_F\varphi$, and it is in this choice that the absence of the order property is used. Indeed, as we show in the proof of Theorem 2.2.1, if every finite F failed, then the model M would contain ladders of every length.

Theorem 2.2.1. *Assume that $\varphi(x, y)$ does not have the order property, let M be a model of T, and let $p \in S_\varphi(M)$ be given. Then there is a finite set $F \subseteq M$ such that the formula $d_F\varphi$ defined before the theorem is a φ-definition of p.*

Proof. For $b \in M$ we shall write $S_b = \{a \in M : M \models \varphi(a, b)\}$. The only property of p we use is that p is finitely satisfiable in M (see Lemma 1.1.2).

☺ **Claim.** There is a finite $F \subseteq M$ such that $S_b \cap F \not\subseteq S_{b'}$ for all $b \in D_p$ and $b' \in M \setminus D_p$.

Granting the claim, we check that $d_F\varphi$ defines p; to this end, let $b \in M$ be given.

(a) If $b \in D_p$, then the set $W := S_b \cap F$ belongs to $\mathcal{W}_F$ and satisfies $W \subseteq S_b$, so we have $M \models \bigwedge_{a\in W} \varphi(a,b)$. Hence the formula $d_F\varphi(b)$ holds in M.

(b) If $b \notin D_p$ and $W \in \mathcal{W}_F$, with $W = S_{b_0} \cap F$ for some $b_0 \in D_p$, then the claim gives $W \not\subseteq S_b$. Thus some $a \in W$ has $M \models \neg\varphi(a,b)$, and hence the conjunction $\bigwedge_{a\in W} \varphi(a,b)$ fails in M. As this holds for every $W \in \mathcal{W}_F$, we conclude that $M \models \neg d_F\varphi(b)$.

☺ **Proof of the claim.** Suppose that the claim fails. Then for every finite $F \subseteq M$ there are $b \in D_p$ and $b' \in M \setminus D_p$ with $S_b \cap F \subseteq S_{b'}$, that is, with $M \models \varphi(a,b) \to \varphi(a,b')$ for all $a \in F$. Letting the failure of the claim and the finite satisfiability of p take turns, we build a configuration in which Ramsey's theorem will find a ladder (see Figure 2.2). We choose recursively, for $i < \omega$, elements $b_i \in D_p$, $b'_i \in M \setminus D_p$ and $a_i \in M$ as follows.

(a) The failure of the claim, applied to $F = \{a_j : j < i\}$, provides $b_i \in D_p$ and $b'_i \in M \setminus D_p$ with

$$M \models \varphi(a_j, b_i) \to \varphi(a_j, b'_i) \qquad \text{for all } j < i. \tag{2.1}$$

(b) Since $\varphi(x, b_k) \in p$ and $\neg\varphi(x, b'_k) \in p$ for $k \le i$, Lemma 1.1.2 gives $a_i \in M$ with

$$M \models \varphi(a_i, b_k) \wedge \neg\varphi(a_i, b'_k) \qquad \text{for all } k \le i. \tag{2.2}$$

The pairs $j < i$ are colored by the truth values $\big(\varphi(a_j,b_i), \varphi(a_j,b'_i)\big) \in \{0,1\}^2$. By (2.1), the color $(1,0)$ never occurs. Ramsey's theorem now provides an infinite set $H = \{h_0 < h_1 < \cdots\} \subseteq \omega$ on which all pairs have the same color c, and two cases remain. In each of them, let $s, t < \omega$ be given.

☺ **Case $c \in \{(0,1),(1,1)\}$.** If $s < t$, then homogeneity gives $M \models \varphi(a_{h_s}, b'_{h_t})$; if $s \ge t$, then condition (2.2) yields $M \models \neg\varphi(a_{h_s}, b'_{h_t})$. Thus the formula $\varphi(a_{h_s}, b'_{h_t})$ holds if and only if $s < t$, and the shifted parameters $\beta_t = b'_{h_{t+1}}$ satisfy $M \models \varphi(a_{h_s}, \beta_t) \iff s \le t$.

☺ **Case $c = (0,0)$.** This time homogeneity forces $M \models \neg\varphi(a_{h_s}, b_{h_t})$ for $s < t$, and condition (2.2) forces $M \models \varphi(a_{h_s}, b_{h_t})$ for $s \ge t$. Thus the formula $\varphi(a_{h_s}, b_{h_t})$ holds if and only if $s \ge t$. For a given $n \ge 1$, reversing the first n indices by letting $\alpha_s = a_{h_{n-1-s}}$ and $\beta_t = b_{h_{n-1-t}}$ gives $M \models \varphi(\alpha_s, \beta_t) \iff s \le t$ for $s, t < n$.

In both cases the model M contains an n-ladder for every n, so the formula φ has the order property, a contradiction. This proves the claim and the theorem. ☺

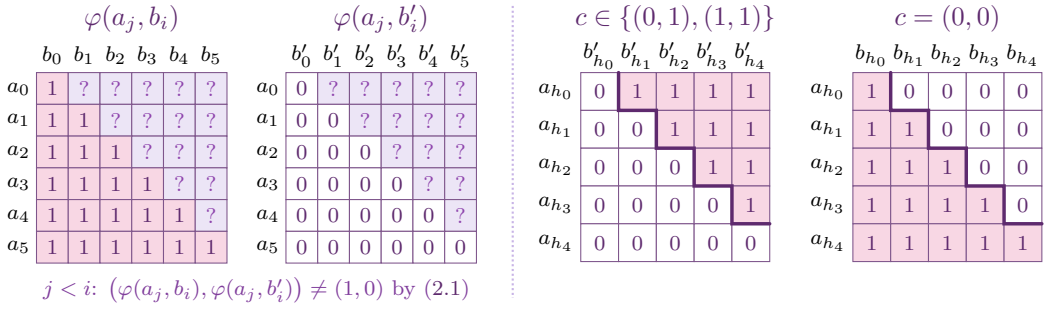


FIGURE 2.2. The configuration built in the proof of the claim in Theorem 2.2.1: the tables forced by (2.2) (left) and the two homogeneous cases on the set H given by Ramsey's theorem (right).

Since the model M and the type p in Theorem 2.2.1 are arbitrary, the theorem gives the implication (1) ⇒ (2) of Theorem 1.1.6. We record it, together with its consequence for quantifier-free formulas.

Corollary 2.2.2. *If φ does not have the order property, then for every model M of T, every $p \in S_\varphi(M)$ is definable by a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$. In particular, if φ is quantifier-free, then every such type has a quantifier-free φ-definition.*

The proof of Theorem 2.2.1 is qualitative: it applies Ramsey's theorem for pairs to an infinite sequence, inside an argument by contradiction. A finite argument of Ramsey type, adapted to binary trees rather than to sequences, appears in Section 3.1, where it extracts ladders of prescribed length directly from trees of explicit height.

2.3 From the order property to tall trees

To prove (3) ⇒ (1) of Theorem 1.1.6, we need a supply of many φ-types over a small model, and binary trees (see Definition 2.1.2) are the natural source. The first step, taken in this section, is to grow tall trees out of long ladders, which is (2) ⇒ (3) of Theorem 2.1.3. The idea is to place the nodes and the leaves of a binary tree on a line, crosswise. In this arrangement, the leaves above the right edge out of a node lie to the left of the node, and the leaves above the left edge lie to the right of the node. The crossing is dictated by the ladder: an element of the first sequence of the ladder and an element of the second satisfy φ exactly when the first lies to the left of the second. In Proposition 2.3.1 below we show that a ladder read along this line then has exactly the pattern of a tree.

Proposition 2.3.1. *Let $n < \omega$ be given. If a model M of T contains an N-ladder for φ, where $N = 2^{n+1} - 1$, then it contains a tree of height n for φ. Hence, if φ has the order property, then the formula φ encodes binary trees of arbitrarily large finite height.*

Proof. The position of a sequence $\sigma \in 2^{\leq n}$ on the line is the number

$$d(\sigma) = \sum_{k<|\sigma|} (1 - \sigma(k))\, 2^{-(k+1)} + 2^{-(|\sigma|+1)} \in (0, 1).$$

The binary expansion of $d(\sigma)$ is $(1 - \sigma(0), \dots, 1 - \sigma(|\sigma| - 1), 1)$, which ends in 1. Thus the map d is injective on $2^{\leq n}$, and the image V of d has exactly $|2^{\leq n}| = 2^{n+1} - 1 = N$ elements.

☺ **Claim.** If $\sigma \in 2^{<n}$, $u \in \{0, 1\}$ and $\eta \in 2^n$ satisfy $\sigma^\frown\langle u\rangle \sqsubseteq \eta$, then the inequality $d(\eta) < d(\sigma)$ holds if and only if $u = 1$.

Indeed, let $k = |\sigma|$; since $\eta|k = \sigma$ and $\eta(k) = u$, we have

$$d(\eta) - d(\sigma) = -u\, 2^{-(k+1)} + R, \qquad R = \sum_{k<j<n} (1 - \eta(j))\, 2^{-(j+1)} + 2^{-(n+1)}.$$

Here the number R satisfies $0 < R \leq \left(2^{-(k+1)} - 2^{-n}\right) + 2^{-(n+1)} < 2^{-(k+1)}$, and hence the difference $d(\eta) - d(\sigma)$ is negative if $u = 1$ and positive if $u = 0$, which proves the claim.

Now let $a_1, \dots, a_N, b_1, \dots, b_N \in M$ form an N-ladder, that is, satisfy $\varphi(a_i, b_j) \iff i \leq j$, and let $e\colon V \to \{1, \dots, N\}$ be the order isomorphism. We index the ladder by the tree, setting $b_\sigma := b_{e(d(\sigma))}$ for $\sigma \in 2^{<n}$ and $a_\eta := a_{e(d(\eta))}$ for $\eta \in 2^n$. If $\sigma^\frown\langle u\rangle \sqsubseteq \eta$, then the points $d(\eta)$ and $d(\sigma)$ are distinct, because d is injective, and hence

$$M \models \varphi(a_\eta, b_\sigma) \quad \text{if and only if} \quad e(d(\eta)) \leq e(d(\sigma)) \quad \text{if and only if} \quad d(\eta) < d(\sigma) \quad \text{if and only if} \quad u = 1.$$

Thus the families (b_σ) and (a_η) form a tree of height n in M. The second statement follows, since a formula with the order property has, for every N, an N-ladder in some model of T. ☺

The construction for $n = 2$ is drawn in Figure 2.3.

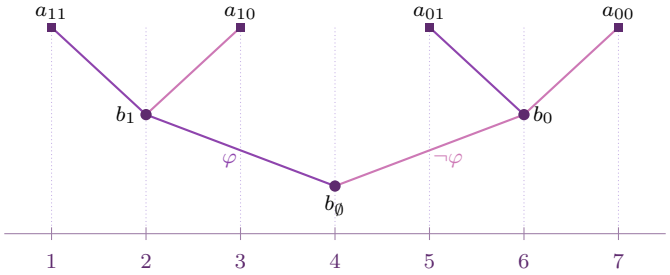


FIGURE 2.3. The proof of Proposition 2.3.1 for $n = 2$: a tree of height 2, growing upward on the positions $1, \dots, 7$ of a ladder of length 7, with the φ-edges in purple and the $\neg\varphi$-edges in pink.

2.4 Tall trees produce non-definable types

Tall trees produce more φ-types over a small model than there are formulas to define them, and this proves (3) $\Rightarrow$ (4) of Theorem 2.1.3. The count itself is carried out in Proposition 2.4.2 below. To prepare for it, in Lemma 2.4.1 below we use compactness to turn finite trees into an infinite tree of parameters, which has no leaves but whose branches will give the types. When L is countable, the count is easiest to see over a countable model, and this case is the picture to keep in mind. A tree of parameters of height ω has $2^{\aleph_0}$ branches, each of which gives a φ-type over a countable model, and only countably many formulas are available to define these types. In Section 2.5 the same tree reappears as a Cantor scheme of clopen sets in the space of φ-types over a countable model. Through this scheme, the space of φ-types maps onto the Cantor space.

Lemma 2.4.1 (Compactness). *Suppose that φ encodes binary trees of arbitrarily large finite height, and let μ be an infinite cardinal. Then there are a model M' of T and tuples $(b_s)_{s\in 2^{<\mu}}$ in M' such that for every $\eta \in 2^\mu$ the set*

$$\Sigma_\eta(x) = \{\varphi(x, b_{\eta|\alpha})^{\eta(\alpha)} : \alpha < \mu\}$$

is finitely satisfiable in M'.

Proof. In the language expanded by new tuples of constants c_s ($s \in 2^{<\mu}$), let Γ be T together with the sentences

$$\gamma_{\eta,F} := \exists x \bigwedge_{\alpha\in F} \varphi(x, c_{\eta|\alpha})^{\eta(\alpha)} \qquad (\eta \in 2^\mu,\ F \subseteq \mu \text{ finite and nonempty}).$$

By compactness, we need only show that every finite $\Gamma_0 \subseteq \Gamma$ has a model.

Let γ_{η_i,F_i} ($i < m$) be the sentences of this form in Γ_0 (if $m = 0$ there is nothing to do). Only finitely many nodes of $2^{<\mu}$ are mentioned in these sentences, and the plan is to embed them into a finite tree, preserving the direction in which each branch leaves each node. Let S be the finite set of all $\eta_i|\alpha$ and $\eta_i|(\alpha+1)$ with $i < m$ and $\alpha \in F_i$. These sequences lie in

$2^{<\mu}$, because the infinite cardinal μ is a limit ordinal. Let $S' = \{s \wedge s' : s, s' \in S\}$, where $s \wedge s'$ is the longest common initial segment. The set S' is finite and closed under $\wedge$, because in a tree the meet $s \wedge s' \wedge s''$ equals one of the pairwise meets. Thus the set S' has a least element r, and every $s \in S' \setminus \{r\}$ has a longest predecessor $s^- \in S'$ with $s^- \sqsubset s$.

We define $f\colon S' \to 2^{<N}$, where $N = |S'|+1$, by recursion on length, setting $f(r) = \emptyset$ and $f(s) = f(s^-)^\frown\langle s(|s^-|)\rangle$. Thus the sequence $f(s)$ records, for each predecessor t of s in S', the direction $s(|t|)$ in which s leaves t. In particular, every value of f has length less than N.

☺ **Claim.** If $s, s' \in S'$ and $s \sqsubset s'$, then we have $f(s)^\frown\langle s'(|s|)\rangle \sqsubseteq f(s')$.

The proof is by induction on $|s'|$. If $s'^- = s$, this is the definition of f. Otherwise, the maximality of s'^- gives $s \sqsubset s'^- \sqsubset s'$, and the induction hypothesis yields $f(s)^\frown\langle s'^-(|s|)\rangle \sqsubseteq f(s'^-) \sqsubseteq f(s')$. It remains only to note that $s'^-(|s|) = s'(|s|)$, because s'^- is an initial segment of s' longer than s.

By hypothesis, some model M_0 of T contains a tree $(b_\sigma)_{\sigma\in 2^{<N}}$, $(a_\nu)_{\nu\in 2^N}$ of height N. In M_0 we interpret c_s as $b_{f(s)}$ for $s \in S'$, and arbitrarily otherwise. Fix $i < m$, let $\alpha^* = \max F_i$ and $s^* = \eta_i|(\alpha^*+1) \in S'$. We pick any $\nu \in 2^N$ extending $f(s^*)$. Each $\alpha \in F_i$ satisfies $\eta_i|\alpha \sqsubset s^*$ and $s^*(\alpha) = \eta_i(\alpha)$, and so the claim gives $f(\eta_i|\alpha)^\frown\langle\eta_i(\alpha)\rangle \sqsubseteq f(s^*) \sqsubseteq \nu$. By the definition of a tree, it follows that $M_0 \models \varphi(a_\nu, b_{f(\eta_i|\alpha)})^{\eta_i(\alpha)}$. Hence the tuple a_ν witnesses γ_{η_i,F_i}, and the structure M_0 with this interpretation is a model of Γ_0.

Thus every finite subset of Γ has a model, and by compactness so does Γ. Let M' be the L-reduct of a model of Γ, and for each $s \in 2^{<\mu}$ let b_s be the interpretation of c_s in this model; then M' and the tuples b_s are as required. ☺

The count in the next proposition compares the number of φ-types with the number of formulas available to define them, and the number of L-formulas is $|L| + \aleph_0$.

Proposition 2.4.2. *Let $\kappa = |L| + \aleph_0$. If φ encodes binary trees of arbitrarily large finite height, then for every cardinal $\lambda \geq \kappa$ there is a model M of T with $|M| \leq \lambda$ and $|S_\varphi(M)| > \lambda$. In particular, there are a model M of T with $|M| \leq \kappa$ and a type $p \in S_\varphi(M)$ such that p is not definable.*

Proof. We need a tree with at most λ nodes and more than λ branches. Let μ be the least cardinal with $2^\mu > \lambda$. Then the cardinal μ is infinite, because λ is, and satisfies $\mu \leq \lambda$, because $2^\lambda > \lambda$. Since $2^\nu \leq \lambda$ for every $\nu < \mu$, we have $|2^{<\mu}| = \sum_{\nu<\mu} 2^\nu \leq \mu \cdot \lambda = \lambda$.

For this μ, Lemma 2.4.1 supplies a model M' and tuples $(b_s)_{s\in 2^{<\mu}}$. By the downward Löwenheim–Skolem theorem, there is $M \preceq M'$ with $|M| \leq \lambda$ containing all b_s. For each $\eta \in 2^\mu$, every finite part of the set Σ_η of Lemma 2.4.1 is satisfiable in M', hence in M, since the formula $\exists x \bigwedge \ldots$ is an $L(M)$-sentence. Thus each set Σ_η is consistent with the elementary diagram of M and extends to some $p_\eta \in S_\varphi(M)$.

If $\eta \neq \eta'$, let α be least with $\eta(\alpha) \neq \eta'(\alpha)$. The types p_η and $p_{\eta'}$ then disagree on $\varphi(x, b_s)$, where $s = \eta|\alpha = \eta'|\alpha$. Hence the space $S_\varphi(M)$ has at least 2^μ elements, a cardinal greater than λ.

For the last statement, we take $\lambda = \kappa$. A definable type is determined by any φ-definition of the type, and there are only $|L(M)| \leq \kappa$ formulas over M. Thus at most κ types in $S_\varphi(M)$ are definable, and some $p \in S_\varphi(M)$ is not. ☺

Together, Propositions 2.3.1 and 2.4.2 give the remaining implication of the definability theorem.

Proof of (3) ⇒ (1) *in Theorem 1.1.6.* If φ has the order property, then by Proposition 2.3.1 the formula φ encodes binary trees of arbitrarily large finite height. In that case Proposition 2.4.2 provides a φ-type over some model of T that is not definable. ☺

This completes the proof of Theorem 1.1.6, and with it that of Theorem 2.1.3. The equivalence of (2) and (3) in Theorem 2.1.3 is the form in which trees are used later (see Section 2.5), so we record it separately.

Theorem 2.4.3. *The formula φ has the order property if and only if the formula encodes binary trees of arbitrarily large finite height.*

Proof. The forward direction is the second statement of Proposition 2.3.1. Conversely, if φ encodes tall trees, then some φ-type is not definable (see Proposition 2.4.2), so the formula φ has the order property by (1) ⇒ (3) of Theorem 1.1.6. ☺

2.5 Types over countable models: a perfect set dichotomy

In Section 2.4 we detected instability by counting types: an unstable formula has, over some model, more types than there are formulas to define them. Over countable models the count has a topological counterpart: the shape of the type space separates the stable formulas from the unstable ones just as sharply.

Throughout this section the language L is countable. For a model M of T, the space $S_\varphi(M)$ is the Stone space of the Boolean algebra $\mathcal{B}_\varphi(M)$ generated by the sets $\varphi(M, b)$ with $b \in M^{|y|}$ (see Section 1.1). If M is countable, then the algebra $\mathcal{B}_\varphi(M)$ is countable, and therefore the space $S_\varphi(M)$ is compact, metrizable and zero-dimensional. Recall that a space is *scattered* if every nonempty subset contains a point isolated in that subset. By the Cantor–Bendixson theorem, a compact metrizable space either is countable, and then scattered, or contains a homeomorphic copy of the Cantor space 2^ω. In Theorem 2.5.1 below we show that stability decides which alternative occurs.

Theorem 2.5.1. *Assume that L is countable. The following are equivalent:*

(1) *The formula φ is stable.*

(2) *For every countable model M of T, the space $S_\varphi(M)$ is countable.*

(3) *For every countable model M of T, the space $S_\varphi(M)$ is scattered.*

Moreover, if φ is unstable, then for some countable model M of T the space $S_\varphi(M)$ has a closed subset homeomorphic to 2^ω.

Proof. (1) ⇒ (2). Let M be a countable model of T. By Theorem 1.1.6, every $p \in S_\varphi(M)$ has a φ-definition, which is an $L(M)$-formula, and the definition determines p. Since there are only countably many $L(M)$-formulas, the space $S_\varphi(M)$ is countable.

(2) ⇔ (3). This is the Cantor–Bendixson theorem for compact metrizable spaces.

(2) ⇒ (1) and the last statement. Suppose that φ is unstable; by Theorem 2.4.3, the formula φ encodes binary trees of arbitrarily large finite height. Hence Lemma 2.4.1, applied with $\mu = \omega$, gives a model M' of T and tuples $(b_\sigma)_{\sigma \in 2^{<\omega}}$ in M' such that every set $\Sigma_\eta(x)$ with $\eta \in 2^\omega$ is finitely satisfiable in M'. The downward Löwenheim–Skolem theorem then provides a countable model $M \preceq M'$ containing every b_σ, and by elementarity every $\Sigma_\eta(x)$ is finitely satisfiable in M. For $\sigma \in 2^{<\omega}$, let C_σ be the clopen set of types $p \in S_\varphi(M)$ that contain $\varphi(x, b_{\sigma|\alpha})^{\sigma(\alpha)}$ for every $\alpha < |\sigma|$. Each C_σ is nonempty, because the formulas

in question form a finitely satisfiable subset of Σ_η for any η extending σ. Moreover, the set C_σ is the disjoint union of $C_{\sigma^\frown\langle 0\rangle}$ and $C_{\sigma^\frown\langle 1\rangle}$, which disagree on $\varphi(x, b_\sigma)$. Thus the sets C_σ form a Cantor scheme. Since $C_\emptyset = S_\varphi(M)$, for each $n < \omega$ the sets C_σ with $|\sigma| = n$ partition $S_\varphi(M)$, and sending p to the unique η with $p \in C_{\eta|n}$ for every n defines a continuous map from $S_\varphi(M)$ to 2^ω. The map is onto, because each $\bigcap_n C_{\eta|n}$ is a decreasing intersection of nonempty closed sets in a compact space. Hence the space $S_\varphi(M)$ is uncountable, and (2) fails. Finally, being compact and metrizable, the space $S_\varphi(M)$ contains a closed copy of 2^ω by the Cantor–Bendixson theorem, which proves the last statement. ☺

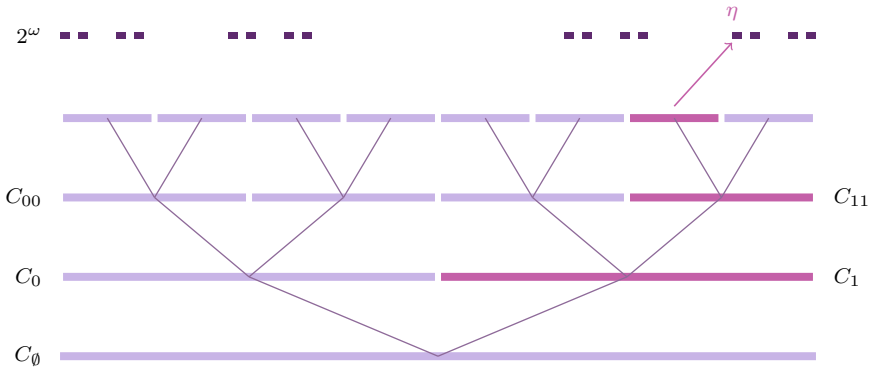


FIGURE 2.4. The Cantor scheme of clopen sets C_σ in the proof of Theorem 2.5.1, growing upward, with the pink bars $C_1 \supseteq C_{11} \supseteq C_{110} \supseteq \cdots$ following one branch η to its point in 2^ω, drawn at the top as a Cantor set.

The parameters that Lemma 2.4.1 provides for $\mu = \omega$ thus index a Cantor scheme of clopen sets in a type space (see Figure 2.4). The rank introduced in Section 3.2 below is a finite combinatorial cousin of the Cantor–Bendixson rank: it measures how many times in succession a finite conjunction of instances of φ and their negations can be split in two by instances of φ. Indeed, in Lemma 3.2.2(iii) we show that, for every $m < \omega$, a conjunction that splits into two pieces of rank at least m has rank at least $m + 1$.

2.6 EXERCISES

The exercises follow the sections of the chapter. Throughout, we write T_E for the theory of an equivalence relation E with infinitely many classes, all of them infinite, as in the exercises of Section 1.3. We also write θ for the damped ladder of those exercises, the function $\theta\colon \mathbb{N} \times \mathbb{N} \to [0, 1]$ given by $\theta(m, n) = 2^{-m-1}$ if $m \leq n$ and $\theta(m, n) = 0$ if $m > n$. For every finite 0–1 sequence σ we define $d(\sigma) = \sum_{k<|\sigma|}(1 - \sigma(k))\, 2^{-(k+1)} + 2^{-(|\sigma|+1)}$, as in the proof of Proposition 2.3.1. In the proof the sequence σ ranges over $2^{\leq n}$, but the number $d(\sigma)$ depends only on σ, not on n. The mark $(*)$ indicates the exercises that ask for more than a direct application of the text.

1. *(Section 2.1.)* Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$.

(a) Take the nodes $b_\emptyset = 4$, $b_1 = 2$ and $b_0 = 6$ and the leaves $a_{11} = 1$, $a_{10} = 3$, $a_{01} = 5$ and $a_{00} = 7$. Check that these elements form a tree of height 2 for φ, and draw it in the manner of Figure 2.1.

(b) Find a tree of height 3 for φ whose fifteen elements are the integers $1, \ldots, 15$. (Hint: place the root at 8, and place the leaves above the right edge out of each node to the left of that node.)

(c) Let $(I, <)$ be a linear order, let $n \geq 1$ be given, and let (b_σ), (a_η) be a tree of height n for $x < y$ in $(I, <)$. Show that, for every $\sigma \in 2^{<n}$ and all $\eta, \eta' \in 2^n$ with $\sigma^\frown\langle 1\rangle \sqsubseteq \eta$ and $\sigma^\frown\langle 0\rangle \sqsubseteq \eta'$, we have $a_\eta < b_\sigma \leq a_{\eta'}$. Deduce that the leaves are pairwise distinct, and that $a_{\nu'} < a_\nu$ whenever $\nu, \nu' \in 2^n$ and ν precedes ν' in the lexicographic order. Thus the arrangement of part (a), in which the leaves turning right lie to the left, is forced.

2. *(Section 2.1.)*

(a) Let $n < \omega$ be given, and let (b_σ), (a_η) be a tree of height n for φ in a model M of T. For a finite 0–1 sequence σ, let $\bar{\sigma}$ be the sequence of the same length with $\bar{\sigma}(k) = 1 - \sigma(k)$ for every $k < |\sigma|$. Show that the families $(b_{\bar{\sigma}})$ and $(a_{\bar{\eta}})$ form a tree of height n for $\neg\varphi$ in M. Conclude that φ and $\neg\varphi$ admit trees of the same heights in M.

(b) Let M be an infinite set, regarded as a structure in the empty language, and let $\varphi(x, y)$ be the formula $x = y$. Show that M contains a tree of height 1 for φ and no tree of height 2. (Hint: in a tree of height 2, the leaf a_{11} equals both $b_\emptyset$ and b_1.)

(c) Let M be a model of T_E. Show that the largest height of a tree for xEy in M is 1.

(d) In the setting of part (b), show that the largest m such that M contains an m-ladder is 1 for the formula $x = y$ and 2 for the formula $x \neq y$. Thus the length of the longest ladder may change when φ is replaced by $\neg\varphi$, whereas by part (a) the height of the tallest tree does not.

3. *(Section 2.1.)* Let K be a finite field, let T be the theory of infinite vector spaces over K, and let $n \geq 1$ be given. As in the exercises of Section 1.3, let $\varphi(x_1, \ldots, x_n; y)$ be the formula which says that y lies in the span of $x_1, \ldots, x_n$. Let M be a model of T. Show that the largest height of a tree for φ in M is n. (Hint: in a tree of height h, for $\sigma \in 2^{<h}$, let U_σ be the span of the nodes b_τ with $\tau^\frown\langle 1\rangle \sqsubseteq \sigma$. Show that $b_\sigma \notin U_\sigma$ by looking at a leaf above $\sigma^\frown\langle 0\rangle$, and follow the branch that always turns right. For the construction, take the nodes linearly independent.) Compare with the exercises of Section 1.3, where the longest ladder for φ in M has length $n + 1$.

4. *(Section 2.2.)* Let M be a model of T, let $p \in S_\varphi(M)$ be given, and recall that $D_p = \{b \in M : \varphi(x, b) \in p\}$. As in Section 2.2, for every finite set $F \subseteq M$ let $\mathcal{W}_F$ be the family of the sets $\{a \in F : M \models \varphi(a, b)\}$ with $b \in D_p$, and let $d_F\varphi(y)$ be the formula $\bigvee_{W \in \mathcal{W}_F} \bigwedge_{a \in W} \varphi(a, y)$.

(a) Let $F \subseteq F'$ be finite, and show that every $b \in M$ with $M \models d_{F'}\varphi(b)$ also satisfies $M \models d_F\varphi(b)$. Deduce that, if $d_F\varphi$ defines p, then so does $d_{F'}\varphi$.

(b) Show that $d_F\varphi$ defines p if and only if F has the property stated in the claim in the proof of Theorem 2.2.1.

(c) Let M be an infinite set in the empty language, and let $\varphi(x, y)$ be the formula $x = y$. Let $c \in M$ be given, and let p be the type with $D_p = \{c\}$. Compute $\mathcal{W}_F$ and $d_F\varphi$ for

every finite $F \subseteq M$, and show that $d_F\varphi$ defines p if and only if $c \in F$. Do the same for the type with $D_p = \emptyset$.

(d) Let M be a model of T_E, let $\varphi(x, y)$ be the formula xEy, and let p be a type with D_p an E-class C. Show that $\mathcal{W}_F = \{F \cap C\}$, and that $d_F\varphi$ defines p if and only if F meets C. Determine the least size of a set F that works.

(e) In the setting of Exercise 3, let $p \in S_\varphi(M)$ be given. Show that there is a single n-tuple $c \in M^n$ such that $d_{\{c\}}\varphi$ is the formula $\varphi(c, y)$ and defines p. (Hint: the set D_p is a subspace of M of dimension at most n (see the exercises of Section 1.3).)

5. *(Section 2.2.)* Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$. Let r be an irrational real number, and let $p \in S_\varphi(\mathbb{Q})$ be the type with $D_p = \{b \in \mathbb{Q} : b > r\}$.

(a) Let $F \subseteq \mathbb{Q}$ be finite, and write $F_r = \{a \in F : a < r\}$. Show that every member of $\mathcal{W}_F$ contains F_r, and that $F_r \in \mathcal{W}_F$. Deduce that the formula $d_F\varphi(y)$ is equivalent in $(\mathbb{Q}, <)$ to $\bigwedge_{a \in F_r} a < y$, which defines the set $\{b : b > \max F_r\}$ when F_r is nonempty and the set $\mathbb{Q}$ otherwise.

(b) Conclude that no finite $F \subseteq \mathbb{Q}$ makes $d_F\varphi$ a definition of p.

(c) By part (b), the claim in the proof of Theorem 2.2.1 fails for p. Let $(c_i)_{i<\omega}$ be a strictly increasing sequence of rationals below r, and let c be a rational above r. Show that the choices $b_i = c$, $b'_i = c_i$ and $a_i = c_i$ satisfy conditions (2.1) and (2.2) of the proof. Determine the color of each pair, and the ladder that the proof extracts.

Compare with the exercises of Section 1.3 on $(\mathbb{Q}, <)$, where the same type is shown not to be definable at all.

6. *(Section 2.3.)* Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$, and let $a_i = i$ and $b_j = j + \frac{1}{2}$ for $1 \leq i, j \leq 7$.

(a) Show that these elements form a 7-ladder for φ.

(b) Carry out the proof of Proposition 2.3.1 with $n = 2$ and this ladder. Compute $d(\sigma)$ for $\sigma \in 2^{\leq 2}$, the index $e(d(\sigma))$, and the elements b_σ and a_η that the proof assigns. Here, as in the proof, the map e is the order isomorphism from the image of d on $2^{\leq 2}$ onto $\{1, \ldots, 7\}$. Check the tree conditions directly, and compare with Figure 2.3.

(c) Let $n < \omega$ be given, and let V be the image of d on $2^{\leq n}$. Show that $V = \{k\, 2^{-(n+1)} : 0 < k < 2^{n+1}\}$, and deduce that the order isomorphism $e \colon V \to \{1, \ldots, 2^{n+1} - 1\}$ satisfies $e(d(\sigma)) = 2^{n+1} d(\sigma)$. Show that this index is odd if σ is a leaf and even if σ is a node, and that the root receives the index 2^n. Identify the elements of the ladder that the proof leaves unused.

7. $(*)$ *(Section 2.3.)* By Proposition 2.3.1, a ladder of length $2^{n+1} - 1$ can be turned into a tree of height n, for every $n \geq 1$. This exercise shows that, for every such n, length 2^n suffices, and that length $2^n - 2$ does not suffice in general.

(a) Let $n \geq 1$ be given, and let $a_1, \ldots, a_{2^n}$, $b_1, \ldots, b_{2^n}$ be a 2^n-ladder for φ in a model M of T. With $k_\sigma = 2^{n+1} d(\sigma)$ for $\sigma \in 2^{\leq n}$, show that the elements $b_\sigma := b_{k_\sigma/2}$ for $\sigma \in 2^{<n}$ and $a_\eta := a_{(k_\eta+1)/2}$ for $\eta \in 2^n$ form a tree of height n in M. (Hint: use Exercise 6(c), and note that $k_\eta < k_\sigma$ if and only if $k_\eta + 1 \leq k_\sigma$.)

(b) Let $k \in \mathbb{N}$ be given, and let R_k be the binary relation on $\mathbb{N}$ given by $R_k(m, m')$ if and only if $m \leq m'$ and $m \leq k$. Let T_k be the theory of $(\mathbb{N}, R_k)$, with $\varphi(x, y)$ the formula $R_k(x, y)$. Show that T_k implies the following two sentences: exactly $k+1$ elements x satisfy $R_k(x, x)$, and every x with $\neg R_k(x, x)$ satisfies $\neg R_k(x, y)$ for all y.

(c) Show that every model of T_k contains a $(k+1)$-ladder and no $(k+2)$-ladder for φ. Show that a model of T_k contains a tree of height h for φ if and only if $2^h \leq k+2$. (Hint: two distinct leaves of a tree disagree on some formula $\varphi(x, b_\sigma)$. By part (b), a leaf a with $\neg R_k(a, a)$ satisfies none of the formulas $\varphi(x, b_\sigma)$, and hence at most $k+2$ leaves are available. For the construction, work in $(\mathbb{N}, R_k)$, let $a_\eta = (k_\eta - 1)/2$ and $b_\sigma = k_\sigma/2 - 1$ with $n = h$, and note that the existence of a tree of height h is expressed by a sentence.)

(d) Let $N, n \geq 1$ be given, and deduce that the structure $(\mathbb{N}, R_{N-1})$ contains an N-ladder for φ and no tree of height n when $N < 2^n - 1$. Thus the length of the ladder needed for a tree of height n lies between $2^n - 1$ and 2^n.

(e) Show that $R_k(m, m')$ holds if and only if $\theta(m, m') \geq 2^{-k-1}$, and that $\theta(m, m') > 0$ holds if and only if $m \leq m'$. Conclude that the formula R_k, which defines the level set $\{\theta \geq 2^{-k-1}\}$ of the damped ladder, is stable in T_k and that the heights of trees for R_k tend to infinity with k. Conclude also that the formula $\theta > 0$ has the order property. Explain why these three conclusions are consistent with the stability of θ (see the exercises of Section 1.3).

8. *(Section 2.4.)* Let M be a model of T. We say that M contains a *tree of height* ω for φ if there are tuples $(b_\sigma)_{\sigma \in 2^{<\omega}}$ in M such that, for every $\eta \in 2^\omega$, the set $\Sigma_\eta(x) = \{\varphi(x, b_{\eta|k})^{\eta(k)} : k < \omega\}$ is finitely satisfiable in M. The sets Σ_η and their finite satisfiability are as in Lemma 2.4.1 with $\mu = \omega$.

(a) Show that, if M contains a tree of height ω, then the model M contains a tree of height n for every $n < \omega$, and the formula φ is unstable. (Hint: for $\eta \in 2^n$, choose $\eta' \in 2^\omega$ extending η and let $a_\eta \in M$ realize the finite part $\{\varphi(x, b_{\eta|k})^{\eta(k)} : k < n\}$ of $\Sigma_{\eta'}$; for instability, use Theorem 2.4.3.)

(b) Assume that L is countable, and show that φ is unstable if and only if some countable model of T contains a tree of height ω for φ. (Hint: Lemma 2.4.1 and the downward Löwenheim–Skolem theorem.)

(c) Show that, if M is countable and contains a tree of height ω, then $|S_\varphi(M)| = 2^{\aleph_0}$. Conclude that a countable model with only countably many φ-types contains no tree of height ω (for the converse, see Exercise 15).

9. *(Section 2.4.)* Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$, and let $b_\sigma = d(\sigma)$ for $\sigma \in 2^{<\omega}$.

(a) Show that the set $\{b_\sigma : \sigma \in 2^{<\omega}\}$ is the set of dyadic rationals in the interval $(0, 1)$, which is countable and dense in $(0, 1)$. (Hint: Exercise 6(c).)

(b) Show that the elements b_σ form a tree of height ω in $\mathbb{Q}$ (see Exercise 8), without any appeal to compactness. (Hint: for $\eta \in 2^\omega$ and $n < \omega$, the claim in the proof of Proposition 2.3.1 shows that $d(\eta|n)$ satisfies $\varphi(x, b_{\eta|k})^{\eta(k)}$ for all $k < n$.)

(c) Deduce that $|S_\varphi(\mathbb{Q})| = 2^{\aleph_0}$, and that already the instances $\varphi(x, b)$ with b a dyadic rational in $(0, 1)$ separate $2^{\aleph_0}$ types. Conclude that some φ-type over $\mathbb{Q}$ is not definable, without using the description of definable sets of $(\mathbb{Q}, <)$.

(d) For $\eta \in 2^\omega$, let $r_\eta = \sum_{k<\omega}(1 - \eta(k))\, 2^{-(k+1)}$, and recall from Exercise 8 the set $\Sigma_\eta = \{\varphi(x, b_{\eta|k})^{\eta(k)} : k < \omega\}$. Assume that η is not eventually constant. Show that $r_\eta < b_{\eta|k}$ if $\eta(k) = 1$ and $r_\eta > b_{\eta|k}$ if $\eta(k) = 0$, and that every $p \in S_\varphi(\mathbb{Q})$ containing Σ_η contains $x < b$ for each rational $b > r_\eta$ and $\neg(x < b)$ for each rational $b < r_\eta$. Determine the types that contain Σ_η when r_η is irrational.

10. $(*)$ *(Section 2.4.)* This exercise shows that the passage to a new model in Lemma 2.4.1 cannot be avoided. For $N \geq 1$, let C_N be a chain of N elements. Let M be the disjoint union of the chains C_N, ordered by letting $c < c'$ hold exactly when c and c' lie in the same chain and c precedes c' there. Let $\varphi(x, y)$ be the formula $x < y$, and let T be the theory of M.

(a) Show that M contains a tree of height n for φ for every $n < \omega$. (Hint: Exercise 1.)

(b) Let $(b_\sigma)_{\sigma \in 2^{<\omega}}$ be a tree of height ω in M (see Exercise 8). Show that every node b_σ with $\sigma \neq \emptyset$ and $\sigma(0) = 1$ lies in the chain of $b_\emptyset$. Show also that, for every $n < \omega$, the 2^n leaves a_η with $\eta(0) = 1$ of the tree of height $n + 1$ given by Exercise 8(a) are pairwise distinct and lie in the chain of $b_\emptyset$. Conclude that M contains no tree of height ω.

Since M contains trees of every finite height, Lemma 2.4.1 can be applied to the elementary diagram of M in place of T. The lemma then gives an elementary extension M' of M that contains a tree of height ω. Show that, in such an M', the set of elements comparable with the root $b_\emptyset$ of such a tree is infinite, although in M every such set is finite.

11. *(Section 2.4.)* This exercise concerns the choice of μ in the proof of Proposition 2.4.2. Assume that L is countable, and hence that the cardinal $\kappa = |L| + \aleph_0$ of the proposition equals $\aleph_0$.

(a) Show that, for $\lambda = \aleph_0$, the proof takes $\mu = \aleph_0$ and produces a countable model M with $|S_\varphi(M)| = 2^{\aleph_0}$.

(b) Let $\lambda = \aleph_1$, and assume that $2^{\aleph_0} > \aleph_1$. Show that the proof takes $\mu = \aleph_0$, and that the set $2^{<\omega_1}$ has more than $\aleph_1$ elements. Thus, with $\mu = \lambda$, the bound $|2^{<\mu}| \leq \lambda$ used in the proof would fail.

(c) Let $\lambda = \aleph_1$, and assume that $2^{\aleph_0} = \aleph_1$. Show that the proof takes $\mu = \aleph_1$, and that $|2^{<\omega_1}| = \aleph_1$.

(d) Assume that φ encodes binary trees of arbitrarily large finite height. Deduce from the proposition that, for every cardinal $\lambda \geq \kappa$, some model M of T with $|M| \leq \lambda$ carries a φ-type that is not definable.

12. *(Section 2.5.)* This exercise computes Cantor–Bendixson ranks in two simple stable examples, and shows why Theorem 2.5.1 is confined to countable models. For a topological space X, let X' be the set of points of X that are not isolated in X. Define $X^{(0)} = X$, $X^{(\alpha+1)} = (X^{(\alpha)})'$, and $X^{(\delta)} = \bigcap_{\alpha<\delta} X^{(\alpha)}$ for limit δ. The *Cantor–Bendixson rank* of a point $p \in X$ is the ordinal α with $p \in X^{(\alpha)} \setminus X^{(\alpha+1)}$, when such an ordinal exists. Recall

that X is scattered if every nonempty subset of X has a point that is isolated in that subset.

(a) Let M be a countably infinite set in the empty language, and let $\varphi(x, y)$ be the formula $x = y$. Using the description of $S_\varphi(M)$ as a one-point compactification in the exercises of Section 1.3, show that the realized types have rank 0, that the remaining type has rank 1, and that $S_\varphi(M)^{(2)} = \emptyset$.

(b) Do the same for a countable model of T_E and the formula xEy.

(c) Let M be an uncountable set in the empty language, with φ the formula $x = y$. Show that $S_\varphi(M)$ is uncountable and scattered. Explain why this does not contradict Theorem 2.5.1.

13. $(*)$ *(Section 2.5.)* Work in the setting of Exercise 3, and let M be a model of T. By the exercises of Section 1.3, the map $p \mapsto D_p$ is a bijection from $S_\varphi(M)$ onto the set of subspaces of M of dimension at most n. Show by induction on $j \leq n$ that $S_\varphi(M)^{(j)}$ consists of the types p with $\dim D_p \leq n - j$, and conclude that the Cantor–Bendixson rank of the type with $D_p = \{0\}$ is n and that $S_\varphi(M)^{(n+1)} = \emptyset$. (Hint: let p be a type with $\dim D_p = k$. The conjunction of the formulas $\varphi(x, c)$ with $c \in D_p$ defines a neighborhood of p in which every other type q satisfies $D_q \supsetneq D_p$. When $k < n-j$, a basic neighborhood of p is determined by finitely many formulas, among them finitely many $\neg\varphi(x, b)$ with $b \notin D_p$. Using that K and D_p are finite and M is infinite, find infinitely many vectors $e \in M$ for which the subspaces $D_p + Ke$ have dimension $k + 1$ and avoid these b.) Compare with Exercise 3: for this formula, the largest Cantor–Bendixson rank of a point of $S_\varphi(M)$ equals the largest height of a tree for φ.

14. *(Section 2.5.)* This exercise exhibits a copy of 2^ω inside $S_\varphi(\mathbb{Q})$ explicitly, and compares it with the map onto 2^ω in the proof of Theorem 2.5.1. Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$.

(a) For $\eta \in 2^\omega$, let $s_\eta = \sum_{k<\omega}(1 + \eta(k))\, 10^{-(k+1)!}$, and show that the map $\eta \mapsto s_\eta$ is continuous and injective, and that every s_η is irrational. (Hint: the decimal expansion of s_η has its nonzero digits exactly at the positions $(k + 1)!$.)

(b) Let $p_\eta \in S_\varphi(\mathbb{Q})$ be the type with $D_{p_\eta} = \{b \in \mathbb{Q} : b > s_\eta\}$. Show that the map $\eta \mapsto p_\eta$ is a homeomorphism of 2^ω onto a closed subset of $S_\varphi(\mathbb{Q})$.

(c) In the proof of Theorem 2.5.1, take $M = \mathbb{Q}$ and use the tree of Exercise 9 in place of the one obtained there. Show that the map from $S_\varphi(\mathbb{Q})$ onto 2^ω in the proof sends the two types with $D_p = \{b : b \geq \frac{1}{3}\}$ and $D_p = \{b : b > \frac{1}{3}\}$ to the same sequence. (Hint: $\frac{1}{3}$ has binary expansion $0.0101\ldots$.) Thus the map need not be injective, and so need not be a homeomorphism, which is why the proof appeals to the Cantor–Bendixson theorem.

15. $(*)$ *(Section 2.5.)* Assume that L is countable, and let M be a countable model of T. Show that $S_\varphi(M)$ is uncountable if and only if M contains a tree of height ω for φ (see Exercise 8). (Hint: by the Cantor–Bendixson theorem, an uncountable compact metrizable space contains a nonempty closed subset P without isolated points. Choose the nodes b_σ by recursion on $|\sigma|$ in such a way that the clopen set of types containing $\varphi(x, b_{\sigma|k})^{\sigma(k)}$ for all $k < |\sigma|$ meets P. Two distinct types in P differ on some instance $\varphi(x, b)$.) Deduce from Exercise 8(a) a second proof of the implication (1) $\Rightarrow$ (2) of

Theorem 2.5.1. This second proof shows that, if some countable model has uncountably many φ-types, then the formula φ encodes binary trees of arbitrarily large finite height, and hence is unstable. If Corollary 3.1.6 is used in place of Theorem 2.4.3, the second proof avoids type definability altogether.

2.7 Historical remarks

- **Section 2.2.** Definability of φ-types over models for formulas without the order property is due to Shelah [**She71**, **She90**]. The sharper form proved here, in which the definition is a positive Boolean combination of instances of φ itself, goes back to Hrushovski and Pillay [**HP94**]. See also Chapter 1 of Pillay's book on geometric stability theory [**Pil96**]. Pillay [**Pil17**, Remark 2.4] derives the positive form directly from Grothendieck's double limit criterion (see Theorem A.1.3 below) by a compactness argument. Positivity matters in applications: a quantifier-free formula receives quantifier-free definitions.
- **Section 2.3.** Binary trees enter stability theory through Shelah's local rank $R(-, \varphi, 2)$. In Lemma 2.7 of his 1971 paper [**She71**], Shelah computes the rank of a type from the consistency of families of formulas indexed by the branches of binary trees of finite height. The formulas for which this rank is infinite are thus exactly those that encode binary trees of arbitrarily large finite height (see Definition 2.1.2). Shelah showed that they are the formulas with the order property [**She71**, **She90**]. Quantitatively, each passage costs a logarithm. A ladder of length exponential in n yields a tree of height n (see Proposition 2.3.1). In the other direction, a tree yields a ladder whose length is only logarithmic in the height of the tree (see Section 3.1).
- **Section 2.4.** The construction of this section is the local form of Shelah's counting argument [**She71**]. A tree of parameters of height ω (see Exercise 8 of Section 2.6) gives $2^{\aleph_0}$ φ-types over a countable set of parameters. When the language is countable, so many types cannot all be definable. The same count is also the origin of the *stability spectrum*, the class of cardinals λ for which a theory is λ-stable. Being λ-stable means having at most λ types over every set of size λ. A theory with the order property has more than λ types over some set of size λ, for every cardinal $\lambda \geq |T|$, and hence the theory is λ-stable for no λ [**She90**]. For dense linear orders the number of types over a set of size λ is governed by Dedekind cuts. It is consistent that, for some uncountable λ, every linear order of size λ has fewer than 2^λ cuts, and hence the count cannot be raised to 2^λ in general.
- **Section 2.5.** The dichotomy behind this section is older than model theory. Cantor introduced derived sets in 1872, in his work on trigonometric series. In 1883 the young Swedish mathematician Ivar Bendixson proved that every uncountable closed set is the union of a perfect set and a countable set. Bendixson's proof appeared in a letter to Cantor published in *Acta Mathematica*. Morley [**Mor65**] carried the same analysis into model theory in 1965, in his proof of the categoricity theorem. He applied a derivative of Cantor–Bendixson type to spaces of types, and called a theory totally transcendental when the derivation exhausts them. For a countable theory, the type spaces over countable sets are then countable, and hence scattered. The rank that grew out of his analysis, Morley rank, is the Cantor–Bendixson rank in type spaces over $\aleph_0$-saturated models.

Chapter

3

More stability, trees, and definability

Roadmap

In Chapter 2 we showed that tall binary trees and long ladders occur together, and that in their absence every type is definable. In this chapter we make both results effective: we bound the height of a tree that forces a ladder of given length, and we read off the definition of a type from a rank. In Section 3.1 a Ramsey lemma for trees (see Lemma 3.1.3) shows that a binary tree of sufficiently large finite height in a model contains a ladder of any given length in that model. The passage from trees to ladders comes with an explicit bound on the height required (see Proposition 3.1.5 and Corollary 3.1.6). In Section 3.2 a rank measures the height of the binary trees that can be grown above a formula (see Lemma 3.2.2). The definition of a type is obtained from a formula of least rank in that type (see Proposition 3.2.3). With stability thus understood from inside a single formula, in Chapter 4 we turn to a weaker prohibition, under which ladders are allowed.

The notational conventions in this chapter are those of Chapter 2.

3.1 Trees give the order property: a quantitative proof by Ramsey's theorem

In Chapter 2 we derived Theorem 2.4.3 from the definability theorem; here we prove directly, and quantitatively, that tall trees force the order property. The proof is pure finite combinatorics: it is the 2-valued case of a Ramsey-type argument due to Hodges, in the presentation of Conant and Terry [**CT26**]. Finding a ladder inside a tree requires passing to smaller trees embedded in it, and the first definition says what such an embedding is. We recall the notation of Section 2.1: for a positive integer s, the sets $2^{<s}$ and 2^s consist of the 0–1 sequences of length less than s and of length s, and $2^{\leq s} = 2^{<s} \cup 2^s$. For a sequence ρ and $u \in \{0,1\}$, the relation $\rho^\frown\langle u\rangle \sqsubseteq \tau$ means that the extension of ρ by the entry u is an initial segment of τ.

Definition 3.1.1. Let $s, t \geq 1$ be given.

(a) A *node embedding* is a map $\iota\colon 2^{<s} \to 2^{<t}$ such that, for all $\rho, \tau \in 2^{<s}$ and $u \in \{0,1\}$, the relation $\rho^\frown\langle u\rangle \sqsubseteq \tau$ implies $\iota(\rho)^\frown\langle u\rangle \sqsubseteq \iota(\tau)$.

(b) A *full embedding* is a map $\iota\colon 2^{\leq s} \to 2^{\leq t}$ sending $2^{<s}$ into $2^{<t}$ and 2^s into 2^t, with the same property for all $\rho, \tau \in 2^{\leq s}$.

Lemma 3.1.2. *Let $s, t \geq 1$ be given. Every node embedding $\iota\colon 2^{<s} \to 2^{<t}$ extends to a full embedding. If (b_σ), (a_η) is a tree of height t for φ in a model M of T and ι is a full embedding $2^{\leq s} \to 2^{\leq t}$, then the families $(b_{\iota(\sigma)} : \sigma \in 2^{<s})$ and $(a_{\iota(\eta)} : \eta \in 2^s)$ form a tree of height s in M.*

Proof. To extend ι, let $\iota(\eta)$, for each $\eta \in 2^s$, be any element of 2^t extending $\iota(\eta|(s-1))^\frown\langle\eta(s-1)\rangle$. It remains to be shown that $\rho^\frown\langle u\rangle \sqsubseteq \eta$ implies $\iota(\rho)^\frown\langle u\rangle \sqsubseteq \iota(\eta)$, so suppose that $\rho^\frown\langle u\rangle \sqsubseteq \eta$. If $\rho = \eta|(s-1)$, the requirement holds by construction. Otherwise we have $\rho^\frown\langle u\rangle \sqsubseteq \eta|(s-1)$, and therefore $\iota(\rho)^\frown\langle u\rangle \sqsubseteq \iota(\eta|(s-1)) \sqsubseteq \iota(\eta)$. Pairs with $\tau \in 2^{<s}$ are handled by the node embedding itself, and no element of $2^{\leq s}$ properly extends a leaf, so the extended map is a full embedding. The second statement is immediate, since whether $\varphi(a_\eta, b_\sigma)$ holds depends only on the direction u with $\sigma^\frown\langle u\rangle \sqsubseteq \eta$, and full embeddings preserve this direction. ☺

To find a ladder, the proof of Proposition 3.1.5 below splits the nodes of a tree into two classes, and it needs a tall tree embedded in one of the classes. The next lemma, a Ramsey theorem for node embeddings, provides such an embedding; an instance of it is drawn in Figure 3.1. In its proof and in the rest of this chapter, for $u \in \{0,1\}$ and a finite 0–1 sequence σ, we write $\langle u\rangle^\frown\sigma$ for the sequence obtained by prefixing u to σ.

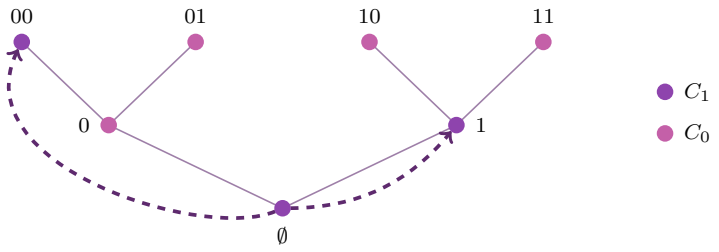


FIGURE 3.1. The nodes $2^{<3}$, growing upward, colored with two colors C_0 and C_1, and a node embedding $\iota\colon 2^{<2} \to 2^{<3}$ into C_1 (dashed arrows) that witnesses Lemma 3.1.3 for $t_0 = t_1 = 2$ and $t = 3$.

Lemma 3.1.3 (Ramsey lemma for trees). *Let $t_0, t_1 \geq 1$ be given, let $t = t_0 + t_1 - 1$, and suppose that $2^{<t} = C_0 \cup C_1$. Then for some $u \in \{0,1\}$ there is a node embedding $\iota\colon 2^{<t_u} \to 2^{<t}$ with image contained in C_u.*

Proof. We proceed by induction on t, looking at the class of the root and at the two copies of $2^{<t-1}$ that sit above the root, one through each child. Either one copy carries a node embedding of the required full height into the class not containing the root, or both copies carry node embeddings into the class of the root. In the latter case, these embeddings fall one level short of the height required of that class, and the root joins them into one. If $t = 1$, then we have $t_0 = t_1 = 1$, and the map $\emptyset \mapsto \emptyset$ works for the u with $\emptyset \in C_u$. Now suppose that $t > 1$, and let w be an index with $\emptyset \in C_w$. If $t_w = 1$, the map $\emptyset \mapsto \emptyset$ works, so assume that $t_w \geq 2$. For $v \in \{0,1\}$ and $i \in \{0,1\}$ let $C_i^v = \{\sigma \in 2^{<t-1} : \langle v\rangle^\frown\sigma \in C_i\}$. Then the union

$C_0^v \cup C_1^v$ covers $2^{<t-1}$, and we have $(t_w - 1) + t_{1-w} - 1 = t - 1$. The induction hypothesis, applied to the pair C_0^v, C_1^v with $t_w - 1$ in place of t_w, shows that for each v either

(i) there is a node embedding $\chi\colon 2^{<t_{1-w}} \to C_{1-w}^v$, in which case the map $\sigma \mapsto \langle v\rangle^\frown\chi(\sigma)$ is a node embedding $2^{<t_{1-w}} \to 2^{<t}$ with image in C_{1-w}, and we are done; or

(ii) there is a node embedding $\chi_v\colon 2^{<t_w-1} \to C_w^v$.

It remains to treat the case in which (ii) holds for both v. In this case the map $\iota\colon 2^{<t_w} \to 2^{<t}$ is given by $\iota(\emptyset) = \emptyset$ and $\iota(\langle v\rangle^\frown\sigma) = \langle v\rangle^\frown\chi_v(\sigma)$. The image of ι lies in C_w, because $\emptyset \in C_w$ and each χ_v takes values in C_w^v. Suppose that $\rho^\frown\langle u\rangle \sqsubseteq \tau$. If $\rho = \emptyset$, then the sequence τ has the form $\langle u\rangle^\frown\sigma$, whence $\emptyset^\frown\langle u\rangle \sqsubseteq \iota(\tau)$. If $\rho = \langle v\rangle^\frown\rho'$, then we have $\tau = \langle v\rangle^\frown\tau'$ with $\rho'^\frown\langle u\rangle \sqsubseteq \tau'$, and therefore $\chi_v(\rho')^\frown\langle u\rangle \sqsubseteq \chi_v(\tau')$. In both cases the relation $\iota(\rho)^\frown\langle u\rangle \sqsubseteq \iota(\tau)$ holds, so the map ι is a node embedding. ☺

The ladders found inside a tree come in two orientations, one for each of the two classes in which the Ramsey lemma may find its embedding. The induction in the proof of Proposition 3.1.5 keeps track of both orientations.

Definition 3.1.4. Let $t \geq 1$ be given, and let $(b_\sigma)_{\sigma\in 2^{<t}}$, $(a_\eta)_{\eta\in 2^t}$ be a tree for φ in a model M of T. For $k \geq 1$, an *up-ladder of length k* in this tree consists of nodes $\sigma_1, \ldots, \sigma_k$ and leaves $\eta_1, \ldots, \eta_k$ such that, for all $1 \leq p, q \leq k$, we have $M \models \varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$. For $\ell \geq 1$, a *down-ladder of length ℓ* is defined in the same way, with ℓ in place of k and "$p > q$" in place of "$p \leq q$".

The height needed to force a ladder is governed by a familiar recursion. Define $B(k, \ell)$ for $k, \ell \geq 1$ by $B(k, 1) = B(1, \ell) = 1$ and $B(k, \ell) = B(k-1, \ell) + B(k, \ell-1) + 1$ for $k, \ell \geq 2$. By induction on $k + \ell$, using Pascal's rule $\binom{k+\ell-3}{k-2} + \binom{k+\ell-3}{k-1} = \binom{k+\ell-2}{k-1}$, we have $B(k, \ell) = 2\binom{k+\ell-2}{k-1} - 1$. The numbers $B(k, \ell) + 1$ satisfy, with equality, the Erdős–Szekeres recursion for the Ramsey numbers $R(k, \ell)$, and the bound mirrors $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$. The proof below follows the Erdős–Szekeres pattern as well: we fix a leaf η_* and split the nodes according to whether a_{η_*} satisfies the corresponding instance of φ. The Ramsey lemma for trees (see Lemma 3.1.3) finds a large subtree whose nodes all lie on one side of the split, and induction finds a ladder inside the subtree. The pair formed by η_* and the root of the subtree extends this ladder by one rung (see Figure 3.2).

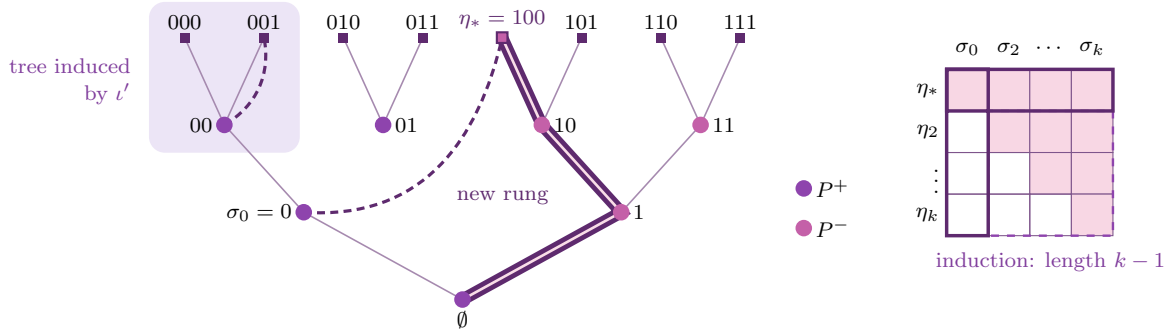


FIGURE 3.2. Case + in the proof of Proposition 3.1.5 for $k = \ell = 2$, in a tree (b_σ), (a_η) of height $B(2, 2) = 3$ for φ, with the new rung (σ_0, η_*) (left) and the table of $\varphi(a_{\eta_p}, b_{\sigma_q})$ gaining a first row and a first column (right).

Proposition 3.1.5. *For all $k, \ell \geq 1$, every tree of height $B(k, \ell)$ for φ in a model of T contains an up-ladder of length k or a down-ladder of length ℓ.*

Proof. We argue by induction on $k+\ell$. In a tree of height 1, the root $\emptyset$ with the leaf $\langle 1\rangle$ is an up-ladder of length 1, and the root with the leaf $\langle 0\rangle$ is a down-ladder of length 1. This settles the case $k=1$ or $\ell=1$.

Let $k,\ell\geq 2$ be given, let $t=B(k,\ell)$, and let (b_σ), (a_η) be a tree of height t in a model M of T. Fix a leaf $\eta_*\in 2^t$, which divides the nodes of $2^{<t}$ into the two sets

$$P^+=\{\sigma\in 2^{<t}: M\models\varphi(a_{\eta_*},b_\sigma)\},\qquad P^-=2^{<t}\setminus P^+.$$

Let $t^+=B(k-1,\ell)+1$ and $t^-=B(k,\ell-1)+1$; then $t^++t^--1=t$. By Lemma 3.1.3, one of the two cases below occurs.

Case +: there is a node embedding $\iota\colon 2^{<t^+}\to 2^{<t}$ with image in P^+. The map ι extends to a full embedding $\hat\iota$ (see Lemma 3.1.2), and we set $\iota'(\tau)=\hat\iota(\langle 0\rangle^\frown\tau)$ for $\tau\in 2^{\leq t^+-1}$. Then the map ι' is a full embedding $2^{\leq t^+-1}\to 2^{\leq t}$, where $t^+-1=B(k-1,\ell)$. Writing σ_0 for $\iota(\emptyset)$, we have $\sigma_0{}^\frown\langle 0\rangle\sqsubseteq\iota'(\tau)$ for every τ, since $\emptyset{}^\frown\langle 0\rangle\sqsubseteq\langle 0\rangle^\frown\tau$. By Lemma 3.1.2, the embedding ι' induces a tree of height $B(k-1,\ell)$ in M. Applied to the induced tree, the induction hypothesis gives either a down-ladder of length ℓ, which is then a down-ladder in the original tree, or an up-ladder $(\sigma'_p,\eta'_p)_{p=1}^{k-1}$. In the latter case, let

$$(\sigma_1,\eta_1)=(\sigma_0,\eta_*),\qquad(\sigma_{p+1},\eta_{p+1})=(\iota'(\sigma'_p),\iota'(\eta'_p))\quad(1\leq p<k).$$

We check that $\varphi(a_{\eta_p},b_{\sigma_q})\iff p\leq q$ for $1\leq p,q\leq k$.

(a) For $p=1$, every σ_q is in the image of ι, which lies in P^+: for $q\geq 2$ we have $\sigma_q=\hat\iota(\langle 0\rangle^\frown\sigma'_{q-1})$ with $\langle 0\rangle^\frown\sigma'_{q-1}\in 2^{<t^+}$, where $\hat\iota$ agrees with ι. Hence the formula $\varphi(a_{\eta_*},b_{\sigma_q})$ holds, and indeed $1\leq q$.

(b) For $p\geq 2$ and $q=1$, the formula $\neg\varphi(a_{\eta_p},b_{\sigma_0})$ holds because the sequence η_p extends $\sigma_0{}^\frown\langle 0\rangle$, and indeed $p>1$.

(c) For $p,q\geq 2$: this is the up-ladder property of the smaller tree.

Thus the pairs $(\sigma_p,\eta_p)_{p=1}^k$ form an up-ladder of length k.

Case −: there is a node embedding $\iota\colon 2^{<t^-}\to 2^{<t}$ with image in P^-. We proceed symmetrically, with $\hat\iota$ and $\sigma_0=\iota(\emptyset)$ as in Case +, and with $\iota'(\tau)=\hat\iota(\langle 1\rangle^\frown\tau)$ for $\tau\in 2^{\leq t^--1}$. Now $t^--1=B(k,\ell-1)$, and the map ι' is a full embedding $2^{\leq t^--1}\to 2^{\leq t}$ whose values all extend $\sigma_0{}^\frown\langle 1\rangle$. By induction the induced tree contains an up-ladder of length k, and we are done, or a down-ladder $(\sigma'_p,\eta'_p)_{p=1}^{\ell-1}$. As before, we prepend (σ_0,η_*) to this ladder.

(a) For $p=1$, the formula $\neg\varphi(a_{\eta_*},b_{\sigma_q})$ holds for all q, because the nodes lie in P^-, and indeed $1\not> q$.

(b) For $p\geq 2$ and $q=1$, the formula $\varphi(a_{\eta_p},b_{\sigma_0})$ holds, because $\sigma_0{}^\frown\langle 1\rangle\sqsubseteq\eta_p$, and indeed $p>1$.

(c) For $p,q\geq 2$: this is the down-ladder property of the smaller tree.

Thus the pairs $(\sigma_p,\eta_p)_{p=1}^\ell$ form a down-ladder of length ℓ, and the proof is complete. ☺

Corollary 3.1.6. *Let $n\geq 1$ be given. If a model M of T contains a tree of height $B(n+1,n+1)$ for φ, then it contains an n-ladder for φ.*

Proof. By Proposition 3.1.5, the model M contains an up-ladder or a down-ladder of length $n+1$. An up-ladder of length $n+1$ is an $(n+1)$-ladder, and so contains an n-ladder. A down-ladder of length $n+1$ yields an n-ladder by Remark 1.1.5. ☺

3.2 The classical proof of definability, via ranks

In Proposition 3.2.3 below we give the classical argument that, in the absence of tall trees, every φ-type is definable. The idea is to measure each finite piece of a type by the height of the tallest tree that can be grown above the piece. The type is then defined by a piece of minimal height. Together with Corollary 3.1.6, which turns tall trees into long ladders, the argument gives a second route to (1) $\Rightarrow$ (3) in Theorem 1.1.6. The definitions obtained in this way contain quantifiers and negations of instances of φ (see Lemma 3.2.2(iv)). Their virtue is that they are read off from a single formula of the type, the one of least rank; the positive Boolean definitions of (2) come from the argument of Section 2.2.

Until the end of the proof of Proposition 3.2.3, we assume that there is $n < \omega$ such that no model of T contains a tree of height n for φ. We also fix a model M of T. By a *φ-conjunction over M* we mean a finite conjunction $\theta(x)$ of formulas $\varphi(x,b)^u$ with $b \in M$ and $u \in \{0,1\}$.

Definition 3.2.1. For a φ-conjunction θ over M and $m < \omega$, we write $R(\theta) \geq m$ if there are $b_\sigma \in M$ ($\sigma \in 2^{<m}$) such that for every $\eta \in 2^m$, we have

$$M \models \exists x \Big(\theta(x) \wedge \bigwedge_{k<m} \varphi(x, b_{\eta|k})^{\eta(k)} \Big).$$

We set $R(\theta) = -1$ when θ is not satisfiable in M. When θ is satisfiable in M, we write $R(\theta)$ for the largest m with $R(\theta) \geq m$.

For $m = 0$ the family of nodes is empty, and the inequality $R(\theta) \geq 0$ holds if and only if $M \models \exists x\, \theta(x)$. A witness for $R(\theta) \geq m$, restricted to the nodes of length less than m', is a witness for $R(\theta) \geq m'$ whenever $m' \leq m$. The relation $R(\theta) \geq m$ is therefore downward closed in m, and when $R(\theta)$ is defined, the relation holds exactly for $m \leq R(\theta)$, in agreement with the numerical reading of the notation. In Lemma 3.2.2(i) below we show that this largest m exists.

Lemma 3.2.2. *Let θ be a φ-conjunction over M, and let $m < \omega$ be given.*

(i) *The relation $R(\theta) \geq n$ does not hold. In particular, whenever θ is satisfiable in M, the rank $R(\theta)$ is well defined and $R(\theta) < n$.*

(ii) *If χ is a φ-conjunction over M and $\theta' = \theta \wedge \chi$, then the ranks satisfy $R(\theta') \leq R(\theta)$.*

(iii) *If $b \in M$ satisfies $R(\theta \wedge \varphi(x,b)^u) \geq m$ for both $u \in \{0,1\}$, then we have $R(\theta) \geq m+1$.*

(iv) *There is an $L(M)$-formula $\psi_{\theta,m}(y)$ such that $R(\theta \wedge \varphi(x,b)) \geq m \iff M \models \psi_{\theta,m}(b)$ for all $b \in M$.*

Proof. (i) If $R(\theta) \geq n$, witnessed by (b_σ), then choosing a witness $a_\eta \in M$ for each $\eta \in 2^n$ gives a tree of height n in M, which the standing assumption forbids.

(ii) A witness for $R(\theta') \geq m$ is a witness for $R(\theta) \geq m$.

(iii) For each $u \in \{0,1\}$, let $(b^u_\sigma)_{\sigma \in 2^{<m}}$ witness $R(\theta \wedge \varphi(x,b)^u) \geq m$. The two witnesses and the tuple b combine into a single family, with $b_\emptyset = b$ and $b_{\langle u \rangle^\frown \sigma} = b^u_\sigma$. A branch of length $m+1$ has the form $\langle u \rangle^\frown \eta$ with $\eta \in 2^m$, and the corresponding formula is $\theta \wedge \varphi(x,b)^u \wedge \bigwedge_{k<m} \varphi(x, b^u_{\eta|k})^{\eta(k)}$, which is satisfiable in M. Hence the family $(b_\sigma)_{\sigma \in 2^{<m+1}}$ witnesses $R(\theta) \geq m+1$.

(iv) The required formula is

$$\psi_{\theta,m}(y) := \exists (z_\sigma)_{\sigma \in 2^{<m}} \bigwedge_{\eta \in 2^m} \exists x \Big(\theta(x) \wedge \varphi(x,y) \wedge \bigwedge_{k<m} \varphi(x, z_{\eta|k})^{\eta(k)} \Big),$$

whose parameters are those of θ. ☺

Proposition 3.2.3. *Let M be a model of T. If for some $n < \omega$ no model of T contains a tree of height n for φ, then every $p \in S_\varphi(M)$ is definable.*

Proof. A piece of p of least rank cannot be split by an instance of φ into two pieces of that rank (see Lemma 3.2.2(iii)). The type p is therefore recovered by asking, for each parameter $b \in M$, whether $\varphi(x, b)$ cuts out the piece that keeps the rank. Let Φ_p be the set of finite conjunctions of members of p. Each $\theta \in \Phi_p$ is satisfiable in M, because the $L(M)$-sentence $\exists x\, \theta$ is consistent with the elementary diagram of M, hence true in M. By Lemma 3.2.2(i), the ranks of the members of Φ_p are natural numbers smaller than n; we choose $\theta \in \Phi_p$ of minimal rank $m = R(\theta)$. We show that $\psi_{\theta,m}$ is a φ-definition of p. By Lemma 3.2.2(iv), it suffices to show that for every $b \in M$, we have

$$\varphi(x, b) \in p \quad \text{if and only if} \quad R(\theta \wedge \varphi(x, b)) \geq m.$$

If $\varphi(x, b) \in p$, then the conjunction $\theta \wedge \varphi(x, b)$ belongs to Φ_p, so we have $R(\theta \wedge \varphi(x, b)) \geq m$ by minimality. Conversely, suppose that $\varphi(x, b) \notin p$. Then the formula $\neg\varphi(x, b)$ belongs to p, and minimality again forces $R(\theta \wedge \neg\varphi(x, b)) \geq m$. It follows that $R(\theta \wedge \varphi(x, b)) < m$, since otherwise Lemma 3.2.2(iii) would give $R(\theta) \geq m + 1$ (see Figure 3.3), contrary to $R(\theta) = m$. This proves the equivalence, and with it the proposition. ☺

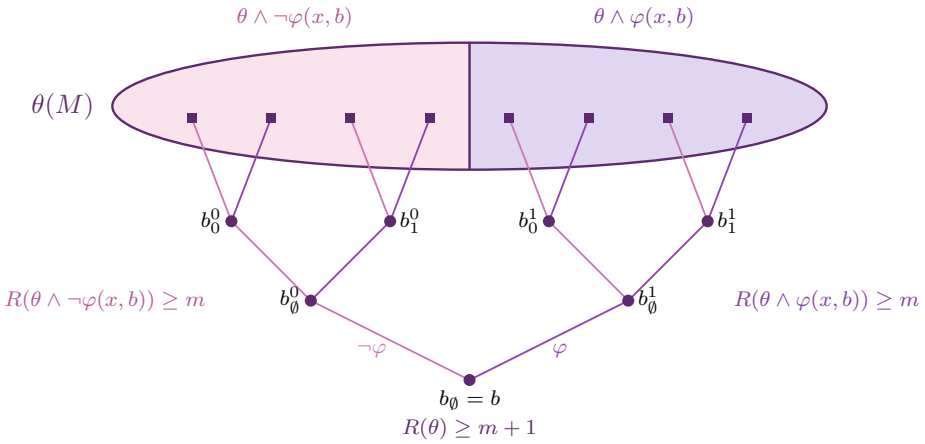


FIGURE 3.3. The gluing in Lemma 3.2.2(iii), drawn for $m = 2$: two trees of height m over the halves into which $\varphi(x, b)$ splits $\theta(M)$, joined at the root b into a tree of height $m + 1$.

We close the section by comparing the arguments of this chapter with those of Chapter 2. The comparison leads to the matter of where the parameters of a definition can be found.

Remark 3.2.4.

(a) The passages between ladders and trees are effective. Let $n \geq 1$ be given. Any tree of height $B(n+1, n+1) = 2\binom{2n}{n} - 1$ contains an up-ladder or a down-ladder of length $n+1$ (see Proposition 3.1.5). In other words, the tree contains $n+1$ nodes and $n+1$ leaves that form a ladder in one of two orientations. Conversely, a ladder of length $2^{n+1} - 1$ yields a tree of height n (see Proposition 2.3.1).

(b) The proof of Theorem 2.2.1 uses only that the type p in its statement is finitely satisfiable in the model M (see Lemma 1.1.2). Consequently, the theorem applies to any φ-type over a set A that is finitely satisfiable in A (see Definition 3.2.5(a) below), and the definition that it gives has its parameters in a finite subset F of A. The rank argument of Proposition 3.2.3, for its part, produces a definition whose parameters are those of the formula θ of minimal rank chosen in its proof.

(c) For a type that is not finitely satisfiable in its parameter set, even when φ is stable, the natural conclusion is a definition with finitely many conjugates over the parameter set, and a definition over the parameter set itself need not exist. For this reason the definability theorems are stated for types over models. The definition and the example that follow make this precise (see Definition 3.2.5 and Example 3.2.6).

Definition 3.2.5. Let N be a model of T, and let $A \subseteq N$ and $p \in S_\varphi(N)$ be given.

(a) A *complete φ-type over A* is defined as in Definition 1.1.1, with b ranging over A instead of M and the elementary diagram of N in place of that of M. Such a type is *finitely satisfiable in A* if every finite subset of the type is realized by an element of A.

(b) The type p is *definable over A* if p has a φ-definition that is an $L(A)$-formula.

(c) The type p is *definable almost over A* if p has a φ-definition of the form $\psi(y, c)$, with $\psi(y, z)$ an L-formula and c a tuple of N, such that, in every elementary extension N' of N, only finitely many distinct subsets of N' are of the form $\psi(N', c')$ with $c' \in N'$ realizing the same complete type over A as c.

In other words, a type is definable almost over A when some definition of it has only finitely many conjugates over A. A type definable over A is definable almost over A: take c to enumerate the parameters from A; then $c' = c$ is the only possibility. The converse does not hold, as an equivalence relation with two classes shows.

Example 3.2.6. Let T be the theory of an equivalence relation E with exactly two classes, both infinite, let $\varphi(x, y)$ be $E(x, y)$, and let $A = \emptyset$. Let N be a countable model of T with classes C_1 and C_2, and let $p \in S_\varphi(N)$ be the φ-type saying that x lies in C_1. In other words, the type p contains $E(x, b)$ if and only if $b \in C_1$. For any $c \in C_1$, the formula $E(y, c)$ is a φ-definition of p. Both classes are countably infinite, so for any two elements of N some automorphism of N sends one to the other. All elements of N therefore have the same complete type over $\emptyset$. In every elementary extension N' of N, the relation E still has exactly two classes, and these classes are the only sets of the form $E(N', c')$ with $c' \in N'$. Hence the type p is definable almost over $\emptyset$. The type p is not definable over $\emptyset$: since all elements of N have the same complete type over $\emptyset$, every L-formula $\psi(y)$ without parameters defines $\emptyset$ or N, never C_1. As the remark above leads us to expect, finite satisfiability in $\emptyset$ is not available here: the set $\emptyset$ contains no element to realize the formulas of p. The formula $E(x, y)$ is nevertheless stable, since it has no 2-ladder: the relations $a_1 E b_1$, $a_1 E b_2$ and $a_2 E b_2$ together force $a_2 E b_1$.

Remark 3.2.7. The condition in Definition 3.2.5(c) counts sets, not parameters. In an elementary extension N' of N, two tuples c' and c'' define the same set exactly when they are equivalent under the equivalence relation $E_\psi(z, z')$ given by $\forall y\,(\psi(y, z) \leftrightarrow \psi(y, z'))$. Let T^{eq} be the theory obtained from T by adding, for each $\emptyset$-definable equivalence relation, a sort for its classes and a function symbol for the map that sends each tuple to its class. The

class of c under E_ψ is an element of a model of T^{eq}, called the *canonical parameter* of the definition $\psi(y,c)$. By compactness, the condition of Definition 3.2.5(c) holds as soon as it holds in one $|A|^+$-saturated elementary extension of N. The condition says that the canonical parameter is *algebraic* over A in T^{eq}, that is, the canonical parameter satisfies a formula of T^{eq} with parameters in A that has only finitely many solutions. Thus p is definable almost over A if and only if p has a φ-definition in T^{eq} whose parameters are algebraic over A (see Chapter 1 of Pillay's book on geometric stability theory [**Pil96**] and Tent and Ziegler [**TZ12**, Section 8.4]). Real parameters do not suffice: in the theory of Example 3.2.6, no element of a model is algebraic over $\emptyset$, while the canonical parameter of $E(y,c)$, the class of c, has only two conjugates over $\emptyset$.

3.3 Exercises

The exercises follow the sections of the chapter. As in the exercises of Section 2.6, we write T_E for the theory of an equivalence relation E with infinitely many classes, all of them infinite. The mark $(*)$ indicates the exercises that ask for more than a direct application of the text.

1. *(Section 3.1.)* Let $s,t \geq 1$ be given, and let $\iota\colon 2^{<s} \to 2^{<t}$ be a node embedding.

(a) Show that $|\iota(\sigma)| \geq |\sigma|$ for every $\sigma \in 2^{<s}$. Show that the map ι is injective and sends incomparable sequences to incomparable sequences, and that, for all $\rho,\tau \in 2^{<s}$ and $u \in \{0,1\}$, we have $\rho^\frown\langle u\rangle \sqsubseteq \tau$ if and only if $\iota(\rho)^\frown\langle u\rangle \sqsubseteq \iota(\tau)$. (Hint: if ρ and τ are incomparable, let π be their longest common initial segment, and apply the definition to π. For the converse implication, treat separately the cases $\tau \sqsubseteq \rho$, $\rho^\frown\langle 1-u\rangle \sqsubseteq \tau$, and ρ,τ incomparable.)

(b) Show that there is no node embedding $2^{<s} \to 2^{<t}$ when $t < s$, and that the identity is the only node embedding $2^{<s} \to 2^{<s}$.

(c) Show that the map sending σ to $\langle 0\rangle^\frown\sigma$ is a node embedding $2^{<s} \to 2^{<s+1}$. Let δ be the map that repeats every entry, given by $\delta(\sigma) = \langle \sigma(0),\sigma(0),\sigma(1),\sigma(1),\dots\rangle$ with $|\delta(\sigma)| = 2|\sigma|$; show that δ is a node embedding $2^{<s} \to 2^{<2s-1}$. Show that, for $s \geq 2$, the map sending σ to $\sigma^\frown\langle 0\rangle$ is not a node embedding $2^{<s} \to 2^{<s+1}$.

(d) For $s,t \geq 1$, let $N(s,t)$ be the number of node embeddings $2^{<s} \to 2^{<t}$, and define $N(s,0) = 0$. Show that $N(1,t) = 2^t - 1$ and that

$$N(s,t) = \sum_{j<t} 2^j\, N(s-1,t-j-1)^2 \qquad (s \geq 2).$$

(Hint: classify the node embeddings by $j = |\iota(\emptyset)|$, and write $\iota(\langle v\rangle^\frown\sigma) = \iota(\emptyset)^\frown\langle v\rangle^\frown\chi_v(\sigma)$, as in the proof of Lemma 3.1.3. Check that χ_0 and χ_1 are arbitrary node embeddings $2^{<s-1} \to 2^{<t-j-1}$.) Deduce that $N(2,3) = 11$, $N(2,4) = 71$ and $N(3,4) = 123$, and list the eleven node embeddings $2^{<2} \to 2^{<3}$.

2. *(Section 3.1.)*

(a) Let $s,t \geq 1$ be given, let $\iota\colon 2^{<s} \to 2^{<t}$ be a node embedding, and let $\hat\iota\colon 2^{\leq s} \to 2^{\leq t}$ be a map that extends ι and sends 2^s into 2^t. Show that the map $\hat\iota$ is a full embedding if and only if $\iota(\eta|(s-1))^\frown\langle\eta(s-1)\rangle \sqsubseteq \hat\iota(\eta)$ for every $\eta \in 2^s$. Deduce that the number of full embeddings extending ι is $\prod_{\eta\in 2^s} 2^{\,t-1-|\iota(\eta|(s-1))|}$.

(b) Let $\iota\colon 2^{<2} \to 2^{<3}$ be the node embedding of Figure 3.1, given by $\iota(\emptyset) = \emptyset$, $\iota(0) = 00$ and $\iota(1) = 1$. Show that ι has exactly four extensions to a full embedding $2^{\leq 2} \to 2^{\leq 3}$, and list them.

(c) Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$. For $\sigma \in 2^{\leq 3}$ let $b_\sigma = 16\, d(\sigma)$ if $|\sigma| < 3$ and $a_\sigma = 16\, d(\sigma)$ if $|\sigma| = 3$, with d as in the proof of Proposition 2.3.1. Explicitly, the nodes are $b_\emptyset = 8$, $b_0 = 12$, $b_1 = 4$, $b_{00} = 14$, $b_{01} = 10$, $b_{10} = 6$, $b_{11} = 2$, and the leaves are $a_{000} = 15$, $a_{001} = 13$, $a_{010} = 11$, $a_{011} = 9$, $a_{100} = 7$, $a_{101} = 5$, $a_{110} = 3$, $a_{111} = 1$. Show, by the claim in the proof of Proposition 2.3.1 or directly, that these elements form a tree of height 3 for φ. Let $\hat{\iota}$ be the extension of the node embedding of part (b) with $\hat{\iota}(00) = 000$, $\hat{\iota}(01) = 001$, $\hat{\iota}(10) = 100$ and $\hat{\iota}(11) = 110$. Compute the tree of height 2 given by Lemma 3.1.2, and check the tree conditions directly. Show that, if the value $\hat{\iota}(00)$ is replaced by 010, the resulting families are no longer a tree.

3. *(Section 3.1.)* For a finite 0–1 sequence σ and $v \in \{0, 1\}$, let $\#_v(\sigma)$ be the number of $k < |\sigma|$ with $\sigma(k) = v$, and for $k < \omega$ let 0^k and 1^k be the constant sequences of length k.

(a) In the coloring of Figure 3.1, where $C_1 = \{\emptyset, 1, 00\}$ and $C_0 = \{0, 01, 10, 11\}$, show that the embedding drawn there is the only node embedding $2^{<2} \to 2^{<3}$ with image in C_1. Show also that no node embedding $2^{<2} \to 2^{<3}$ has image in C_0.

(b) Let $s, t \geq 1$ be given, and let $\iota\colon 2^{<s} \to 2^{<t}$ be a node embedding. Show that $\#_v(\iota(\sigma)) \geq \#_v(\sigma)$ for every $\sigma \in 2^{<s}$ and $v \in \{0, 1\}$.

(c) Let $t_0, t_1 \geq 1$ be given with $t_0 + t_1 \geq 3$, write $t = t_0 + t_1 - 2$, and let $C_0 = \{\sigma \in 2^{<t} : \#_0(\sigma) \leq t_0 - 2\}$ and $C_1 = 2^{<t} \setminus C_0$. Show that no node embedding $2^{<t_0} \to 2^{<t}$ has image in C_0, and that no node embedding $2^{<t_1} \to 2^{<t}$ has image in C_1. (Hint: consider the images of 0^{t_0-1} and of 1^{t_1-1}, and note that every $\sigma \in C_1$ satisfies $\#_1(\sigma) \leq t_1 - 2$.) Conclude that the number $t_0 + t_1 - 1$ in Lemma 3.1.3 cannot be replaced by a smaller one.

4. *(Section 3.1.)* Let $B(k, \ell)$ be defined by the recursion of Section 3.1.

(a) Compute $B(k, \ell)$ for $1 \leq k, \ell \leq 5$ from the recursion, and check the values against the closed form $2\binom{k+\ell-2}{k-1} - 1$. Compute the bound $B(n+1, n+1) = 2\binom{2n}{n} - 1$ of Corollary 3.1.6 for $1 \leq n \leq 5$. The values are 3, 11, 39, 139 and 503.

(b) Show that the numbers $B(k, \ell)+1$ satisfy Pascal's rule $B(k, \ell)+1 = \big(B(k-1, \ell)+1\big) + \big(B(k, \ell-1)+1\big)$ for $k, \ell \geq 2$, with the value 2 when $k = 1$ or $\ell = 1$, and derive the closed form from this without guessing it. Recall that the Ramsey numbers satisfy $R(k, 1) = R(1, \ell) = 1$ and the Erdős–Szekeres inequality $R(k, \ell) \leq R(k-1, \ell) + R(k, \ell-1)$. Show that the numbers $\binom{k+\ell-2}{k-1}$ satisfy the corresponding equality with the same boundary values, and deduce that $B(k, \ell) + 1$ is exactly twice the Erdős–Szekeres bound for $R(k, \ell)$. Locate the summand 1 of the recursion for B in the proof of Proposition 3.1.5.

(c) Let $n \geq 1$ be given. Show that every tree of height $B(n, n+1)$ for φ in a model M of T yields an n-ladder for φ in M, and that $B(n, n+1) = \binom{2n}{n} - 1$. Thus the height in Corollary 3.1.6 may be lowered to less than half of its value. For $1 \leq n \leq 5$ the new values are 1, 5, 19, 69 and 251.

(d) Show that $4^n/(2n+1) \leq \binom{2n}{n} < 4^n$ for $n \geq 1$. Deduce that a tree of height $h \geq 4$ for φ in a model M of T yields an n-ladder in M with $n = \lfloor \log_4 h \rfloor$, and that the bound $B(n+1, n+1)$ grows exponentially in n. Compare with Proposition 2.3.1, by which a ladder of length m produces a tree of height $\lfloor \log_2(m+1) \rfloor - 1$: in both directions the passage loses a logarithm.

5. *(Section 3.1.)* Work with the tree of height 3 for $x < y$ in $(\mathbb{Q}, <)$ of Exercise 2(c), and carry out the proof of Proposition 3.1.5 with $k = \ell = 2$, for which $B(2,2) = 3$ and $t^+ = t^- = 2$. As in that proof, for a fixed leaf $\eta_* \in 2^3$ we write P^+ for the set of nodes $\sigma \in 2^{<3}$ for which a_{η_*} satisfies $\varphi(x, b_\sigma)$, and P^- for its complement $2^{<3} \setminus P^+$. In parts (a) and (b), whenever the proof needs a node embedding $2^{<2} \to 2^{<3}$, take the inclusion map, and extend it to the full embedding $\hat{\iota}$ given by $\hat{\iota}(\eta) = \eta^\frown\langle 0\rangle$ for $\eta \in 2^2$.

(a) Take $\eta_* = 111$. Show that $P^+ = 2^{<3}$, and hence that the proof is in Case $+$. Compute the map ι' and the induced tree of height $B(1,2) = 1$, whose node is $b_0 = 12$ and whose leaves are $a_{000} = 15$ and $a_{010} = 11$. Find the up-ladder of length 1 given by the base case of the induction, and show that the proof returns the up-ladder with $\sigma_1 = \emptyset$, $\eta_1 = 111$, $\sigma_2 = 0$ and $\eta_2 = 010$. Check directly that the elements $a_{111} = 1$, $a_{010} = 11$ and $b_\emptyset = 8$, $b_0 = 12$ form a 2-ladder.

(b) Take $\eta_* = 000$. Show that $P^+ = \emptyset$, whence the proof is in Case $-$, and that it returns the down-ladder with $\sigma_1 = \emptyset$, $\eta_1 = 000$, $\sigma_2 = 1$ and $\eta_2 = 100$, with elements 15, 7 and 8, 4. Check this down-ladder directly, and determine the ladder that Remark 1.1.5 extracts from it.

(c) For each of the eight leaves η_*, determine the set P^+, decide whether there is a node embedding $2^{<2} \to 2^{<3}$ with image in P^+, and decide whether there is one with image in P^-. Show that the first exists exactly for $\eta_* \in \{011, 100, 101, 110, 111\}$ and the second exactly for $\eta_* \in \{000, 001, 010, 011, 100\}$, in accordance with Lemma 3.1.3.

6. $(*)$ *(Section 3.1.)* For $m \geq 2$, let $H(m)$ be the least h with the following property: for every theory T, every formula $\varphi(x, y)$ and every model M of T, if M contains a tree of height h for φ, then the model M contains an m-ladder for φ. By Exercise 4(c), we have $H(m) \leq \binom{2m}{m} - 1$.

(a) Let $n \geq 1$ be given, let T be the theory of infinite sets in the empty language, and let $\varphi(x_1, \ldots, x_n; y)$ be the formula $\bigvee_{i=1}^n y = x_i$. This formula says that y is an entry of the tuple x. Show that every model of T contains an n-ladder for φ and no $(n+1)$-ladder. (Hint: in an $(n+1)$-ladder, the elements $b_1, \ldots, b_{n+1}$ are pairwise distinct entries of a_1.)

(b) In the setting of part (a), show that every model of T contains a tree of height n for φ and no tree of height $n+1$. (Hint: as in Exercise 3, write 1^k for the constant sequence of length k with value 1. In a tree of height h, let $j < k < h$ be given, and look at a leaf above $1^k\frown\langle 0\rangle$ to see that $b_{1^j} \neq b_{1^k}$; all the nodes b_{1^k} are entries of a_{1^h}. For the construction, take the nodes pairwise distinct, and let the leaf a_η list the nodes b_σ with $\sigma\frown\langle 1\rangle \sqsubseteq \eta$, repeated as needed. If there are no such nodes, let a_η list an element distinct from all the nodes.) For $n = 1$, compare with Exercise 2(b) of Section 2.6.

(c) Deduce that $H(m) \geq m$. Let (b_σ), (a_η) be a tree of height 2 for an arbitrary formula φ in a model M. Show that the elements a_{11}, a_{10} and b_1, $b_\emptyset$, in this order, form a 2-ladder. Conclude that $H(2) = 2$, whereas Exercise 4(c) gives the bound 5 and

Corollary 3.1.6 the bound 11. (A finite search by computer shows that $H(3) = 3$ and $H(4) \geq 5$.)

(d) In the setting of Exercise 3 of Section 2.6, the span formula over a finite field has trees of height n, and its longest ladders have length $n+1$ (see the exercises of Section 1.3). Show that this example yields the inequality $H(n+2) \geq n+1$, which is weaker than the one of part (c). Recall that the structures $(\mathbb{N}, R_k)$ of Exercise 7 of Section 2.6 contain $(k+1)$-ladders, no $(k+2)$-ladder, and no tree of height $\lfloor \log_2(k+2) \rfloor + 1$. Show that these structures yield only the inequality $H(m) \geq \lfloor \log_2 m \rfloor + 1$, which for $m \geq 3$ is weaker than the one of part (c), and explain why: their ladders are long compared with their trees. Show that the structures $(\mathbb{N}, R_k)$ bear instead on the converse passage: for $n \geq 2$, a ladder of length $2^n - 2$ need not yield a tree of height n. Thus Proposition 2.3.1 needs ladders of length at least $2^n - 1$ in general (see Exercise 7(d) of Section 2.6).

7. *(Section 3.2.)* Let M be a model of T. The condition $R(\theta) \geq m$ makes sense, by the same definition, for every $L(M)$-formula $\theta(x)$, and without the standing assumption of Section 3.2. In this and the following exercises we use it in this generality. We write $R(\theta) = -1$ if θ is not satisfiable in M, $R(\theta) = \infty$ if $R(\theta) \geq m$ for every m, and R^M for R when the model must be indicated. As in the proof of Proposition 3.2.3, for $p \in S_\varphi(M)$ we write Φ_p for the set of finite conjunctions of members of p.

(a) Show that $R(\theta) \geq m$ if and only if M contains a tree of height m for φ whose leaves all satisfy θ. In particular, the rank $R(x = x)$ is the largest height of a tree for φ in M, or ∞ if there is no largest height.

(b) Show that, for each m, the condition $R(\theta) \geq m$ is expressed in M by an $L(M)$-sentence whose parameters are those of θ. Deduce that $R^M(\theta) = R^N(\theta)$ whenever $M \preceq N$. Show that $R(\theta) \leq R(\theta')$ whenever $\theta(M) \subseteq \theta'(M)$; in particular, the rank $R(\theta)$ depends only on the set $\theta(M)$, and Lemma 3.2.2(ii) holds for every $L(M)$-formula χ.

(c) Show the converse of Lemma 3.2.2(iii): if $R(\theta) \geq m+1$, then some $b \in M$ satisfies $R(\theta \wedge \varphi(x,b)^u) \geq m$ for both $u \in \{0,1\}$. Deduce that, when $R(x = x)$ is finite, every θ satisfiable in M satisfies

$$R(\theta) = 1 + \max_{b \in M} \min_{u \in \{0,1\}} R(\theta \wedge \varphi(x,b)^u).$$

8. *(Section 3.2.)* Let M be an infinite set in the empty language, and let $\varphi(x,y)$ be the formula $x = y$. By Exercise 2(b) of Section 2.6, the standing assumption of Section 3.2 holds with $n = 2$. Use the conventions of Exercise 7.

(a) Show that $R(x = x) = 1$, and that a φ-conjunction θ over M that is satisfiable in M has $R(\theta) = 0$ if it has a conjunct of the form $x = c$ and $R(\theta) = 1$ otherwise.

(b) Let $a, a' \in M$ be distinct, and let θ be the formula $x = a \vee x = a'$. Show that $R(\theta) = 1$, although $R(x = a) = R(x = a') = 0$. Show that, for $b = a$, both formulas $\theta \wedge \varphi(x,b)^u$ have rank 0, and check Lemma 3.2.2(iii) in this case. Thus a single instance of φ can lower the rank on both sides.

(c) For $c \in M$, let p_c be the type with $D_{p_c} = \{c\}$, and let p_∞ be the type with $D_{p_\infty} = \emptyset$ (see Exercise 1 of Section 1.3). Show that the least rank of a member of Φ_{p_c} is 0 and

that of a member of Φ_{p_∞} is 1. For θ the formula $x = c$, write out the formula $\psi_{\theta,0}(y)$ of Lemma 3.2.2(iv) and show that it is equivalent to $y = c$. For θ the formula $x \neq c$, which belongs to p_∞, show that

$$\psi_{\theta,1}(y) = \exists z\big(\exists x\,(x \neq c \wedge x = y \wedge x \neq z) \wedge \exists x\,(x \neq c \wedge x = y \wedge x = z)\big),$$

and that this formula is equivalent in M to $y \neq y$. Compare with the definition $d_F\varphi$ of Exercise 4(c) of Section 2.6, which for p_∞ is the empty disjunction, and compare the least ranks with the Cantor–Bendixson ranks of Exercise 12(a) of Section 2.6.

(d) Do the same for a model M of T_E and the formula xEy. Show that $R(x = x) = 1$, and that a satisfiable φ-conjunction has rank 0 if it has a conjunct of the form xEc and rank 1 otherwise. For θ the formula xEc, show that the formula $\psi_{\theta,0}(y)$ is $\exists x\,(xEc \wedge xEy)$, which is equivalent to yEc in M. For θ the formula $\neg xEc$, show that the formula $\psi_{\theta,1}(y)$ is equivalent to $y \neq y$. Compare with Exercises 4(d) and 12(b) of Section 2.6.

Thus in both settings the definitions produced by Proposition 3.2.3 contain a quantifier and, for the type whose least rank is 1, the negation of an instance of φ. Each of these definitions is nevertheless equivalent to a positive Boolean combination of instances of φ, as Theorem 2.2.1 predicts.

9. (∗) *(Section 3.2.)* Work in the setting of Exercise 3 of Section 2.6: let K be a finite field, and let T be the theory of infinite vector spaces over K. Let $n \geq 1$ be given, let $\varphi(x_1, \ldots, x_n; y)$ say that y lies in the span of $x_1, \ldots, x_n$, and let M be a model of T. Use the conventions of Exercise 7.

(a) Let $r, s < \omega$ and $c_0, \ldots, c_{r-1}, d_0, \ldots, d_{s-1} \in M$ be given, let θ be the φ-conjunction $\bigwedge_{i<r} \varphi(x, c_i) \wedge \bigwedge_{j<s} \neg\varphi(x, d_j)$, and let U be the span of $c_0, \ldots, c_{r-1}$. Show that θ is satisfiable in M if and only if $\dim U \leq n$ and no d_j lies in U, and that in this case $R(\theta) = n - \dim U$. (Hint: recall from Exercise 3 of this section that 1^k is the constant sequence of length k with value 1. For the upper bound, let (b_σ), (a_η) be a tree of height h whose leaves satisfy θ. For every $k < h$, show by looking at a leaf above $1^k{}^\frown\langle 0\rangle$ that the node b_{1^k} does not lie in the span of U and the nodes b_{1^j} with $j < k$. Then count the dimension of the span of a_{1^h}. For the lower bound, note that the model M has infinite dimension, and take the nodes linearly independent over the span of U and the d_j.)

(b) Deduce that $R(x = x) = n$, the largest height of a tree found in Exercise 3 of Section 2.6. Thus no model of T contains a tree of height $n + 1$ for φ, the standing assumption of Section 3.2 holds with the bound $n+1$, and the inequality $R(\theta) < n+1$ of Lemma 3.2.2(i) is attained.

(c) Let $p \in S_\varphi(M)$ be given; then D_p is a subspace of M of dimension at most n (see the exercises of Section 1.3). Show that the least rank of a member of Φ_p is $n - \dim D_p$, and that it is attained by $\theta_p = \varphi(x, 0) \wedge \bigwedge_{c\in C} \varphi(x, c)$ for any basis C of D_p. Show that this least rank equals the Cantor–Bendixson rank of p found in Exercise 13 of Section 2.6.

(d) Show directly from part (a) that $\psi_{\theta_p,m}(y)$, with $m = n - \dim D_p$, defines D_p in M. For $n = 1$ and $D_p = \{0\}$, write out $\psi_{\theta_p,1}(y)$ and show that it is equivalent to $y = 0$. Compare with the definition $\varphi(c, y)$ by a single instance in Exercise 4(e) of Section 2.6.

10. *(Section 3.2.)* Work in $(\mathbb{Q}, <)$, with $\varphi(x, y)$ the formula $x < y$, which has trees of every finite height in $\mathbb{Q}$. Use the conventions of Exercise 7.

(a) Show that every φ-conjunction θ over $\mathbb{Q}$ that is satisfiable in $\mathbb{Q}$ defines a set containing an open interval, and that $R(\theta) = \infty$. (Hint: the numbers $2^{m+1}d(\sigma)$ with $\sigma \in 2^{\leq m}$, for d as in the proof of Proposition 2.3.1, form a tree of height m. A tree for $x < y$ is carried to a tree by every order-preserving map.)

(b) Let r be an irrational real number, and let p be the type with $D_p = \{b \in \mathbb{Q} : b > r\}$. Show that, for every $\theta \in \Phi_p$ and every m, there is a rational $b < r$ with $R(\theta \wedge \varphi(x, b)) \geq m$, and conclude that the formula $\psi_{\theta,m}$ does not define p. Identify the step of the proof of Proposition 3.2.3 that fails. Recall from Exercise 3 of Section 1.3 that p has no definition at all.

11. $(*)$ *(Section 3.2.)* Use the conventions of Exercise 7.

(a) Show that the following are equivalent: for some $n < \omega$, no model of T contains a tree of height n for φ; for every model M of T, the rank $R^M(x = x)$ is finite. (Hint: for the nontrivial implication, use Lemma 2.4.1 with $\mu = \omega$.) Show that, if T is complete, these conditions are also equivalent to the finiteness of $R^M(x = x)$ for a single model M of T. Show that completeness cannot be dropped, by taking for T the theory of linear orders and for φ the formula $x < y$.

(b) Let M be the disjoint union of finite chains of Exercise 10 of Section 2.6, with $\varphi(x, y)$ the formula $x < y$. Show that $R(x = x) = \infty$ in M, although M contains no tree of height ω. Thus infinite rank is witnessed by finite trees.

(c) Let $k \in \mathbb{N}$ be given, and let T_k be as in Exercise 7 of Section 2.6. Show that $R(x = x) = \lfloor \log_2(k + 2) \rfloor$ in every model of T_k, and therefore that the standing assumption of Section 3.2 holds with $n = \lfloor \log_2(k + 2) \rfloor + 1$, although every model of T_k contains a $(k + 1)$-ladder.

3.4 Historical remarks

Section 3.1. Shelah had shown that a formula encodes tall trees exactly when it has the order property, but his proof passed through uncountable cardinals. Hodges [**Hod81**] set out in 1981 to free this finitary statement from them, in a short paper titled *Encoding orders and trees in binary relations*. The Ramsey-type argument that extracts ladders from trees is his, and it appears again in his book [**Hod93**]. It proves directly, without definability of types, that tall trees force the order property, and it comes with an explicit bound on the height required. Conant and Terry [**CT26**] improved the bounds and extended the argument from $\{0, 1\}$-valued formulas to real-valued functions; we follow their version. The applications of stability to finite combinatorics, among them the regularity lemma of Malliaris and Shelah for stable graphs, require bounds of this kind.

Section 3.2. In Section 3.2, for a φ-conjunction θ over a model M, the rank $R(\theta)$ is the largest height of a tree for φ in M whose leaves all satisfy θ (see Exercise 7 of Section 3.3). This rank is Shelah's local rank $R(\theta, \varphi, 2)$ [**She71**, **She90**]. The proof that a type is defined through any of its formulas of minimal rank is the classical route to definability of types. Older than the combinatorial and analytic proofs given elsewhere in this book, it is the one found in most textbooks, for instance those of Pillay [**Pil96**] and of Tent and Ziegler [**TZ12**].

The definitions it produces involve quantifiers; those of Section 2.2 are positive Boolean combinations of instances of φ, and add no quantifiers to those of φ.

Chapter

4

From stability to NIP

Roadmap

NIP is the second dividing line. Shelah introduced the independence property in 1971, in the paper in which he defined the order property, and in the same year Vapnik and Chervonenkis found the same configuration, under the name of shattering, in probability theory. The configuration forbidden by NIP is the shattered set, and a shattered set contains a ladder of its own size. A formula has the independence property when it shatters arbitrarily large finite sets of parameters, that is, when on arbitrarily large finite sets of parameters its instances realize every pattern of truth values. Otherwise the formula is NIP (negation of the independence property — see Definition 4.1.1). In Section 4.1 we show that a ladder is a special case of this configuration, whence stable formulas are NIP (see Proposition 4.1.2), and we also give the basic examples. The order of the rationals is unstable, and yet NIP, since its instances realize only a few of the possible patterns (see Example 4.1.3). The other basic example, the edge relation of the random graph, has the independence property (see Example 4.1.4). An NIP formula has only polynomially many types over finite sets of parameters, by the Sauer–Shelah lemma (see the exercises in Section 4.3). In Section 4.2 we return to bounded real-valued functions, as in Section 1.2. A function is NIP exactly when every sequence of its instances has a subsequence that converges pointwise on the space of types (see Theorem 4.2.4). In this setting an independent sequence produces a copy of the Stone–Čech compactification of the natural numbers (see Proposition 4.2.5). The random graph, which has the independence property, will reappear in Chapter 5 as the model case of the third dividing line.

We treat formulas in Section 4.1, with the conventions of Section 1.1, and bounded functions in Section 4.2, with those of Section 1.2.

4.1 The independence property

A ladder (see Definition 1.1.4) is a special pattern, a staircase. Each tuple on the left side of the ladder satisfies φ with exactly those tuples on the right that stand at its own position or beyond. The independence property asks instead that *every* pattern be realized. What the name calls independent are the instances of φ. When every pattern is realized on finitely many parameters, the instances of φ at these parameters define in the model an independent family of sets in the sense of set theory. In such a family, every intersection of some of the sets with the complements of the others is nonempty.

Definition 4.1.1. Let M be a model of T.

(a) For a natural number n, tuples $b_1, \ldots, b_n$ of M are *shattered by φ in M* if for every $S \subseteq \{1, \ldots, n\}$ there is a tuple a_S of M such that, for all $1 \leq i \leq n$, we have

$$M \models \varphi(a_S, b_i) \quad \text{if and only if} \quad i \in S.$$

(b) The formula φ has the *independence property* if for every natural number n some model of T contains n tuples shattered by φ. Otherwise the formula φ is *NIP*; the letters abbreviate *negation of the independence property*. NIP formulas are also called *dependent*, the name that Shelah has come to prefer.

(c) The *VC dimension* of φ is the largest n such that some model of T contains n tuples shattered by φ. This dimension is 0 if no model of T contains a single tuple shattered by φ, and ∞ if there is no largest such n.

A theory is *NIP* if every formula is NIP.

Thus the formula φ is NIP if and only if the VC dimension of φ is finite. The notion of shattering comes from probability theory, where Vapnik and Chervonenkis introduced it in their study of the uniform convergence of empirical frequencies, and it has since become basic in learning theory. In both fields, shattering is defined for a family of sets as follows. A family $\mathcal{F}$ of subsets of a set X *shatters* a set $Z \subseteq X$ if every subset of Z is of the form $Y \cap Z$ with $Y \in \mathcal{F}$. To connect the two uses, let M be a model of T, and write $\varphi(a, M) = \{b \in M : M \models \varphi(a, b)\}$ for $a \in M$. If $b_1, \ldots, b_n$ are shattered by φ in M, then every subset of $\{b_1, \ldots, b_n\}$ is cut out by one of the sets $\varphi(a, M)$. In other words, the family $\{\varphi(a, M) : a \in M\}$ shatters $\{b_1, \ldots, b_n\}$ in the sense just defined.

As for ladders, every subfamily of a shattered family of tuples is itself shattered. By compactness, therefore, the formula φ has the independence property if and only if some model M of T contains an infinite sequence $(b_i)_{i<\omega}$ on which φ realizes every pattern. Explicitly, for every $S \subseteq \omega$ there is a tuple a_S of M with $M \models \varphi(a_S, b_i)$ if and only if $i \in S$.

Proposition 4.1.2. *If φ does not have the order property of Definition 1.1.4, then the formula φ is NIP.*

Proof. We prove the contrapositive, by finding a ladder among the patterns that shattering provides. Let n be a natural number, and suppose that tuples $b_1, \ldots, b_n$ are shattered by φ in a model M of T, witnessed by the tuples a_S. For $1 \leq i \leq n$, let $S_i = \{j : i \leq j \leq n\}$ and $a_i = a_{S_i}$. We then have $M \models \varphi(a_i, b_j)$ if and only if $j \in S_i$, that is, if and only if $i \leq j$. Thus the suffixes S_i cut out a staircase pattern, and the tuples $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ form an n-ladder. Since n is arbitrary, the formula φ has the order property whenever it has the independence property. ☺

The case $n = 3$ of the proof is pictured in Figure 4.1.

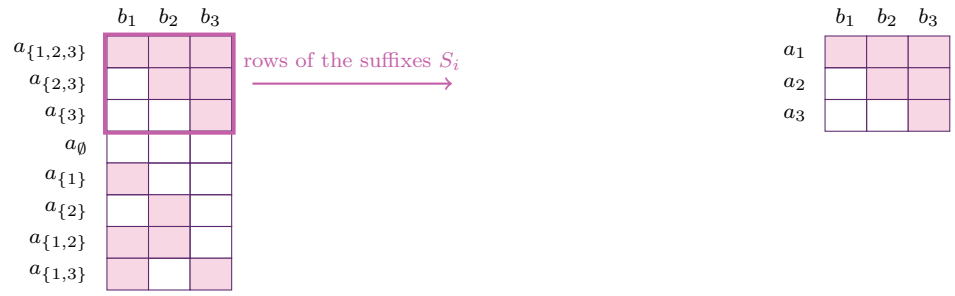


FIGURE 4.1. The proof of Proposition 4.1.2 for $n = 3$: the rows a_S for shattered tuples b_1, b_2, b_3, with pink cells where $\varphi(a_S, b_j)$ holds (left), and the 3-ladder formed by the framed rows of the suffixes S_1, S_2, S_3 (right).

The converse of Proposition 4.1.2 does not hold: the order of the rationals, whose cuts led to the order property in Section 1.1, is unstable and NIP.

Example 4.1.3. Let T be the theory of $(\mathbb{Q}, <)$ and let $\varphi(x, y)$ be $x < y$. Then the elements $a_i = i$ and $b_j = j + \frac{1}{2}$, for natural numbers i and j, satisfy $\varphi(a_i, b_j)$ if and only if $i \leq j$, so the formula φ has the order property and is unstable. However, the formula φ is NIP, with VC dimension 1. Indeed, let $b_1 \leq b_2$ be elements of a model. The pattern $\{b_1\}$ requires some a with $a < b_1$ and $a \not< b_2$, that is, some a with $b_2 \leq a < b_1$, which is impossible. Hence no two elements are shattered, while a single element b is shattered by any $a < b$ together with any $a \geq b$. The theory of $(\mathbb{Q}, <)$ is NIP as well, and not only the formula $x < y$. By quantifier elimination (see Fact 5.5.2(b) and Example 5.5.3(2) below), every formula $\psi(x, y)$ is equivalent modulo T to a Boolean combination of formulas $u < v$ and $u = v$. Here each of u and v is a variable of x or of y. Each of these formulas is NIP. Suppose first that one of u, v belongs to x and the other to y. For $u < v$, the argument just given, applied to the order or to its reverse, shows that no two tuples are shattered. For $u = v$, no two tuples are shattered either, since the patterns $\{1\}$ and $\{1, 2\}$ cannot both be realized. When both variables belong to the same side, no two tuples are shattered at all. By Exercise 7(c) of Section 4.3, Boolean combinations of NIP formulas are NIP.

The random graph lies at the other extreme: its edge relation realizes every pattern on every finite set of vertices (see Figure 4.2).

Example 4.1.4. Let T be the theory of the random graph. A graph has the *extension property* if, for all disjoint finite sets U, V of vertices, some vertex outside $U \cup V$ is adjacent to every vertex of U and to no vertex of V. The models of T are the graphs with the extension property. Let $\varphi(x, y)$ be the edge relation $E(x, y)$. Any n distinct vertices $b_1, \ldots, b_n$ are shattered: for $S \subseteq \{1, \ldots, n\}$ apply the extension property to $U = \{b_i : i \in S\}$ and $V = \{b_i : i \notin S\}$. Thus the relation E has the independence property.

The independence property of $\varphi(x, y)$ concerns tuples shattered on the y-side. To shatter on the x-side, we exchange the roles of the variables and write $\varphi^{\mathrm{op}}(y, x)$ for $\varphi(x, y)$. In

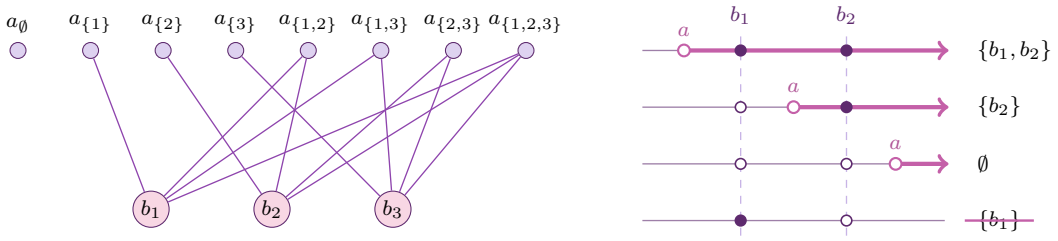


FIGURE 4.2. Shattering in the random graph, where each witness a_S with $S \subseteq \{1, 2, 3\}$ is joined to b_i exactly when $i \in S$ (left), and the rays $\{y : a < y\}$ in $(\mathbb{Q}, <)$, none of which realizes the pattern $\{b_1\}$ (right).

Proposition 4.1.5 below we show that the choice of side does not matter: 2^n tuples shattered on one side yield n tuples shattered on the other.

Proposition 4.1.5. *The formula φ is NIP if and only if φ^{op} is NIP. More precisely, for every natural number n, if a model of T contains 2^n tuples shattered by φ, then the same model contains n tuples shattered by φ^{op}.*

Proof. The idea is to index the shattered tuples by sets in such a way that membership itself becomes the pattern, and then to read the pattern from the other side. Let M be a model of T containing 2^n tuples shattered by φ, indexed as b_I with $I \subseteq \{1, \ldots, n\}$, and let the tuples a_S with $S \subseteq \mathcal{P}(\{1, \ldots, n\})$ witness that they are shattered by φ. For $1 \leq i \leq n$, the tuple $a_i = a_{S_i}$ is the witness for the set $S_i = \{I : i \in I\}$. We then have $M \models \varphi(a_i, b_I)$ if and only if $I \in S_i$, that is, if and only if $i \in I$. Thus for every $I \subseteq \{1, \ldots, n\}$ the tuple b_I satisfies $\varphi^{\mathrm{op}}(b_I, a_i)$ exactly for $i \in I$, and the tuples $a_1, \ldots, a_n$ are shattered by φ^{op}. Applied to φ and to φ^{op}, whose opposite is φ, the argument gives the equivalence. ☺

Remark 4.1.6. The structures of Examples 4.1.3 and 4.1.4 are unstable for different reasons, and by a theorem of Shelah these two reasons are the only ones. For a model M of T, a formula $\psi(x, y)$ and a tuple b of M, write $\psi(M, b) = \{a \in M : M \models \psi(a, b)\}$. The formula ψ has the *strict order property* if for every natural number n some model M of T contains tuples $b_1, \ldots, b_n$ with $\psi(M, b_1) \subsetneq \psi(M, b_2) \subsetneq \cdots \subsetneq \psi(M, b_n)$. The formula $x < y$ of Example 4.1.3 is the basic case. Shelah's dichotomy states that a theory is unstable if and only if some formula has the independence property or some formula has the strict order property [**She90**, Chapter II]. Thus an unstable theory encodes arbitrary subsets, as the random graph does, or an infinite order, or both; in an unstable NIP theory only the second reason occurs.

4.2 The real-valued case

As in Section 1.2, formulas now give way to bounded functions on arbitrary sets. The goal is an analytic form of NIP. In Theorem 4.2.4 below we show that a function is NIP exactly when every sequence of its instances, viewed as functions on its space of left types, has a pointwise convergent subsequence. The definition is adapted to arbitrary sets in two ways. The independent configuration is asked for directly, as an infinite sequence on which every finite pattern is realized, and the values on the two sides of a pattern are separated by a uniform gap.

Definition 4.2.1. Let A and B be sets, and let $\varphi\colon A \times B \to \mathbb{R}$ be a bounded function. A sequence $(b_n)_{n<\omega}$ in B is *independent for* φ if there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $a \in A$ with

$$\varphi(a, b_n) \geq s \text{ for } n \in P \qquad \text{and} \qquad \varphi(a, b_n) \leq r \text{ for } n \in Q.$$

The function φ has the *independence property* if some sequence in B is independent for φ, and is *NIP* otherwise.

For a $\{0,1\}$-valued function, any $0 < r < s < 1$ may be used, and a sequence is independent exactly when each finite part of the sequence is shattered in the sense of Definition 4.1.1. By compactness, a formula $\varphi(x, y)$ is NIP in the sense of Definition 4.1.1 if and only if the corresponding function on $M^{|x|} \times M^{|y|}$ is NIP for every model M of T.

As in the $\{0,1\}$-valued case (see Proposition 4.1.2), stability implies NIP. The proof there turned a shattered set into a ladder, a staircase being one of the patterns that shattering provides. In the proof of Proposition 4.2.2 below, the ladder appears among the types: ultrafilter limits of witnesses give left φ-types (see Definition 1.2.7) that form a ladder against the terms of the independent sequence. The separate continuity of the extension $\hat{\varphi}$ of Proposition 1.2.8 then rules the ladder out.

Proposition 4.2.2. *Let A and B be sets. Every stable function $\varphi\colon A \times B \to [0,1]$ (see Definition 1.2.5) is NIP.*

Proof. Recall from Definition 1.2.7 that the space $\mathrm{LS}(\varphi)$ of left φ-types is the closure in $[0,1]^B$ of the functions $\mathrm{ltp}_\varphi(a) = \varphi(a, \cdot)$ with $a \in A$. Likewise, the space $\mathrm{RS}(\varphi)$ of right φ-types is the closure in $[0,1]^A$ of the functions $\mathrm{rtp}_\varphi(b) = \varphi(\cdot, b)$ with $b \in B$. Since φ is stable, we may let $\hat{\varphi}\colon \mathrm{LS}(\varphi) \times \mathrm{RS}(\varphi) \to [0,1]$ be the separately continuous extension of φ constructed in the proof of Proposition 1.2.8. Taking a constant family in (1.2) gives $\hat{\varphi}(p, \mathrm{rtp}_\varphi(b)) = p(b)$ for all $p \in \mathrm{LS}(\varphi)$ and $b \in B$.

Suppose, toward a contradiction, that a sequence $(b_n)_{n<\omega}$ in B is independent for φ, with reals $r < s$. For natural numbers $m < N$ we choose $a_{m,N} \in A$ with

$$\varphi(a_{m,N}, b_n) \geq s \text{ for } m \leq n < N \qquad \text{and} \qquad \varphi(a_{m,N}, b_n) \leq r \text{ for } n < m,$$

and let $a_{m,N}$ be arbitrary for $N \leq m$. Fix a nonprincipal ultrafilter $\mathcal{U}$ on ω, with which we form the limit types

$$p_m = \lim_{N,\mathcal{U}} \mathrm{ltp}_\varphi(a_{m,N}) \in \mathrm{LS}(\varphi).$$

For fixed n the displayed inequalities hold for all $N > \max(m, n)$, so we have

$$p_m(b_n) \geq s \text{ if } n \geq m, \qquad p_m(b_n) \leq r \text{ if } n < m. \tag{4.1}$$

Now let $q = \lim_{n,\mathcal{U}} \mathrm{rtp}_\varphi(b_n) \in \mathrm{RS}(\varphi)$ and $p = \lim_{m,\mathcal{U}} p_m \in \mathrm{LS}(\varphi)$. We compute $\hat{\varphi}(p, q)$ in two ways.

First, for fixed m, the continuity of $\hat{\varphi}(p_m, \cdot)$ yields $\hat{\varphi}(p_m, q) = \lim_{n,\mathcal{U}} p_m(b_n)$, a limit that is at least s because $p_m(b_n) \geq s$ for all $n \geq m$ by (4.1). The continuity of $\hat{\varphi}(\cdot, q)$ now gives $\hat{\varphi}(p, q) = \lim_{m,\mathcal{U}} \hat{\varphi}(p_m, q) \geq s$.

Second, limits in $\mathrm{LS}(\varphi)$ are taken coordinatewise, and $p_m(b_n) \leq r$ for all $m > n$, whence

$$p(b_n) = \lim_{m,\mathcal{U}} p_m(b_n) \leq r \quad \text{for every } n.$$

By continuity of $\hat{\varphi}(p, \cdot)$, we obtain

$$\hat{\varphi}(p, q) = \lim_{n,\mathcal{U}} \hat{\varphi}(p, \mathrm{rtp}_\varphi(b_n)) = \lim_{n,\mathcal{U}} p(b_n) \leq r.$$

Hence we have $s \leq \hat{\varphi}(p, q) \leq r$, contradicting $r < s$. ☺

Stability was characterized in Section 1.2 by the exchange of iterated limits. NIP, which is weaker by Proposition 4.2.2, is characterized instead by the pointwise convergence of subsequences of the evaluation maps $e_b\colon \mathrm{LS}(\varphi) \to [0,1]$ of Subsection 1.2.4, given by $e_b(p) = p(b)$ for $b \in B$. The analytic input is Rosenthal's dichotomy: a bounded sequence of functions has either a pointwise convergent subsequence or a subsequence that behaves like an independent sequence.

Fact 4.2.3 (Rosenthal). *Let X be a set, and let $(f_n)_{n<\omega}$ be a uniformly bounded sequence of real-valued functions on X. Then the sequence (f_n) has a subsequence that converges pointwise on X, or a subsequence (f_{n_k}) for which there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $x \in X$ with $f_{n_k}(x) \geq s$ for $k \in P$ and $f_{n_k}(x) \leq r$ for $k \in Q$.*

Theorem 4.2.4. *Let A and B be sets, and let $\varphi\colon A \times B \to [0,1]$ be a function. The following are equivalent:*

(1) *The function φ is NIP.*

(2) *For every sequence $(b_n)_{n<\omega}$ in B, the sequence of the evaluation maps $e_{b_n}\colon p \mapsto p(b_n)$ has a subsequence that converges pointwise on $\mathrm{LS}(\varphi)$.*

Proof. The idea is that the evaluation maps along an independent sequence have no pointwise convergent subsequence. On the other hand, by Rosenthal's dichotomy, every sequence of evaluation maps without a pointwise convergent subsequence has a subsequence that is independent on the space of types. The remaining work is to pass between types and the elements of A whose types approximate them.

(2) $\Rightarrow$ (1). We show that an independent sequence violates condition (2). Suppose that a sequence $(b_n)_{n<\omega}$ in B is independent for φ, with reals $r < s$, and let (b_{n_k}) be a subsequence. For each $K < \omega$, independence provides $a_K \in A$ with $\varphi(a_K, b_{n_k}) \geq s$ for even $k < K$ and $\varphi(a_K, b_{n_k}) \leq r$ for odd $k < K$. Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω, and let p be the point $\lim_{K,\mathcal{U}} \mathrm{ltp}_\varphi(a_K)$ of $\mathrm{LS}(\varphi)$. For each fixed k the inequality for k holds for all $K > k$, so the value $e_{b_{n_k}}(p) = p(b_{n_k})$ is at least s for even k and at most r for odd k. Hence the sequence $(e_{b_{n_k}}(p))_k$ does not converge. Since the subsequence was arbitrary, condition (2) fails.

(1) $\Rightarrow$ (2). Conversely, suppose that condition (2) fails for a sequence $(b_n)_{n<\omega}$ in B; we find an independent subsequence. Since no subsequence of (e_{b_n}) converges pointwise on $\mathrm{LS}(\varphi)$, the second alternative of Fact 4.2.3 holds for $X = \mathrm{LS}(\varphi)$ and $f_n = e_{b_n}$. That alternative provides a subsequence (b_{n_k}) and reals $r < s$ such that for all disjoint finite $P, Q \subseteq \omega$ there is $p \in \mathrm{LS}(\varphi)$ with $p(b_{n_k}) \geq s$ for $k \in P$ and $p(b_{n_k}) \leq r$ for $k \in Q$. To leave room for approximation, fix r' and s' with $r < r' < s' < s$. Since p is a pointwise limit of left types $\mathrm{ltp}_\varphi(a)$ and only finitely many coordinates are involved, some $a \in A$ satisfies $\varphi(a, b_{n_k}) > s'$ for $k \in P$ and $\varphi(a, b_{n_k}) < r'$ for $k \in Q$. Thus the sequence (b_{n_k}) is independent for φ, with $r' < s'$, and the function φ is not NIP. ☺

In Proposition 4.2.5 below we show that the independence property has a striking topological form: an independent sequence produces a copy of $\beta\mathbb{N}$.

Proposition 4.2.5. *Let A and B be sets, let $\varphi\colon A \times B \to [0,1]$ be a function, and let $(b_n)_{n<\omega}$ be a sequence in B that is independent for φ. Then the map $\mathcal{U} \mapsto \lim_{n,\mathcal{U}} e_{b_n}$ is a homeomorphism from $\beta\mathbb{N}$ onto a closed subset of the pointwise closure of $\{e_b : b \in B\}$ in $[0,1]^{\mathrm{LS}(\varphi)}$.*

Proof. The idea is that independence lets a single point of $\mathrm{LS}(\varphi)$ separate any set of indices from its complement. The values of the separating point at the b_n lie above one threshold

for the indices in the set and below another for those outside it. Two distinct ultrafilters, which disagree on some set of indices, therefore give limits that differ at such a point. The map $n \mapsto e_{b_n}$ from $\mathbb{N}$ into the compact space $[0,1]^{\mathrm{LS}(\varphi)}$ extends to the continuous map $\mathcal{U} \mapsto \lim_{n,\mathcal{U}} e_{b_n}$ on $\beta\mathbb{N}$, whose image lies in the pointwise closure of $\{e_b : b \in B\}$. Since $\beta\mathbb{N}$ is compact and $[0,1]^{\mathrm{LS}(\varphi)}$ is Hausdorff, a continuous injection of $\beta\mathbb{N}$ into $[0,1]^{\mathrm{LS}(\varphi)}$ is a homeomorphism onto its image, which is compact and therefore closed. It remains to be shown that the map is injective. Let $r < s$ witness the independence of (b_n), and let $\mathcal{U} \neq \mathcal{V}$ be ultrafilters on $\mathbb{N}$; since they differ, some $X \in \mathcal{U}$ has $\mathbb{N} \setminus X \in \mathcal{V}$. For each finite $F \subseteq \mathbb{N}$ there is $a_F \in A$ with $\varphi(a_F, b_n) \geq s$ for $n \in F \cap X$ and $\varphi(a_F, b_n) \leq r$ for $n \in F \setminus X$. Let $\mathcal{D}$ be an ultrafilter on the finite subsets of $\mathbb{N}$ that contains $\{G : G \supseteq F\}$ for every finite F, and let $p = \lim_{F,\mathcal{D}} \mathrm{ltp}_\varphi(a_F) \in \mathrm{LS}(\varphi)$. Then we have $p(b_n) \geq s$ for $n \in X$ and $p(b_n) \leq r$ for $n \notin X$. Hence the limit of $e_{b_n}(p)$ along $\mathcal{U}$ is at least s, and the limit along $\mathcal{V}$ is at most r. The two limits of (e_{b_n}) therefore differ at p, and the map is injective. ☺

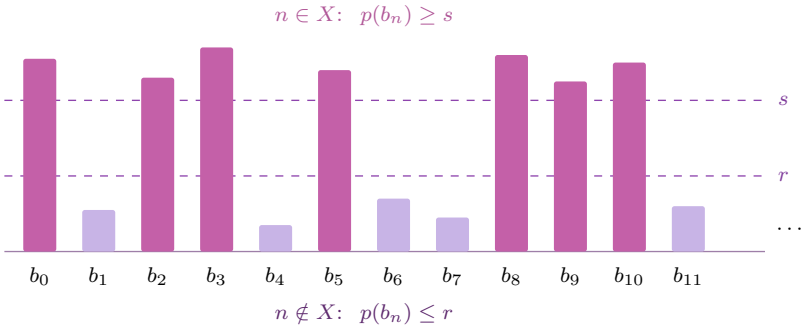


FIGURE 4.3. The proof of Proposition 4.2.5: the values $p(b_n)$ of a single point p of $\mathrm{LS}(\varphi)$, above s for n in a prescribed set $X \subseteq \mathbb{N}$ (pink bars) and below r off X.

Thus the evaluation maps e_b, which are the instances of φ viewed as continuous functions on $\mathrm{LS}(\varphi)$, face a dichotomy. If φ is NIP, every sequence of instances has a pointwise convergent subsequence (see Theorem 4.2.4). If φ has the independence property, the closure of the instances contains a copy of $\beta\mathbb{N}$, a compact space in which no sequence of distinct points converges. The dichotomy is the topological face of Rosenthal's ℓ_1 theorem [**Ros74**]. In the theory of Bourgain, Fremlin and Talagrand [**BFT78**], the same dichotomy governs the pointwise closure of a pointwise bounded sequence of continuous functions on a Polish space. Either the closure is a *Rosenthal compactum*, a compact space of Baire class 1 functions on a Polish space, or it contains a copy of $\beta\mathbb{N}$. Simon [**Sim15b**] develops the NIP side of this picture for spaces of types. Stable functions sit at the rigid end of the spectrum: their type spaces are Eberlein compacta (see Corollary A.1.8 below).

Remark 4.2.6. We can also derive Proposition 4.2.2, with heavier machinery, from Theorem 4.2.4. By Grothendieck's double limit criterion, in the form of Proposition A.1.7 below, stability of φ makes $\{e_b : b \in B\}$ relatively weakly compact in $C(\mathrm{LS}(\varphi))$. By Corollary A.1.6 below, which rests on the Eberlein–Šmulian theorem (see Fact A.1.5), every sequence (e_{b_n}) then has a pointwise convergent subsequence. Thus condition (2) of Theorem 4.2.4 holds, and the function φ is NIP by the implication (2) $\Rightarrow$ (1), which does not use Rosenthal's dichotomy. The direct proof of Proposition 4.2.2 avoids both Grothendieck's criterion and the Eberlein–Šmulian theorem.

Remark 4.2.7. By Theorem 1.2.9, the stable functions are exactly the restrictions of the pairing between a reflexive Banach space E and its dual E^*. The independence property and the strict order property have Banach space counterparts of the same kind. Khanaki [**Kha15**] proved that φ is stable, NIP or NSOP according as a Banach space naturally associated with φ is reflexive, contains no copy of ℓ_1 or is weakly sequentially complete. Here NSOP means the absence of the strict order property of Remark 4.1.6; a Banach space is weakly sequentially complete when every weakly Cauchy sequence in it converges weakly.

4.3 Exercises

The exercises follow the sections of the chapter, and the mark $(*)$ singles out those that ask for more than a direct application of the text. Throughout, we write T_{RG} for the theory of the random graph (see Example 4.1.4). A family of subsets of a set shatters a subset in the sense of learning theory, recalled after Definition 4.1.1.

1. *(Section 4.1.)* Let T be the theory of $(\mathbb{Q}, <)$, and let $\varphi(x_1x_2, y)$ be the formula $x_1 < y \wedge y < x_2$. For a pair $a = (a_1, a_2)$ in a model M, the set $\varphi(a, M)$ is the open interval with endpoints a_1 and a_2, and it is empty when $a_2 \leq a_1$. Show that the formula φ and its opposite φ^{op}, obtained by exchanging the roles of the variables, both have VC dimension 2 (see Definition 4.1.1). (Hint: for φ^{op}, let three pairs $a_i = (l_i, r_i)$, for $i = 1, 2, 3$, be shattered by φ^{op}. Show that for some index k the interval $\varphi(a_k, M)$ is contained in the union of the other two, which rules out the pattern $\{k\}$. To find k, choose i with l_i least and j with r_j greatest; when $i \neq j$, use the pattern $\{i, j\}$.) Show finally that the formula φ has the order property (see Definition 1.1.4).

2. *(Section 4.1.)* Let T be the theory of an equivalence relation E with infinitely many classes, all of them infinite. Show that the formula $E(x, y)$ has VC dimension 1 and that no model of T contains a 2-ladder for it. For $k \geq 1$, let $\psi_k(x_1 \dots x_k, y)$ be the formula $\bigvee_{l=1}^{k} E(x_l, y)$. Show that the VC dimension of ψ_k is k, and that no model of T contains a $(k+1)$-ladder for ψ_k. Thus for every k there is a stable formula of VC dimension exactly k. (Hint: in a $(k+1)$-ladder $a_1, \dots, a_{k+1}$ and $b_1, \dots, b_{k+1}$ for ψ_k, the elements $b_1, \dots, b_{k+1}$ lie in pairwise distinct classes, and each of these classes contains a coordinate of a_1.)

3. *(Section 4.1.)* Let T be the theory of $(\mathbb{Q}, <)$. For $R \subseteq \{<, =, >\}$, let $\varphi_R(x, y)$ be the disjunction of the formulas $x < y$, $x = y$ and $y < x$ that correspond to the members of R, the empty disjunction being $\bot$. Show that every quantifier-free formula $\varphi(x, y)$, with x and y single variables, is equivalent modulo T to some φ_R. Show that every φ_R has VC dimension at most 1. (Hint: for $b_1 < b_2$ there are only five positions of an element a relative to b_1 and b_2. List the pairs of truth values of $\varphi_R(a, b_1)$ and $\varphi_R(a, b_2)$.) Show that φ_R is stable if and only if R contains both or neither of the symbols $<$ and $>$. Thus four of the eight formulas are unstable and NIP, as in Example 4.1.3.

4. *(Section 4.1.)* Work in T_{RG}, and let $\varphi(x, y)$ be a quantifier-free formula in the language of graphs, with x and y single variables. Show that φ is equivalent modulo T_{RG} to a formula $(x = y \wedge \alpha) \vee (x \neq y \wedge \beta)$, where α is $\top$ or $\bot$ and β is one of $\top$, $\bot$, $E(x, y)$ and $\neg E(x, y)$. Show that the formula φ is NIP if and only if β is $\top$ or $\bot$. Equivalently, show that φ is NIP if and only if it is equivalent modulo T_{RG} to a Boolean combination of the

formula $x = y$. (Hint: when β is $\neg E(x, y)$, argue as in Example 4.1.4 with the sets U and V exchanged.)

5. *(Section 4.1.)* Let T be the theory of $(\mathbb{N}, |)$, where $x \mid y$ means that x divides y, and let $\varphi(x, y)$ be the formula $x \mid y$. For distinct primes $p_1, \ldots, p_n$, show that the elements $b_i = \prod_{j \neq i} p_j$, for $1 \leq i \leq n$, are shattered by φ in $\mathbb{N}$. Conclude that the formula φ has the independence property. Show directly, without Proposition 4.1.5, that the elements $p_1, \ldots, p_n$ are shattered by φ^{op} in $\mathbb{N}$. Show also that the formula φ has the strict order property of Remark 4.1.6. Thus the structure $(\mathbb{N}, |)$ is unstable for both of the reasons named in Remark 4.1.6.

6. *(Section 4.1.)* Let $m \geq 1$ be given, write $X = \{1, \ldots, 2^m\}$, and let M be the finite structure whose universe is the disjoint union of X and $\mathcal{P}(X)$. The structure M has one binary relation R such that $R(c, d)$ holds if and only if $c \in X$, $d \in \mathcal{P}(X)$ and $c \in d$. Let T be the complete theory of M, and let $\varphi(x, y)$ be the formula $R(x, y)$. Since M is finite, every model of T is isomorphic to M. Show that the VC dimension of φ^{op} is 2^m and that the VC dimension of φ is m. (Hint: for the lower bound, label the elements of X by the subsets of $\{1, \ldots, m\}$. For the upper bound, count the witnesses a_S.) Conclude that the bound of Proposition 4.1.5 is optimal: a model may contain 2^m tuples shattered by one formula and no $m + 1$ tuples shattered by its opposite.

7. $(*)$ *(Section 4.1.)* Part (a) is the Sauer–Shelah lemma, mentioned in the historical remarks; parts (b) and (c) draw its consequences for NIP formulas.

(a) Let X be a set with m elements, let $d \geq 0$ be given, and let $\mathcal{F}$ be a family of subsets of X that shatters no subset of X with $d + 1$ elements. Show that $|\mathcal{F}| \leq \sum_{i=0}^{d} \binom{m}{i}$. (Hint: argue by induction on m. Fix $x \in X$, let $\mathcal{F}'$ be the family of the sets $Y \setminus \{x\}$ with $Y \in \mathcal{F}$, and let $\mathcal{F}''$ be the family of the sets $Y \subseteq X \setminus \{x\}$ with $Y \in \mathcal{F}$ and $Y \cup \{x\} \in \mathcal{F}$. Show that $|\mathcal{F}| = |\mathcal{F}'| + |\mathcal{F}''|$ and that the family $\mathcal{F}''$ shatters no subset with d elements. For $d = 0$ the last condition means that $\mathcal{F}''$ is empty.)

(b) Suppose that the formula $\varphi(x, y)$ has finite VC dimension d, let M be a model of T, and let $b_1, \ldots, b_m$ be tuples of M. Show that, as a ranges over M, the sets $\{i : M \models \varphi(a, b_i)\}$ take at most $\sum_{i=0}^{d} \binom{m}{i} \leq (m + 1)^d$ values.

(c) Suppose that the formulas $\varphi(x, y)$ and $\psi(x, y)$ are NIP. Show that the formulas $\neg\varphi$, $\varphi \wedge \psi$ and $\varphi \vee \psi$ are NIP. (Hint: for the conjunction, let d and e be bounds for the VC dimensions of φ and ψ. Count with (b) the patterns that $\varphi \wedge \psi$ realizes on m tuples, and compare $(m + 1)^{d+e}$ with 2^m for large m.)

8. *(Section 4.2.)* For $n \geq 1$, let P_n be the set of pairs (n, i) with $1 \leq i \leq n$, and let Q_n be the set of pairs (n, Y) with $Y \subseteq \{1, \ldots, n\}$. Let M be the structure whose universe is the disjoint union of the sets P_n and Q_n, with one binary relation R. The relation R holds of $((n, Y), (n, i))$ if and only if $i \in Y$, and of no other pair. Let T be the complete theory of M, and let $\varphi(x, y)$ be the formula $R(x, y)$. Show that the formula φ has the independence property in the sense of Definition 4.1.1. Show that the corresponding function $\varphi \colon M \times M \to \{0, 1\}$ is NIP in the sense of Definition 4.2.1. (Hint: all terms of an independent sequence lie in a single set P_n and are pairwise distinct.) Show finally, by compactness, that some model of T contains a sequence independent for φ. Thus,

in the equivalence stated after Definition 4.2.1, the words "for every model" cannot be replaced by "for some model".

9. *(Section 4.2.)* Let $\varphi\colon A \times \mathbb{N} \to [0,1]$ be a function such that, for every $a \in A$, the sequence $(\varphi(a,n))_n$ is monotone, and in the same direction for all a.

(a) Show from Definition 4.2.1 that the function φ is NIP. (Hint: two terms of an independent sequence are distinct, and one pattern of size two already fails.)

(b) Verify condition (2) of Theorem 4.2.4 directly. (Hint: every $p \in \mathrm{LS}(\varphi)$ is monotone on $\mathbb{N}$, and every sequence in $\mathbb{N}$ has a subsequence that is constant or strictly increasing.)

(c) Apply (a) to the function $\varphi(x,n) = x^n$ on $[0,1] \times \mathbb{N}$ of Example A.1.4 below and to the function of Example 1.2.12. Show that both functions are unstable in the sense of Definition 1.2.5.

(d) Let A be a topological space, let B be a set, and let $\varphi\colon A \times B \to [0,1]$ be an NIP function such that every function $\varphi(\cdot,b)$ is continuous on A. Show that for every sequence (b_n) in B some subsequence of the functions $\varphi(\cdot,b_n)$ converges pointwise on A, and that its limit is therefore of the first Baire class. Show that for x^n on $[0,1]$ the limit is not continuous.

10. *(Section 4.2.)* Let $\varphi\colon [0,1] \times \mathbb{N} \to [0,1]$ be given by $\varphi(t,k) = \frac{1}{2}\bigl(1 + \sin(2\pi\, 8^k t)\bigr)$. Show that the sequence $(k)_{k<\omega}$ is independent for φ, with $r = \frac{1}{2} - \frac{\sqrt{2}}{4}$ and $s = \frac{1}{2} + \frac{\sqrt{2}}{4}$. (Hint: write $t = \sum_{j\geq 1} d_j 8^{-j}$ with digits $d_j \in \{0,2,6\}$. The value $\sin(2\pi\, 8^k t)$ is at least $\frac{\sqrt{2}}{2}$ when $d_{k+1} = 2$, and at most $-\frac{\sqrt{2}}{2}$ when $d_{k+1} = 6$.) Show also that no subsequence of the continuous functions $t \mapsto \varphi(t,k)$ converges pointwise on $[0,1]$. (Hint: for a subsequence (k_l), choose t with $d_{k_l+1} = 2$ for even l, with $d_{k_l+1} = 6$ for odd l, and with all other digits 0. Independence alone would not suffice, by Exercise 12(a).) Deduce from Proposition 4.2.5 that the pointwise closure of $\{e_k : k \in \mathbb{N}\}$ in $[0,1]^{\mathrm{LS}(\varphi)}$ contains a copy of $\beta\mathbb{N}$.

11. $(*)$ *(Section 4.2.)* Let $\varphi\colon [0,2\pi] \times \mathbb{N} \to [0,1]$ be given by $\varphi(x,n) = \frac{1}{2}(1 + \sin nx)$. Show that $\int_0^{2\pi} (\sin nx - \sin n'x)^2\, dx = 2\pi$ for distinct $n, n' \geq 1$. Deduce from the dominated convergence theorem that no subsequence of the functions $\varphi(\cdot,n)$ converges pointwise on $[0,2\pi]$. Apply Fact 4.2.3 with $X = [0,2\pi]$, not $X = \mathrm{LS}(\varphi)$, to conclude that some subsequence of $(n)_{n<\omega}$ is independent for φ. Deduce that the function $(x,y) \mapsto \frac{1}{2}(1 + \sin xy)$ on $\mathbb{R} \times \mathbb{R}$ has the independence property. Compare Exercise 10, which gives an explicit independent sequence.

12. *(Section 4.2.)* Let A be the set of finite subsets of $\mathbb{N}$, and let $\varphi\colon A \times \mathbb{N} \to \{0,1\}$ be given by $\varphi(F,n) = 1$ if $n \in F$ and $\varphi(F,n) = 0$ otherwise.

(a) Show that the sequence $(n)_{n<\omega}$ is independent for φ. Show that for every sequence (n_k) of distinct natural numbers the functions $\varphi(\cdot,n_k)$ converge pointwise to 0 on A. Thus condition (2) of Theorem 4.2.4 cannot be weakened to convergence at the realized types $\mathrm{ltp}_\varphi(F)$.

(b) Show that $\mathrm{LS}(\varphi)$ is the set of all functions from $\mathbb{N}$ to $\{0,1\}$, which we identify with $\mathcal{P}(\mathbb{N})$, and that $e_n(Z) = 1$ if and only if $n \in Z$.

(c) Let $\mathcal{U}$ be an ultrafilter on $\mathbb{N}$. Show that the function $\lim_{n,\mathcal{U}} e_n$ on $\mathcal{P}(\mathbb{N})$ takes the value 1 at Z if and only if $Z \in \mathcal{U}$. Conclude that the map of Proposition 4.2.5 sends each ultrafilter to its own characteristic function, and that it is a homeomorphism from $\beta\mathbb{N}$ onto the pointwise closure of $\{e_n : n \in \mathbb{N}\}$.

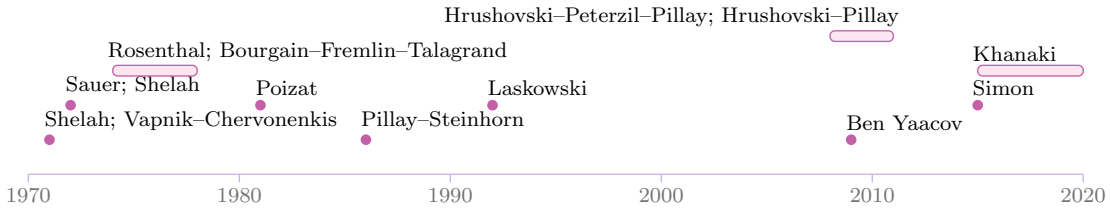
(d) Show that the function φ is unstable in the sense of Definition 1.2.1. (Hint: consider $\neg\varphi$.)

13. $(*)$ *(Section 4.2.)* Let A and φ be as in Exercise 12, and let $\psi\colon \mathbb{N} \times A \to \{0,1\}$ be given by $\psi(n, F) = \varphi(F, n)$. Show that the function ψ is NIP. (Hint: in an independent sequence (F_k), the finite set F_0 would have to contain 2^N distinct elements for every N.) Show that the function ψ has the order property. Thus, for functions on arbitrary sets, the independence property of a function does not pass to its opposite, in contrast with Proposition 4.1.5. Show, however, that for every k some sets $F_1, \ldots, F_k \in A$ have all 2^k patterns realized by elements of $\mathbb{N}$. The asymmetry thus comes from the requirement that a single infinite sequence carry all finite patterns.

4.4 Historical remarks

Section 4.1. The independence property was introduced by Shelah [She71], in the same paper as the order property. His theorem that an unstable theory has the independence property or the strict order property appears in Chapter II of his *Classification Theory* [She90]. Independently of Shelah, Vapnik and Chervonenkis [VC71] introduced shattering and the VC dimension in probability theory, in their study of the uniform convergence of empirical frequencies. Shattering and the VC dimension later became basic in learning theory. The polynomial bound on the number of traces (the subsets cut out on an n-element set) of a family of finite VC dimension is now known as the Sauer–Shelah lemma. Its history is crowded: Vapnik and Chervonenkis announced it in a short note of 1968 [VC68] and proved it in their paper of 1971. Sauer [Sau72] proved the lemma independently, in answer to a question that, as he records in his paper, Erdős had transmitted to him in Nice. Shelah [She72], again independently, found the bound in his work on stability and credited it also to Perles. Laskowski [Las92] observed that a formula is NIP exactly when the family of sets it defines has finite VC dimension. With this observation, two lines of research that had grown up for some twenty years in ignorance of each other were seen to have drawn the same dividing line.

Section 4.2. Rosenthal's dichotomy lies at the heart of his ℓ_1 theorem [Ros74]. The ℓ_1 theorem states that a bounded sequence in a Banach space has either a weakly Cauchy subsequence or a subsequence equivalent to the unit vector basis of ℓ_1. The systematic study of pointwise compact sets of Baire class 1 functions is due to Rosenthal [Ros77] and to Bourgain, Fremlin and Talagrand [BFT78]. The analysts' theory reached model theory only decades later, notably through Simon's study of the spaces of φ-types of NIP formulas as Rosenthal compacta [Sim15b], and through the work of Khanaki [Kha20, Kha15]. The timeline below follows the combinatorial and the analytic strands side by side.



1971 **Shelah; Vapnik–Chervonenkis.** Shelah [She71] introduces the independence property. Independently, Vapnik and Chervonenkis [VC71] introduce shattering and what is now called VC dimension, in the study of uniform convergence of empirical frequencies.

1972 **Sauer; Shelah.** Sauer [Sau72] and Shelah [She72] independently prove the Sauer–Shelah lemma, the polynomial bound on the number of traces of a set system of finite VC dimension. Sauer answers a question of Erdős; Shelah credits also Perles. Vapnik and Chervonenkis had announced the lemma in 1968 [VC68] and proved it in 1971 [VC71].

1974–1978 **Rosenthal; Bourgain–Fremlin–Talagrand.** Rosenthal proves his ℓ_1 theorem [Ros74] and studies pointwise compact sets of Baire class 1 functions [Ros77]. Bourgain, Fremlin and Talagrand [BFT78] develop the theory of these "Rosenthal compacta".

1981 **Poizat.** Poizat [Poi81] characterizes countable NIP theories by a bound on the number of types, and develops the theory of unstable theories without the independence property.

1986 **Pillay–Steinhorn.** Pillay and Steinhorn [PS86] develop the theory of o-minimal structures, following van den Dries. These structures are NIP and are the main source of unstable NIP theories.

1992 **Laskowski.** Laskowski [Las92] identifies NIP formulas with formulas defining families of finite VC dimension.

2008–2011 **Hrushovski–Peterzil–Pillay; Hrushovski–Pillay.** The study of invariant and generically stable measures in NIP theories [HPP08, HP11], motivated by Pillay's conjecture on definable groups in o-minimal structures, turns NIP into a central topic of model theory.

2009 **Ben Yaacov.** Ben Yaacov [BY09] extends VC classes, the families of sets of finite VC dimension, to $[0,1]$-valued functions, the real-valued analogue of NIP used in Section 4.2.

2015 **Simon.** Simon [Sim15b] applies the theory of Rosenthal compacta to spaces of φ-types of NIP formulas, and publishes the monograph [Sim15a] on NIP theories.

2015–2020 **Khanaki.** Khanaki [Kha20, Kha15] relates stability, NIP and NSOP to three properties of Banach spaces (see Remark 4.2.7).

Chapter
5

From stability to simplicity

Roadmap

Simplicity is the third dividing line, and it forbids a tree in which the parameters branch into incompatible alternatives. Shelah introduced dividing and the tree property in *Classification Theory* (1978), and named the simple theories in 1980. A formula divides over a set when its parameter can be moved along a sequence indiscernible over the set to instances that are uniformly inconsistent: for some integer $k \geq 2$, any k of the instances are jointly inconsistent (see Definition 5.2.3). The tree property is dividing iterated along a tree with infinitely many successors at each node, and a complete theory is simple when no formula has it (see Definition 5.2.7). In Section 5.1 we fix the conventions on monster models, and in Section 5.2 we define dividing and the tree property. In Section 5.3 we show that stable theories are simple (see Proposition 5.3.1) and that a theory is stable exactly when it is simple and NIP (see Theorem 5.3.4). Hence the three dividing lines meet in stability. We also show that in a simple theory each type of a finite tuple is nondividing over some subset of its domain of cardinality at most that of the theory (see Theorem 5.3.6). In Section 5.4 we compute dividing in the random graph and prove that its theory is simple, although it has the independence property (see Theorems 5.4.5 and 5.4.6). In Section 5.5 we prove the same for the Fraïssé limits in a finite relational language with disjoint 2- and 3-amalgamation (see Theorem 5.5.9). The rational order is not simple, by Proposition 5.3.2 of Section 5.3, and in Section 5.5 we show that the Henson graph is not simple either (see Example 5.5.10). In Section 5.6 we show that stable real-valued functions do not have the 2-tree property (see Theorem 5.6.4). In the random graph a relation of independence can already be seen by hand (see Theorem 5.4.9); in Chapters 6 and 7 we obtain such relations in general.

5.1 Conventions

In this section we fix the setting for the rest of the book. This setting consists of a complete theory, a monster model of the theory in which all our sets and tuples live, and the notation for types over subsets of the monster model. Dividing, the central notion of this

chapter (see Definition 5.2.3), moves the parameters of a formula along sequences of tuples that have the same type over a fixed set. We want all such sequences at hand in a single model, whose automorphisms over the fixed set carry any term of such a sequence to any other.

Throughout, we work with a *complete* theory T in the language L, and we write $\kappa = |T| = |L| + \aleph_0$. We work inside a monster model $\mathbb{M}$ of T: a $\bar{\kappa}$-saturated and strongly $\bar{\kappa}$-homogeneous model, for a cardinal $\bar{\kappa}$ larger than every cardinal that our constructions require. In the proof of Proposition 5.3.1, for instance, the cardinal $\bar{\kappa}$ must exceed a cardinal defined there. A set is *small* if its cardinality is less than $\bar{\kappa}$. Letters A, B, C denote small subsets of $\mathbb{M}$; tuples may be infinite, but are always small. We freely identify a set with an enumerating tuple. A *model* is a small elementary substructure $M \preceq \mathbb{M}$. Every model of T of size $<\bar{\kappa}$ embeds elementarily in $\mathbb{M}$, and every type over a small set is realized in $\mathbb{M}$.

We write $\mathrm{tp}(a/A)$ for the complete type of a over A, $S_x(A)$ for the set of complete types over A in the variables x, $a \equiv_A a'$ for $\mathrm{tp}(a/A) = \mathrm{tp}(a'/A)$, and MB for $M \cup B$. By strong homogeneity, we have $a \equiv_A a'$ if and only if $\sigma(a) = a'$ for some $\sigma \in \mathrm{Aut}(\mathbb{M}/A)$.

Until now, stability has been a property of single formulas. We now define it for the theory T, together with a sufficient condition that counts types; in Section 5.5 we use this condition to recognize stable Fraïssé limits.

Definition 5.1.1.

(a) The theory T is *stable* if no L-formula $\varphi(x, y)$ has the order property (see Definition 1.1.4).

(b) For an infinite cardinal λ, the theory T is *λ-stable* if $|S_x(A)| \leq \lambda$ whenever $|A| \leq \lambda$ and x is a finite tuple of variables.

Lemma 5.1.2. *If T is λ-stable for some $\lambda \geq \kappa$, then the theory T is stable.*

Proof. We show that a formula with the order property produces too many types. Suppose that $\varphi(x, y)$ has the order property. By Proposition 2.3.1, the formula φ encodes binary trees of arbitrarily large finite height, and Proposition 2.4.2 then provides a model N of T with $|N| \leq \lambda$ and more than λ complete φ-types over N. Since T is complete, we may assume $N \preceq \mathbb{M}$. Every φ-type over N is consistent with the elementary diagram of N, and therefore extends to a complete type in $S_x(N)$; distinct φ-types have distinct extensions. Hence the space $S_x(N)$ has more than $\lambda \geq |N|$ points, and the theory T is not λ-stable. ☺

5.2 Dividing and the tree property

Dividing makes precise when a formula says something new about a set, and the tree property iterates dividing along a tree with infinitely many successors at each node. In this section we define both, and with them the simple theories, those in which no formula has the tree property.

Three conventions precede the definitions of this section. A small set of formulas with parameters in $\mathbb{M}$ is *consistent* if it is realized in $\mathbb{M}$. A set written $\{\psi_i : i \in I\}$ is regarded as indexed by I: distinct indices count as distinct members, even when the formulas coincide. Finally, we often identify a tuple with the set of its coordinates. Thus, for tuples a, b and a set C, the inclusion $a \cap b \subseteq C$ says that every element that is a coordinate of both a and b belongs to C.

A formula $\varphi(x, b)$ with a tuple b of parameters may say about x something that a small set C already knows, or something new; two examples show the difference. In a model of

the theory of $(\mathbb{Q}, <)$, for elements $b_1 < b_2$, the formula $b_1 < x < b_2$ is one of infinitely many formulas $b_1^i < x < b_2^i$ with $b_2^i < b_1^{i+1}$ for all $i < \omega$. These formulas look alike over $\emptyset$ and are pairwise contradictory. In the random graph, the formula $E(x, b)$, where b is a vertex, looks like $E(x, b')$ for every other vertex b'. Any finite number of these formulas nevertheless hold together, because the extension property of Example 4.1.4 provides a common neighbor. The first formula *divides* over $\emptyset$, and the second does not (see Example 5.2.6 and Theorem 5.4.5). Indiscernible sequences make "look alike" precise.

Definition 5.2.1. Let C be a small set. A sequence $(b_i)_{i<\omega}$ of tuples of the same length is *indiscernible over* C if for every $n \geq 1$ and all $i_1 < \cdots < i_n$ and $j_1 < \cdots < j_n$ in ω, we have

$$(b_{i_1}, \ldots, b_{i_n}) \equiv_C (b_{j_1}, \ldots, b_{j_n}).$$

Every infinite subsequence of a sequence indiscernible over C is indiscernible over C, and a sequence indiscernible over C is indiscernible over every subset of C. Taking $n = 1$ in the definition, we see that all the terms of a sequence indiscernible over C have the same type over C. Leibniz held that things that cannot be told apart are one and the same. Indiscernible sequences live at the edge of his principle: their terms may be distinct, and yet no formula over C tells one increasing tuple of them from another.

Indiscernible sequences are plentiful. Every sequence leaves a trace, the set of formulas that hold of all its increasing tuples. Formally, fix a sequence $(b_i)_{i<\omega}$ and a small set C. The *Ehrenfeucht–Mostowski type* of (b_i) over C is the set of $L(C)$-formulas $\psi(y_1, \ldots, y_n)$ with $n \geq 1$ such that $\mathbb{M} \models \psi(b_{i_1}, \ldots, b_{i_n})$ for all $i_1 < \cdots < i_n$ in ω. In Lemma 5.2.2 below we show that Ramsey's theorem, together with compactness, produces an indiscernible sequence whose trace contains the trace of the given one.

Lemma 5.2.2 (Extraction of indiscernible sequences). *Let C be a small set, and let $(b_i)_{i<\omega}$ be tuples of the same length. Then there is a sequence $(b'_i)_{i<\omega}$, indiscernible over C, that realizes the Ehrenfeucht–Mostowski type of (b_i) over C: for every formula $\psi(y_1, \ldots, y_n)$ of the Ehrenfeucht–Mostowski type and all $j_1 < \cdots < j_n$, we have $\mathbb{M} \models \psi(b'_{j_1}, \ldots, b'_{j_n})$. In particular, if all the b_i have the same type over C, then every b'_j has the same type over C as b_0.*

Proof. The idea is to apply Ramsey's theorem to finitely many formulas at a time, and to conclude by compactness. Let Σ be the set of $L(C)$-formulas in variables y_i $(i < \omega)$ consisting of

(a) the formulas $\psi(y_{j_1}, \ldots, y_{j_n})$, for every ψ in the Ehrenfeucht–Mostowski type of (b_i) over C and all $j_1 < \cdots < j_n$, and

(b) the formulas $\theta(y_{i_1}, \ldots, y_{i_n}) \leftrightarrow \theta(y_{j_1}, \ldots, y_{j_n})$, for every $L(C)$-formula θ and all $i_1 < \cdots < i_n$ and $j_1 < \cdots < j_n$.

A realization of Σ is a sequence as required: the formulas of type (b) say that it is indiscernible over C, and those of type (a) that it realizes the Ehrenfeucht–Mostowski type. Since Σ has at most $|C| + \aleph_0$ parameters, by saturation it suffices to show that every finite $\Sigma_0 \subseteq \Sigma$ is satisfiable.

Let $\theta_1, \ldots, \theta_m$ be the formulas that occur in the formulas of type (b) in Σ_0, and let n be the largest number of variables among them. If no formula of type (b) occurs in Σ_0, we take $H = \omega$ below. We color each set $\{i_1 < \cdots < i_n\} \subseteq \omega$ by the sequence of truth values of $\theta_l(b_{i_1}, \ldots, b_{i_{n_l}})$, $l = 1, \ldots, m$, where $n_l \leq n$ is the number of variables of θ_l. There are at most 2^m colors, so Ramsey's theorem gives an infinite set $H \subseteq \omega$ all of whose n-element subsets have the same color. Let $h_0 < h_1 < \cdots$ enumerate H, and let each y_i be interpreted as b_{h_i}.

Every increasing n_l-tuple from H is the beginning of an increasing n-tuple from H, because H is infinite. The truth value of θ_l on increasing n_l-tuples from H therefore does not depend on the tuple, and the formulas of type (b) in Σ_0 hold. The formulas of type (a) hold as well, because they hold of all increasing tuples of the original sequence. For the last assertion, every formula of $\mathrm{tp}(b_0/C)$, in one variable y_1, belongs to the Ehrenfeucht–Mostowski type. ☺

In the random graph, and in the Fraïssé limits in a finite relational language studied in Section 5.5, quantifier elimination gives a sharper conclusion over finite sets, for sequences of finite tuples: an infinite subsequence of the given sequence is itself indiscernible (see Lemmas 5.4.4 and 5.5.7).

Indiscernible sequences allow us to say exactly when a formula tells something new about a set: when the parameter of the formula can be moved along an indiscernible sequence to instances that contradict one another.

Definition 5.2.3. Let C be a small set, let $\varphi(x, y)$ be an L-formula, and let b be a tuple of the length of y.

(a) The formula $\varphi(x, b)$ *divides over* C if there is a sequence $(b_i)_{i<\omega}$, indiscernible over C, with $b_0 = b$, such that the set $\{\varphi(x, b_i) : i < \omega\}$ is inconsistent.

(b) For $k \geq 2$, an indexed set $\{\psi_i : i \in I\}$ of formulas is *k-inconsistent* if the set $\{\psi_{i_1}, \ldots, \psi_{i_k}\}$ is inconsistent for all distinct $i_1, \ldots, i_k \in I$, and $\varphi(x, b)$ *k-divides over* C if the sequence in (a) can be chosen so that $\{\varphi(x, b_i) : i < \omega\}$ is k-inconsistent.

(c) Let $B \supseteq C$ be small. A type $p \in S_x(B)$ *divides over* C if some formula of p divides over C.

Clause (c) makes sense because every $L(B)$-formula has the form $\varphi(x, b)$ with $\varphi \in L$ and b a finite tuple from B. In the next lemma we show that dividing is k-dividing for a suitable k; thus the notions of (a) and (b) agree.

Lemma 5.2.4. *Let C be a small set, let $\varphi(x, y)$ be an L-formula, let $(b_i)_{i<\omega}$ be indiscernible over C, and suppose that $\{\varphi(x, b_i) : i < \omega\}$ is inconsistent. Then there is $k \geq 2$ such that $\{\varphi(x, b_i) : i < \omega\}$ is k-inconsistent. In particular, a formula divides over C if and only if it k-divides over C for some $k \geq 2$.*

Proof. By compactness and saturation, some finite subset $\{\varphi(x, b_{i_1}), \ldots, \varphi(x, b_{i_k})\}$ with $i_1 < \cdots < i_k$ is inconsistent. Since adding formulas keeps this subset inconsistent, we may assume $k \geq 2$. Thus no element of $\mathbb{M}$ satisfies all the formulas $\varphi(x, b_{i_l})$, $1 \leq l \leq k$. For $j_1 < \cdots < j_k$, indiscernibility gives $(b_{j_1}, \ldots, b_{j_k}) \equiv_C (b_{i_1}, \ldots, b_{i_k})$, and hence no element satisfies all the formulas $\varphi(x, b_{j_l})$ either. The set $\{\varphi(x, b_i) : i < \omega\}$, indexed by $i < \omega$, is therefore k-inconsistent. ☺

Remark 5.2.5. Let $\varphi(x, y)$ be an L-formula, let b be a tuple, and let $k \geq 2$ be an integer. If $C \subseteq C'$ are small sets and $\varphi(x, b)$ k-divides over C', then the formula $\varphi(x, b)$ k-divides over C, because a sequence indiscernible over C' is indiscernible over C. Consequently, if $C \subseteq C' \subseteq B$ and $p \in S_x(B)$ divides over C', then the type p divides over C. The terms of the sequence in Definition 5.2.3 all have the type of b over C. Dividing is therefore a strengthening of Shelah's original condition, which asks only for infinitely many realizations b_i of $\mathrm{tp}(b/C)$ such that the set $\{\varphi(x, b_i)\}$ is k-inconsistent. The two conditions are in fact equivalent (see Tent and Ziegler [**TZ12**, Chapter 7]), but we do not need this equivalence.

Example 5.2.6. Let T be the theory of $(\mathbb{Q}, <)$, let $b_1 < b_2$ be elements of $\mathbb{M}$, and let $\theta(x; y_1, y_2)$ be the formula $y_1 < x < y_2$. Choose pairs $b^i = (b^i_1, b^i_2)$ for $i < \omega$, with $b^0 = (b_1, b_2)$, such that $b^i_1 < b^i_2$ and $b^i_2 < b^{i+1}_1$ for every i. By quantifier elimination for dense linear orders without endpoints (see Fact 5.5.2(b) and Example 5.5.3(2) below, or Hodges [**Hod93**, Theorem 7.4.1]), the type over $\emptyset$ of a tuple is determined by the order between its coordinates. For every $n \geq 1$, moreover, any two increasing n-tuples of the sequence (b^i) are ordered alike. Hence the sequence (b^i) is indiscernible over $\emptyset$. The intervals (b^i_1, b^i_2) are pairwise disjoint, so the set $\{\theta(x, b^i) : i < \omega\}$ is 2-inconsistent, and $\theta(x, b_1, b_2)$ 2-divides over $\emptyset$.

In Example 5.2.6 we can go on: inside the interval (b_1, b_2) there are infinitely many disjoint subintervals, inside each of these infinitely many more, and so on. Choosing one interval at each stage, we obtain a consistent descending sequence of formulas; at each stage, the available choices are pairwise inconsistent. The tree property abstracts this picture. To state it, we use the notation of Chapter 2 for sequences, with ω in place of 2. The set $\omega^{<\omega}$ consists of the finite sequences of natural numbers, and the set ω^ω of the infinite ones. We write $|\sigma|$ for the length of σ and $\sigma^\frown\langle i\rangle$ for the extension of σ by the entry i. The initial segment of η of length n is denoted $\eta|n$, and we write $\sigma \sqsubseteq \tau$ when σ is an initial segment of τ.

Definition 5.2.7. Let $\varphi(x, y)$ be an L-formula, and let $k \geq 2$ be given. The formula φ has the *k-tree property* if there are tuples b_σ, for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, such that the following hold:

(a) for every $\eta \in \omega^\omega$, the set $\{\varphi(x, b_{\eta|n}) : 1 \leq n < \omega\}$ is consistent;

(b) for every $\sigma \in \omega^{<\omega}$, the set $\{\varphi(x, b_{\sigma^\frown\langle i\rangle}) : i < \omega\}$ is k-inconsistent.

The formula φ has the *tree property* if it has the k-tree property for some $k \geq 2$. The theory T is *simple* if no L-formula has the tree property.

Along each branch the formulas are consistent; among the immediate successors of a node, called *siblings*, they are k-inconsistent. Each step up a branch therefore chooses one of infinitely many incompatible options. A simple theory is one in which no single formula can make an ω-sequence of such choices. Shelah's original name for the simple theories, *treeless*, said precisely this. The trees of Chapter 2 were binary, with parameters at the nodes and elements at the leaves. The trees here branch infinitely, carry parameters at every node, and have no leaves (see Figure 5.1).

In the next remark we show that, by compactness, trees of finite height suffice. It is in this form that we verify the tree property below (see Proposition 5.3.2, Lemma 5.3.5, and Example 5.5.10): there, common solutions are found for only finitely many formulas of a branch at a time.

Remark 5.2.8. Let $\varphi(x, y)$ be an L-formula, and let $k \geq 2$ be given. Suppose that for every $n \geq 1$ there are tuples b_σ, for $\sigma \in \omega^{\leq n}$ with $\sigma \neq \emptyset$, such that the set $\{\varphi(x, b_{\eta|m}) : 1 \leq m \leq n\}$ is consistent for every $\eta \in \omega^n$ and condition (b) of Definition 5.2.7 holds for every $\sigma \in \omega^{<n}$. Then the formula φ has the k-tree property. Indeed, by compactness a branch $\{\varphi(x, b_{\eta|m}) : 1 \leq m < \omega\}$ is consistent if and only if each of its finite initial segments is. Hence the k-tree property is expressed, in variables y_σ for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, by the set of formulas

$$\exists x \bigwedge_{m=1}^{n} \varphi(x, y_{\eta|m}) \quad (\eta \in \omega^\omega,\ n \geq 1), \qquad \neg\exists x \bigwedge_{l=1}^{k} \varphi(x, y_{\sigma^\frown\langle i_l\rangle}) \quad (\sigma \in \omega^{<\omega},\ i_1 < \cdots < i_k).$$

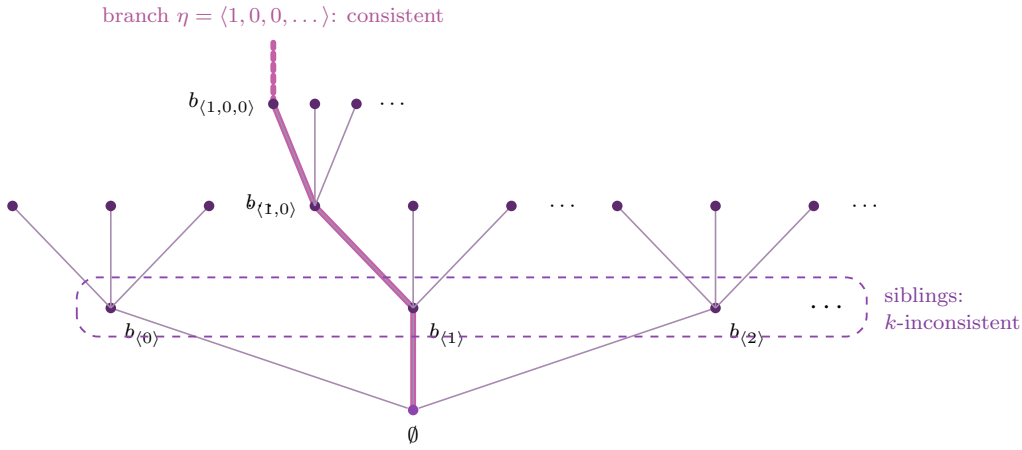


FIGURE 5.1. The k-tree property (see Definition 5.2.7): a tree growing upward with the parameter b_σ at each node σ, a branch η along which the formulas are consistent, and the k-inconsistent successors of the root (framed).

A finite subset of this set mentions only nodes of length at most some n, and the given family of height n satisfies it. By compactness and saturation, the whole set is realized in $\mathbb{M}$.

5.3 Simplicity between stability and NIP

Simplicity now takes its place in the classification. We show that every stable theory is simple, that a theory with the strict order property is not, and that the simple NIP theories are exactly the stable ones. We close with the first structural consequence of simplicity: every type of a finite tuple is nondividing over a subset of its domain of cardinality no greater than that of the theory.

Proposition 5.3.1. *If T is stable, then the theory T is simple.*

Proof. The idea is to count branches. A tree with λ successors at every node, where λ is an infinite cardinal, has $\lambda^{\aleph_0}$ branches. If the siblings at each node are k-inconsistent for some integer $k \geq 2$, a single type accepts fewer than k successors at each node. Each type can therefore follow at most $2^{\aleph_0}$ branches, and a stable formula has too few types to follow all the branches.

Suppose that $\varphi(x, y)$ has the k-tree property. We work with the cardinal $\lambda = \sup_n \lambda_n$, where $\lambda_0 = 2^\kappa$ and $\lambda_{n+1} = 2^{\lambda_n}$. This cardinal satisfies $\lambda \geq 2^\kappa$, the set $\lambda^{<\omega}$ has cardinality λ, and the cofinality of λ is ω; hence $\lambda^{\aleph_0} > \lambda$ by König's theorem.

☺ **Step 1: a wide tree.** There are tuples b_σ, for $\sigma \in \lambda^{<\omega}$ with $\sigma \neq \emptyset$, satisfying the two conditions of Definition 5.2.7 with λ in place of ω. Indeed, the conditions are expressed by formulas as in Remark 5.2.8, and a finite set of them mentions finitely many nodes. Let $J \subseteq \lambda$ be the finite set of entries of these nodes, and let $e \colon J \to \omega$ be injective. The map sending σ to the sequence $e \circ \sigma$ preserves initial segments and the relation of being siblings, so the tuples $b_{e \circ \sigma}$ of the given tree satisfy the finite set. By compactness and saturation, the wide tree exists in $\mathbb{M}$, and the downward Löwenheim–Skolem theorem provides a model $M \preceq \mathbb{M}$ with $|M| = \lambda$ that contains all the tuples b_σ.

☺ **Step 2: few types.** Recall that $S_\varphi(M)$ denotes the set of complete φ-types over M (see Definition 1.1.1). By Theorem 1.1.6, every $p \in S_\varphi(M)$ has a φ-definition that is a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$. The type p is determined by the set $D_p = \{b \in M : \varphi(x, b) \in p\}$, hence by its definition. There are at most $|M| + \aleph_0 = \lambda$ positive Boolean combinations of instances $\varphi(a, y)$ with $a \in M$, whence $|S_\varphi(M)| \leq \lambda$.

☺ **Step 3: many branches.** For each $\eta \in \lambda^\omega$, we choose a_η realizing $\{\varphi(x, b_{\eta|n}) : n \geq 1\}$ and write p_η for the φ-type $\mathrm{tp}_\varphi(a_\eta/M)$ of a_η over M. Fix $p \in S_\varphi(M)$, and consider, for each node σ, the set $\mathrm{Acc}_\sigma = \{\alpha < \lambda : \varphi(x, b_{\sigma^\frown\langle\alpha\rangle}) \in p\}$ of successors accepted by p. Each set Acc_σ has fewer than k elements. Otherwise the type p would contain k siblings, whose conjunction is inconsistent in $\mathbb{M}$ and hence in M, whereas p is consistent with the elementary diagram of M. If $p_\eta = p$, then every n satisfies $\eta(n) \in \mathrm{Acc}_{\eta|n}$. Hence the map sending such an η to the sequence of positions of $\eta(n)$ in $\mathrm{Acc}_{\eta|n}$ is injective into $(k-1)^\omega$, and at most $2^{\aleph_0}$ branches η satisfy $p_\eta = p$. It follows that

$$\lambda^{\aleph_0} = |\lambda^\omega| \leq |S_\varphi(M)| \cdot 2^{\aleph_0} \leq \lambda,$$

because $\lambda \geq 2^\kappa \geq 2^{\aleph_0}$; this contradicts $\lambda^{\aleph_0} > \lambda$. ☺

Simplicity is thus weaker than stability. In Proposition 5.3.2 below we show that a definable infinite order already carries the tree property: the strict order property (see Remark 4.1.6) produces the picture drawn after Example 5.2.6, of nested intervals with infinitely many disjoint subintervals inside each one.

Proposition 5.3.2. *If some formula has the strict order property of Remark 4.1.6, then the theory T is not simple.*

Proof. The idea is to replace the nested intervals of Example 5.2.6 by the bands of a chain of definable sets (see Figure 5.2). Let $\psi(x, y)$ have the strict order property, and let $\psi(\mathbb{M}, c)$ denote the set of solutions of $\psi(x, c)$ in $\mathbb{M}$. By compactness and saturation, there are tuples c_q, for $q \in \mathbb{Q}$, with $\psi(\mathbb{M}, c_q) \subsetneq \psi(\mathbb{M}, c_r)$ whenever $q < r$. Indeed, a finite set of these conditions mentions finitely many rationals $q_1 < \cdots < q_n$ and is satisfied by a strictly increasing chain of length n. Such a chain exists in some model of T, and hence in $\mathbb{M}$. Let $\theta(x; y_1, y_2)$ be the formula $\psi(x, y_2) \wedge \neg\psi(x, y_1)$, and for rationals $q < r$ let

$$\Delta(q, r) = \psi(\mathbb{M}, c_r) \setminus \psi(\mathbb{M}, c_q),$$

the set of solutions of $\theta(x; c_q, c_r)$, which is nonempty because the inclusion $\psi(\mathbb{M}, c_q) \subsetneq \psi(\mathbb{M}, c_r)$ is strict. Two properties of these sets follow from monotonicity of the chain:

(i) if $q \leq q' < r' \leq r$, then the set $\Delta(q', r')$ is contained in $\Delta(q, r)$;

(ii) if $r \leq q' < r'$, then the sets $\Delta(q, r)$ and $\Delta(q', r')$ are disjoint, since $\Delta(q, r) \subseteq \psi(\mathbb{M}, c_r) \subseteq \psi(\mathbb{M}, c_{q'})$ and $\Delta(q', r')$ is disjoint from $\psi(\mathbb{M}, c_{q'})$.

We assign rationals $q_\sigma < r_\sigma$ to the nodes $\sigma \in \omega^{<\omega}$ by recursion, starting from $(q_\emptyset, r_\emptyset) = (0, 1)$ and defining, with $d = r_\sigma - q_\sigma$,

$$q_{\sigma^\frown\langle i\rangle} = q_\sigma + d\,(1 - 2^{-i}), \qquad r_{\sigma^\frown\langle i\rangle} = q_\sigma + d\,(1 - 2^{-i-1}).$$

These numbers satisfy $q_\sigma \leq q_{\sigma^\frown\langle i\rangle} < r_{\sigma^\frown\langle i\rangle} \leq r_\sigma$, as well as $r_{\sigma^\frown\langle i\rangle} \leq q_{\sigma^\frown\langle j\rangle}$ for $i < j$. The parameters of the tree are the pairs $b_\sigma = (c_{q_\sigma}, c_{r_\sigma})$ for $\sigma \neq \emptyset$. For a branch η and $n \geq 1$, property (i) gives $\Delta(q_{\eta|n}, r_{\eta|n}) \subseteq \Delta(q_{\eta|m}, r_{\eta|m})$ for $m \leq n$, so any element of the nonempty set $\Delta(q_{\eta|n}, r_{\eta|n})$ satisfies $\theta(x, b_{\eta|m})$ for all $m \leq n$. Every finite part of the branch $\{\theta(x, b_{\eta|m}) : 1 \leq m < \omega\}$ is therefore consistent, and by compactness and saturation so is

the whole branch (see Remark 5.2.8). By property (ii), the sets attached to two siblings are disjoint. Thus the formula θ has the 2-tree property. ☺



FIGURE 5.2. The proof of Proposition 5.3.2: the rational intervals $(q_\sigma, r_\sigma]$ carried by the nodes σ, drawn at the height of the length of σ, with one branch in pink.

The two propositions combine with Shelah's dichotomy, quoted in Remark 4.1.6, to characterize the stable theories as the simple NIP ones.

Fact 5.3.3 (Shelah [She90, Chapter II]). *If T is unstable, then some formula has the independence property or some formula has the strict order property.*

Theorem 5.3.4. *A complete theory is stable if and only if it is simple and NIP.*

Proof. If T is stable, then the theory T is simple by Proposition 5.3.1, and every formula is NIP by Proposition 4.1.2. Conversely, if T is unstable, then by Fact 5.3.3 either some formula has the independence property or some formula has the strict order property. In the first case the theory is not NIP, and in the second it is not simple by Proposition 5.3.2. ☺

Thus simplicity and NIP are the two halves of stability: an unstable theory has a formula with the independence property or a formula with the strict order property; simplicity excludes the latter, and NIP excludes the former. Neither half contains the other: the random graph is simple and not NIP (see Theorem 5.4.6 and Example 4.1.4), and the rational order is NIP and not simple (see Example 4.1.3 and Proposition 5.3.2). In Figure 5.3, the Fraïssé limits studied in Section 5.5 take their places in the four regions determined by simplicity and NIP.

The first consequence of simplicity concerns dividing. In a stable theory, the type of a finite tuple over a model is the unique independent extension of its restriction to some submodel of size at most κ. We establish this local character for stable theories in Chapter 6. In simple theories it takes the form of a statement about dividing: in Theorem 5.3.6 below we show that in a simple theory every type of a finite tuple is nondividing over a subset of its domain of size at most κ. The proof runs the construction of the tree property backwards: a long sequence of dividing formulas is turned into a tree. We build the tree one level at a time, and use automorphisms to copy, above each node of the top level, a sequence that witnesses dividing.

Lemma 5.3.5. *Let $\varphi(x, y)$ be an L-formula, and let $k \geq 2$ be given.*

(1) *Let $n \geq 1$ be an integer, and let $a_0, \ldots, a_{n-1}$ be tuples such that $\{\varphi(x, a_m) : m < n\}$ is consistent and, for every $m < n$, the formula $\varphi(x, a_m)$ k-divides over $A_m = \{a_l : l < m\}$. Then there are tuples b_σ, for $\sigma \in \omega^{\leq n}$ with $\sigma \neq \emptyset$, such that*

(i) for every $\eta \in \omega^n$, we have $(b_{\eta|1}, \ldots, b_{\eta|n}) \equiv (a_0, \ldots, a_{n-1})$, and

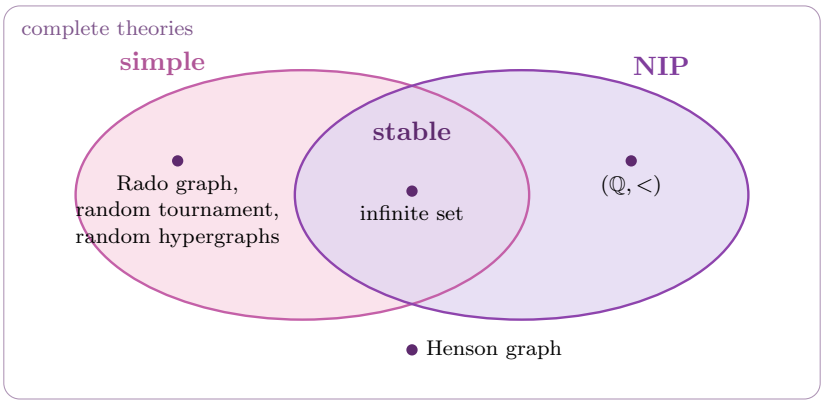


FIGURE 5.3. The stable theories as the simple NIP theories (see Theorem 5.3.4), with the pure infinite set, the Rado graph, the random tournament, the random hypergraphs, the rational order and the Henson graph in their regions.

(ii) for every $\sigma \in \omega^{<n}$, the set $\{\varphi(x, b_{\sigma^\frown\langle i\rangle}) : i < \omega\}$ is k-inconsistent.

In particular, these tuples form a tree of height n as in Remark 5.2.8.

(2) *If for every $n \geq 1$ there are tuples $a_0, \ldots, a_{n-1}$ as in (1), possibly depending on n, then the formula φ has the k-tree property. This holds in particular if $(a_n)_{n<\omega}$ is a sequence such that $\{\varphi(x, a_n) : n < \omega\}$ is consistent and, for every n, the formula $\varphi(x, a_n)$ k-divides over $A_n = \{a_m : m < n\}$.*

Proof. (1) For each $m < n$, fix a sequence $(a^i_m)_{i<\omega}$, indiscernible over A_m, with $a^0_m = a_m$ and $\{\varphi(x, a^i_m) : i < \omega\}$ k-inconsistent. We build the tuples level by level: for $j = 1, \ldots, n$ we find b_σ for $\sigma \in \omega^{\leq j}$, $\sigma \neq \emptyset$, satisfying (i) and (ii) with j in place of n and $(a_0, \ldots, a_{j-1})$ in place of $(a_0, \ldots, a_{n-1})$. Condition (i) gives condition (a) of Definition 5.2.7 for the branches of length j, because $\{\varphi(x, a_m) : m < j\}$ is consistent. For $j = 1$, the tuples $b_{\langle i\rangle} = a^i_0$ serve: condition (i) holds because all terms of an indiscernible sequence have the same type, and condition (ii) holds by the choice of (a^i_0).

Suppose that the tuples have been built for some $j < n$. For each $\eta \in \omega^j$, by (i) and strong homogeneity there is $f_\eta \in \mathrm{Aut}(\mathbb{M})$ with $f_\eta(a_0, \ldots, a_{j-1}) = (b_{\eta|1}, \ldots, b_{\eta|j})$. The tuples already built are kept, and the new ones are $b_{\eta^\frown\langle i\rangle} = f_\eta(a^i_j)$ for $i < \omega$. Since $a^i_j \equiv_{A_j} a_j$, we have

$$(b_{\eta|1}, \ldots, b_{\eta|j}, b_{\eta^\frown\langle i\rangle}) = f_\eta(a_0, \ldots, a_{j-1}, a^i_j) \equiv (a_0, \ldots, a_{j-1}, a^i_j) \equiv (a_0, \ldots, a_j),$$

which is (i) for $j + 1$. The set $\{\varphi(x, b_{\eta^\frown\langle i\rangle}) : i < \omega\}$ is the image under f_η of a k-inconsistent set, and hence is k-inconsistent; this is (ii) at the new level, and (ii) at the old levels is unchanged.

(2) By (1), for every $n \geq 1$ there is a tree of height n whose branches are consistent and whose siblings are k-inconsistent, and Remark 5.2.8 gives the k-tree property. The last assertion follows from (1), applied to the initial segments $a_0, \ldots, a_{n-1}$ of the sequence. ☺

Part (2) yields the local character of dividing at once. We need the finite form (1) as well: in Chapter 7 the tuples $a_0, \ldots, a_{n-1}$ are produced afresh for each n.

Theorem 5.3.6 (Local character of dividing). *Assume that T is simple, and let a finite tuple a and a small set B be given. Then there is $A \subseteq B$ with $|A| \leq \kappa$ such that* $\mathrm{tp}(a/B)$ *does not divide over A.*

Proof. Suppose not: then $\mathrm{tp}(a/B)$ divides over every subset of B of size at most κ. We choose, by recursion on $\alpha < \kappa^+$, finite tuples b_α from B, formulas $\varphi_\alpha(x, y) \in L$ and integers $k_\alpha \geq 2$. At stage α, the tuples chosen so far form the set $B_\alpha = \bigcup_{\beta<\alpha} b_\beta$, a subset of B of size at most κ. By assumption, the type $\mathrm{tp}(a/B)$ divides over B_α, so some formula $\varphi_\alpha(x, b_\alpha)$ of $\mathrm{tp}(a/B)$ with $\varphi_\alpha \in L$ and b_α from B divides over B_α. By Lemma 5.2.4, this formula k_α-divides over B_α for some $k_\alpha \geq 2$.

There are only κ pairs (φ, k) with $\varphi \in L$ and $k < \omega$, and the cardinal κ^+ is regular; hence some pair (φ, k) equals $(\varphi_\alpha, k_\alpha)$ for unboundedly many α. We list such indices as $\alpha_0 < \alpha_1 < \cdots$ and write $a_n = b_{\alpha_n}$. The set $A_n = \{a_m : m < n\}$ is contained in B_{α_n}, so the formula $\varphi(x, a_n)$ k-divides over A_n by Remark 5.2.5. The formulas $\varphi(x, a_n)$ all belong to $\mathrm{tp}(a/B)$, and so they form a consistent set. By Lemma 5.3.5(2), the formula φ has the k-tree property, contradicting simplicity. ☺

Remark 5.3.7. The converse of Theorem 5.3.6 holds as well: if every type over a model is nondividing over a submodel of size at most κ, then the theory T is simple (see Corollary 7.6.4). The version of this converse over arbitrary sets goes back to Chapter III of Shelah's *Classification Theory* [**She90**]. Kim and Pillay [**KP97**], building on Kim [**Kim98**], characterized the simple theories by an abstract relation of independence, as we do for the stable theories in Chapter 6. The Kim–Pillay theorem is the subject of Chapter 7. In Sections 5.4 and 5.5 we prove the corresponding statements directly, from the axioms, for the random graph and for a class of Fraïssé limits.

5.4 The Rado graph

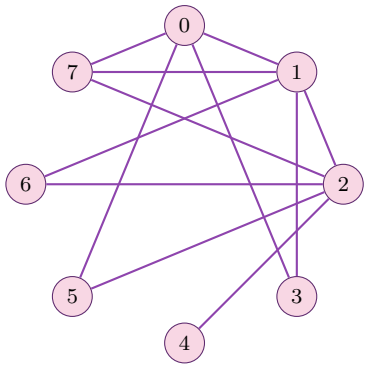
The random graph entered Chapter 4 as the basic example of a formula with the independence property (see Example 4.1.4), and in this section it becomes the model case of a simple unstable theory. We compute dividing in the random graph from the extension property alone. Two conclusions follow from the computation: the random graph is simple, and its natural independence relation satisfies every axiom of independence except uniqueness, with the independence theorem in its place.

Let T_{RG} be the theory of Example 4.1.4. The axioms of T_{RG} say that the edge relation E is irreflexive and symmetric, and they add one axiom for each pair m, n of natural numbers. For all disjoint sets U, V of vertices with $|U| = m$ and $|V| = n$, the axiom for m, n provides a vertex outside $U \cup V$ adjacent to every vertex of U and to no vertex of V. These last axioms are the *extension axioms*. A countable model is easy to write down.

Example 5.4.1. Let the vertices be the natural numbers, and for $i < j$ join i and j if the i-th binary digit of j is 1 (digits counted from the right, starting at 0). This graph satisfies the extension axioms. To see this, let U, V be disjoint finite sets of natural numbers, choose N larger than every element of $U \cup V$, and define

$$z = 2^N + \sum_{u \in U} 2^u.$$

Since $z \geq 2^N > N$, the number z is larger than every element of $U \cup V$, and in particular lies outside $U \cup V$. The binary digits of z are 1 at the positions $u \in U$ and 0 at the positions $v \in V$. Consequently, the element z is adjacent to every vertex of U and to no vertex of V. The graph is called the *Rado graph* $\mathcal{R}$ (see Figure 5.4).



j	binary	neighbors $i < j$
1	001	$\{0\}$
2	010	$\{1\}$
3	011	$\{0, 1\}$
4	100	$\{2\}$
5	101	$\{0, 2\}$
6	110	$\{1, 2\}$
7	111	$\{0, 1, 2\}$

FIGURE 5.4. The vertices $0, \ldots, 7$ of the Rado graph of Example 5.4.1, where for $i < j$ the vertex j is joined to i exactly when the binary expansion of j has digit 1 in position i, with a table of the expansions.

Fact 5.4.2. *The theory T_{RG} is complete, $\aleph_0$-categorical, and admits elimination of quantifiers; its countable model is the Rado graph (see Hodges* [**Hod93**, Theorem 7.4.1]*, or Fact 5.5.2 below).*

From now until the end of this section, the theory T is T_{RG}. By quantifier elimination, the type of a tuple over a set is determined by the equalities and edges that it satisfies with parameters from the set. We use this in the following form: for small tuples a, a' and a small set C,

(5.1) $a \equiv_C a'$ if and only if the map that fixes C pointwise and sends each coordinate of a to the corresponding coordinate of a' is a well-defined bijection from $C \cup a$ onto $C \cup a'$ that preserves edges and non-edges.

The computation of dividing rests on two tools. The first comes from the extension axioms: any small graph that agrees with $\mathbb{M}$ on a given small set embeds into $\mathbb{M}$ over the set, where an *embedding* is an injective map between graphs that preserves edges and non-edges.

Lemma 5.4.3. *Let C be a small set of vertices, and let X be a graph whose vertex set is the disjoint union of C and a small set X_0. Assume that the restriction of X to C is the graph induced by $\mathbb{M}$ on C. Then there is an embedding $g\colon X \to \mathbb{M}$ with $g(c) = c$ for every $c \in C$.*

Proof. Introduce a variable x_v for each $v \in X_0$, and let Σ be the set of formulas $x_v \neq x_w$, $x_v \neq c$, $E(x_v, x_w)^u$ and $E(x_v, c)^u$ (for $v \neq w$ in X_0 and $c \in C$) that describe X. The superscript $u \in \{0, 1\}$ records whether the edge is present in X. A realization of Σ is an embedding of the required kind. Since Σ is a small set of formulas over C, by saturation it suffices to show that every finite subset of Σ is satisfiable. A finite subset mentions finite sets $C_0 \subseteq C$ and $\{v_1, \ldots, v_m\} \subseteq X_0$, and the images $w_1, \ldots, w_m \in \mathbb{M}$ are chosen one at a time. If $w_1, \ldots, w_{l-1}$ have been chosen, we apply the extension axiom to the set U of elements of $C_0 \cup \{w_1, \ldots, w_{l-1}\}$ that must be adjacent to w_l, and to the set V of the others. The vertex

w_l so obtained lies outside $C_0 \cup \{w_1, \dots, w_{l-1}\}$. Hence the new vertices are distinct from each other and from C_0, and the elements $w_1, \dots, w_m$ satisfy the given finite subset of Σ. ☺

The second tool is Ramsey's theorem, in the sharper form announced after Lemma 5.2.2: over a finite set, every sequence of tuples of fixed length has an infinite subsequence that is itself indiscernible.

Lemma 5.4.4. *Let D be a finite set of vertices, and let $(b_i)_{i<\omega}$ be tuples of the same finite length. Then there is an infinite set $H \subseteq \omega$ such that $(b_i)_{i\in H}$ is indiscernible over D.*

Proof. The *diagram* over D of a tuple is the set of equalities and edges, and of their negations, that hold among its coordinates and elements of D. Since D is finite and the tuples have fixed length, there are finitely many possible diagrams over D of one tuple b_i, and of a pair (b_i, b_j). By the pigeonhole principle, all b_i with i in some infinite set have the same diagram over D. If we color each pair $i < j$ of this set by the diagram of (b_i, b_j) over D, Ramsey's theorem gives an infinite homogeneous set H. Each equality or edge involves at most two vertices, and therefore at most two of the tuples. Consequently, for $i_1 < \cdots < i_n$ in H, the diagram of $(b_{i_1}, \dots, b_{i_n})$ over D is determined by the diagrams of the b_{i_l} and of the pairs (b_{i_l}, b_{i_m}) with $l < m$. The diagrams of the b_{i_l} and of these pairs do not depend on the choice of $i_1 < \cdots < i_n$ in H. By (5.1), the sequence $(b_i)_{i\in H}$ is indiscernible over D. ☺

We are now in a position to compute dividing. As the example of $E(x, b)$ at the start of Section 5.2 suggests, a type over $C \cup B$ divides over C exactly when the type equates one of the variables with an element of B outside C.

Theorem 5.4.5 (Dividing in the Rado graph). *Let a be a finite tuple, and let C and B be small sets. Then the type $\mathrm{tp}(a/C \cup B)$ divides over C if and only if $a \cap B \not\subseteq C$.*

Proof. Suppose first that some coordinate $d = a_j$ of a lies in $B \setminus C$. The idea is to find a sequence of distinct vertices, indiscernible over C and beginning with d, along which the copies of the equation $x_j = d$ are pairwise inconsistent. The formula $x_j = d$ belongs to $\mathrm{tp}(a/C \cup B)$. Consider the graph on $C \cup \{d'_i : i < \omega\}$ in which each new vertex d'_i is adjacent to $c \in C$ exactly when d is, and no two new vertices are adjacent. Applied to this graph, Lemma 5.4.3 yields distinct vertices $d''_i \notin C$ with these adjacencies. By (5.1), for every $n \geq 1$ any two increasing n-tuples of (d''_i) have the same type over C, and the vertices d''_0 and d satisfy $d''_0 \equiv_C d$. Strong homogeneity then provides $f \in \mathrm{Aut}(\mathbb{M}/C)$ with $f(d''_0) = d$. The sequence $(f(d''_i))_{i<\omega}$ is indiscernible over C, begins with d, and consists of distinct vertices, so the set $\{x_j = f(d''_i) : i < \omega\}$ is 2-inconsistent. Hence the type $\mathrm{tp}(a/C \cup B)$ divides over C.

Conversely, suppose that $a \cap B \subseteq C$. Let $\varphi(x, b)$ be a formula of $\mathrm{tp}(a/C \cup B)$, with $\varphi \in L$ and b a finite tuple from $C \cup B$, and let $(b_i)_{i<\omega}$ be indiscernible over C with $b_0 = b$. The idea is to join a fresh copy of a to each b_i exactly as a is joined to b. The result is a single tuple a'' with $a''b_i \equiv_C ab$ for all i; such a tuple satisfies every $\varphi(x, b_i)$, and so the formula $\varphi(x, b)$ does not divide over C. For this to be well defined, coordinates shared by different terms of the sequence must play the same role in b. Here $e[p]$ denotes the p-th coordinate of a tuple e.

☺ **Claim.** If $i \neq j$ and $b_i[p] = b_j[q]$, then also $b[p] = b[q]$, and all terms of the sequence have the same p-th coordinate and the same q-th coordinate.

Exchanging the roles of (i, p) and (j, q) if necessary, we may assume that $i < j$, and we take any index $l > j$. Since $(b_i, b_j) \equiv_C (b_i, b_l)$, we have $b_i[p] = b_l[q]$, and the equivalence $(b_i, b_j) \equiv_C (b_j, b_l)$ yields $b_j[p] = b_l[q]$. Combining the two equalities, we obtain $b_j[p] = b_i[p] = b_j[q]$, and since $b_j \equiv_C b$, it follows that $b[p] = b[q]$. Moreover, since $b_i[p] = b_j[p]$ with $i < j$,

indiscernibility gives $b_{i'}[p] = b_{j'}[p]$ for all $i' < j'$; thus the p-th coordinate is constant. Finally, the relations $b_{i'} \equiv_C b$ and $b[p] = b[q]$ give $b_{i'}[p] = b_{i'}[q]$ for every i', so the q-th coordinate is constant as well. This proves the claim (see Figure 5.5).

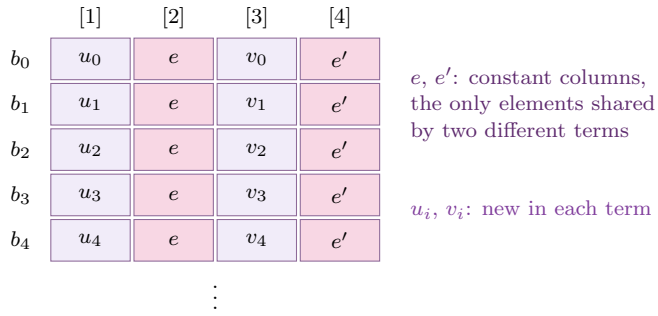


FIGURE 5.5. The claim in the proof of Theorem 5.4.5: the terms of a sequence indiscernible over C (rows) and their coordinates (columns), in which an element outside C that occurs in two different terms occupies a whole column.

With the claim in hand, we build the fresh copy of a. Let I be the set of indices v with $a_v \notin C$. For $v \in I$ we have $a_v \notin b$, because $b \subseteq C \cup B$ and $a \cap B \subseteq C$. Let $V = \bigcup_i b_i \setminus C$, and let X be the graph on $C \cup V$ together with fresh vertices a'_v for $v \in I$, where $a'_v = a'_w$ exactly when $a_v = a_w$, with the following edges:

(i) on $C \cup V$, the edges of $\mathbb{M}$;

(ii) the vertex a'_v is adjacent to $c \in C$ if and only if a_v is adjacent to c, and adjacent to a'_w if and only if a_v is adjacent to a_w;

(iii) if $e = b_i[p] \in V$, then the element a'_v is adjacent to e if and only if a_v is adjacent to $b[p]$.

Clause (iii) does not depend on the choice of i and p with $e = b_i[p]$. Indeed, if also $e = b_j[q]$, then the coordinates $b[p]$ and $b[q]$ coincide, by the claim when $i \neq j$ and because $b_i \equiv_C b$ when $i = j$. By Lemma 5.4.3, there is an embedding g of X into $\mathbb{M}$ fixing $C \cup V$. Let a'' be the tuple with $a''_v = g(a'_v)$ for $v \in I$ and $a''_v = a_v$ for $v \notin I$. Given an index i, consider the map that fixes C and sends a to a'', and b to b_i, coordinatewise. On $C \cup b$ this map is the bijection given by $b \equiv_C b_i$; the coordinates a_v with $v \in I$ lie outside $C \cup b$, and their images lie outside $C \cup V$. Hence the map is a well defined bijection from $C \cup a \cup b$ onto $C \cup a'' \cup b_i$. By (i)–(iii) this map preserves edges and non-edges, and therefore $a''b_i \equiv_C ab$ by (5.1). As i was arbitrary, the tuple a'' is the one we sought. ☺

By Theorem 5.4.5, each dividing step uses up a coordinate of a, and a tuple of length n can therefore divide at most n times in succession. This count is the heart of the proof that the random graph is simple.

Theorem 5.4.6. *Let a be a tuple of length n, and let $D_0 \subseteq D_1 \subseteq \cdots \subseteq D_m$ be small sets such that $\mathrm{tp}(a/D_{j+1})$ divides over D_j for every $j < m$. Then the length m of the chain is at most n. The theory T_{RG} is simple.*

Proof. By Theorem 5.4.5, applied with $C = D_j$ and $B = D_{j+1}$, the tuple a meets D_{j+1} in a coordinate outside D_j. Thus the sets $a \cap D_0 \subsetneq a \cap D_1 \subsetneq \cdots \subsetneq a \cap D_m$ increase strictly, and the tuple a has at least m distinct coordinates, whence $m \leq n$.

For the second assertion, suppose that $\varphi(x, y)$ has the k-tree property, witnessed by tuples b_σ, and let n be the length of x. The idea is to find, along a branch, $n+1$ dividing steps, which the first part forbids. For this we prune the tree in such a way that siblings become indiscernible over the nodes below them. By recursion, we define a set $P \subseteq \omega^{\leq n+1}$ of nodes, to which the empty sequence belongs. If $\sigma \in P$ has length at most n, let $D_\sigma = \bigcup_{1 \leq l \leq |\sigma|} b_{\sigma|l}$, a finite set (empty when $\sigma = \emptyset$). In that case we choose by Lemma 5.4.4 an infinite set $H_\sigma \subseteq \omega$ such that $(b_{\sigma^\frown\langle i\rangle})_{i \in H_\sigma}$ is indiscernible over D_σ, and we admit the sequences $\sigma^\frown\langle i\rangle$ with $i \in H_\sigma$ to P. We fix $\eta \in \omega^{n+1}$ with $\eta|l \in P$ for every $l \leq n+1$ and an element a realizing $\{\varphi(x, b_{\eta|l}) : 1 \leq l \leq n+1\}$, and we abbreviate $D_{\eta|l}$ as D_l for $l \leq n+1$, where $D_{\eta|n+1}$ is defined by the same formula.

Given $l \leq n$, we write $\sigma = \eta|l$; the terms $b_{\sigma^\frown\langle i\rangle}$ with $i \in H_\sigma$ and $i \geq \eta(l)$ form a sequence indiscernible over D_l that begins with $b_{\eta|l+1}$, and the corresponding formulas are k-inconsistent. Hence the formula $\varphi(x, b_{\eta|l+1})$ divides over D_l, and, since this formula belongs to $\mathrm{tp}(a/D_{l+1})$, the type $\mathrm{tp}(a/D_{l+1})$ divides over D_l. The chain $D_0 \subseteq \cdots \subseteq D_{n+1}$ thus has $n+1$ dividing steps, while by the first part of the theorem a chain of dividing steps for a tuple of length n has at most n of them (see Figure 5.6). This contradiction shows that no formula has the tree property, and the theory T_{RG} is simple. ☺

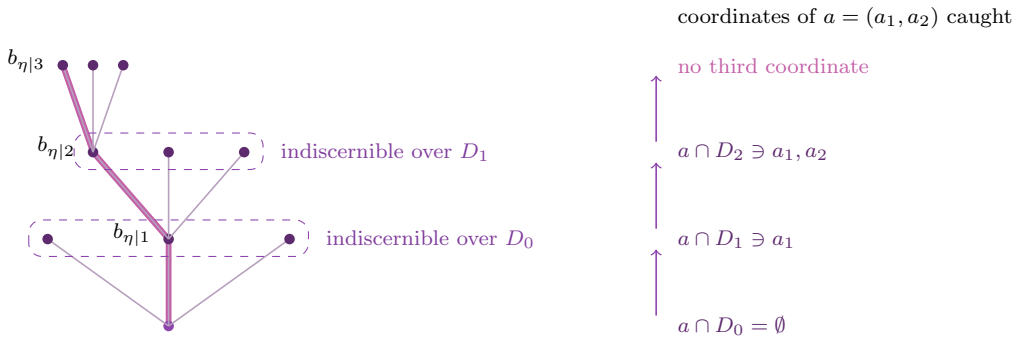


FIGURE 5.6. The proof of Theorem 5.4.6 for a pair $x = (x_1, x_2)$, that is, with $n = 2$: a pruned tree whose branch (pink) catches a new coordinate of the realization a at each dividing step.

Simplicity settled, we turn to the second aim of this section, the independence relation of the random graph. An independence relation over a model M is a relation $A \mathop{\smile\!\!\!\!\!\!|}_M B$, read "$A$ is independent from B over M", and the following axioms are the yardstick for such relations. In Chapter 6 we show that a theory is stable exactly when it carries a relation satisfying all these axioms (see Theorem 6.2.2). In Theorem 5.4.9 below we show that the random graph carries a relation satisfying all of them but one.

The axioms are easier to accept once their purpose is clear. A relation deserves the name of independence only if it expresses the thought that B tells us nothing about A beyond what M already tells us. Several requirements follow at once from this thought. Whether A is independent from B over M should depend only on how A, B and M sit in the monster model, and not on how they are named (INV). What is independent from a set should be independent from the parts of the set (MON), and independence should be decided by finite parts alone (FC). If B tells us nothing new about A, then the set A should tell us nothing new about B (SYM). Every type over M should be realizable in a position of independence

from any given set (EX). Enlarging the base within B should not destroy independence (BMON), and independence over a chain of bases should compose (TR). A finite tuple should be independent from a model over some small submodel (LOC), in order that independence can always be witnessed by a small base. Finally, if B tells us nothing new about A, there is nothing to choose between two ways of placing A independently from B, and so the independent extension should be unique (UN). This last requirement separates stability from simplicity: the random graph meets all the others, and in place of uniqueness it has the independence theorem (see Theorem 5.4.9). The axioms below are these requirements, made exact (see Figure 5.7).

Definition 5.4.7. An *independence relation (over models)* is a relation $A \mathop{\smile\!\!\!\!\!\!|}_M B$ between small sets A, B and models M satisfying the following axioms, for all small A, A', B, B' and all models M, N.

(INV) *Invariance.* The relation $A \mathop{\smile\!\!\!\!\!\!|}_M B$ implies $\sigma(A) \mathop{\smile\!\!\!\!\!\!|}_{\sigma(M)} \sigma(B)$ for every $\sigma \in \mathrm{Aut}(\mathbb{M})$.

(MON) *Monotonicity.* If $A \mathop{\smile\!\!\!\!\!\!|}_M B$, $A' \subseteq A$ and $B' \subseteq B$, then also $A' \mathop{\smile\!\!\!\!\!\!|}_M B'$.

(BMON) *Base monotonicity.* If $A \mathop{\smile\!\!\!\!\!\!|}_M B$ and N is a model with $M \subseteq N \subseteq B$, then the relation also holds over N, that is, $A \mathop{\smile\!\!\!\!\!\!|}_N B$.

(TR) *Transitivity.* If $M \subseteq N$, $A \mathop{\smile\!\!\!\!\!\!|}_M N$ and $A \mathop{\smile\!\!\!\!\!\!|}_N B$, then the two relations combine to give $A \mathop{\smile\!\!\!\!\!\!|}_M NB$.

(FC) *Finite character.* The relation $A \mathop{\smile\!\!\!\!\!\!|}_M B$ holds if and only if $a \mathop{\smile\!\!\!\!\!\!|}_M b$ for all finite tuples a from A and b from B.

(SYM) *Symmetry.* The relation $A \mathop{\smile\!\!\!\!\!\!|}_M B$ holds if and only if $B \mathop{\smile\!\!\!\!\!\!|}_M A$.

(EX) *Existence (extension).* For all A, M, B there is $A' \equiv_M A$ with $A' \mathop{\smile\!\!\!\!\!\!|}_M B$.

(UN) *Uniqueness (stationarity).* If $A \equiv_M A'$, $A \mathop{\smile\!\!\!\!\!\!|}_M B$ and $A' \mathop{\smile\!\!\!\!\!\!|}_M B$, then the sets A and A' have the same type over MB.

(LOC) *Local character.* For every finite tuple a and every model N there is a model $M \subseteq N$ with $|M| \leq \kappa$ and $a \mathop{\smile\!\!\!\!\!\!|}_M N$.

For the random graph, the computation of Theorem 5.4.5 suggests a candidate: a tuple should be independent from a set when it shares with it nothing outside the base.

Definition 5.4.8. For small sets A, B, C, write $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}_C B$ if $A \cap B \subseteq C$. We call $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ *trivial independence*; the superscript distinguishes it from the abstract relations $\mathop{\smile\!\!\!\!\!\!|}$ of Definition 5.4.7.

By Theorem 5.4.5, a finite tuple a satisfies $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}_C B$ if and only if $\mathrm{tp}(a/C \cup B)$ does not divide over C. We compare the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ with the axioms of Definition 5.4.7, read with arbitrary models M, N of T_{RG} as bases. We add one further axiom, *anti-reflexivity* (AR): if $a \mathop{\smile\!\!\!\!\!\!|}_M a$ for a finite tuple a, then every coordinate of a lies in M. This axiom reappears for stable theories in Theorem 6.2.2.

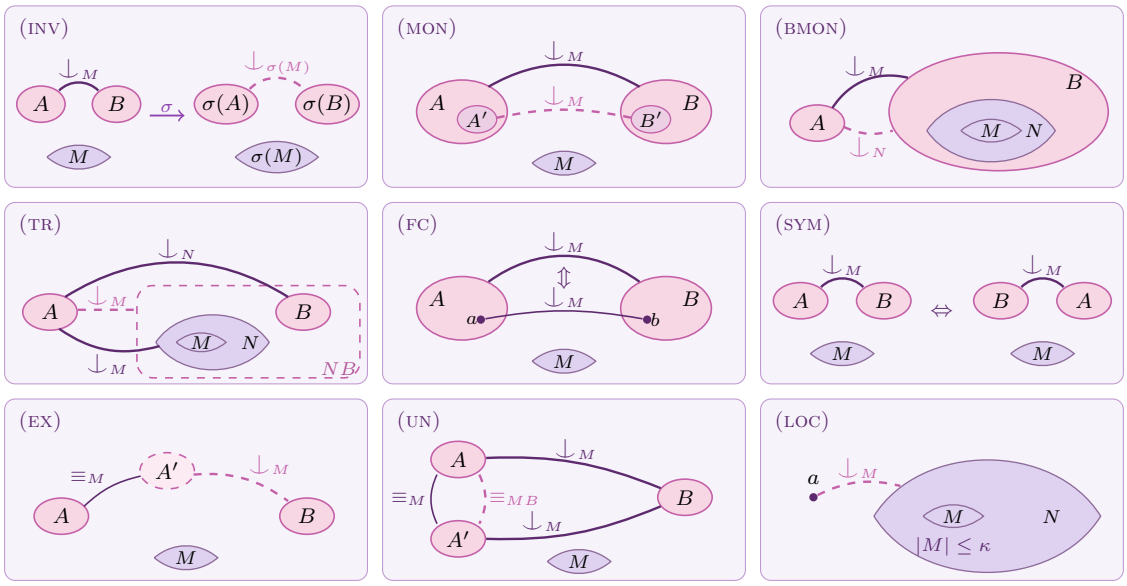


FIGURE 5.7. The axioms of Definition 5.4.7, one panel for each, with hypotheses drawn solid in purple and conclusions dashed in pink.

Theorem 5.4.9 (Independence in the Rado graph). *The following hold in* T_{RG}*:*

(a) *The relation* $\downarrow^{\mathrm{t}}$ *satisfies* (INV), (MON), (BMON), (TR), (FC), (SYM), (EX), (LOC) *and* (AR).

(b) *The relation* $\downarrow^{\mathrm{t}}$ *does not satisfy* (UN).

(c) (Independence theorem.) *Let* C *be a small set, and let* b_1, b_2, c_1, c_2 *be finite tuples. If* $b_1 \downarrow^{\mathrm{t}}_C b_2$, $c_1 \downarrow^{\mathrm{t}}_C b_1$, $c_2 \downarrow^{\mathrm{t}}_C b_2$ *and* $c_1 \equiv_C c_2$*, then there is* c *with* $c \equiv_{Cb_1} c_1$, $c \equiv_{Cb_2} c_2$ *and* $c \downarrow^{\mathrm{t}}_C b_1 b_2$.

Proof. (a) Invariance, monotonicity, finite character, symmetry and anti-reflexivity are immediate from the definition. *Base monotonicity*: if $A \cap B \subseteq M$ and $M \subseteq N \subseteq B$, then also $A \cap B \subseteq N$. *Transitivity*: if $M \subseteq N$, $A \cap N \subseteq M$ and $A \cap B \subseteq N$, then the inclusions $A \cap B \subseteq A \cap N \subseteq M$ give $A \cap (N \cup B) \subseteq M$. *Local character*: given a finite tuple a and a model N, the downward Löwenheim–Skolem theorem gives a model $M \preceq N$ with $a \cap N \subseteq M$ and $|M| \leq \kappa$, and then the relation $a \downarrow^{\mathrm{t}}_M N$ holds. *Existence*: given A, M and B, enumerate A as a tuple $(a_v)_v$, and let X be the graph on $M \cup B$ together with fresh vertices a'_v, one for each coordinate a_v outside M (equal coordinates receiving equal vertices). The edges on $M \cup B$ are those of $\mathbb{M}$; the vertices a'_v are joined to each other and to M as the a_v are, and to no vertex of $B \setminus M$. By Lemma 5.4.3, the graph X embeds into $\mathbb{M}$ over $M \cup B$, and the resulting tuple A' satisfies $A' \equiv_M A$ by (5.1) and $A' \cap B \subseteq M$.

(b) Let M be a model, and let b be a vertex not in M. By Lemma 5.4.3 there are vertices $a, a' \notin M \cup \{b\}$, adjacent to no vertex of M, with a adjacent to b and a' not. These vertices satisfy $a \equiv_M a'$ by (5.1), $a \downarrow^{\mathrm{t}}_M b$ and $a' \downarrow^{\mathrm{t}}_M b$, but $a \not\equiv_{Mb} a'$, since only a is adjacent to b.

(c) Let I be the set of indices v with $(c_1)_v \notin C$; since $c_1 \equiv_C c_2$, it is also the set of indices with $(c_2)_v \notin C$, and the coordinates $(c_1)_v$ and $(c_2)_v$ coincide for $v \notin I$. Let X be the graph on $C \cup b_1 \cup b_2$, with the edges of $\mathbb{M}$, together with fresh vertices c'_v for $v \in I$ (equal coordinates

receiving equal vertices). In X, the new vertices are joined to each other and to $C \cup b_1$ as the coordinates of c_1 are, and to $b_2 \setminus C$ as the coordinates of c_2 are. The two prescriptions overlap only on C, because $b_1 \cap b_2 \subseteq C$, and on C (and among the new vertices) they agree, because $c_1 \equiv_C c_2$. By Lemma 5.4.3, there is an embedding g of X into $\mathbb{M}$ fixing $C \cup b_1 \cup b_2$. Let c be the tuple with $c_v = g(c'_v)$ for $v \in I$ and $c_v = (c_1)_v$ for $v \notin I$. The coordinates $(c_1)_v$ with $v \in I$ lie outside $C \cup b_1$, since $c_1 \cap b_1 \subseteq C$, and the coordinates c_v with $v \in I$ lie outside $C \cup b_1 \cup b_2$. Hence the map fixing $C \cup b_1$ and sending c_1 to c is a bijection preserving edges and non-edges, and the tuple c satisfies $c \equiv_{Cb_1} c_1$ by (5.1). In the same way, we have $c \equiv_{Cb_2} c_2$. Finally, the coordinates of c outside C are the c_v with $v \in I$, which avoid $b_1 \cup b_2$, and therefore $c \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}_{C} b_1 b_2$. ☺

The smallest interesting case of the construction in the proof of (c) appears in Figure 5.8.

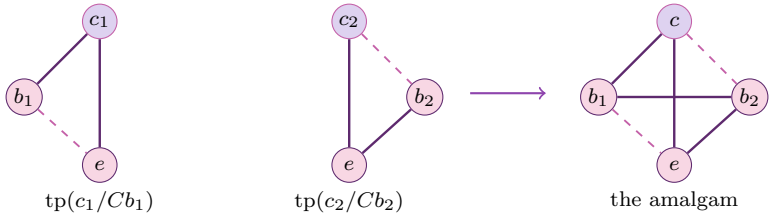


FIGURE 5.8. The independence theorem in the Rado graph, with $C = \{e\}$ (see Theorem 5.4.9(c)), where solid lines are edges and dashed lines non-edges.

Stability and simplicity, side by side

In a stable theory, as we show in Chapter 6, every type over a model M has exactly one independent extension to $M \cup B$. This extension is the heir, obtained by applying the definitions of the type, one for each formula, to parameters from $M \cup B$. In the Rado graph, a type over M that is not realized in M has more than one independent extension to $M \cup \{b\}$, where b is a vertex outside M: there is one for each way of joining the new vertices to b (see Theorem 5.4.9(b)). In place of uniqueness stands the independence theorem (see Theorem 5.4.9(c)): independent extensions over two independent sets can always be realized together. Dividing is measured by coordinates (see Theorem 5.4.5), each dividing step uses up one coordinate, and so no formula can have the tree property (see Theorem 5.4.6).

5.5 Fraïssé limits

In this section we isolate the two properties of the class of finite graphs on which the arguments of Section 5.4 depend, and we prove that every Fraïssé limit of a class with these properties is simple. In the proofs of Theorems 5.4.5 and 5.4.9, new vertices were joined to old ones as prescribed by several types at once, and Lemma 5.4.3 embedded the resulting graph into the monster model. Since every finite graph embeds into the random graph, the prescriptions only had to be consistent with one another. In a Fraïssé limit only the structures of a given class embed, and the structure assembled from the prescriptions must belong to the class. In the proof of the independence theorem, three structures are given, each on the

base together with two of three tuples (the new tuple and the two parameters). The task of completing these three structures to a single structure on the base together with all three tuples is a 3-amalgamation problem.

Throughout this section, the language L is finite and relational. For an L-structure N, the *age* of N is the class of finite L-structures that embed into N.

Definition 5.5.1. A *Fraïssé class* is a nonempty class $\mathcal{K}$ of finite L-structures, closed under isomorphism, with the following properties:

(a) *hereditary property*: every substructure of a structure in $\mathcal{K}$ belongs to $\mathcal{K}$;

(b) *joint embedding property*: any two structures in $\mathcal{K}$ embed into a common structure in $\mathcal{K}$;

(c) *amalgamation property*: if $A, B_1, B_2 \in \mathcal{K}$ and $f_l \colon A \to B_l$, for $l = 1, 2$, are embeddings, then there are $B \in \mathcal{K}$ and embeddings $g_l \colon B_l \to B$ with $g_1 \circ f_1 = g_2 \circ f_2$.

Fraïssé's theorem, stated below, describes the limit of such a class by two properties, which we define first. An L-structure N is *ultrahomogeneous* if every isomorphism between finite substructures of N extends to an automorphism of N. Given a Fraïssé class $\mathcal{K}$, consider structures $A \subseteq B$ in $\mathcal{K}$ with $|B \setminus A| = 1$, where A may be empty. A structure N has the *extension property* for $\mathcal{K}$ if, for all such A and B, every embedding of A into N extends to an embedding of B into N.

Fact 5.5.2 (Fraïssé [Fra54]; see [Hod93, Theorems 7.1.2 and 7.4.1]). *Let $\mathcal{K}$ be a Fraïssé class.*

(a) *There is, up to isomorphism, a unique countable ultrahomogeneous L-structure $M_{\mathcal{K}}$ whose age is $\mathcal{K}$.*

(b) *The theory $T_{\mathcal{K}}$ of $M_{\mathcal{K}}$ is $\aleph_0$-categorical and admits elimination of quantifiers.*

(c) *An L-structure N is a model of $T_{\mathcal{K}}$ if and only if every finite substructure of N belongs to $\mathcal{K}$ and N has the extension property for $\mathcal{K}$.*

The structure $M_{\mathcal{K}}$ is the *Fraïssé limit* of $\mathcal{K}$.

Example 5.5.3. The following are Fraïssé classes; their amalgamation property is checked by interleaving for linear orders and by taking unions for the others (orienting the new pairs, for tournaments).

(1) The finite sets, in the language with equality only. The limit is a countably infinite set, and the theory of the limit is stable: by quantifier elimination, for every $n \geq 1$, the type of an n-tuple over a set A is determined by the equalities among its coordinates and with elements of A. Consequently, there are at most $|A| + \aleph_0$ such types. Hence the theory is λ-stable for every infinite λ, and it is stable by Lemma 5.1.2.

(2) The finite linear orders. The limit is $(\mathbb{Q}, <)$.

(3) The finite graphs. The limit is the Rado graph, and the theory $T_{\mathcal{K}}$ is T_{RG}.

(4) The finite tournaments (complete graphs with an orientation of each edge). The limit is the *random tournament*.

(5) For $r \geq 2$, the finite r-uniform hypergraphs. The limit is the *random r-hypergraph*.

(6) The finite triangle-free graphs. The limit is the *Henson graph* $\mathcal{H}$.

In the proofs of Section 5.4, the new vertices were always fresh: no new vertex was identified with an old one. In Definition 5.5.4 below, we ask the same of amalgamation in $\mathcal{K}$. To state the definition, we treat structures as sets with relations and say that a structure D *extends* a structure B if B is a substructure of D.

Definition 5.5.4. Let $\mathcal{K}$ be a Fraïssé class.

(a) The class $\mathcal{K}$ has *disjoint* 2-*amalgamation* if whenever $B_1, B_2 \in \mathcal{K}$ have a common substructure A with $B_1 \cap B_2 = A$, some structure in $\mathcal{K}$ on $B_1 \cup B_2$ extends B_1 and B_2.

(b) The class $\mathcal{K}$ has *disjoint* 3-*amalgamation* if the following holds. Let B_1, B_2, B_3 be structures with a common substructure $A \in \mathcal{K}$ and with $B_l \cap B_m = A$ for distinct $l, m \in \{1, 2, 3\}$, and let $B_{12}, B_{13}, B_{23} \in \mathcal{K}$ be structures on $B_1 \cup B_2$, $B_1 \cup B_3$, $B_2 \cup B_3$ such that B_{lm} extends B_l and B_m for $1 \leq l < m \leq 3$. Then some structure in $\mathcal{K}$ on $B_1 \cup B_2 \cup B_3$ extends B_{12}, B_{13} and B_{23}.

In a 3-amalgamation problem, every tuple contained in the union of two of the sets B_l already has its relations prescribed by one of the structures B_{lm}. Only the relations on tuples that meet all three of the sets $B_l \setminus A$ remain free. Both the general shape of a 3-amalgamation problem and two problems without solution are sketched in Figure 5.9.

Example 5.5.5. All classes of Example 5.5.3 have disjoint 2-amalgamation. For sets, graphs, triangle-free graphs and hypergraphs, take the union of the two structures without relations between $B_1 \setminus A$ and $B_2 \setminus A$. In the case of triangle-free graphs, a triangle in the union that is contained neither in B_1 nor in B_2 would need two adjacent vertices, one in each of $B_1 \setminus A$ and $B_2 \setminus A$. For tournaments, orient the new pairs from B_1 to B_2. For linear orders, place the elements of $B_1 \setminus A$ before those of $B_2 \setminus A$ within each interval determined by A.

The classes (1), (3), (4) and (5) have disjoint 3-amalgamation. For sets, graphs and tournaments, every relation involves at most two elements. Hence in a 3-amalgamation problem every relation is already prescribed, and the union of B_{12}, B_{13} and B_{23} is a solution. For r-uniform hypergraphs, declare that no hyperedge meets all three of the sets $B_l \setminus A$.

The classes (2) and (6) do not have disjoint 3-amalgamation. Take $A = \emptyset$ and one-element sets $B_1 = \{x\}$, $B_2 = \{y\}$, $B_3 = \{z\}$. For linear orders, let the structures B_{12}, B_{23} and B_{13} say that $x < y$, $y < z$ and $z < x$ respectively: no linear order extends all three. For triangle-free graphs, let each B_{lm} be an edge: the only graph on $\{x, y, z\}$ extending them is a triangle.

For the rest of this section, we fix a Fraïssé class $\mathcal{K}$, write $T = T_{\mathcal{K}}$, and let $\mathbb{M}$ be a monster model of T. By quantifier elimination, the equivalence (5.1) holds with "preserves edges and non-edges" replaced by "is an isomorphism of substructures"; we refer to this version as (5.1) as well. The two tools of Section 5.4, the embedding of graphs over a set (see Lemma 5.4.3) and the extraction of indiscernible subsequences by Ramsey's theorem (see Lemma 5.4.4), generalize directly.

Lemma 5.5.6. *Let C be a small subset of $\mathbb{M}$, and let X be an L-structure whose universe is the disjoint union of C and a small set X_0. Assume that the restriction of X to C is the substructure of $\mathbb{M}$ on C, and that the finite substructures of X all belong to $\mathcal{K}$. Then there is an embedding $g\colon X \to \mathbb{M}$ with $g(c) = c$ for every $c \in C$.*

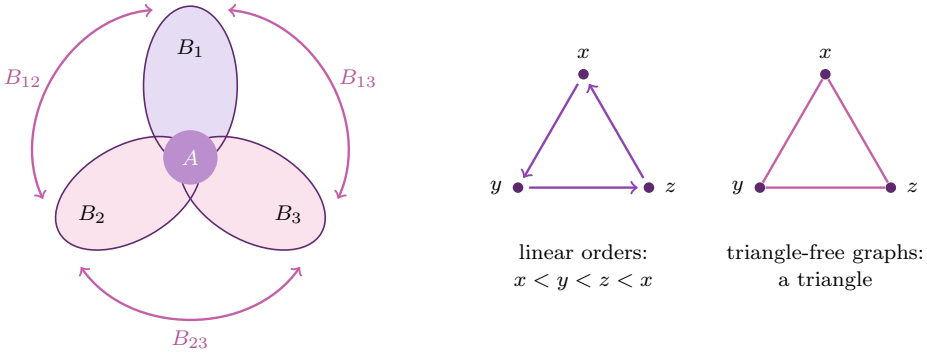


FIGURE 5.9. Left, a disjoint 3-amalgamation problem (see Definition 5.5.4(b)); right, two problems without solution, with $A = \emptyset$ and one point in each B_l (see Example 5.5.5).

Proof. The argument is that of the proof of Lemma 5.4.3, with the atomic formulas of L in place of edges. By saturation, it suffices to embed over C_0, for all finite sets $C_0 \subseteq C$ and $\{v_1, \ldots, v_m\} \subseteq X_0$, the substructure of X on $C_0 \cup \{v_1, \ldots, v_m\}$. The substructures of X on the sets $C_0 \cup \{v_1, \ldots, v_l\}$ with $l \leq m$ belong to $\mathcal{K}$, and each of them extends the previous one by one element. Applying the extension property of Fact 5.5.2(c) m times, starting from the inclusion of C_0 into $\mathbb{M}$, gives the required embedding. ☺

Lemma 5.5.7. *Let D be a finite subset of $\mathbb{M}$, and let $(b_i)_{i<\omega}$ be tuples of the same finite length. Then there is an infinite set $H \subseteq \omega$ such that $(b_i)_{i\in H}$ is indiscernible over D.*

Proof. Let r be the larger of 2 and the maximal arity of a symbol of L. We argue as in the proof of Lemma 5.4.4, coloring each r-element set $i_1 < \cdots < i_r$ by the diagram over D of $(b_{i_1}, \ldots, b_{i_r})$. Here the diagram of a tuple over D is the set of atomic formulas with parameters from D, and of their negations, satisfied by the tuple. Ramsey's theorem gives an infinite homogeneous set H. Each atomic formula involves at most r of the tuples. If it involves the tuples $b_{j_1}, \ldots, b_{j_s}$, with $j_1 < \cdots < j_s$ in H and $s \leq r$, then its truth value is read off from the diagram of any increasing r-tuple from H that begins with $j_1, \ldots, j_s$. This truth value therefore depends only on the position of the tuples involved. Hence, for every $n \geq 1$, all increasing n-tuples from H have the same diagram over D, and the sequence is indiscernible over D by (5.1). ☺

The first half of the proof of Theorem 5.4.5 needed an indiscernible sequence of distinct conjugates of a vertex, and in the Rado graph such a sequence was written down by hand. In general we obtain it from disjoint 2-amalgamation and the extraction of indiscernible sequences (see Lemma 5.2.2).

Lemma 5.5.8. *Assume that $\mathcal{K}$ has disjoint 2-amalgamation. Let C be a small set, and let d be an element of $\mathbb{M}$ not in C. Then there is a sequence $(d_i)_{i<\omega}$ of distinct elements, indiscernible over C, with $d_0 = d$.*

Proof. First, there are infinitely many distinct realizations of $\mathrm{tp}(d/C)$. By quantifier elimination and saturation, it suffices to find, for every finite $C_0 \subseteq C$ and every $n \geq 1$, some n distinct elements with the same diagram over C_0 as d. Let Y be the substructure of $\mathbb{M}$ on $C_0 \cup \{d\}$. By $n-1$ applications of disjoint 2-amalgamation, there is a structure in $\mathcal{K}$ on $C_0 \cup \{d^1, \ldots, d^n\}$, with new elements d^l, in which each $C_0 \cup \{d^l\}$ is a copy of Y over C_0. By

Lemma 5.5.6, this structure embeds into $\mathbb{M}$ over C_0, and the images of $d^1, \dots, d^n$ are the required elements. Let $(d'_i)_{i<\omega}$ be distinct realizations of $\mathrm{tp}(d/C)$.

Next, we apply Lemma 5.2.2 over C to the sequence (d'_i). The Ehrenfeucht–Mostowski type of this sequence over C contains every formula of $\mathrm{tp}(d/C)$, in one variable, and the formula $y_1 \neq y_2$. Hence Lemma 5.2.2 gives a sequence (d''_i) of distinct realizations of $\mathrm{tp}(d/C)$, indiscernible over C. An automorphism of $\mathbb{M}$ over C sending d''_0 to d carries (d''_i) to the required sequence. ☺

In Theorem 5.5.9 below we show that, with these lemmas, the arguments of Section 5.4 go through for Fraïssé limits. In these limits, disjoint 2-amalgamation supplies the conjugates that make a type divide, and disjoint 3-amalgamation supplies the common realizations that keep it from dividing.

Theorem 5.5.9. *Let $\mathcal{K}$ be a Fraïssé class in a finite relational language with disjoint 2-amalgamation and disjoint 3-amalgamation, and let $T = T_{\mathcal{K}}$.*

(1) *For a finite tuple a and small sets C, B, the type $\mathrm{tp}(a/C \cup B)$ divides over C if and only if $a \cap B \not\subseteq C$.*

(2) *If a is a tuple of length n and $D_0 \subseteq \cdots \subseteq D_m$ are small sets such that $\mathrm{tp}(a/D_{j+1})$ divides over D_j for every $j < m$, then the length m of the chain is at most n. The theory T is simple.*

(3) *The relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ of Definition 5.4.8 satisfies all the axioms of Definition 5.4.7 except possibly (UN), together with (AR) and the independence theorem stated in Theorem 5.4.9(c).*

Proof. (1) If $a \cap B \not\subseteq C$, the argument of the first part of the proof of Theorem 5.4.5 applies, with Lemma 5.5.8 in place of the explicit sequence (d''_i).

Conversely, suppose that $a \cap B \subseteq C$. Let $\varphi(x, b)$ belong to $\mathrm{tp}(a/C \cup B)$, with b a finite tuple from $C \cup B$, and let $(b_i)_{i<\omega}$ be indiscernible over C with $b_0 = b$. Let $\Sigma(x)$ be the set of formulas $\psi(x, b_i)$, for $i < \omega$ and $\psi(x, y)$ an $L(C)$-formula with $\psi(x, b) \in \mathrm{tp}(a/C \cup b)$. A realization a'' of Σ satisfies $a''b_i \equiv_C ab$ for every i, and in particular satisfies every $\varphi(x, b_i)$. By saturation, it therefore suffices to show that every finite subset Σ_0 of Σ is realized. Here, unlike in the Rado graph, the structure to be embedded is not determined by the prescriptions, and 3-amalgamation supplies it.

The set Σ_0 involves finitely many indices $i_1 < \cdots < i_h$ and finitely many formulas, which by quantifier elimination we may take to be quantifier-free with parameters from a finite set $C_0 \subseteq C$. Enlarging C_0, we may assume that it contains $a \cap C$ and $b \cap C$. Let F be the set of elements not in C that belong to two distinct terms of the sequence. By the claim in the proof of Theorem 5.4.5, which used only indiscernibility, each element of F is a coordinate that is constant along the sequence. It follows that $F \subseteq b$ and $F \subseteq b_i$ for every i. Since $b_i \cap C = b \cap C$ for every i, we obtain $b_i \cap b_j \subseteq C_0 \cup F$ for $i \neq j$. The amalgamation below takes place over the set $A = C_0 \cup F$.

Let I be the set of indices v with $a_v \notin C$. No coordinate a_v with $v \in I$ lies in b, because b is a tuple from $C \cup B$ and $a \cap B \subseteq C$. We introduce fresh elements a'_v for $v \in I$, equal exactly when the a_v are, and let a' be the tuple with these coordinates and with $a'_v = a_v$ for $v \notin I$. For $l \leq h$, let π_l be the map from $A \cup b_{i_l} \cup a'$ onto $A \cup b \cup a$ that fixes A, sends b_{i_l} to b coordinatewise, and sends a' to a. The map π_l is a well defined bijection, as the following three observations show. First, a coordinate $b_{i_l}[p]$ lies in A if and only if $b[p]$ does,

and in that case the two coordinates are equal. For coordinates in C_0 both assertions hold because $b_{i_l} \equiv_C b$, and for coordinates in F they hold because these coordinates are constant. Second, the map is injective on b_{i_l} because $b_{i_l} \equiv_C b$. Third, the coordinates a_v with $v \in I$ lie outside $A \cup b$, because $a \cap b \subseteq C$ and $F \subseteq b$. Let Z_l be the structure on $A \cup b_{i_l} \cup a'$ that makes π_l an isomorphism onto the substructure of $\mathbb{M}$ on $A \cup b \cup a$. Then the structure Z_l belongs to $\mathcal{K}$, its restriction to $A \cup a'$ does not depend on l, and its restriction to $A \cup b_{i_l}$ is the substructure of $\mathbb{M}$. The last assertion holds because, on $A \cup b_{i_l}$, the map π_l is the restriction of an automorphism of $\mathbb{M}$ over C that sends b_{i_l} to b and fixes F.

By induction on $l \leq h$, we build $Y_l \in \mathcal{K}$ on $A \cup a' \cup b_{i_1} \cup \cdots \cup b_{i_l}$ extending $Z_1, \ldots, Z_l$ and the substructure of $\mathbb{M}$ on $A \cup b_{i_1} \cup \cdots \cup b_{i_l}$; the induction begins with $Y_1 = Z_1$. For $l \geq 2$, disjoint 3-amalgamation over A is applied to

$$B_1 = A \cup a', \qquad B_2 = A \cup b_{i_1} \cup \cdots \cup b_{i_{l-1}}, \qquad B_3 = A \cup b_{i_l},$$

with $B_{12} = Y_{l-1}$, $B_{13} = Z_l$, and B_{23} the substructure of $\mathbb{M}$ on $B_2 \cup B_3$. The sets B_1, B_2, B_3 meet pairwise in A: the elements a'_v with $v \in I$ are fresh, and we have $b_{i_l} \cap b_{i_j} \subseteq A$. The three structures agree on B_1, B_2 and B_3, by the properties of the Z_l and by the induction hypothesis on Y_{l-1}. A solution is the required Y_l (see Figure 5.10).

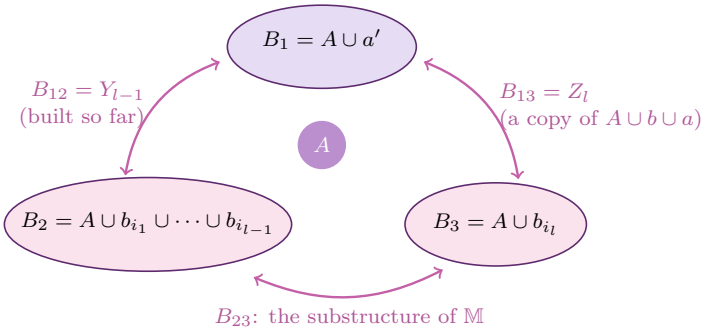


FIGURE 5.10. The inductive step in the proof of Theorem 5.5.9(1), in which disjoint 3-amalgamation over A glues Y_{l-1}, Z_l and the substructure of $\mathbb{M}$ into Y_l.

By Lemma 5.5.6, applied to Y_h over the finite set $A \cup b_{i_1} \cup \cdots \cup b_{i_h}$, there is an embedding g of Y_h into $\mathbb{M}$ fixing this set. Let a'' be the tuple $g(a')$. For each l, the map $g \circ \pi_l^{-1}$ is an isomorphism from the substructure on $A \cup b \cup a$ onto the substructure on $A \cup b_{i_l} \cup a''$ that fixes C_0, sends b to b_{i_l}, and sends a to a''. Hence the tuples $a''b_{i_l}$ and ab satisfy the same quantifier-free formulas over C_0, and therefore the tuple a'' realizes Σ_0, which completes the proof of (1).

(2) The first statement follows from (1) as in the proof of Theorem 5.4.6, and the second from the first as there, with Lemma 5.5.7 in place of Lemma 5.4.4.

(3) The axioms other than existence are checked as in the proof of Theorem 5.4.9(a). Existence and the independence theorem are proved as in Theorem 5.4.9(a) and (c), with compactness reducing each to finitely many formulas over a finite part C_0 of the base. In both proofs, the structure to be embedded is supplied by amalgamation. For existence, given A, M and B, let A_0 and B_0 be finite parts of A and B. Let $C_0 \subseteq M$ be a finite set containing $A_0 \cap M$ and $B_0 \cap M$. We apply disjoint 2-amalgamation over the substructure on C_0 to the substructure on $C_0 \cup B_0$ and to a copy over C_0 of the substructure on $C_0 \cup A_0$ whose new elements are disjoint from B_0. For the independence theorem, we take C_0 to contain

$(b_1 \cup b_2 \cup c_1) \cap C$, and we let c' be a fresh copy of the coordinates of c_1 outside C. We apply disjoint 3-amalgamation over the substructure on C_0 to

$$B_1 = C_0 \cup c', \qquad B_2 = C_0 \cup b_1, \qquad B_3 = C_0 \cup b_2,$$

with B_{12} making $C_0 \cup c' \cup b_1$ a copy of $C_0 \cup c_1 \cup b_1$, and with B_{13} making $C_0 \cup c' \cup b_2$ a copy of $C_0 \cup c_2 \cup b_2$. The third structure, B_{23}, is the substructure of $\mathbb{M}$. The sets B_1, B_2, B_3 meet pairwise in C_0 because $b_1 \cap b_2 \subseteq C$, and the structures B_{12} and B_{13} agree on B_1 because $c_1 \equiv_C c_2$. We embed the solution over $C_0 \cup b_1 \cup b_2$ by Lemma 5.5.6 and conclude by saturation. ☺

By Example 5.5.5, Theorem 5.5.9 applies to the Rado graph, the random tournament and the random r-hypergraphs. The theorem applies as well to the pure infinite set, which is stable. The class of finite linear orders has disjoint 2-amalgamation but not disjoint 3-amalgamation, and the theory of its limit, the rational order, is not simple (see Proposition 5.3.2). The Henson graph, which we treat in the next example, shows that disjoint 3-amalgamation is essential even when the forbidden configuration is a triangle rather than an order.

Example 5.5.10. Let $\mathcal{K}$ be the class of finite triangle-free graphs; its limit is the Henson graph $\mathcal{H}$, whose theory is $T_{\mathcal{K}}$. We show that the formula $\varphi(x; y_1, y_2)$, defined as $E(x, y_1) \wedge E(x, y_2)$, has the 2-tree property. The idea has two halves. The parameters of any two siblings are joined by an edge between one coordinate of each, so that a common solution of the two formulas would close a triangle; the coordinates along a branch are kept pairwise nonadjacent, so that a common neighbor may be added to them without creating a triangle.

Let G be the graph with vertices u_σ, v_σ, for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, whose edges are the pairs $\{u_{\sigma^\frown\langle i\rangle}, v_{\sigma^\frown\langle j\rangle}\}$ with $\sigma \in \omega^{<\omega}$ and $i < j$. Every edge joins a vertex u_τ to a vertex $v_{\tau'}$, so the graph G is bipartite and in particular triangle-free. By Lemma 5.5.6 with $C = \emptyset$, we may assume that G is an induced subgraph of $\mathbb{M}$, and we write $b_\sigma = (u_\sigma, v_\sigma)$.

Siblings are 2-inconsistent. For $i < j$, a solution of $\varphi(x, b_{\sigma^\frown\langle i\rangle}) \wedge \varphi(x, b_{\sigma^\frown\langle j\rangle})$ would be adjacent to $u_{\sigma^\frown\langle i\rangle}$ and to $v_{\sigma^\frown\langle j\rangle}$, which are adjacent to each other, and would form a triangle with them (see Figure 5.11). This is impossible, because every finite substructure of a model of $T_{\mathcal{K}}$ is triangle-free.

Branches are consistent. Two distinct nodes of a branch have different lengths and so are never siblings. Hence, for $\eta \in \omega^\omega$ and $n \geq 1$, the $2n$ vertices $u_{\eta|m}, v_{\eta|m}$ $(1 \leq m \leq n)$ are pairwise nonadjacent. The graph obtained from them by adding one vertex adjacent to all of them is triangle-free, and Lemma 5.5.6 embeds it over these vertices; the new vertex satisfies $\varphi(x, b_{\eta|m})$ for $1 \leq m \leq n$. By Remark 5.2.8, condition (a) of Definition 5.2.7 follows.

The formula φ therefore witnesses that the theory $T_{\mathcal{K}}$ is not simple. The theory also has the independence property. To see this, take an integer $n \geq 1$, pairwise nonadjacent vertices $w_1, \ldots, w_n$ and a set $S \subseteq \{1, \ldots, n\}$; adding a vertex adjacent exactly to the w_l with $l \in S$ creates no triangle. Hence the edge relation shatters $\{w_1, \ldots, w_n\}$.

Remark 5.5.11. The proof of Theorem 5.5.9(2) shows that a type of an n-tuple admits at most n successive dividing steps. Two notions of simplicity theory measure such chains (see the monographs of Wagner [**Wag00**] and Casanovas [**Cas11**]). A simple theory is *supersimple* if, for every finite tuple a and every set B, there is a finite subset B_0 of B such that $\mathrm{tp}(a/B)$

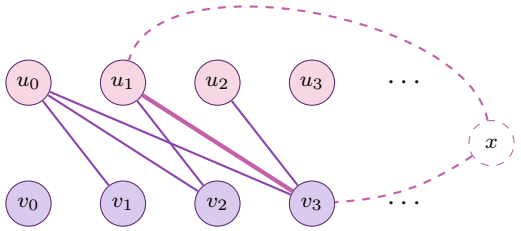


FIGURE 5.11. The siblings $b_i = (u_i, v_i)$ of one node in Example 5.5.10, with u_i joined to v_j exactly when $i < j$, and the triangle x, u_1, v_3 (pink) that a common solution of $\varphi(x, b_1)$ and $\varphi(x, b_3)$ would close.

does not divide over B_0. The *SU-rank* is defined by induction: every type has SU-rank at least 0; the SU-rank of a type p over A is at least $\alpha + 1$ if p has an extension of SU-rank at least α that divides over A; and, for a limit ordinal λ, the SU-rank of p is at least λ if the SU-rank of p is at least α for every $\alpha < \lambda$. In this language, the theory T is supersimple, and every type of a single element has SU-rank at most 1. Disjoint amalgamation in higher dimensions has further consequences: Kruckman [**Kru19**] studied Fraïssé classes with disjoint n-amalgamation for every n, and showed that their limits have pseudofinite theories. A theory is pseudofinite when each of its sentences has a finite model. On the other side of the line, the class of finite triangle-free graphs is the basic example of a *free amalgamation* class. In a free amalgamation class, the union of two structures over a common substructure, with no new relations, is always an amalgam. Shelah showed that the Henson graph has the property SOP_3 of his hierarchy of strict order properties, but not SOP_4 [**She96**, Claim 2.8]. Conant [**Con17**] developed the model theory of free amalgamation theories in general, and showed in particular that they are NSOP_4: none of them has SOP_4.

5.6 The real-valued case

At the end of Chapters 1 and 4 we passed from formulas to bounded functions on arbitrary sets, and in this section the tree property makes the same passage. For a formula, compactness turns consistent finite parts of branches into consistent branches, and the proof that stable theories are simple counts definitions (see Proposition 5.3.1). For a function on arbitrary sets, a tree carries its own witnesses, branch by branch, and the role of a definition is taken by an approximation by polynomials in instances (see Theorem 1.2.11). In Theorem 5.6.4 below we show that approximations can be counted as definitions are, provided that they are counted up to a tolerance. The tolerance is supplied by a gap between the values that the tree requires on its branches and the values that it forbids among siblings. We treat the 2-tree property of Definition 5.2.7, in which siblings are pairwise incompatible.

For finite sequences we use the notation recalled before Definition 5.2.7, in particular $|\sigma|$ for the length of σ and $\sigma \sqsubseteq \tau$ for the relation of being an initial segment.

Definition 5.6.1. Let A, B be sets, let $\varphi\colon A \times B \to [0,1]$ be a function, and let $r < s$ be reals. The function φ has the 2-*tree property with gap* (r, s) if there are $b_\sigma \in B$, for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, such that the following hold:

(a) for every $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, there is $a \in A$ with $\varphi(a, b_{\sigma|m}) \geq s$ for $1 \leq m \leq |\sigma|$;

(b) for every $\tau \in \omega^{<\omega}$ and all $i \neq j$, there is no $a \in A$ with $\varphi(a, b_{\tau^\frown\langle i\rangle}) > r$ and $\varphi(a, b_{\tau^\frown\langle j\rangle}) > r$.

The function φ has the 2-*tree property* if it has the 2-tree property with gap (r, s) for some reals $r < s$.

Condition (a) asks for witnesses to the finite parts of the branches, which is the form in which branches are given on arbitrary sets. Only gaps with $0 \leq r < s \leq 1$ matter for a $[0, 1]$-valued function: a gap with $s > 1$ violates condition (a), and a gap with $r < 0$ violates condition (b), because condition (a) provides elements of A and every value of φ exceeds r.

For a formula, the two definitions agree, whatever the gap. Fix reals $0 \leq r < s \leq 1$, and observe that a value of a $\{0, 1\}$-valued function is at least s, or larger than r, exactly when it is 1. A formula $\varphi(x, y)$ has the 2-tree property of Definition 5.2.7 if and only if, for some model M, the function φ on $M^{|x|} \times M^{|y|}$ has the 2-tree property with gap (r, s). Indeed, suppose that the tuples b_σ witness the 2-tree property in $\mathbb{M}$. By the downward Löwenheim–Skolem theorem, some model $M \preceq \mathbb{M}$ contains all of them. Since M is an elementary submodel of $\mathbb{M}$, the model M contains witnesses for the finite parts of the branches, and pairs of siblings remain inconsistent in M. For the converse, recall that every model of T embeds elementarily in $\mathbb{M}$. A tree as in Definition 5.6.1 in a model $M \preceq \mathbb{M}$ is a tree in $\mathbb{M}$ whose finite branches are consistent and whose siblings are 2-inconsistent. It follows from Remark 5.2.8 that the formula φ has the 2-tree property.

The proof that a stable function has no tree as in Definition 5.6.1 counts branches. They are most easily counted in a binary tree of infinite height, the real-valued form of the trees of Definition 2.1.2. In Lemma 5.6.3 below we show that every witness of the 2-tree property contains such a binary tree.

Definition 5.6.2. Let A and B be sets, let $\varphi\colon A \times B \to [0, 1]$ be a function, and let $r < s$ be reals. A *binary tree of infinite height for φ with gap (r, s)* consists of families $(b_\sigma : \sigma \in 2^{<\omega})$ in B and $(a_\tau : \tau \in 2^{<\omega})$ in A such that, for all $\sigma, \tau \in 2^{<\omega}$,

$$\sigma^\frown\langle 1\rangle \sqsubseteq \tau \implies \varphi(a_\tau, b_\sigma) \geq s, \qquad \sigma^\frown\langle 0\rangle \sqsubseteq \tau \implies \varphi(a_\tau, b_\sigma) \leq r.$$

Lemma 5.6.3. *Let A and B be sets, and let $r < s$ be reals. If a function $\varphi\colon A \times B \to [0, 1]$ has the 2-tree property with gap (r, s), then it has a binary tree of infinite height with gap (r, s).*

Proof. The idea is to keep two children of each node and discard the others. Let (b_σ) witness the 2-tree property with gap (r, s). The map $f\colon 2^{<\omega} \to \omega^{<\omega}$ is defined by recursion: $f(\emptyset) = \emptyset$, $f(\sigma^\frown\langle 1\rangle) = f(\sigma)^\frown\langle 0\rangle$ and $f(\sigma^\frown\langle 0\rangle) = f(\sigma)^\frown\langle 1\rangle$. Then the map f preserves initial segments, and the nodes $f(\sigma^\frown\langle 0\rangle)$ and $f(\sigma^\frown\langle 1\rangle)$ are siblings. We take $b'_\sigma = b_{f(\sigma^\frown\langle 1\rangle)}$ for $\sigma \in 2^{<\omega}$, and for $\tau \in 2^{<\omega}$ let $a'_\tau \in A$ satisfy $\varphi(a'_\tau, b_{f(\tau)|m}) \geq s$ for $1 \leq m \leq |f(\tau)|$, as condition (a) allows (for $\tau = \emptyset$ any element of A will do).

Let $\sigma, \tau \in 2^{<\omega}$ be given. If $\sigma^\frown\langle 1\rangle \sqsubseteq \tau$, then also $f(\sigma^\frown\langle 1\rangle) \sqsubseteq f(\tau)$, so the element b'_σ lies on the branch below $f(\tau)$ and satisfies $\varphi(a'_\tau, b'_\sigma) \geq s$. If $\sigma^\frown\langle 0\rangle \sqsubseteq \tau$, then likewise $f(\sigma^\frown\langle 0\rangle) \sqsubseteq f(\tau)$, whence $\varphi(a'_\tau, b_{f(\sigma^\frown\langle 0\rangle)}) \geq s > r$. Since the nodes $f(\sigma^\frown\langle 0\rangle)$ and $f(\sigma^\frown\langle 1\rangle)$ are siblings, condition (b) now gives $\varphi(a'_\tau, b'_\sigma) \leq r$. Thus the families (b'_σ) and (a'_τ) form a binary tree of infinite height with gap (r, s). ☺

The embedding f of the proof of Lemma 5.6.3 is drawn in Figure 5.12. At the node $f(\sigma)$, the binary tree keeps two children of the ω-branching tree. The child 0 carries the parameter b'_σ of the binary node σ, and a leaf that turns right at σ passes through this child. A leaf that turns left passes through the child 1 instead and takes values at least s there. By the incompatibility of siblings, such a leaf therefore takes values at most r at the child 0.

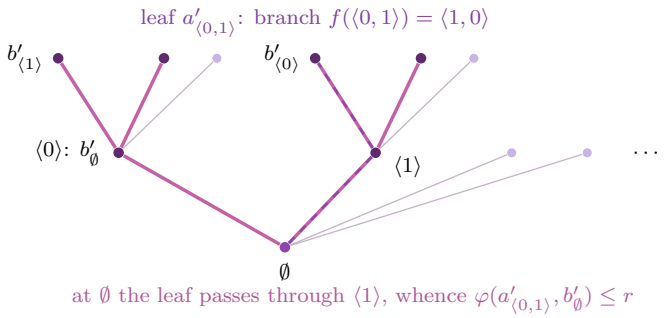


FIGURE 5.12. The embedding f of the proof of Lemma 5.6.3: the binary tree inside the ω-branching tree on the children 0 and 1 of the nodes it uses (pink), with the other children (pale) discarded.

Theorem 5.6.4. *Let A and B be sets. A stable function $\varphi\colon A \times B \to [0,1]$ (see Definition 1.2.5) has no binary tree of infinite height with a gap. In particular, a stable function does not have the 2-tree property.*

Proof. The idea is to count: a binary tree of infinite height has $2^{\aleph_0}$ branches, and the types along distinct branches differ by at least $s - r$ at some node. A stable function, on the other hand, has only countably many approximate definitions over a countable set, and two branches with nearly the same definition cannot differ by $s - r$ anywhere.

Suppose that (b_σ) and (a_τ) form a binary tree of infinite height for φ with gap (r, s); throughout, we write ε for $(s - r)/5$. Let $A_0 = \{a_\tau : \tau \in 2^{<\omega}\}$ and $B_0 = \{b_\sigma : \sigma \in 2^{<\omega}\}$, and let φ_0 be the restriction of φ to $A_0 \times B_0$. Every instance of Definition 1.2.5 for φ_0 is an instance for φ, so the function φ_0 is stable.

☺ **Step 1: types along branches.** Fix a nonprincipal ultrafilter $\mathcal{U}$ on ω, and for $\eta \in 2^\omega$ let p_η be the type $\lim_{n,\mathcal{U}} \operatorname{ltp}_{\varphi_0}(a_{\eta|n})$ in the space $\mathrm{LS}(\varphi_0)$ of left φ_0-types (see Definition 1.2.7). Let $\sigma \sqsubseteq \eta$ be given. For every $n > |\sigma|$ we have $\sigma^\frown\langle\eta(|\sigma|)\rangle \sqsubseteq \eta|n$, so the value $\varphi(a_{\eta|n}, b_\sigma)$ is at least s if $\eta(|\sigma|) = 1$ and at most r if $\eta(|\sigma|) = 0$, and the same bound holds for $p_\eta(b_\sigma)$. If $\eta \neq \eta'$ and σ is the longest common initial segment of η and η', then the values $\eta(|\sigma|)$ and $\eta'(|\sigma|)$ differ, and therefore

$$|p_\eta(b_\sigma) - p_{\eta'}(b_\sigma)| \geq s - r. \tag{5.2}$$

☺ **Step 2: approximate definitions.** Let $\hat{\varphi}_0$ be the separately continuous function of Proposition 1.2.8 for φ_0, defined on the product of $\mathrm{LS}(\varphi_0)$ with the space $\mathrm{RS}(\varphi_0)$ of right φ_0-types. As in the proof of Proposition 4.2.2, we have $\hat{\varphi}_0(p, \operatorname{rtp}_{\varphi_0}(b)) = p(b)$ for all $p \in \mathrm{LS}(\varphi_0)$ and $b \in B_0$, where $\operatorname{rtp}_{\varphi_0}(b)$ is the right φ_0-type of b. Let $\eta \in 2^\omega$ be given. By Theorem 1.2.11, in the form given by its proof, there are $a_1, \ldots, a_k \in A_0$ and a real polynomial P with $|\hat{\varphi}_0(p_\eta, q) - P(q(a_1), \ldots, q(a_k))| \leq \varepsilon$ for all $q \in \mathrm{RS}(\varphi_0)$. Taking $q = \operatorname{rtp}_{\varphi_0}(b)$ gives

$$\big|p_\eta(b) - P\big(\varphi(a_1, b), \ldots, \varphi(a_k, b)\big)\big| \leq \varepsilon \qquad \text{for all } b \in B_0.$$

Each monomial is bounded by 1 on $[0,1]^k$, so changing the coefficients of P by small enough amounts gives a polynomial P_η with rational coefficients such that $|P - P_\eta| \leq \varepsilon$ on $[0,1]^k$. We attach to η the tuple $\bar{a}_\eta = (a_1, \ldots, a_k)$. Then every $b \in B_0$ satisfies $|p_\eta(b) - P_\eta(\varphi(a_1, b), \ldots, \varphi(a_k, b))| \leq 2\varepsilon$.

☺ **Step 3: counting.** The set A_0 is countable and there are countably many polynomials with rational coefficients, so there are only countably many pairs $(\bar{a}_\eta, P_\eta)$, while the set 2^ω is uncountable. Hence there are $\eta \neq \eta'$ with $(\bar{a}_\eta, P_\eta) = (\bar{a}_{\eta'}, P_{\eta'})$, and then the functions p_η and $p_{\eta'}$ differ by at most $4\varepsilon < s - r$ on B_0. This contradicts (5.2) and proves the first statement. The second statement follows from the first by Lemma 5.6.3. ☺

For formulas, Theorem 5.6.4 gives a second proof of Proposition 5.3.1 for the 2-tree property. If $\varphi(x, y)$ is stable, then it is stable on every model (see Remark 1.2.2), so the corresponding $\{0, 1\}$-valued function on every model is stable in the sense of Definition 1.2.5 (see Proposition 1.2.4). By Theorem 5.6.4, this function does not have the 2-tree property with any gap, and by the paragraph following Definition 5.6.1 the formula φ does not have the 2-tree property. The two proofs count different things: definitions over a model of large cardinality in the proof of Proposition 5.3.1, approximate definitions over a countable set here. The second count is the real-valued form of the counting in the proof of (1) ⇒ (2) in Theorem 2.5.1. Indeed, a binary tree of infinite height is a Cantor scheme of types, and stability leaves room only for countably many types up to a tolerance.

The edge relation of the random graph separates the two notions in the second statement of Theorem 5.6.4: it is unstable, and it does not have the 2-tree property. Let M be a model of T_{RG}. By the extension axioms (see Example 4.1.4), every sequence of distinct vertices of M is independent for the edge relation in the sense of Definition 4.2.1. The edge relation is therefore not NIP, and by Proposition 4.2.2 the edge relation is not stable. On the other hand, the random graph is simple (see Theorem 5.4.6), so by the paragraph following Definition 5.6.1 the edge relation does not have the 2-tree property on any model.

5.7 Exercises

The exercises follow the sections of the chapter. Throughout, we write N_E for the structure $(\omega \times \omega, E)$ in which $(m, n)\, E\, (m', n')$ holds if and only if $m = m'$, and T_E for its theory. Thus the relation E is an equivalence relation with infinitely many classes, all of them infinite. The mark $(*)$ indicates the exercises that ask for more than a direct application of the text.

1. *(Section 5.1.)* Assume that T_E admits elimination of quantifiers (see Exercise 10). Let A be a small set and let $n \geq 1$ be given. Show that the type over A of an n-tuple a is determined by three kinds of data. The first specifies which coordinates of a are equal and which are E-equivalent. The second specifies which coordinates are equal to an element of A, and to which one. The third specifies which coordinates are E-equivalent to an element of A, and to which E-class of elements of A. Deduce that $|S_x(A)| \leq |A| + \aleph_0$, that T_E is λ-stable for every infinite λ, and that T_E is stable and simple (see Lemma 5.1.2 and Proposition 5.3.1). Show, in contrast, that the theory of $(\mathbb{Q}, <)$ has $2^{\aleph_0}$ complete 1-types over a countable model, and deduce from Lemma 5.1.2 that it is λ-stable for no infinite λ. (Hint: for the last assertion, find n-ladders for $x < y$; see Definition 1.1.4.)

2. *(Section 5.2.)* Let C be a small set and b a tuple. Show that the formula $x = b$ divides over C if and only if $\mathrm{tp}(b/C)$ has infinitely many realizations, and that it then 2-divides over C. (Hint: if $(b_i)_{i<\omega}$ is indiscernible over C and $b_i = b_j$ for some $i < j$, then the sequence is constant. For the converse, apply Lemma 5.2.2 to distinct realizations of $\mathrm{tp}(b/C)$, and note that the formula $y_1 \neq y_2$ belongs to their Ehrenfeucht–Mostowski type.) The next example shows that the condition $b \notin C$ alone does not make the

formula $x = b$ divide over C. Let T be the theory of a structure (N, E) in which E is an equivalence relation with infinitely many classes, each with exactly two elements. Let c and b be distinct E-equivalent elements. Show that $x = b$ does not divide over $\{c\}$, although $b \notin \{c\}$.

3. *(Section 5.2.)* Let C be a small set, let $k \geq 2$ be an integer, let $\varphi(x, y)$ and $\psi(x, z)$ be L-formulas, and let b and c be tuples such that $\mathbb{M} \models \forall x\,(\psi(x, c) \to \varphi(x, b))$. Show that if $\varphi(x, b)$ k-divides over C, then the formula $\psi(x, c)$ k-divides over C. (Hint: let $(b_i)_{i<\omega}$ witness k-dividing, choose c_i with $b_i c_i \equiv_C bc$, and apply Lemma 5.2.2 over C to the sequence $(b_i c_i)_{i<\omega}$; the formulas $\forall x\,(\psi(x, z_1) \to \varphi(x, y_1))$ and $\neg\exists x \bigwedge_{l=1}^{k} \varphi(x, y_l)$ belong to its Ehrenfeucht–Mostowski type.) Deduce that if $\varphi(x, b)$ divides over C, then so does $\varphi(x, b) \wedge \chi(x, d)$ for every L-formula $\chi(x, w)$ and every tuple d.

4. *(Section 5.2.)* Work in T_E, and assume that T_E admits elimination of quantifiers (see Exercise 10). Let C be a small set, and let b be an element.

(a) Show that the formula xEb divides over C if and only if b is E-equivalent to no element of C, and that it then 2-divides over C. In particular, the formula xEb divides over $\emptyset$. (Hint: for one direction, choose $b_1, b_2, \ldots$ in distinct E-classes that meet neither C nor the class of b, and show that $(b, b_1, b_2, \ldots)$ is indiscernible over C.)

(b) Show that the formula $x = b$ divides over C if and only if $b \notin C$ (see Exercise 2).

(c) Let a be an element and B a small set. Show that $\mathrm{tp}(a/C \cup B)$ divides over C if and only if either $a \in B \setminus C$, or the element a is E-equivalent to some element of B and to no element of C. (Hint: in the remaining cases, consider a formula of $\mathrm{tp}(a/C \cup B)$ with parameters from a finite tuple d, and a sequence $(d_i)_{i<\omega}$ indiscernible over C with $d_0 = d$. Given such a formula and such a sequence, find a single element a'' with $a'' d_i \equiv_C ad$ for every i. Distinguish the cases $a \in C$, $a \notin C \cup B$ with a equivalent to an element of C, and $a \notin C \cup B$ with a equivalent to no element of $C \cup B$.)

Compare (c) with Theorem 5.4.5: in T_E a type may divide over C although $a \cap B \subseteq C$.

5. *(Section 5.2.)* Work in the theory of $(\mathbb{Q}, <)$, let C be a small set, and let $b_1 < b_2$ be elements. Show that the formula $b_1 < x < b_2$ divides over C if and only if no element c of C satisfies $b_1 \leq c \leq b_2$, and that it then 2-divides over C. Show also that the formulas $x < b_1$ and $b_1 < x$ divide over no small set. (Hint: for the first assertion, follow Example 5.2.6, choosing the pairs b^i in the cut of C that contains b_1 and b_2. For the converse, let $c \in C$ satisfy $b_1 \leq c \leq b_2$, and consider a sequence indiscernible over C that begins with (b_1, b_2). If $c = b_1$, the first coordinate of every term of the sequence equals c, and the case $c = b_2$ is symmetric; if $b_1 < c < b_2$, every interval of the sequence contains c.)

6. *(Section 5.3.)* Work in the theory of $(\mathbb{Q}, <)$, and let $\theta(x; y_1, y_2)$ be the formula $y_1 < x < y_2$. Find pairs $a_n = (a_n^1, a_n^2)$, for $n < \omega$, such that $\{\theta(x, a_n) : n < \omega\}$ is consistent and the formula $\theta(x, a_n)$ 2-divides over the set A_n of coordinates of the pairs a_m with $m < n$ (see Exercise 5). Deduce from Lemma 5.3.5 that θ has the 2-tree property, and compare the discussion following Example 5.2.6. Deduce also that there are an element a and small sets $D_0 \subseteq D_1 \subseteq \cdots$ such that $\mathrm{tp}(a/D_{n+1})$ divides over D_n for every n; compare Theorem 5.4.6. Finally, show that the formula $x < y$ has the k-tree property

for no $k \geq 2$, although it has the order property. Thus, in a theory that is not simple, a formula with the order property need not have the tree property.

7. *(Section 5.3.)* Let T be the theory of the structure $(\omega^{<\omega}, \sqsubseteq)$, where $\sqsubseteq$ is the relation of being an initial segment. Show that the formula $y \sqsubseteq x$ has the 2-tree property, with $b_\sigma = \sigma$ for every nonempty $\sigma \in \omega^{<\omega}$. (Hint: the structure $(\omega^{<\omega}, \sqsubseteq)$ embeds elementarily in $\mathbb{M}$; two siblings have no common extension, a fact expressed by a sentence with parameters.) Show that the formula $x \sqsubseteq y$ has the k-tree property for no $k \geq 2$. (Hint: the theory T says that some element is an initial segment of every element.) Deduce that T is neither simple nor stable.

8. *(Section 5.4.)* Work in T_{RG}. Let C be a small set, let b be a finite tuple, and let $\varphi(x, y)$ be an L-formula in which x is a single variable. Show that $\varphi(x, b)$ divides over C if and only if every solution of $\varphi(x, b)$ in $\mathbb{M}$ is a coordinate of b that does not belong to C. (Hint: if some solution a is not such a coordinate, then the intersection $a \cap b$ is contained in C; use Theorem 5.4.5. For the converse, use Lemma 5.4.3 and (5.1) to build a sequence indiscernible over C that begins with b and whose terms are pairwise disjoint outside C. The sequence is built as in the first part of the proof of Theorem 5.4.5.) Deduce that the formulas $E(x, b_1)$ and $\neg E(x, b_1)$, for a single vertex b_1, divide over no small set.

9. *(Section 5.4.)* Work in T_{RG}. Let a be a finite tuple and B a small set. Show that $\mathrm{tp}(a/B)$ does not divide over the finite set $a \cap B$. Thus, in Theorem 5.3.6 for T_{RG}, the set A can be chosen finite (see Remark 5.5.11). Show also that the bound of Theorem 5.4.6 is attained: for every $n \geq 1$, give a tuple a of length n and sets $D_0 \subseteq \cdots \subseteq D_n$ such that $\mathrm{tp}(a/D_{j+1})$ divides over D_j for every $j < n$.

10. $(*)$ *(Section 5.5.)* Let $\mathcal{K}_E$ be the class of finite structures (X, E) in which E is an equivalence relation.

(a) Show that $\mathcal{K}_E$ is a Fraïssé class with disjoint 2-amalgamation. (Hint: on $B_1 \cup B_2$, make an element of $B_1 \setminus A$ equivalent to an element of $B_2 \setminus A$ exactly when both are equivalent to a common element of A.)

(b) Show that N_E is ultrahomogeneous and has age $\mathcal{K}_E$. Deduce from Fact 5.5.2 that $T_E = T_{\mathcal{K}_E}$ and that T_E admits elimination of quantifiers.

(c) Show that $\mathcal{K}_E$ does not have disjoint 3-amalgamation. (Hint: take $A = \emptyset$ and one-element sets B_1, B_2, B_3.)

(d) Conclude from Exercise 1 that the theory T_E is simple, so that the hypotheses of Theorem 5.5.9 are sufficient for simplicity without being necessary. Deduce from Exercise 4(c) that, for $\mathcal{K}_E$, the description of dividing in conclusion (1) of Theorem 5.5.9 requires disjoint 3-amalgamation.

(e) Consider the class of finite structures (X, E_1, E_2) in which E_1 and E_2 are equivalence relations. Show that this class is a Fraïssé class with disjoint 2-amalgamation and without disjoint 3-amalgamation, and that the theory of its limit is stable. (Hint: count types as in Exercise 1, using Fact 5.5.2(b).)

11. *(Section 5.5.)* Let L consist of one unary relation symbol P, and let $\mathcal{K}$ be the class of finite L-structures in which at most one element satisfies P. Show that $\mathcal{K}$ is a Fraïssé class whose limit is a countably infinite set with exactly one element in P, and that the

class has disjoint 3-amalgamation but not disjoint 2-amalgamation. Let d be the element of P in $\mathbb{M}$. Show that no sequence of distinct elements beginning with d is indiscernible over $\emptyset$. Show also that $\mathrm{tp}(d/\{d\})$ does not divide over $\emptyset$ (every sequence indiscernible over $\emptyset$ that begins with d is constant). Thus disjoint 2-amalgamation is essential both in Lemma 5.5.8 and in Theorem 5.5.9(1), although the class $\mathcal{K}$ has disjoint 3-amalgamation.

12. $(*)$ *(Section 5.5.)* Let L consist of one ternary relation symbol R, and let $\mathcal{K}$ be the class of finite 3-uniform hypergraphs (see Example 5.5.3(5)) with no *tetrahedron*. Here a 3-uniform hypergraph is an L-structure in which R holds only of triples of distinct elements and is invariant under permutations of its arguments. A tetrahedron is a set of four distinct elements of which every three form a hyperedge.

(a) Show that $\mathcal{K}$ is a Fraïssé class with disjoint 2-amalgamation. (Hint: take the union, with no new hyperedges, as in Example 5.5.5.)

(b) Show that $\mathcal{K}$ has disjoint 3-amalgamation. (Hint: declare that no hyperedge meets all three of the sets $B_l \setminus A$. A tetrahedron that meets all three sets contains a triple that meets all three.)

(c) Deduce from Theorem 5.5.9 that $T_{\mathcal{K}}$ is simple. Show that the formula $R(x; y_1, y_2)$ has the independence property (see Definition 4.1.1), and deduce that $T_{\mathcal{K}}$ is not stable.

13. $(*)$ *(Section 5.5.)* Let $\mathcal{K}$ be the class of finite graphs with no four pairwise adjacent vertices.

(a) Show that $\mathcal{K}$ is a Fraïssé class with disjoint 2-amalgamation, and that it does not have disjoint 3-amalgamation. (Hint: take $A = \emptyset$, $B_1 = \{x\}$, $B_2 = \{y\}$ and $B_3 = \{z, w\}$ with z adjacent to w, and let B_{12}, B_{13} and B_{23} be complete graphs.)

(b) Let $\psi(x; y_1, y_2, y_3)$ be the formula $E(x, y_1) \wedge E(x, y_2) \wedge E(x, y_3)$. Show that ψ has the 2-tree property in $T_{\mathcal{K}}$, and conclude that $T_{\mathcal{K}}$ is not simple. (Hint: adapt Example 5.5.10 by taking vertices $u_\sigma, v_\sigma, w_\sigma$ for nonempty $\sigma \in \omega^{<\omega}$, joining v_σ to w_σ, and joining $u_{\sigma^\frown\langle i\rangle}$ to $v_{\sigma^\frown\langle j\rangle}$ and to $w_{\sigma^\frown\langle j\rangle}$ whenever $i < j$. Show that this graph has no four pairwise adjacent vertices, and let $b_\sigma = (u_\sigma, v_\sigma, w_\sigma)$.)

14. *(Section 5.6.)* Consider the sets $A = \omega^{<\omega}$ and $B = \omega^{<\omega} \setminus \{\emptyset\}$. Define $\varphi\colon A \times B \to [0, 1]$ by $\varphi(a, b) = 2/3$ if b is an initial segment of a, by $\varphi(a, b) = 0$ if a is a proper initial segment of b, and by $\varphi(a, b) = 1/3$ if a and b are incomparable.

(a) Show that, for reals $r < s$, the function φ has the 2-tree property with gap (r, s) (see Definition 5.6.1) if and only if $1/3 \leq r < s \leq 2/3$. (Hint: for $r < 1/3$, use condition (b) with $\tau = \emptyset$ and an element a of length 1.)

(b) Fix a gap (r, s) as in (a), and apply to the tree $b_\sigma = \sigma$ the construction in the proof of Lemma 5.6.3, taking $a'_\tau = f(\tau)$, where $f\colon 2^{<\omega} \to \omega^{<\omega}$ is the embedding constructed in the proof. Compute the parameters $b'_\emptyset$, $b'_{\langle 0\rangle}$, $b'_{\langle 1\rangle}$ and the elements $a'_{\langle 0\rangle}$, $a'_{\langle 1\rangle}$ of the resulting binary tree. Then check the two inequalities of Definition 5.6.2 for $\sigma = \emptyset$.

(c) Show directly from Definition 1.2.5 that φ is not stable, as Theorem 5.6.4 predicts. (Hint: let a_m and b_n be the sequences of zeros of lengths m and $n + 1$, and let $\mathcal{U}$ and $\mathcal{V}$ be nonprincipal.)

5.8 Historical remarks

Section 5.1. Working inside a single large saturated and strongly homogeneous model goes back to Shelah [She90], who wrote $\mathfrak{C}$ for this model. The name *monster model* became standard later. The notion of ω-stability comes from Morley's categoricity theorem [Mor65]. The study of λ-stability for arbitrary λ, including the fact that stability is equivalent to λ-stability for all λ with $\lambda^{|T|} = \lambda$, is due to Shelah [She71, She90]. In Lemma 5.1.2 we prove one direction of this equivalence in a stronger form: stability already follows from λ-stability for a single $\lambda \geq |T|$.

Section 5.2. Dividing, forking and the tree property were introduced by Shelah in *Classification Theory* ([She90, Chapter III]; first edition 1978). Forking, a notion built from dividing, is the subject of Section 7.3. The simple theories, those in which no formula has the tree property, were named and studied in his paper [She80]. Indiscernible sequences go back to Ehrenfeucht and Mostowski [EM56] and were made a standard tool of classification theory by Morley [Mor65] and Shelah [She90]. After Shelah's paper of 1980, the simple unstable theories received little attention for some fifteen years. Two examples brought them back: the pseudofinite fields, and the difference fields of Chatzidakis and Hrushovski [CH99]. Hrushovski showed that the theories of pseudofinite fields are simple and unstable, in work circulated as a manuscript from 1991 and published in 2002 [Hru02]. Here a field is *pseudofinite* if it is infinite and satisfies every sentence true in all finite fields, and a *difference field* is a field with a distinguished automorphism.

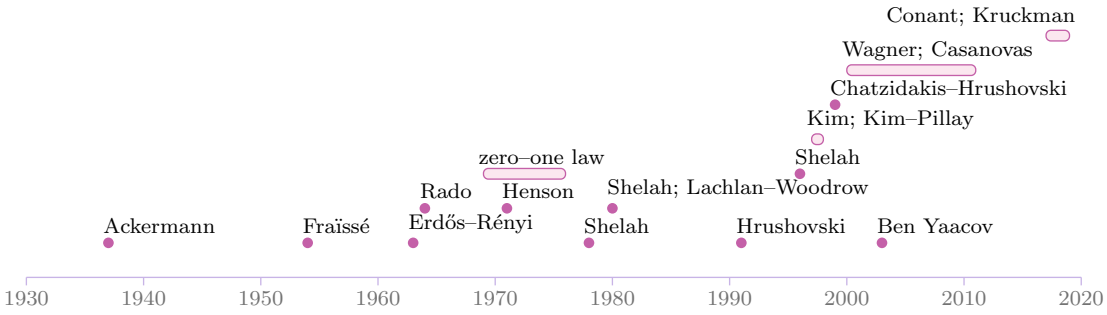
Section 5.3. That stable theories are simple, and that the strict order property implies the tree property, go back to Shelah [She90, Chapter III], [She80]. The dichotomy of Fact 5.3.3 is proved in Chapter II of *Classification Theory* [She90]. Kim [Kim98] proved that forking and dividing agree in simple theories and that forking independence is symmetric. Kim and Pillay [KP97] abstracted the independence theorem from Hrushovski's work on pseudofinite fields and proved it in every simple theory. They also showed that, together with the other properties of forking independence, the independence theorem characterizes the simple theories. The title of this chapter is borrowed from their survey [KP98]. In their monographs, Wagner [Wag00] and Casanovas [Cas11] develop the theory in full.

Section 5.4. For his proof of the consistency of set theory without the axiom of infinity, Ackermann [Ack37] devised a coding of the hereditarily finite sets by natural numbers. The graph of Example 5.4.1 is implicit in this coding: the relation "the i-th binary digit of j is 1" encodes membership, and symmetrizing this relation gives the Rado graph. Erdős and Rényi [ER63] proved that a random countable graph, with each edge chosen independently with probability $\frac{1}{2}$, is isomorphic with probability 1 to a fixed graph. Their paper is titled *Asymmetric graphs*, and the graph its authors found is as homogeneous as a graph can be. Rado [Rad64] gave the explicit construction of Example 5.4.1 and proved that the graph is universal for countable graphs. The theory T_{RG} is also the almost sure theory of finite graphs: a first-order sentence holds in almost all finite graphs if and only if it holds in the Rado graph. This equivalence follows from the zero–one law of Glebskiĭ, Kogan, Liogon'kiĭ and Talanov [GKLT69] and of Fagin [Fag76]. Cameron [Cam97] collects the many descriptions of the Rado graph in his survey.

Section 5.5. Fraïssé [Fra54] introduced ages and proved Fact 5.5.2(a), generalizing Cantor's characterization of $(\mathbb{Q}, <)$ as the unique countable dense linear order without endpoints. Henson [Hen71] constructed, for each $n \geq 3$, the countable ultrahomogeneous graph that is

universal for the countable K_n-free graphs, those in which no n vertices are pairwise adjacent. Lachlan and Woodrow [LW80] proved that the countably infinite ultrahomogeneous graphs are, up to complementation, the disjoint unions of complete graphs of equal size, the Rado graph, and the Henson graphs. Kruckman [Kru19] made disjoint n-amalgamation a systematic tool for the theories of Fraïssé limits.

Section 5.6. Simplicity for structures with real-valued relations was developed by Ben Yaacov [BY03] in the framework of compact abstract theories, which includes continuous logic. The argument of Theorem 5.6.4 is elementary by comparison: it combines the polynomial approximation of Theorem 1.2.11 with the counting of branches in the proof of Theorem 2.5.1.



1937 **Ackermann.** Ackermann [Ack37] codes the hereditarily finite sets by natural numbers through their binary digits; symmetrized, the coding gives the Rado graph.

1954 **Fraïssé.** Fraïssé [Fra54] introduces ages and proves that every Fraïssé class has a unique countable ultrahomogeneous limit.

1963 **Erdős–Rényi.** Erdős and Rényi [ER63] show that the random countable graph is unique up to isomorphism with probability 1.

1964 **Rado.** Rado [Rad64] gives the explicit construction of Example 5.4.1 and proves its universality.

1969–1976 **zero–one law.** Glebskiĭ, Kogan, Liogon'kiĭ and Talanov [GKLT69] prove the zero–one law for first-order sentences about finite graphs, and Fagin [Fag76] proves it independently. The almost sure theory of finite graphs is T_{RG}.

1971 **Henson.** Henson [Hen71] constructs, for each $n \geq 3$, the countable ultrahomogeneous graph that omits the complete graph K_n on n vertices and is universal for the countable graphs that omit it.

1978 **Shelah.** In the first edition of *Classification Theory* (see the second edition [She90]), Shelah introduces forking, dividing and the tree property.

1980 **Shelah; Lachlan–Woodrow.** Shelah [She80] names and studies the simple theories, those without the tree property. Lachlan and Woodrow [LW80] classify the countable ultrahomogeneous graphs.

1991 **Hrushovski.** In his manuscript on pseudofinite fields, published in 2002 [Hru02], Hrushovski shows that their theories are simple and proves an independence theorem for them.

1996 **Shelah.** Shelah [She96] introduces the hierarchy of properties SOP_n; the Henson graph has SOP_3.

1997–1998 **Kim; Kim–Pillay.** Kim [Kim98] proves that forking equals dividing in simple theories and that forking is symmetric. Kim and Pillay [KP97] prove the independence theorem; their survey [KP98] is titled *From stability to simplicity*.

1999 **Chatzidakis–Hrushovski.** Chatzidakis and Hrushovski [CH99] develop the model theory of difference fields, a central example of a simple unstable theory.

2000–2011 **Wagner; Casanovas.** Wagner [Wag00] and Casanovas [Cas11] publish their monographs on simple theories.

2003 **Ben Yaacov.** Ben Yaacov [BY03] extends simplicity to compact abstract theories, a framework that includes continuous logic.

2017–2019 **Conant; Kruckman.** Conant [Con17] develops free amalgamation theories; Kruckman [Kru19] studies disjoint n-amalgamation and pseudofinite countably categorical theories.

Chapter
6

Stability and notions of independence

Roadmap

In the first five chapters we describe the dividing lines by the configurations they forbid. In the last two we describe stability and simplicity by the independence relations that the absence of these configurations produces. In this chapter we do so for stability, through the heirs and coheirs with which Lascar and Poizat presented forking in stable theories in 1979. In Section 6.1 we use the definability of types to extend a type over a model to any larger set. This extension, the heir, evaluates the definitions of the type at the new parameters (see Definition 6.1.2 and Lemma 6.1.3), and in a stable theory the heir is the unique independent extension of the type. The symmetry of the independence relation defined by heirs rests on Harrington's lemma, by which definitions of types over a model commute (see Lemma 6.1.4). In Subsection 6.1.1 we show that the lemma is an exchange of iterated ultrafilter limits, as in Section 1.2. In Section 6.2 we prove that a theory is stable exactly when it carries an independence relation over models satisfying the axioms of Definition 5.4.7, uniqueness included (see Theorem 6.2.2). The direction from such a relation to stability is a count of types (see Lemma 5.1.2). In Section 6.3 we record which axioms suffice for stability, and how the relation extends, as nonforking independence, to arbitrary base sets.

We keep the conventions of Section 5.1, and we measure independence against the axioms of Definition 5.4.7.

6.1 Definitions and heirs

In a stable theory every type over a model has a definition scheme, one definition for each formula (see Lemma 6.1.1). A definition can be evaluated at parameters outside the model, so the scheme extends the type to larger sets of parameters; the result is the heir (see Definition 6.1.2). The main lemma of the section is the fact that will make independence symmetric: definitions over a model commute (see Lemma 6.1.4). In Subsection 6.1.1 we

show that this commutation is the exchange of iterated ultrafilter limits that defines stability in Section 1.2.

Lemma 6.1.1. *Assume that T is stable, let M be a model, and let $p \in S_x(M)$ be given, where x may be infinite. For every $L(M)$-formula $\varphi(x', y)$, with x' a finite subtuple of x and y finite, there is an $L(M)$-formula $d_p\varphi(y)$ such that for all $b \in M$, we have*

$$\varphi(x', b) \in p \quad \text{if and only if} \quad M \models d_p\varphi(b).$$

Any two formulas with this property are equivalent in $\mathbb{M}$.

Proof. The idea is to regard the parameters of φ from M as further parameter variables and to apply the definability of types to the resulting L-formula. The parameters are then substituted back into the definition so obtained. Let $\varphi_0 \in L$ and $m \in M$ be such that $\varphi(x', y) = \varphi_0(x', y, m)$. As in Section 1.1, we write $\varphi_0^1 = \varphi_0$ and $\varphi_0^0 = \neg\varphi_0$. The set of formulas $\varphi_0(x', c)^u$, with $c \in M$ and $u \in \{0, 1\}$, that belong to p is a complete φ_0-type over M in the sense of Definition 1.1.1. Since φ_0 does not have the order property, Theorem 1.1.6 shows that this φ_0-type has a definition $\psi(z)$ over M, which is a positive Boolean combination of instances of φ_0. Here the tuple z corresponds to the parameter variables y followed by those replaced by m. The formula $d_p\varphi(y) := \psi(y, m)$ then has the required property. For uniqueness, suppose that $\psi_1(y)$ and $\psi_2(y)$ both have this property. Then the model M satisfies the $L(M)$-sentence $\forall y\,(\psi_1 \leftrightarrow \psi_2)$, and since $M \preceq \mathbb{M}$, so does $\mathbb{M}$. ☺

The family $(d_p\varphi)_\varphi$ given by Lemma 6.1.1, determined up to equivalence in $\mathbb{M}$, is called the *definition scheme* of p.

Definition 6.1.2. Assume that T is stable. Let M be a model, let $p \in S_x(M)$ be given, and let B be small. Define

$$p|B := \{\varphi(x', b) :\ \varphi(x', y) \in L(M),\ b \in B,\ \mathbb{M} \models d_p\varphi(b)\}.$$

Here x' ranges over the finite subtuples of x. The set $p|B$ is called the *heir* of p over MB.

The heir $p|B$ is a set of $L(MB)$-formulas, and every $L(MB)$-formula has the form $\varphi(x', b)$ with $\varphi \in L(M)$ and $b \in B$. The construction is pictured in Figure 6.1. In the next lemma we show that the heir is in fact a complete type extending p, and that heirs are compatible with enlarging the parameter set and the base model.

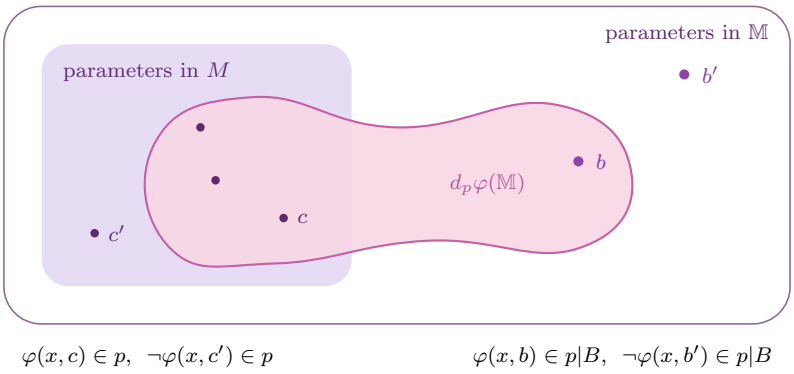


FIGURE 6.1. Heirs (see Definition 6.1.2) of a type $p \in S_x(M)$ over a model M, drawn for one $L(M)$-formula $\varphi(x, y)$ and parameters b and b' from a small set B.

Lemma 6.1.3. *Assume that T is stable, let M be a model, let $p \in S_x(M)$ be given, and let B and D be small sets. Then the following hold:*

(a) *The set $p|B$ belongs to $S_x(MB)$ and extends p.*

(b) *If $B \subseteq D$, then the type $p|B$ is the restriction of $p|D$ to MB.*

(c) *If N is a model with $M \subseteq N$, then the type $p|N$ is defined by the definitions of p: for every $L(M)$-formula $\varphi_0(x', y, z)$, with x' a finite subtuple of x, and every tuple n from N of the length of z, the formula $d_p\varphi_0(y, n)$ is a definition of $p|N$ for $\varphi_0(x', y, n)$. Moreover, we have $(p|N)|B = p|NB$.*

Proof. Completeness and consistency in (a) rest on one observation: a first-order statement about the definitions of p that holds in M is an $L(M)$-sentence. The statement therefore holds in $\mathbb{M}$, where it can be applied to parameters from B.

(a) *Completeness.* Let $\varphi \in L(M)$ be given. Since p is complete, for every $c \in M$ exactly one of $\varphi(x', c)$, $\neg\varphi(x', c)$ lies in p. Hence we have $M \models \forall y \big(d_p(\neg\varphi)(y) \leftrightarrow \neg d_p\varphi(y)\big)$, and the same holds in $\mathbb{M}$. Thus for $b \in B$ exactly one of $\varphi(x', b)$, $\neg\varphi(x', b)$ belongs to $p|B$.

Consistency. Let k be a positive integer, let $\varphi_i(x_i, b_i) \in p|B$ be given for $i < k$, and let $\chi(\bar{x}; y_0, \dots, y_{k-1})$ be the conjunction $\bigwedge_{i<k} \varphi_i(x_i, y_i)$, where $\bar{x}$ lists the variables in the x_i. For $\bar{c} \in M$, the formula $\chi(\bar{x}, \bar{c})$ is in p if and only if each $\varphi_i(x_i, c_i)$ is, and in that case the formula $\chi(\bar{x}, \bar{c})$ is satisfiable in M. Therefore we have

$$M \models \forall \bar{y}\Big(\bigwedge_{i<k} d_p\varphi_i(y_i) \;\to\; d_p\chi(\bar{y})\Big) \quad \text{and} \quad M \models \forall \bar{y}\big(d_p\chi(\bar{y}) \to \exists \bar{x}\, \chi(\bar{x}, \bar{y})\big).$$

Both sentences transfer to $\mathbb{M}$, where, applied to $\bar{b} = (b_0, \dots, b_{k-1})$, they give $\mathbb{M} \models \exists \bar{x} \bigwedge_i \varphi_i(x_i, b_i)$. By compactness, the set $p|B$ is consistent.

Extension. For $c \in M$ we have $\varphi(x', c) \in p \iff M \models d_p\varphi(c)$, which holds if and only if $\varphi(x', c) \in p|B$.

(b) This is immediate from the definition.

(c) By (a), the type $p|N$ extends p. For $n \in N$, the formula $\varphi(x', n)$ belongs to $p|N$ if and only if $\mathbb{M} \models d_p\varphi(n)$, so the formula $d_p\varphi$ is a definition of $p|N$ for $\varphi \in L(M)$. For an $L(N)$-formula $\varphi(x', y) = \varphi_0(x', y, n)$ with $\varphi_0 \in L(M)$, the formula $d_p\varphi_0(y, n)$ defines $p|N$. By the uniqueness clause of Lemma 6.1.1, the definition scheme of $p|N$ is thus given, up to equivalence, by that of p. Now for $b \in B$ and $n \in N$, we have

$$\varphi_0(x', b, n) \in (p|N)|B \quad \text{if and only if} \quad \mathbb{M} \models d_p\varphi_0(b, n) \quad \text{if and only if} \quad \varphi_0(x', b, n) \in p|NB.$$

The fulcrum of the chapter is the following lemma: definitions of types over a model commute. Apart from the existence of definition schemes (see Lemma 6.1.1), the lemma is the only consequence of the absence of the order property that we use in this chapter. To state it, fix a model M, types $p \in S_x(M)$ and $q \in S_y(M)$ with x and y finite, and a formula $\varphi(x, y) \in L(M)$. As before, we write $d_p\varphi(y)$ for the definition of p for $\varphi(x, y)$. For the definition of q for φ, regarded as a formula in the variables y with parameter variables x, we write $d_q\varphi(x)$.

Lemma 6.1.4 (Harrington's lemma). *Assume that T is stable, let M be a model, and let $p \in S_x(M)$, $q \in S_y(M)$ and $\varphi(x, y) \in L(M)$ be given, where x and y are finite. Then we have*

$$d_p\varphi(y) \in q \quad \textit{if and only if} \quad d_q\varphi(x) \in p.$$

Proof. The idea is to realize p and q alternately, in such a way that each pair (a_i, b_j) consults one definition below the diagonal and the other on and above the diagonal. If the two definitions disagreed, the result would be a ladder. Recall that a ladder for a formula is a pair of sequences such that the formula holds at their i-th and j-th terms, respectively, exactly when $i \leq j$ (see Definition 1.1.4). We choose recursively $a_i, b_i \in \mathbb{M}$ $(i < \omega)$ such that a_i realizes $p|\{b_j : j < i\}$ and b_i realizes $q|\{a_j : j \leq i\}$. This is possible by Lemma 6.1.3(a) and saturation. In particular, each a_i realizes p and each b_i realizes q.

If $j < i$, then since $a_i \models p|\{b_j\}$ and $b_j \models q$, we have

$$\mathbb{M} \models \varphi(a_i, b_j) \quad \text{if and only if} \quad \mathbb{M} \models d_p\varphi(b_j) \quad \text{if and only if} \quad d_p\varphi(y) \in q.$$

If $i \leq j$, then since $b_j \models q|\{a_i\}$ and $a_i \models p$, we have

$$\mathbb{M} \models \varphi(a_i, b_j) \quad \text{if and only if} \quad \mathbb{M} \models d_q\varphi(a_i) \quad \text{if and only if} \quad d_q\varphi(x) \in p.$$

Suppose that the two right-hand sides differ. The formula $\varphi(x, y)$ has the form $\varphi_0(x, y, m)$ with $\varphi_0(x, y, z) \in L$ and m a tuple from M of the length of z. Appending m to each b_j produces tuples $b'_j = b_j m$. Then either $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i \leq j$ for all i, j, or $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i > j$ for all i, j. In the first case the formula $\varphi_0(x; y, z)$ has the order property directly, and in the second by Remark 1.1.5. Either way this contradicts the stability of T, and the lemma follows. ☺

The configuration built in the proof is drawn in Figure 6.2.

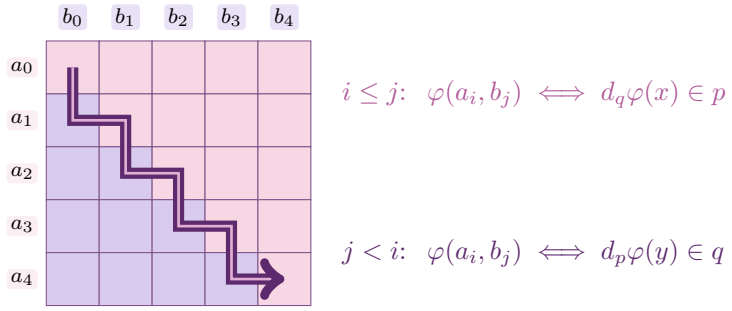


FIGURE 6.2. The proof of Lemma 6.1.4, for types $p \in S_x(M)$ and $q \in S_y(M)$ over a model M and a formula $\varphi(x, y) \in L(M)$, with the elements realized in the order $a_0, b_0, a_1, b_1, \ldots$ along the pink path.

6.1.1 Harrington's lemma as limit exchange

Harrington's lemma is the exchange of iterated ultrafilter limits of Section 1.2, read in the language of types. We now make this precise, and in doing so obtain a second proof of the lemma that does not build a ladder by hand.

The first step is to represent types over a model M by ultrafilters on tuples from M. Let x and y be finite tuples of variables. Every type $p \in S_x(M)$ is finitely satisfiable in M: if $\psi(x) \in p$, then we have $\mathbb{M} \models \exists x\, \psi(x)$, and since ψ is an $L(M)$-formula and $M \preceq \mathbb{M}$, some $a \in M^{|x|}$ satisfies ψ. Hence the sets $\psi(M) = \{a \in M^{|x|} : \mathbb{M} \models \psi(a)\}$, for $\psi \in p$, have the finite intersection property. If $\mathcal{U}$ is an ultrafilter on $M^{|x|}$ that contains all of them, then the type p is the *trace* of $\mathcal{U}$, in the sense that

$$\psi(x) \in p \quad \text{if and only if} \quad \psi(M) \in \mathcal{U} \qquad \text{for every } L(M)\text{-formula } \psi(x).$$

Similarly, every $q \in S_y(M)$ is the trace of an ultrafilter $\mathcal{V}$ on $M^{|y|}$. Fix p, q, $\mathcal{U}$ and $\mathcal{V}$ in this way, and let $\varphi(x,y) \in L(M)$ be given. As in Section 1.2, we identify φ with the $\{0,1\}$-valued function $(a',b') \mapsto \varphi(a',b')$ on $M^{|x|} \times M^{|y|}$. For $b' \in M^{|y|}$, the definition of p and the choice of $\mathcal{U}$ give

$$\mathbb{M} \models d_p\varphi(b') \quad \text{if and only if} \quad \varphi(x,b') \in p \quad \text{if and only if} \quad \varphi(M,b') \in \mathcal{U} \quad \text{if and only if} \quad \lim_{a',\mathcal{U}} \varphi(a',b') = 1.$$

Therefore we have

$$d_p\varphi(y) \in q \quad \text{if and only if} \quad \{b' \in M^{|y|} : \mathbb{M} \models d_p\varphi(b')\} \in \mathcal{V} \quad \text{if and only if} \quad \lim_{b',\mathcal{V}} \lim_{a',\mathcal{U}} \varphi(a',b') = 1,$$

and in the same way we have

$$d_q\varphi(x) \in p \quad \text{if and only if} \quad \lim_{a',\mathcal{U}} \lim_{b',\mathcal{V}} \varphi(a',b') = 1.$$

Thus the two sides of Harrington's lemma state that the two iterated limits of φ along $\mathcal{U}$ and $\mathcal{V}$ equal 1, and the lemma says exactly that these iterated limits agree. No heirs, saturation or realizations enter here: the computation takes place entirely in M.

When T is stable, the limits do agree. To see this, write $\varphi(x,y) = \varphi_0(x,y,m)$, where $\varphi_0(x,y,z)$ is an L-formula and m is a tuple from M that takes the place of z. The formula $\varphi_0(x;y,z)$ does not have the order property, and therefore, by Remark 1.2.2, the formula φ_0 is stable on M. The restriction φ of φ_0 to the parameters of the form (b',m) is stable as well, since a ladder for φ or $\neg\varphi$ would be one for φ_0 or $\neg\varphi_0$. The function φ is then stable in the sense of Definition 1.2.5, by Proposition 1.2.4. By Corollary 1.2.6, applied to the families $(a')_{a' \in M^{|x|}}$ and $(b')_{b' \in M^{|y|}}$, the two iterated limits are equal, and Lemma 6.1.4 is thereby proved again.

The two proofs are the same argument at different levels of abstraction. The alternating realizations (a_i, b_j) in the first proof consult one definition below the diagonal and the other on and above it. In exactly the same way, the sequences of Proposition 1.2.3 display one iterated limit above the diagonal and the other below it. In both cases a disagreement is converted into a ladder.

The computation also explains what the two orders of realization mean. To see this, we extend the trace of $\mathcal{U}$ to larger sets of parameters in a second way. The heir of Definition 6.1.2 evaluates the definitions of a type, which a stable theory provides (see Lemma 6.1.1); the coheir consults the ultrafilter $\mathcal{U}$ itself, and is defined in every theory. For a small set B, the *coheir* of $\mathcal{U}$ over MB is the set

$$q_{\mathcal{U}}|B = \big\{\varphi(x,b) :\ \varphi(x,y) \in L(M),\ b \text{ a finite tuple from } B,\ \{a \in M^{|x|} : \mathbb{M} \models \varphi(a,b)\} \in \mathcal{U}\big\}.$$

We first collect its basic properties, which are used again in Chapter 7, and then return to the two orders of realization.

Lemma 6.1.5 (Coheirs). *Let M be a model, let x be a finite tuple of variables, let $\mathcal{U}$ be an ultrafilter on $M^{|x|}$, and let B and B' be small sets. Then the following hold:*

(a) *The coheir $q_{\mathcal{U}}|B$ of $\mathcal{U}$ over MB is a complete type over MB, and it is finitely satisfiable in M.*

(b) *Every type over MB that is finitely satisfiable in M equals $q_{\mathcal{U}'}|B$ for some ultrafilter $\mathcal{U}'$ on $M^{|x|}$.*

(c) *If $B \subseteq B'$, then we have $q_{\mathcal{U}}|B \subseteq q_{\mathcal{U}}|B'$.*

(d) *The coheir is invariant over M: every $\sigma \in \mathrm{Aut}(\mathbb{M}/M)$ satisfies $\sigma(q_{\mathcal{U}}|B) = q_{\mathcal{U}}|\sigma(B)$.*

Proof. For an $L(MB)$-formula $\varphi(x,b)$, we write $\varphi(M,b) = \{a \in M^{|x|} : \mathbb{M} \models \varphi(a,b)\}$. These sets satisfy $(\neg\varphi)(M,b) = M^{|x|} \setminus \varphi(M,b)$ and $(\varphi \wedge \psi)(M,b) = \varphi(M,b) \cap \psi(M,b)$.

(a) Since $\mathcal{U}$ is an ultrafilter, exactly one of $\varphi(M,b)$ and its complement belongs to $\mathcal{U}$, so exactly one of $\varphi(x,b)$ and $\neg\varphi(x,b)$ belongs to $q_{\mathcal{U}}|B$. If $\varphi_1(x,b_1),\ldots,\varphi_n(x,b_n)$ belong to $q_{\mathcal{U}}|B$, the intersection of the sets $\varphi_l(M,b_l)$ belongs to $\mathcal{U}$ and is therefore nonempty, and any of its elements is a tuple from M satisfying all n formulas. Hence the set $q_{\mathcal{U}}|B$ is finitely satisfiable in M, in particular consistent, and complete.

(b) Let p be a type over MB finitely satisfiable in M. The sets $\varphi(M,b)$, for $\varphi(x,b) \in p$, have the finite intersection property, so some ultrafilter $\mathcal{U}'$ on $M^{|x|}$ contains all of them. The type p is then contained in $q_{\mathcal{U}'}|B$; as both are complete types over MB, they are equal.

(c) This is immediate from the definition.

(d) Let a be a tuple from M. Since σ fixes a and the parameters of φ from M, we have $\mathbb{M} \models \varphi(a,b)$ if and only if $\mathbb{M} \models \varphi(a,\sigma(b))$. Hence the sets $\varphi(M,b)$ and $\varphi(M,\sigma(b))$ coincide, and it follows that $\varphi(x,b) \in q_{\mathcal{U}}|B$ if and only if $\varphi(x,\sigma(b)) \in q_{\mathcal{U}}|\sigma(B)$. ☺

We return to the two orders of realization. If b realizes $q_{\mathcal{V}}|\emptyset$ and a realizes $q_{\mathcal{U}}|\{b\}$, then unwinding the definitions gives

$$\mathbb{M} \models \varphi(a,b) \quad \text{if and only if} \quad \lim_{a',\mathcal{U}} \lim_{b',\mathcal{V}} \varphi(a',b') = 1,$$

and realizing the two coheirs in the opposite order exchanges the two limits. Thus the two orders of realizing coheirs give pairs with the same type over M exactly when, for every $L(M)$-formula $\varphi(x,y)$, the two iterated limits of φ along $\mathcal{U}$ and $\mathcal{V}$ agree. In a stable theory, moreover, the coheir $q_{\mathcal{U}}|B$ is the heir $p|B$ of the trace p of $\mathcal{U}$, and Harrington's lemma shows why. Let $\varphi(x,y) \in L(M)$ be given, let b be a finite tuple from B, and write $r = \mathrm{tp}(b/M)$. The tuples $a \in M^{|x|}$ with $\mathbb{M} \models \varphi(a,b)$ are exactly those that satisfy $d_r\varphi(x)$, the definition of r for φ regarded as a formula in the variables y. Hence the formula $\varphi(x,b)$ belongs to the coheir if and only if $d_r\varphi(x) \in p$, which by Lemma 6.1.4 holds if and only if $d_p\varphi(y) \in r$, that is, if and only if $\mathbb{M} \models d_p\varphi(b)$. The consequences are drawn in parts (b) and (c) of Exercise 8 of Section 6.4. Read in this way, the symmetry of independence proved in Section 6.2 is the commutation of ultrafilter limits: the logical counterpart of exchanging the order of two limits in analysis.

6.2 Independence relations

The axioms of Definition 5.4.7 record how the heir relation behaves. When T is stable the heir relation satisfies all of them, and conversely a relation satisfying just two of them, uniqueness and local character, forces T to be stable (see Theorem 6.2.2). In Chapter 5 we met relations satisfying all the axioms except, possibly, uniqueness (see Theorems 5.4.9 and 5.5.9), and in the random graph a type over a model has more than one independent extension. By the theorem below, uniqueness is the axiom that singles out the stable theories: together with local character, it holds only when T is stable.

In terms of types, the axioms of existence and uniqueness, (EX) and (UN), have a simple meaning. For a relation $\mathop{\smile\!\!\!\!\!\!|}$ satisfying them, a model M and a small set B, they say that every type $p \in S_x(M)$ has exactly one *independent extension* to MB. This extension is the type over MB of any $A \models p$ with $A \mathop{\smile\!\!\!\!\!\!|}_M B$. All the other axioms are statements about

which extensions of p count as independent. In a stable theory the natural candidate for this extension is the heir of Section 6.1, and we take it as the definition.

Definition 6.2.1. Assume that T is stable. For small A, B and a model M we define

$$A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B \quad \text{if and only if} \quad \mathrm{tp}(A/MB) = \mathrm{tp}(A/M)|B.$$

We call $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}$ *heir independence.*

Thus a set is heir independent from B over M when its type over MB is the heir of its type over M, the extension defined by the same definition scheme. For finite tuples the definition becomes a test, formula by formula, which is the form used in the proofs below. Let M be a model, let a and b be finite tuples, and let $p = \mathrm{tp}(a/M)$, a type in the variables x. Then every $L(Mb)$-formula in x has the form $\varphi(x, b)$ with $\varphi(x, y) \in L(M)$. Hence we have

$$a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} b \quad \text{if and only if} \quad \text{for all } \varphi(x, y) \in L(M): \quad \mathbb{M} \models \varphi(a, b) \leftrightarrow d_p\varphi(b). \tag{6.1}$$

Theorem 6.2.2. *For a complete theory T, the following are equivalent:*

(1) *The theory T is stable.*

(2) *There is an independence relation satisfying all the axioms of Definition 5.4.7.*

(3) *There is a relation $A \mathop{\smile\!\!\!\!\!\!|}_{M} B$ satisfying* (UN) *and* (LOC).

Moreover, if T is stable, the relation of Definition 6.2.1 satisfies all the axioms. This relation also satisfies

(AR) Anti-reflexivity. *If $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} a$ for a model M and a finite tuple a, then we have $a \subseteq M$.*

6.2.1 *Proof of* (1) ⇒ (2)

Assume that T is stable, and let $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}$ be the relation of Definition 6.2.1. Throughout, the letters A, B and M denote the sets and the model of the axiom being proved, and we write $p = \mathrm{tp}(A/M)$. In the proofs of (SYM) and (LOC), which concern finite tuples, the letter p names the type of the tuple at hand. Most of the axioms follow by unwinding the definitions with the help of Lemma 6.1.3; symmetry is the one place where Harrington's lemma is needed.

(INV). Let $\sigma \in \mathrm{Aut}(\mathbb{M})$ be given. Applying σ to the parameters of the definition scheme of p gives a definition scheme for $\sigma(p) = \mathrm{tp}(\sigma A/\sigma M)$. Hence σ carries heirs to heirs: $\sigma(p)|\sigma(B) = \sigma(p|B)$. If $\mathrm{tp}(A/MB) = p|B$, it follows that $\mathrm{tp}(\sigma A/\sigma M\, \sigma B) = \sigma(p|B) = \sigma(p)|\sigma(B)$, which is the relation $\sigma A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{\sigma M} \sigma B$.

(MON). Suppose that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$, and let $A' \subseteq A$ and $B' \subseteq B$ be given. The definition scheme of $\mathrm{tp}(A'/M)$ is the part of that of p that involves only variables for A'. Hence the heir $\mathrm{tp}(A'/M)|B'$ is the restriction of $p|B$ to these variables and to MB' (see Lemma 6.1.3(b)), which is $\mathrm{tp}(A'/MB')$ because $\mathrm{tp}(A/MB) = p|B$.

(FC). The condition $\mathrm{tp}(A/MB) = p|B$ says that for every $L(M)$-formula $\varphi(x', y)$ and every $b \in B$ we have $\mathbb{M} \models \varphi(a', b) \leftrightarrow d_p\varphi(b)$, where a' is the finite subtuple of A corresponding to x'. Each instance involves only a finite $a' \subseteq A$ and a finite $b \subseteq B$, and it is exactly an instance of $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} b$. Hence the relation $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$ holds if and only if $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} b$ for all such finite tuples.

(BMON). Let N be a model with $M \subseteq N \subseteq B$, and suppose that $\mathrm{tp}(A/MB) = p|B$. Then Lemma 6.1.3(b) gives $\mathrm{tp}(A/N) = p|N$, and part (c) of the same lemma yields $\mathrm{tp}(A/N)|B = (p|N)|B = p|NB = p|B = \mathrm{tp}(A/NB)$, which means that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{N} B$.

(TR). Let N be a model with $M \subseteq N$, and suppose that $\operatorname{tp}(A/N) = p|N$ and $\operatorname{tp}(A/NB) = \operatorname{tp}(A/N)|B$. By Lemma 6.1.3(c), we have $\operatorname{tp}(A/NB) = (p|N)|B = p|NB$, which means that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} NB$.

(SYM). In view of (FC), it suffices to treat finite tuples a, b. Let $p = \operatorname{tp}(a/M)$ and $q = \operatorname{tp}(b/M)$, and assume that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} b$. For $\varphi(x, y) \in L(M)$, using (6.1) and Lemma 6.1.4, we have

$$\mathbb{M} \models \varphi(a, b) \quad \text{if and only if} \quad \mathbb{M} \models d_p\varphi(b) \quad \text{if and only if} \quad d_p\varphi(y) \in q \quad \text{if and only if} \quad d_q\varphi(x) \in p \quad \text{if and only}$$

By (6.1) with the roles of a and b exchanged, this says that $b \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} a$.

(EX). Saturation and Lemma 6.1.3(a) provide a realization A' of the type $p|B$. This realization satisfies $A' \equiv_M A$ and $A' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$.

(UN). Suppose that $A \equiv_M A'$, and that both sets are independent from B over M. Then both have type p over M, and therefore $\operatorname{tp}(A/MB) = p|B = \operatorname{tp}(A'/MB)$.

(LOC). Let a be a finite tuple, let N be a model, and write $p = \operatorname{tp}(a/N)$. For each L-formula $\varphi(x, y)$ fix a definition $d_p\varphi(y) \in L(N)$, and let $D \subseteq N$ be the set of all parameters used, a set of size at most κ. By the downward Löwenheim–Skolem theorem there is a model $M \preceq N$ with $D \subseteq M$ and $|M| \leq \kappa$, and we write $q = \operatorname{tp}(a/M)$. Each $d_p\varphi$ is an $L(M)$-formula, and for every tuple b from M the formula $\varphi(x, b)$ belongs to q if and only if it belongs to p. Since $M \preceq N$, the formula $\varphi(x, b)$ belongs to p if and only if $M \models d_p\varphi(b)$. Thus the formula $d_p\varphi$ is also a definition of q for φ, and these formulas form a definition scheme for q (an $L(M)$-formula $\varphi_0(x, y, m)$ is handled by $d_p\varphi_0(y, m)$). Hence the heir $q|N$ is defined by the same formulas as p. Both $q|N$ and p are complete types over N, so they coincide: $q|N = p = \operatorname{tp}(a/N)$, which means that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} N$.

(AR). Suppose that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} a$, and let $p = \operatorname{tp}(a/M)$. Assume, toward a contradiction, that some coordinate a_0 of a is not in M. Let $\varphi(x, y)$ be the formula $x_0 = y_0$, where x_0 and y_0 are the corresponding coordinates of x and y. For every tuple c from M the formula $\varphi(x, c)$ is not in p, because $a_0 \neq c_0$. Hence no tuple from M satisfies $d_p\varphi(y)$, and since $M \preceq \mathbb{M}$, no tuple from $\mathbb{M}$ satisfies it either. On the other hand, (6.1) gives $\mathbb{M} \models \varphi(a, a) \leftrightarrow d_p\varphi(a)$, and the formula $\varphi(a, a)$ holds. Thus the tuple a satisfies $d_p\varphi(y)$, a contradiction.

6.2.2 Proof of (2) ⇒ (3) ⇒ (1)

The implication (2) ⇒ (3) is immediate, because a relation satisfying all the axioms satisfies (UN) and (LOC) in particular. For (3) ⇒ (1), the idea is that the axiom (LOC) lets us describe each type over a model N by a type over a small submodel. The axiom (UN) guarantees that this description is faithful, and counting the descriptions then bounds the number of types. Let $\mathop{\smile\!\!\!\!\!\!|}$ satisfy (UN) and (LOC), and write $\lambda = 2^\kappa$; then $\lambda^\kappa = \lambda$. By Lemma 5.1.2, it suffices to show that the theory T is λ-stable.

Let A be a set with $|A| \leq \lambda$, and let x be a finite tuple of variables. The downward Löwenheim–Skolem theorem provides a model $N \supseteq A$ with $|N| \leq \lambda$, and restriction maps $S_x(N)$ onto $S_x(A)$. Thus we need only show that $|S_x(N)| \leq \lambda$.

For each $p \in S_x(N)$ choose a realization a_p. By (LOC) choose a model $M_p \subseteq N$ with $|M_p| \leq \kappa$ and $a_p \mathop{\smile\!\!\!\!\!\!|}_{M_p} N$. We claim that the map

$$p \longmapsto \big(M_p,\ \operatorname{tp}(a_p/M_p)\big)$$

is injective. Indeed, suppose that $M_p = M_{p'} =: M$ and $a_p \equiv_M a_{p'}$. Since $a_p \mathop{\smile\!\!\!\!\!\!|}_M N$ and $a_{p'} \mathop{\smile\!\!\!\!\!\!|}_M N$, the axiom (UN) gives $a_p \equiv_N a_{p'}$, which means that $p = p'$.

It remains to be shown that the map takes at most λ values. The number of subsets of N of size $\leq\kappa$ is at most $\lambda^\kappa = \lambda$, and for each model M among them we have $|S_x(M)| \leq 2^\kappa = \lambda$. Hence there are at most $\lambda \cdot \lambda = \lambda$ types over N. ☺

6.3 Local character, uniqueness, and forking

The chapter ends with two remarks, one looking back at the theorem and one looking ahead. In the first we show how little of Theorem 6.2.2 depends on the bound in (LOC). In the second we explain how heir independence extends, as nonforking, to arbitrary base sets, and over which base sets uniqueness persists.

Remark 6.3.1. The bound $|M| \leq \kappa$ in (LOC) plays no special role in the proof of (3) $\Rightarrow$ (1): it may be weakened to $|M| < \kappa_0$ for any fixed cardinal κ_0, by taking $\lambda = 2^{\kappa_0+\kappa}$, for which $\lambda^{\kappa_0} = \lambda$. Thus stability is equivalent to the existence of a relation for which types over models have unique independent extensions and every type is independent over a submodel of bounded size. When such a relation exists, the theory is stable, and by (1) $\Rightarrow$ (2) heir independence satisfies all the axioms.

Remark 6.3.2. The relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}$ has a counterpart over arbitrary base sets, Shelah's nonforking independence, which we develop in Chapter 7. A type over $B \supseteq C$ *forks* over C if it implies a finite disjunction of formulas each of which divides over C (see Definition 5.2.3). A type that divides over C forks over C, and in Theorem 7.3.3 we show that in a simple theory the two notions agree over models. In a stable theory the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}$ of Definition 6.2.1 is *nonforking independence* restricted to models as bases. Let M be a model, let $B \supseteq M$ be a small set, and let a be a tuple. The following are equivalent: the type $\mathrm{tp}(a/B)$ does not fork over M; this type is the heir of $\mathrm{tp}(a/M)$; this type is definable over M (see Pillay [**Pil96**] or Tent and Ziegler [**TZ12**]). Over arbitrary base sets, nonforking satisfies all the axioms above except uniqueness. Uniqueness holds over sets that are algebraically closed in T^{eq}, the theory of Remark 3.2.7 (Shelah [**She90**]; see also Harnik and Harrington [**HH84**]). Over other sets a type may have several nonforking extensions, as the theory of Example 3.2.6, that of an equivalence relation E with exactly two classes, both infinite, shows over $\emptyset$. Indeed, this theory has only one 1-type over $\emptyset$, and this type has at least two nonforking extensions to a model N with classes C_1 and C_2. Two such extensions are the type of a new element of C_1 and the type of a new element of C_2. For the formula $\varphi(x, y)$ given by $E(x, y)$, the φ-types of these extensions are defined by the conjugate formulas $E(y, c)$ with $c \in C_1$ and with $c \in C_2$. Hence both types are definable almost over $\emptyset$, that is, by a φ-definition with only finitely many conjugates over $\emptyset$ (see Definition 3.2.5), and neither is definable over $\emptyset$. Kim and Pillay [**KP97**] characterized nonforking in a simple theory, and in particular in a stable one (see Chapters 5 and 7). By their theorem, nonforking in a simple theory is the *only* relation on arbitrary small sets satisfying invariance, monotonicity, transitivity, finite character, symmetry, extension, local character and the independence theorem. Developing forking over arbitrary sets requires considerably more machinery; for this reason we have taken models as base sets in this chapter.

6.4 Exercises

The exercises follow the sections of the chapter, and the mark $(*)$ indicates those that ask for more than a direct application of the text.

Throughout, the letter M denotes a model, and we write T_E for the theory of an equivalence relation E with infinitely many classes, all of them infinite. For a small set A, we write $S_1(A)$ for the set of complete types over A in a single variable. As in the text, we write $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}$ for heir independence (see Definition 6.2.1) and (AR) for anti-reflexivity (see Theorem 6.2.2), and the axioms of an independence relation keep their labels from Definition 5.4.7. We use four facts from earlier chapters. The theory T_E admits elimination of quantifiers (see Exercise 10 of Section 5.7), and its types are described in Exercise 1 of the same section. The theory of $(\mathbb{Q}, <)$ admits elimination of quantifiers (see Fact 5.5.2(b) and Example 5.5.3(2)), and this theory is not stable (see Exercise 1 of Section 5.7).

1. *(Section 6.1.)* Assume that T is stable, and let x be a finite tuple of variables. Let $p \in S_x(M)$ be given, let B be a small set, and let $r \in S_x(MB)$ be a type extending p.

(a) Show that $r = p|B$ if and only if r has the following property: for every $L(M)$-formula $\varphi(x, y)$ and every tuple b from B with $\varphi(x, b) \in r$, there is a tuple c from M with $\varphi(x, c) \in p$. (Hint: for one direction, note that $\exists y\, d_p\varphi(y)$ is an $L(M)$-sentence. For the other, apply the property to the formula $\varphi(x, y) \wedge \neg d_p\varphi(y)$.) This property is the definition of an heir given by Lascar and Poizat.

(b) Let a be a finite tuple with $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_M B$, and suppose that some formula $\psi(x) \in \mathrm{tp}(a/MB)$ has only finitely many realizations in $\mathbb{M}$. Show that a is a tuple from M. (Hint: write $\psi(x)$ as $\psi_0(x, b)$ with $\psi_0 \in L(M)$ and b from B. Then apply (a) to the conjunction of $\psi_0(x, y)$ with a formula saying that $\psi_0(x', y)$ has at most k solutions x', where k is the number of realizations of $\psi(x)$. A finite set defined by a formula with parameters from M is contained in M.) Deduce from (MON) that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_M B$ implies $a \cap B \subseteq M$, and obtain (AR) again.

2. *(Section 6.1.)* Let T_∞ be the theory of an infinite set in the empty language, and assume that T_∞ admits elimination of quantifiers.

(a) Let A be a small set. Show that the type over A of a finite tuple is determined by the equalities among its coordinates and between its coordinates and elements of A. Deduce that $|S_x(A)| \leq |A| + \aleph_0$ for finite x, and that T_∞ is stable (see Lemma 5.1.2).

(b) Let $c \in M$ be given, let p_c be the type of c over M, and let $p_\infty \in S_1(M)$ be the type of an element outside M. By (a), this type does not depend on the element. Show that, for the formula $x = y$, the formula $y = c$ is a definition of p_c and the formula $y \neq y$ is a definition of p_∞. Let B be a small set. Deduce that the heir $p_\infty|B$ is the type over MB of an element outside MB, and that the heir $p_c|B$ is the type of c over MB.

(c) Let a be a finite tuple and B a small set. Show that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_M B$ if and only if $a \cap B \subseteq M$. (Hint: use Exercise 1(b) for one direction; for the other, compare a realization of $\mathrm{tp}(a/M)|B$ with a using (a).) Deduce by (FC) that heir independence in T_∞ is trivial independence (see Definition 5.4.8) with models as bases.

3. *(Section 6.1.)* Work in T_E. Let $m \in M$ be given, let a be an element outside M with aEm, and let a' be an element that is E-equivalent to no element of M. Write $p = \mathrm{tp}(a/M)$ and $p' = \mathrm{tp}(a'/M)$, and let B be a small set.

(a) Show that, for the formula $x = y$, the formula $y \neq y$ is a definition of p and of p'. Show that, for the formula xEy, the formula yEm is a definition of p and the formula $y \neq y$ is a definition of p'.

(b) Deduce that the heir $p|B$ is the type over MB of an element outside MB that is E-equivalent to m. Deduce also that the heir $p'|B$ is the type over MB of an element that is E-equivalent to no element of MB. (Hint: by elimination of quantifiers, a complete type over MB in one variable x is determined by the formulas $x = d$ and xEd, with $d \in MB$, that it contains.) Thus the heir of the type of a new element of an old class lies in that class. Likewise, the heir of the type of an element of a new class lies in a class that is new over MB.

(c) Let e be an element. Show that $e \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$ holds if and only if $e \notin B \setminus M$ and, whenever e is E-equivalent to an element of B, the element e is E-equivalent to an element of M. Compare with Exercise 4(c) of Section 5.7, and conclude that $e \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$ holds if and only if $\mathrm{tp}(e/MB)$ does not divide over M.

4. $(*)$ *(Section 6.1.)* Let K be a finite field, and let T be the theory of infinite vector spaces over K. The language of T consists of $+$, $-$, 0 and a unary function symbol for multiplication by each scalar in K (see Exercise 2 of Section 1.3). Assume that T admits elimination of quantifiers. For a set D, write $\langle D \rangle$ for the subspace of $\mathbb{M}$ spanned by D.

(a) Let A be a small set and $a = (a_1, \ldots, a_n)$ a tuple. Show that $\mathrm{tp}(a/A)$ is determined by the set of pairs (λ, w) with $\lambda \in K^n$, $w \in \langle A \rangle$ and $\sum_i \lambda_i a_i = w$. Deduce that $|S_x(A)| \leq |A| + \aleph_0$ for finite x, and that T is stable.

(b) Show that all elements outside M have the same type p_g over M, the *generic type*. Let $\varphi(x, y_1, \ldots, y_k)$ be the formula $x = \mu_1 y_1 + \cdots + \mu_k y_k + m$, with $k \geq 1$, $\mu_j \in K$ and $m \in M$. Show that the formula $y_1 \neq y_1$ is a definition of p_g for φ. Let B be a small set. Deduce that the heir $p_g|B$ is the type over MB of an element outside $\langle MB \rangle$.

(c) Let a be a finite tuple and B a small set. Show that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$ if and only if $\langle Ma \rangle \cap \langle MB \rangle = M$. (Hint: for one direction, apply Exercise 1(a) to a formula $\sum_i \lambda_i x_i = \sum_j \mu_j b_j + m$; for the other, use (a).)

(d) Let a and b be elements. Show that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} b$ holds if and only if $a \in M$, or $b \in M$, or a and b are linearly independent over M. Observe that the conditions in (c) and (d) are visibly symmetric, and derive (AR) from (c).

5. *(Subsection 6.1.1.)* Work in T_E, and let $\varphi(x, y)$ be the formula xEy. By Exercise 1 of Section 5.7, the types in $S_1(M)$ are exactly the types r_c, s_C and t described as follows. For each $c \in M$, the type r_c is that of c over M. For each E-class C of M, the type s_C is that of an element outside M that is E-equivalent to the elements of C. Finally, the type t is that of an element that is E-equivalent to no element of M. Say that r_c *determines* the class of c, that s_C determines C, and that t determines no class.

(a) For each type in $S_1(M)$, find a definition for φ. Since φ is symmetric, the same formulas, with x in place of y, are definitions for φ regarded as a formula in the variable y.

(b) Let $p, q \in S_1(M)$ be given, and verify Harrington's lemma (see Lemma 6.1.4) for p, q and φ directly. To this end, show that $d_p\varphi(y) \in q$ holds if and only if some E-class of M is determined both by p and by q, and that the same holds for $d_q\varphi(x) \in p$.

(c) Let $\mathcal{U}$ and $\mathcal{V}$ be ultrafilters on M with traces p and q. Show that an E-class C of M belongs to $\mathcal{U}$ if and only if p determines C. Show that $\lim_{a',\mathcal{U}} \lim_{b',\mathcal{V}} \varphi(a', b') = 1$ holds if and only if some E-class of M belongs to both $\mathcal{U}$ and $\mathcal{V}$, and that the same holds for the other iterated limit. Thus here both iterated limits depend only on the traces p and q; compare Exercise 7.

6. *(Subsection 6.1.1.)* Work in the theory of $(\mathbb{Q}, <)$, let $\varphi(x, y)$ be the formula $x < y$, and let $p \in S_1(M)$ be the type of an element greater than every element of M. By elimination of quantifiers, this type does not depend on the element.

(a) For every $L(M)$-formula $\chi(x, y)$, let $d_p\chi(y)$ be the formula $\exists z\, \forall t\, (z < t \to \chi(t, y))$. Show that, for every tuple b from M, we have $\chi(x, b) \in p$ if and only if $M \models d_p\chi(b)$. (Hint: by elimination of quantifiers, the truth value of $\chi(t, b)$ is the same for all t greater than the finitely many parameters of $\chi(x, b)$.) Thus the type p has a definition scheme, although the theory is not stable. Show that the proof of Lemma 6.1.3(a) uses stability only through the existence of a definition scheme. Deduce that, for every small set B, the set $p|B$ given by the formula of Definition 6.1.2 is a complete type over MB extending p.

(b) Let b be an element greater than every element of M. Show that $p|\{b\}$ contains the formula $b < x$, and that every type over Mb that extends p and is finitely satisfiable in M contains the formula $x < b$. Let $\mathcal{U}$ be an ultrafilter on M that contains the set $\{c \in M : m < c\}$ for every $m \in M$. Show that $q_{\mathcal{U}}|\emptyset = p$ and that $q_{\mathcal{U}}|\{b\} \neq p|\{b\}$. Thus the heir and the coheir of p differ; compare Exercise 8.

(c) Let $q \in S_1(M)$ be the type p written in the variable y, and let $d_q\varphi(x)$ be the definition of q given by (a) for φ regarded as a formula in the variable y. Show that the formula $d_p\varphi(y)$ is equivalent to $y \neq y$ and that the formula $d_q\varphi(x)$ is equivalent to $x = x$. Conclude that $d_p\varphi(y) \notin q$ and $d_q\varphi(x) \in p$, and hence that the conclusion of Harrington's lemma does not hold for p, q and φ. Compute also the two iterated limits of φ along $\mathcal{U}$ in both variables.

(d) Let b realize q and let a realize $p|\{b\}$; let a' realize p and let b' realize $q|\{a'\}$. Show that $b < a$ and $a' < b'$. Thus the two orders of realization give pairs with different types over M, although in both pairs each coordinate realizes p. Deduce that $\mathrm{tp}(a/Mb) = \mathrm{tp}(a/M)|\{b\}$ but $\mathrm{tp}(b/Ma) \neq \mathrm{tp}(b/M)|\{a\}$, and hence that the relation "$\mathrm{tp}(a/Mb) = \mathrm{tp}(a/M)|\{b\}$", for elements whose types over M have definition schemes, is not symmetric here.

7. *(Subsection 6.1.1.)* Work in the theory of $(\mathbb{Q}, <)$, regard $(\mathbb{Q}, <)$ as a model, and let $\varphi(x, y)$ be the formula $x < y$. Let r be an irrational real number, and let $p_r \in S_1(\mathbb{Q})$ be the type that contains $c < x$ for every rational $c < r$ and $x < c$ for every rational $c > r$.

(a) Show that there are ultrafilters $\mathcal{U}$ and $\mathcal{V}$ on $\mathbb{Q}$, the first containing the set $\{c \in \mathbb{Q} : s < c < r\}$ for every rational $s < r$ and the second containing the set $\{c \in \mathbb{Q} : r < c < s\}$ for every rational $s > r$. Show that $q_{\mathcal{U}}|\emptyset = q_{\mathcal{V}}|\emptyset = p_r$.

(b) Show that $\lim_{a',\mathcal{U}} \lim_{b',\mathcal{U}} \varphi(a',b') = 1$ and $\lim_{b',\mathcal{U}} \lim_{a',\mathcal{U}} \varphi(a',b') = 0$, while $\lim_{a',\mathcal{U}} \lim_{b',\mathcal{V}} \varphi(a',b') = \lim_{b',\mathcal{V}} \lim_{a',\mathcal{U}} \varphi(a',b') = 1$. Thus whether the iterated limits agree depends on the ultrafilters, and not only on their traces; compare Exercise 5(c).

(c) Let b realize p_r and let a realize $q_{\mathcal{U}}|\{b\}$; let a' realize p_r and let b' realize $q_{\mathcal{U}}|\{a'\}$. Show that $a < b$ and $b' < a'$; thus the two orders of realization give pairs with different types over $\mathbb{Q}$. Show also that $q_{\mathcal{U}}|\{b\} \neq q_{\mathcal{V}}|\{b\}$, and conclude that p_r has two different extensions to $\mathbb{Q} \cup \{b\}$ that are finitely satisfiable in $\mathbb{Q}$; compare Exercise 8(b).

8. $(*)$ *(Subsection 6.1.1.)* Assume that T is stable, let x be a finite tuple of variables, let $p \in S_x(M)$ be given, and let B be a small set.

(a) Let $\mathcal{U}$ be an ultrafilter on $M^{|x|}$ with $q_{\mathcal{U}}|\emptyset = p$. Show that $q_{\mathcal{U}}|B = p|B$. (Hint: let $\varphi(x,y) \in L(M)$ and a tuple b from B be given, and write $q = \mathrm{tp}(b/M)$. Show that $\{a \in M^{|x|} : \mathbb{M} \models \varphi(a,b)\}$ is the set of tuples of M that satisfy $d_q\varphi(x)$, and apply Lemma 6.1.4.)

(b) Deduce that $p|B$ is the only type over MB that extends p and is finitely satisfiable in M. (Hint: Lemma 6.1.5(b).)

(c) Let a and b be finite tuples. Deduce that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$ holds if and only if $\mathrm{tp}(a/MB)$ is finitely satisfiable in M, and that $\mathrm{tp}(a/Mb)$ is finitely satisfiable in M if and only if $\mathrm{tp}(b/Ma)$ is. Thus in a stable theory the relation "$\mathrm{tp}(a/Mb)$ is finitely satisfiable in M", called coheir independence, is symmetric; compare Exercise 2 of Section 7.7 and Exercises 6 and 7.

9. *(Section 6.2.)* Work in T_E.

(a) Let A and B be small sets. Show that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$ holds exactly when $A \cap B \subseteq M$ and every element of A that is E-equivalent to an element of B is E-equivalent to an element of M. (Hint: by (FC), it suffices to treat finite tuples a and b. For one direction, apply Exercise 1(a) to the formulas $x_i = y_j$ and $x_i E y_j$. For the other, let a' realize $\mathrm{tp}(a/M)|b$, and compare the data of Exercise 1 of Section 5.7 for a and a' over Mb.)

(b) Using only the description in (a), verify the axioms (SYM), (BMON) and (TR) as well as anti-reflexivity (AR) for $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}$ in T_E.

(c) Verify (UN) directly: let a and a' be tuples with $a \equiv_M a'$, $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$ and $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} B$, and show by Exercise 1 of Section 5.7 that $a \equiv_{MB} a'$.

(d) Let n be a positive integer, let a be an n-tuple, and let N be a model. Let $M \subseteq N$ be a model that contains every coordinate of a lying in N. Suppose also that every coordinate of a that is E-equivalent to an element of N is E-equivalent to an element of M, and show that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} N$. Deduce (LOC), and show that there is a set $F \subseteq N$ with at most n elements such that every countable model M with $F \subseteq M \subseteq N$ satisfies $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_{M} N$.

10. *(Section 6.2.)* Work in the theory of $(\mathbb{Q}, <)$, and consider trivial independence $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ (see Definition 5.4.8) with models as bases.

(a) Show that $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ satisfies (LOC).

(b) Let p be as in Exercise 6, let b be an element greater than every element of M, and let a and a' be elements with $M < a < b < a'$. Show that $a \equiv_M a'$, $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}_M b$ and $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}_M b$, but $a \not\equiv_{Mb} a'$. Thus the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ does not satisfy (UN); by Exercise 6, the type $\mathrm{tp}(a/Mb)$ is the coheir of p, and the type $\mathrm{tp}(a'/Mb)$ is the set $p|\{b\}$.

(c) Let N be a model, let $M \subseteq N$ be a countable model with $M \neq N$, and let $n \in N \setminus M$ be given. Find elements a and a' outside N with $a < n < a'$ and $a \equiv_M a'$. Show that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}_M N$ and $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}_M N$, but $a \not\equiv_N a'$. Thus, if the proof of (3) $\Rightarrow$ (1) in Theorem 6.2.2 is run with $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$, the map $p \mapsto (M_p, \mathrm{tp}(a_p/M_p))$ need not be injective.

(d) Deduce from Theorem 6.2.2 that no relation satisfies both (UN) and (LOC) in this theory.

11. *(Section 6.2.)* Work in T_E with heir independence, and let N be a model. Fix a countable model $M_0 \subseteq N$ and, for every $n \in N$, a countable model M_n with $M_0 \cup \{n\} \subseteq M_n \subseteq N$.

(a) Let $p \in S_1(N)$ be given, let a_p realize p, and show that $a_p \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_M N$ holds for some $M \in \{M_0\} \cup \{M_n : n \in N\}$. (Hint: Exercise 9(d).)

(b) Run the proof of (3) $\Rightarrow$ (1) in Theorem 6.2.2 with this choice of the models M_p. Show that the map $p \mapsto (M_p, \mathrm{tp}(a_p/M_p))$ is injective and that its image has at most $|N|$ elements, and deduce that $|S_1(N)| \leq |N|$. (Hint: each M_n is countable, and Exercise 1 of Section 5.7 bounds $|S_1(M_n)|$ by $\aleph_0$.)

(c) Show directly that $|S_1(N)| = |N|$, and compare the result with the bound $|S_1(N)| \leq 2^{\aleph_0}$ that the proof gives, with $\lambda = 2^{\aleph_0}$, for models N with $|N| \leq 2^{\aleph_0}$.

12. *(Section 6.3.)*

(a) Let κ_0 be an infinite cardinal, and let $\mathop{\smile\!\!\!\!\!\!|}$ be a relation satisfying (UN) and the weakening of (LOC) in which the condition $|M| \leq \kappa$ is replaced by $|M| < \kappa_0$ (see Remark 6.3.1). Write $\lambda = 2^{\kappa_0 + \kappa}$, and show that $\lambda^{\kappa_0} = \lambda$ and that a set of size at most λ has at most λ subsets of size less than κ_0. Show also that $|S_x(M)| \leq \lambda$ for every finite x and every model M with $|M| < \kappa_0$. Complete the proof that T is stable, following the proof of (3) $\Rightarrow$ (1) in Theorem 6.2.2.

(b) Let $A \mathop{\smile\!\!\!\!\!\!|}_M B$ mean that $A \subseteq M$ or $B \subseteq M$. Show that this relation satisfies (UN) in every complete theory, and that it does not satisfy (LOC) in the theory of $(\mathbb{Q}, <)$. (Hint: consider a model N with $|N| > \aleph_0$ and an element outside N.) Together with Exercise 10, conclude that neither (UN) nor (LOC) alone implies stability. Hence both axioms are needed in condition (3) of Theorem 6.2.2.

13. (∗) *(Section 6.3.)* Let T_2 be the theory of an equivalence relation E with exactly two classes, both infinite (see Example 3.2.6), and assume that T_2 admits elimination of quantifiers. Let N be a countable model with classes C_1 and C_2, and, for $i = 1, 2$, let $p_i \in S_1(N)$ be the type of an element outside N that is E-equivalent to the elements of C_i.

(a) Show that T_2 is stable, that $S_1(\emptyset)$ has exactly one element, and that $S_1(N)$ consists of the types realized in N together with p_1 and p_2.

(b) Show that every type realized in N divides over $\emptyset$ (see Exercise 2 of Section 5.7), and that p_1 and p_2 do not divide over $\emptyset$ (see Definition 5.2.3). (Hint: let $(d_k)_{k<\omega}$ be a

sequence of tuples indiscernible over $\emptyset$. Since there are only two classes, corresponding coordinates of the tuples d_k are E-equivalent.) Thus the unique type over $\emptyset$ has exactly two extensions to N that do not divide over $\emptyset$; compare Remark 6.3.2.

(c) Let $M \subseteq N$ be a model. Show that M meets both classes, that the restrictions of p_1 and p_2 to M are different, and that p_i is the heir of its restriction to M. Thus over a model the two extensions are already distinguished by their restrictions.

(d) Let $\mathop{\smile\!\!\!\!\!\!|}$ be a relation between small sets over arbitrary small base sets C that satisfies invariance and existence over $\emptyset$. Here invariance means that $A \mathop{\smile\!\!\!\!\!\!|}_C B$ implies $\sigma(A) \mathop{\smile\!\!\!\!\!\!|}_{\sigma(C)} \sigma(B)$ for every $\sigma \in \mathrm{Aut}(\mathbb{M})$, and existence over $\emptyset$ means that for all A and B there is $A' \equiv_\emptyset A$ with $A' \mathop{\smile\!\!\!\!\!\!|}_\emptyset B$. Show that uniqueness over $\emptyset$ does not hold: there are elements a' and a'' with $a' \equiv_\emptyset a''$, $a' \mathop{\smile\!\!\!\!\!\!|}_\emptyset N$ and $a'' \mathop{\smile\!\!\!\!\!\!|}_\emptyset N$, but $a' \not\equiv_N a''$. (Hint: an automorphism of N that exchanges C_1 and C_2 extends, by strong homogeneity, to an automorphism σ of $\mathbb{M}$ with $\sigma(N) = N$.)

6.5 Historical remarks

Abstract independence. The axiomatic study of independence is older than model theory. Steinitz, in his foundational paper on the algebraic theory of fields [Ste10], introduced transcendence bases and showed that any two transcendence bases have the same cardinality. The argument is an exchange argument of the kind now called the Steinitz exchange lemma, the same one that gives the dimension of a vector space. Van der Waerden, in *Moderne Algebra* [vdW30], observed that linear dependence and algebraic dependence obey the same few rules. From an abstract notion of dependence whose central axiom is exchange, he derived the existence of bases and the invariance of their cardinality. Whitney [Whi35] introduced *matroids*, structures that isolate the abstract properties of linear dependence, and axiomatized them in several equivalent ways: by independent sets, bases, circuits and rank. Nakasawa developed the same theory independently in the mid-1930s. Birkhoff [Bir35] and Mac Lane [ML36] related matroid theory to lattices and to projective geometry.

After a period in which matroids were studied mainly by Rado and Tutte, Rota made them central to combinatorics. In his monograph with Crapo [CR70], matroids appear as *combinatorial geometries*, and a closure operator with the exchange property is called a *pregeometry*, the term still used in model theory. Rota's view was that dependence is a geometric notion with invariants of its own, such as the Möbius function and the characteristic polynomial of the associated lattice.

Model theory met these ideas in the study of uncountably categorical theories. Marsh, in his Dartmouth thesis of 1966, introduced the strongly minimal theories and showed that in them algebraic closure has the exchange property, so algebraic closure is a pregeometry and every model has a well-defined dimension. Baldwin and Lachlan [BL71] carried this dimension to strongly minimal sets in an arbitrary uncountably categorical theory, and with it they gave a new proof of Morley's categoricity theorem [Mor65]. Shelah's forking [She90] carries independence beyond pregeometries. Nonforking independence is a relation between sets rather than a closure operator, and in a stable theory the relation has symmetry and existence, and over models uniqueness, the properties on which arguments about dimension depend. The work of Zilber and Hrushovski turned the geometry of forking into a subject of its own, geometric stability theory [Pil96]. In this chapter we characterize independence by

axioms, as van der Waerden and Whitney did for linear dependence. This point of view is that of Harnik and Harrington, discussed below.

Section 6.1. Heirs and coheirs (*héritiers* and *cohéritiers*) were introduced by Lascar and Poizat [LP79]. They gave an alternative presentation of Shelah's forking in stable theories, based only on definability of types and finite satisfiability. The commutation of the definitions of two types over a model (see Lemma 6.1.4) is due to Harrington; it appears in the paper of Harnik and Harrington [HH84]. It is the key step in the proof that independence in stable theories is symmetric (see the proof of Theorem 6.2.2).

Section 6.2. Nonforking independence was introduced by Shelah [She90]. The characterization of stability through abstract independence notions is due to Harnik and Harrington [HH84]. The main theorem of this chapter, Theorem 6.2.2, is a version of their characterization with models as bases, in which uniqueness (stationarity over models) is the key axiom. Kim and Pillay [KP97] extended the characterization to simple theories, with the independence theorem, the amalgamation of independent extensions of one type to two independent sets, in place of uniqueness (see Chapter 7). The anchor symbol $\perp$ for independence was introduced by Makkai in a lecture in Jerusalem, during the Model Theory Year at the Institute for Advanced Studies (1980–81), and was adopted at once. The symbol appears in print in Makkai's survey [Mak84].

Chapter
7

Simplicity and notions of independence

This chapter is in preparation: its Sections 7.1–7.6, the exercises and the historical remarks are written; the roadmap below indicates what remains.

Roadmap

In this chapter we do for simplicity what we did for stability in Chapter 6, and we explain the independence relation found by hand for the random graph in Chapter 5 (see Theorem 5.4.9). In Section 7.1 we define nondividing independence over models and Morley sequences in an arbitrary theory. In Section 7.2 we prove that in a simple theory any single Morley sequence witnesses dividing (see Theorem 7.2.2). In Section 7.3 we prove that forking equals dividing over models and that nonforking independence is symmetric and transitive (see Theorems 7.3.3 and 7.3.5). In Section 7.4 we prove the independence theorem: two independent extensions of one type over a model, to sets that are independent of each other, have a common independent extension (see Theorem 7.4.7). In Section 7.5 we show that simplicity together with uniqueness of independent extensions is exactly stability, which returns us to Chapter 6 (see Theorem 7.5.2). In Section 7.6 we show that a theory is simple exactly when it carries a relation satisfying the axioms of Definition 5.4.7 other than uniqueness, together with extension and the independence theorem (see Theorem 7.6.5). In a final section, still to be written, we will return to the Rado graph and the Fraïssé limits of Chapter 5.

7.1 Dividing over models

A type that does not divide over a model says nothing about its parameters outside the model that the model could not have foreseen. In precise terms, along any sequence of copies of these parameters that is indiscernible over the model, the type can be realized against all the copies at once. In this section we turn this description into a lemma and draw from it three consequences that hold in every theory. The first is that types finitely satisfiable in a model do not divide over it, whence nondividing extensions always exist. The second is that these finitely satisfiable types generate indiscernible sequences of a special kind, the Morley

sequences. The third is that nondividing is transitive on the left. Apart from local character, simplicity enters only in the next section.

Definition 7.1.1. For small sets A, B, C, write $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{C} B$ if for every finite tuple a from A the type $\mathrm{tp}(a/CB)$ does not divide over C. We call $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ *nondividing independence*.

For a finite tuple b, we write $\mathrm{tp}(a/Cb)$ for $\mathrm{tp}(a/C \cup b)$. A formula of $\mathrm{tp}(a/CB)$ mentions only finitely many elements of B, so it already belongs to $\mathrm{tp}(a/Cb)$ for some finite tuple b from B. Hence the relation $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{C} B$ holds if and only if $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{C} b$ for all finite tuples a from A and b from B; this is finite character, built into the definition.

The definition of dividing (see Definition 5.2.3) concerns L-formulas, but parameters from C cause no difficulty. Let $\psi(x, y) \in L(C)$ be given, and write $\psi(x, y) = \psi_0(x, y, c)$ with $\psi_0 \in L$ and c a tuple from C. We say that the formula $\psi(x, b)$ divides over C when $\psi_0(x, b, c)$ does, in the sense of Definition 5.2.3. A sequence indiscernible over C that begins with (b, c) has the form $((b_i, c))_{i<\omega}$, because each of its terms has the type of (b, c) over C. Conversely, the sequence $((b_i, c))_{i<\omega}$ is indiscernible over C whenever $(b_i)_{i<\omega}$ is. Hence the formula $\psi(x, b)$ divides over C if and only if $\{\psi(x, b_i) : i < \omega\}$ is inconsistent for some sequence $(b_i)_{i<\omega}$, indiscernible over C, with $b_0 = b$.

7.1.1 Nondividing and indiscernible sequences

The definition of dividing quantifies over indiscernible sequences, one formula at a time. The first lemma quantifies over sequences once and for all. By this lemma, a type does not divide exactly when it can be moved, without changing it, to a position from which a given indiscernible sequence still looks indiscernible (see Figure 7.1).

Lemma 7.1.2. *Let C be a small set, let a be a small tuple, and let b be a finite tuple. The following are equivalent:*

(1) *the type $\mathrm{tp}(a/Cb)$ does not divide over C;*

(2) *for every sequence $(b_i)_{i<\omega}$, indiscernible over C, with $b_0 = b$, there is a' with $a' \equiv_{Cb} a$ such that $(b_i)_{i<\omega}$ is indiscernible over $C \cup a'$.*

Proof. We denote by $p(x, y)$ the set of $L(C)$-formulas $\psi(x, y)$ with $\mathbb{M} \models \psi(a, b)$, and by $p(x, b_i)$ the set of formulas $\psi(x, b_i)$ with $\psi \in p$; thus $p(x, b) = \mathrm{tp}(a/Cb)$.

(2) $\Rightarrow$ (1). Let $\psi(x, b)$ be a formula of $\mathrm{tp}(a/Cb)$, with $\psi(x, y) \in L(C)$, and let $(b_i)_{i<\omega}$ be indiscernible over C with $b_0 = b$. By (2) there is $a' \equiv_{Cb} a$ such that the sequence is indiscernible over $C \cup a'$. Since $\mathbb{M} \models \psi(a', b_0)$, indiscernibility over $C \cup a'$ gives $\mathbb{M} \models \psi(a', b_i)$ for every i. Hence the set $\{\psi(x, b_i) : i < \omega\}$ is consistent. By the observation following Definition 7.1.1, the formula $\psi(x, b)$ does not divide over C.

(1) $\Rightarrow$ (2). Let $(b_i)_{i<\omega}$ be indiscernible over C with $b_0 = b$. The proof has two steps: we first realize all the copies $p(x, b_i)$ at once, and then make the sequence indiscernible over the realization by extracting a new sequence and moving it back.

☺ **Step 1.** The set $\Sigma(x) = \bigcup_{i<\omega} p(x, b_i)$ is consistent. Otherwise, by compactness, some finite subset is inconsistent, and such a subset involves finitely many formulas $\psi_1, \ldots, \psi_r$ of p and finitely many indices $i_1, \ldots, i_n$. Let ψ be the conjunction of $\psi_1, \ldots, \psi_r$. Then the formula $\psi(x, b)$ belongs to $p(x, b)$, and the set $\{\psi(x, b_{i_l}) : 1 \leq l \leq n\}$ is inconsistent. Hence the set $\{\psi(x, b_i) : i < \omega\}$ is inconsistent, and therefore the formula $\psi(x, b)$ divides over C, contradicting (1). Let a'' realize Σ.

☺ **Step 2.** We apply Lemma 5.2.2 over $C \cup a''$ to the sequence $(b_i)_{i<\omega}$. The Ehrenfeucht–Mostowski type of this sequence over $C \cup a''$ contains, for every n, the type of $(b_0, \ldots, b_{n-1})$ over C, because the sequence is indiscernible over C. This Ehrenfeucht–Mostowski type also

contains every formula $\psi(a'', y_1)$ with $\psi \in p$, because a'' realizes every $p(x, b_i)$. Consequently, Lemma 5.2.2 gives a sequence $(b'_i)_{i<\omega}$, indiscernible over $C \cup a''$, such that $(b'_0, \dots, b'_{n-1}) \equiv_C (b_0, \dots, b_{n-1})$ for every n and $\mathbb{M} \models \psi(a'', b'_0)$ for every $\psi \in p$. From the first property we obtain $(b'_i)_{i<\omega} \equiv_C (b_i)_{i<\omega}$, and the second gives $a''b'_0 \equiv_C ab$. By strong homogeneity there is $\sigma \in \mathrm{Aut}(\mathbb{M}/C)$ with $\sigma(b'_i) = b_i$ for every i. The required tuple is $a' = \sigma(a'')$: it satisfies $a'b = \sigma(a''b'_0) \equiv_C ab$, which means that $a' \equiv_{Cb} a$, and the sequence $(b_i) = (\sigma(b'_i))$ is indiscernible over $\sigma(C \cup a'') = C \cup a'$. ☺

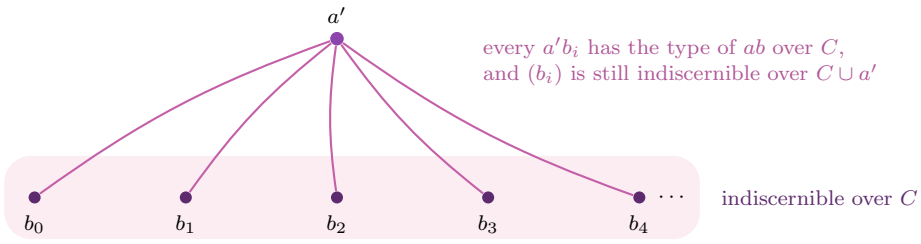


FIGURE 7.1. Lemma 7.1.2: a type $\mathrm{tp}(a/Cb)$ not dividing over C, moved over $C \cup b$ to a tuple a' standing in the same relation to every term of a given indiscernible sequence starting with b.

7.1.2 Coheirs do not divide

Dividing is a statement about many copies of a parameter. A type that is finitely satisfiable in a model is witnessed by single elements of the model, and all the copies of a parameter over the model must treat these elements alike. These two observations make up the whole proof of the next lemma.

Lemma 7.1.3. *Let M be a model, let B be a small set, and let a be a small tuple such that $\mathrm{tp}(a/MB)$ is finitely satisfiable in M. Then $\mathrm{tp}(a/MB)$ does not divide over M, that is, $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$.*

Proof. Let $\psi(x, b)$ be a formula of $\mathrm{tp}(a/MB)$, with $\psi(x, y) \in L(M)$ and b a finite tuple from B, and let $(b_i)_{i<\omega}$ be indiscernible over M with $b_0 = b$. By finite satisfiability there is a tuple m from M with $\mathbb{M} \models \psi(m, b_0)$. The tuple m and the parameters of ψ lie in M, so indiscernibility over M gives $\mathbb{M} \models \psi(m, b_i)$ for every i. Hence the set $\{\psi(x, b_i) : i < \omega\}$ is consistent, and the formula $\psi(x, b)$ does not divide over M. ☺

Every type over a model M is finitely satisfiable in M (see Subsection 6.1.1). In the next proposition we show that such a type extends, for every small set B, to a type over MB that is still finitely satisfiable in M. Together with Lemma 7.1.3, this proposition gives the existence axiom (EX) for $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ in every theory.

Proposition 7.1.4. *Let M be a model, and let A and B be small sets. Then there is $A' \equiv_M A$ such that $\mathrm{tp}(A'/MB)$ is finitely satisfiable in M. In particular, we have $A' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$.*

Proof. The idea is to realize $\mathrm{tp}(A/M)$ while avoiding every formula over MB that no tuple from M satisfies; the type over MB of such a realization is then finitely satisfiable in M. We write π for $\mathrm{tp}(A/M)$, in variables X, and let F be the set of $L(MB)$-formulas $\chi(X)$ that are satisfied by no tuple from M. We claim that $\pi \cup \{\neg\chi : \chi \in F\}$ is consistent. A finite subset is contained in $\{\theta, \neg\chi_1, \dots, \neg\chi_n\}$ for some $\theta \in \pi$ and $\chi_1, \dots, \chi_n \in F$. Since θ is an $L(M)$-formula realized in $\mathbb{M}$ and $M \preceq \mathbb{M}$, some tuple m from M satisfies θ, and this

tuple satisfies each $\neg\chi_l$ because $\chi_l \in F$. By saturation, the set is realized by some A'. This realization has the same type over M as A, and every formula χ of $\mathrm{tp}(A'/MB)$ is satisfied by a tuple from M, since otherwise $\chi \in F$ and $\neg\chi \in \mathrm{tp}(A'/MB)$. Hence the type $\mathrm{tp}(A'/MB)$ is finitely satisfiable in M, and Lemma 7.1.3 gives $A' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$. ☺

For a finite tuple a, the types produced in this way are exactly the coheirs of Lemma 6.1.5. Recall that, for an ultrafilter $\mathcal{U}$ on $M^{|a|}$, the coheir $q_{\mathcal{U}}|B$ is the set of $L(MB)$-formulas whose sets of solutions in $M^{|a|}$ belong to $\mathcal{U}$. A type over MB that is finitely satisfiable in M has the form $q_{\mathcal{U}}|B$ for some ultrafilter $\mathcal{U}$ on $M^{|a|}$ with $q_{\mathcal{U}}|\emptyset = \mathrm{tp}(a/M)$.

7.1.3 Morley sequences

Realizing a coheir over and over, each time over everything realized so far, produces an indiscernible sequence in which every term is independent from the earlier ones (see Lemma 7.1.6 below). In Theorem 7.2.2 below we show that, in a simple theory, whether a formula divides over a model is visible along any single such sequence over the model beginning with its parameter.

Definition 7.1.5. Let M be a model. A *Morley sequence over* M is a sequence $(b_i)_{i<\omega}$, indiscernible over M, such that $b_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} \{b_j : j < i\}$ for every i.

Lemma 7.1.6. *Let M be a model, let b be a finite tuple, and let $\mathcal{U}$ be an ultrafilter on $M^{|b|}$ whose coheir $q_{\mathcal{U}}|\emptyset$ over M (see Lemma 6.1.5) is* $\mathrm{tp}(b/M)$. *Write $b_0 = b$, and for $i \geq 1$ let b_i realize $q_{\mathcal{U}}|\{b_j : j < i\}$. Then the sequence $(b_i)_{i<\omega}$ is a Morley sequence over M with $b_0 = b$.*

The ultrafilter exists by Lemma 6.1.5(b), since $\mathrm{tp}(b/M)$ is finitely satisfiable in M, and the tuples b_i exist by saturation. We write $b_{<i}$ for $(b_j)_{j<i}$.

Proof. Indiscernibility comes from the invariance of coheirs over M (see Lemma 6.1.5(d)). Each term is independent from the earlier ones because coheirs are finitely satisfiable in M, and such types do not divide over M (see Lemma 7.1.3).

For $i \geq 1$ and $i_1 < \cdots < i_n < i$, Lemma 6.1.5(c) gives $q_{\mathcal{U}}|\{b_{i_1}, \ldots, b_{i_n}\} \subseteq q_{\mathcal{U}}|b_{<i} = \mathrm{tp}(b_i/Mb_{<i})$; since the smaller set is a complete type over $M \cup \{b_{i_1}, \ldots, b_{i_n}\}$ by Lemma 6.1.5(a), it equals the type of b_i over that set. In particular, every b_i realizes $q_{\mathcal{U}}|\emptyset = \mathrm{tp}(b/M)$, for $i = 0$ by hypothesis.

We show by induction on n that all increasing n-tuples of the sequence have the same type over M. For $n = 1$ this was just observed. Let $i_1 < \cdots < i_{n+1}$ and $j_1 < \cdots < j_{n+1}$ be given. By the induction hypothesis and strong homogeneity there is $\sigma \in \mathrm{Aut}(\mathbb{M}/M)$ with $\sigma(b_{i_l}) = b_{j_l}$ for $l \leq n$. By Lemma 6.1.5(d), the automorphism σ carries the type $q_{\mathcal{U}}|\{b_{i_1}, \ldots, b_{i_n}\}$ of $b_{i_{n+1}}$ over $M \cup \{b_{i_1}, \ldots, b_{i_n}\}$ to the type $q_{\mathcal{U}}|\{b_{j_1}, \ldots, b_{j_n}\}$ of $b_{j_{n+1}}$ over $M \cup \{b_{j_1}, \ldots, b_{j_n}\}$. It follows that $\sigma(b_{i_{n+1}}) \equiv_{Mb_{j_1} \ldots b_{j_n}} b_{j_{n+1}}$, and therefore $(b_{i_1}, \ldots, b_{i_{n+1}}) \equiv_M (b_{j_1}, \ldots, b_{j_{n+1}})$. Thus the sequence is indiscernible over M.

Finally, for every i the type $\mathrm{tp}(b_i/Mb_{<i})$ is a coheir, hence finitely satisfiable in M by Lemma 6.1.5(a), and Lemma 7.1.3 gives $b_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<i}$. ☺

7.1.4 Transitivity on the left

Nondividing independence can be built up one coordinate block at a time on the left. More precisely, suppose that a first set is independent from a second over a base, and that a third is independent from the second over the base enlarged by the first. In that case the first and the third together are independent from the second over the base. The proof uses the implication (1) ⇒ (2) of Lemma 7.1.2 twice, once for each hypothesis, concludes with the implication (2) ⇒ (1), and requires no hypothesis on T.

Lemma 7.1.7. *Let A, B, C, D be small sets. If $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{C} B$ and $D \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{CA} B$, then $AD \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{C} B$.*

Proof. We regard A as a tuple, and let finite tuples e from D and b from B be given. By finite character it suffices to show that $\mathrm{tp}(Ae/Cb)$ does not divide over C; the types of the finite tuples from $A \cup e$ are restrictions of this type. Every formula of $\mathrm{tp}(A/Cb)$ mentions a finite tuple from A, so the hypothesis $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{C} B$ says that $\mathrm{tp}(A/Cb)$ does not divide over C; likewise $D \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{CA} B$ says that $\mathrm{tp}(e/CAb)$ does not divide over $C \cup A$. We verify condition (2) of Lemma 7.1.2 for the tuple Ae.

Let $(b_i)_{i<\omega}$ be indiscernible over C with $b_0 = b$. By Lemma 7.1.2, applied to A, there is $A' \equiv_{Cb} A$ such that (b_i) is indiscernible over $C \cup A'$. Since $A' \equiv_{Cb} A$, strong homogeneity gives $\sigma \in \mathrm{Aut}(\mathbb{M}/Cb)$ with $\sigma(A') = A$. Then the sequence $(\sigma(b_i))_{i<\omega}$ is indiscernible over $C \cup A$ and begins with $\sigma(b_0) = b$. Applying the same lemma to e over the base $C \cup A$, we obtain $e' \equiv_{CAb} e$ such that $(\sigma(b_i))$ is indiscernible over $C \cup Ae'$. Applying σ^{-1}, which fixes $C \cup b$, we see that (b_i) is indiscernible over $C \cup A'\sigma^{-1}(e')$, and

$$A'\sigma^{-1}(e') = \sigma^{-1}(Ae') \equiv_{Cb} Ae' \equiv_{Cb} Ae.$$

Thus the tuple Ae satisfies condition (2) of Lemma 7.1.2, and the type $\mathrm{tp}(Ae/Cb)$ does not divide over C. ☺

7.1.5 The axioms that hold in every theory

We now measure what we have shown in this section against the axioms of Definition 5.4.7.

Proposition 7.1.8. *In every complete theory, the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$, with models as bases, satisfies* (INV), (MON), (BMON), (FC) *and* (EX). *If T is simple, it also satisfies* (LOC).

Proof. Invariance: an automorphism σ of $\mathbb{M}$ carries sequences indiscernible over C to sequences indiscernible over $\sigma(C)$ and preserves consistency. Hence the formula $\varphi(x, b)$ divides over C if and only if $\varphi(x, \sigma(b))$ divides over $\sigma(C)$. Monotonicity is immediate from the definition, and finite character was observed after Definition 7.1.1. *Base monotonicity*: suppose that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$, let N be a model with $M \subseteq N \subseteq B$, and let a be a finite tuple from A. Since $NB = B = MB$, a formula of $\mathrm{tp}(a/NB)$ that divided over N would divide over M by Remark 5.2.5. *Existence* is the content of Proposition 7.1.4. *Local character*: let a finite tuple a and a model N be given, and recall that $\kappa = |T|$ (see Section 5.1). If T is simple, Theorem 5.3.6 gives $A_0 \subseteq N$ with $|A_0| \leq \kappa$ such that $\mathrm{tp}(a/N)$ does not divide over A_0. By the downward Löwenheim–Skolem theorem there is a model $M \preceq N$ with $A_0 \subseteq M$ and $|M| \leq \kappa$. By Remark 5.2.5 once more, the type $\mathrm{tp}(a/N)$ does not divide over M, which is the relation $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} N$. ☺

Symmetry, transitivity and the independence theorem remain, and these are the properties that rest on simplicity. By a theorem of Kim [**Kim01**], symmetry of forking independence (see Definition 7.3.1 below) already characterizes the simple theories, and so does transitivity. We derive all three from simplicity by way of Kim's lemma, the subject of the next section.

7.2 Kim's lemma

A formula divides when some indiscernible sequence of copies of its parameter makes it inconsistent. The definition asks for one such sequence, and gives no hint where to look for it. In a simple theory there is no need to look: any Morley sequence will do. In such a theory, any sequence of realizations of the type of the parameter in which each term is independent

from the earlier ones will do as well. Both assertions are part of Kim's lemma [**Kim98**], on which the rest of the chapter rests. With the preparations of Section 7.1 the proof is short, and its idea fits in two sentences. Read backwards, the instances of the formula along a Morley sequence form a chain in which each formula divides over the parameters of the earlier ones. Such a chain is turned into a tree by Lemma 5.3.5(1), and a simple theory has no such trees; hence the instances of the formula along the sequence cannot be consistent.

Throughout the section, the letter M denotes a model.

7.2.1 Enlarging the base

The proof of Kim's lemma needs a formula that divides over M to divide also over M together with later terms of the sequence. Enlarging the base can only make dividing harder (see Remark 5.2.5); the next lemma shows that the enlargement costs nothing when the new parameters are independent from the parameter of the formula. Since the proof of the lemma works with k-dividing for a fixed k, we first record that the parameter convention following Definition 7.1.1 applies to k-dividing as well, with the same proof. Explicitly, for $\psi(x,y) \in L(C)$, the formula $\psi(x,b)$ k-divides over C if and only if $\{\psi(x,b_i) : i < \omega\}$ is k-inconsistent for some sequence $(b_i)_{i<\omega}$, indiscernible over C, with $b_0 = b$.

Lemma 7.2.1. *Let C be a small set, let b be a finite tuple, and let e be a small tuple such that $\mathrm{tp}(e/Cb)$ does not divide over C. Let $\varphi(x,y) \in L(C)$ and $k \geq 2$ be given. If $\varphi(x,b)$ k-divides over C, then the formula $\varphi(x,b)$ k-divides over $C \cup e$.*

Proof. Let $(b_i)_{i<\omega}$ be indiscernible over C, with $b_0 = b$ and $\{\varphi(x,b_i) : i < \omega\}$ k-inconsistent. By Lemma 7.1.2 there is $e' \equiv_{Cb} e$ such that (b_i) is indiscernible over $C \cup e'$. As $e' \equiv_{Cb} e$, there is $\sigma \in \mathrm{Aut}(\mathbb{M}/Cb)$ with $\sigma(e') = e$. The sequence $(\sigma(b_i))_{i<\omega}$ is indiscernible over $\sigma(C \cup e') = C \cup e$, and it begins with $\sigma(b) = b$. Since σ fixes the parameters of φ, which lie in C, the set $\{\varphi(x,\sigma(b_i)) : i < \omega\}$ is the image under σ of a k-inconsistent set, and so it is k-inconsistent. Hence the formula $\varphi(x,b)$ k-divides over $C \cup e$. ☺

7.2.2 The lemma

We can now state and prove Kim's lemma. As in Section 7.1, we write $b_{<i}$ for $(b_j)_{j<i}$.

Theorem 7.2.2 (Kim's lemma). *Assume that T is simple. Let M be a model, let b be a finite tuple, and let $\varphi(x,y) \in L(M)$ be a formula.*

(a) *Suppose that $\varphi(x,b)$ divides over M, and let $(b_i)_{i<\omega}$ be a sequence such that $b_i \equiv_M b$ and $b_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M b_{<i}$ for every i. Then the set $\{\varphi(x,b_i) : i < \omega\}$ is inconsistent.*

(b) *The following are equivalent:*

(1) *the formula $\varphi(x,b)$ divides over M;*

(2) *for every Morley sequence $(b_i)_{i<\omega}$ over M with $b_0 = b$, the set $\{\varphi(x,b_i) : i < \omega\}$ is inconsistent;*

(3) *for some Morley sequence $(b_i)_{i<\omega}$ over M with $b_0 = b$, the set $\{\varphi(x,b_i) : i < \omega\}$ is inconsistent.*

Part (a) does not require the sequence to be indiscernible: it asks only that every term have the type of b over M and be independent from the earlier terms. We use this generality in the proof of symmetry (see Theorem 7.3.5). The proof of (a) reads the sequence backwards, because the Morley condition points the wrong way. The Morley condition makes each term

independent from the *earlier* terms, while the tree construction of Lemma 5.3.5 needs each formula to divide over the parameters of the earlier formulas of the chain. Reversing a finite piece of the sequence makes the earlier terms of the chain the later terms of the sequence. Transitivity on the left (see Lemma 7.1.7) shows that these later terms, taken together, are independent from the term at hand; Lemma 7.2.1 then adds them to the base over which the formula divides. An ω-sequence cannot be reversed into an ω-sequence; for this reason the proof works with finite pieces, through part (1) of Lemma 5.3.5, and reaches the tree property through part (2) (see Figure 7.2).

Proof. (a) Suppose, for a contradiction, that $\{\varphi(x, b_i) : i < \omega\}$ is consistent. By Lemma 5.2.4, applied as in the parameter convention, there is $k \geq 2$ such that $\varphi(x, b)$ k-divides over M. Writing $\varphi(x, y) = \varphi_0(x, y, m)$ with $\varphi_0 \in L$ and m a finite tuple from M, we regard φ_0 as a formula $\varphi_0(x; y, z)$ with parameter variables y, z. We show that φ_0 has the k-tree property, which contradicts the simplicity of T. Recall that this property asks for instances of φ_0 indexed by the nodes of the tree $\omega^{<\omega}$, consistent along every branch and k-inconsistent among siblings (see Definition 5.2.7). By Lemma 5.3.5(2), it suffices to find, for every $n \geq 1$, tuples $a_0, \ldots, a_{n-1}$ as in Lemma 5.3.5(1) for the formula φ_0. Let $n \geq 1$ be given.

☺ **Step 1.** For $i < l < n$, we have $b_{i+1} \ldots b_l \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_i$. We argue by induction on l; for $l = i+1$, the hypothesis $b_{i+1} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<i+1}$ gives $b_{i+1} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_i$ by monotonicity. Suppose that the assertion holds for some l with $i < l < n-1$, and let A be the set $\{b_{i+1}, \ldots, b_l\}$. The hypothesis on b_{l+1} says that $\mathrm{tp}(b_{l+1}/Mb_{<l+1})$ does not divide over M. The restriction of this type to $M \cup A \cup b_i$ does not divide over M either. By Remark 5.2.5 applied to $M \subseteq M \cup A \subseteq M \cup A \cup b_i$, this restriction therefore does not divide over $M \cup A$; in other words, the relation $b_{l+1} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{MA} b_i$ holds. Together with $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_i$, Lemma 7.1.7 gives $Ab_{l+1} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_i$, which is the assertion for $l + 1$.

☺ **Step 2.** For $i < n$, the formula $\varphi(x, b_i)$ k-divides over $M \cup \{b_{i+1}, \ldots, b_{n-1}\}$. Since $b_i \equiv_M b$, there is $\sigma \in \mathrm{Aut}(\mathbb{M}/M)$ with $\sigma(b) = b_i$; it fixes the parameters of φ, and it carries a sequence witnessing that $\varphi(x, b)$ k-divides over M to one witnessing that $\varphi(x, b_i)$ k-divides over M. For $i = n - 1$ there is nothing more to show. For $i < n - 1$, Step 1 with $l = n - 1$ says that $\mathrm{tp}(b_{i+1} \ldots b_{n-1}/Mb_i)$ does not divide over M, and Lemma 7.2.1 with $e = b_{i+1} \ldots b_{n-1}$ gives the assertion.

☺ **Step 3.** We reverse the order. The reversed sequence consists of the tuples $a_j = (b_{n-1-j}, m)$ for $j < n$. Then the sets $\{\varphi_0(x, a_j) : j < n\}$ and $\{\varphi(x, b_i) : i < n\}$ coincide, and the latter is consistent. Let $j < n$ be given, and let $A_j = \{a_l : l < j\}$, the set of coordinates of m and of $b_{n-j}, \ldots, b_{n-1}$ (for $j = 0$, the empty set). By Step 2 with $i = n - 1 - j$, the formula $\varphi(x, b_{n-1-j})$ k-divides over $C_j = M \cup \{b_{n-j}, \ldots, b_{n-1}\}$; that is, the formula $\varphi_0(x, a_j)$ k-divides over C_j. Since $A_j \subseteq C_j$, Remark 5.2.5 shows that $\varphi_0(x, a_j)$ k-divides over A_j. Thus the tuples $a_0, \ldots, a_{n-1}$ satisfy the hypotheses of Lemma 5.3.5(1) for the formula φ_0, which completes the proof of (a).

(b) (1) $\Rightarrow$ (2). A Morley sequence over M with $b_0 = b$ is indiscernible over M, so every b_i has the same type over M as b, and satisfies $b_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<i}$ by definition; hence part (a) applies.

(2) $\Rightarrow$ (3). By Lemma 7.1.6 there is a Morley sequence over M beginning with b, and (2) applies to it.

(3) $\Rightarrow$ (1). A Morley sequence over M is indiscernible over M, so this is the definition of dividing, with the parameter convention. ☺

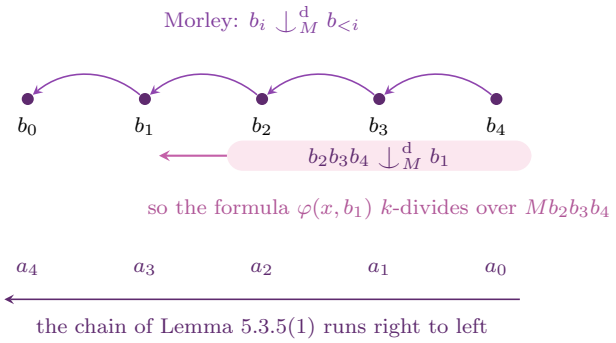


FIGURE 7.2. The proof of Kim's lemma, for a formula $\varphi(x, b)$ that k-divides over M and the first $n = 5$ terms $b_0, \ldots, b_4$ of the sequence.

7.2.3 Without simplicity

In a theory that is not simple, dividing may be visible along one Morley sequence and invisible along another. The theory of $(\mathbb{Q}, <)$ shows how. This theory is not simple, by Proposition 5.3.2: the formula $x < y$ has the strict order property of Remark 4.1.6, since its sets of solutions $\{a : a < b\}$ form strictly increasing chains of every finite length. Quantifier elimination for $(\mathbb{Q}, <)$, applied as in Example 5.2.6 but now with parameters, shows that the type of a tuple over a set D is determined by the order between the coordinates of the tuple and the elements of D. For an element c, write $M < c$ if c is greater than every element of M.

Example 7.2.3. Let T be the theory of $(\mathbb{Q}, <)$, let M be a model, and let $\varphi(x; y_1, y_2)$ be the formula $y_1 < x < y_2$. Let $b = (d_1, d_2)$ be a pair with $M < d_1 < d_2$.
The formula $\varphi(x, b)$ divides over M. Choose pairs $b^i = (b^i_1, b^i_2)$ with $b^0 = b$, with $M < b^i_1 < b^i_2$, and with $b^i_2 < b^{i+1}_1$ for every i. Any two increasing tuples of the same length from the sequence (b^i) have their coordinates ordered in the same way, and all these coordinates are greater than every element of M. By quantifier elimination, the sequence is indiscernible over M. The intervals (b^i_1, b^i_2) are pairwise disjoint, so the formulas $\varphi(x, b^i)$ are 2-inconsistent.
A Morley sequence along which the formula is consistent. Choose pairs $b_i = (b_{i,1}, b_{i,2})$ with $b_0 = b$, and with

$$M < b_{i+1,1} < b_{i,1} \quad \text{and} \quad b_{i,2} < b_{i+1,2}$$

for every i. The intervals $(b_{i,1}, b_{i,2})$ are thus nested, and they grow outward (see Figure 7.3). As before, the sequence (b_i) is indiscernible over M; and every element of the interval (d_1, d_2) satisfies all the formulas $\varphi(x, b_i)$, so these are consistent. It remains to be shown that $b_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<i}$ for $i \geq 1$ (for $i = 0$ there is nothing to prove). Fix $i \geq 1$, and write $c = b_{<i}$, a finite tuple; we verify condition (2) of Lemma 7.1.2 for the type $\mathrm{tp}(b_i/Mc)$.
Let $(c_j)_{j<\omega}$ be indiscernible over M with $c_0 = c$. Every coordinate of c is greater than every element of M, and each c_j has the type of c over M, so the same holds for every coordinate of every c_j. By saturation there is a pair $a' = (a'_1, a'_2)$ with $M < a'_1$, with a'_1 smaller than every coordinate of every c_j, and with a'_2 greater than all these coordinates. The coordinates of a' sit in the same places for every choice of indices $j_1 < \cdots < j_r$: the

first above M and below all the coordinates of $c_{j_1}, \dots, c_{j_r}$, the second above all of them. The order between the coordinates of $c_{j_1}, \dots, c_{j_r}$ and the elements of M does not depend on $j_1 < \cdots < j_r$ either, since (c_j) is indiscernible over M. By quantifier elimination, the sequence (c_j) is indiscernible over $M \cup a'$. Finally, the coordinates of b_i sit exactly where those of a' do: the first lies above M and below every coordinate of c, and the second lies above every coordinate of c. Quantifier elimination therefore gives $a' \equiv_{Mc} b_i$, and Lemma 7.1.2 yields $b_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<i}$; thus the sequence (b_i) is a Morley sequence over M.

The coheir sequence sees the dividing. Let $\mathcal{U}$ be an ultrafilter on M^2 with $q_{\mathcal{U}}|\emptyset = \mathrm{tp}(b/M)$, that is, such that the type of b over M is the coheir of $\mathcal{U}$ over M (see Lemma 6.1.5). With this ultrafilter, let $(b'_i)_{i<\omega}$ be the Morley sequence of Lemma 7.1.6, with $b'_0 = b$ and $b'_i = (b'_{i,1}, b'_{i,2})$. Given $j < i$, we have $M < b'_{j,1}$, since $b'_j \equiv_M b$. Every pair (m_1, m_2) from M therefore satisfies $m_2 < b'_{j,1}$, and the formula $y_2 < b'_{j,1}$ belongs to the coheir $q_{\mathcal{U}}|b'_{<i} = \mathrm{tp}(b'_i/Mb'_{<i})$. Hence the element $b'_{i,2}$ lies below $b'_{j,1}$: each interval lies entirely below the earlier ones, and the formulas $\varphi(x, b'_i)$ are 2-inconsistent, as Kim's lemma predicts in a simple theory.

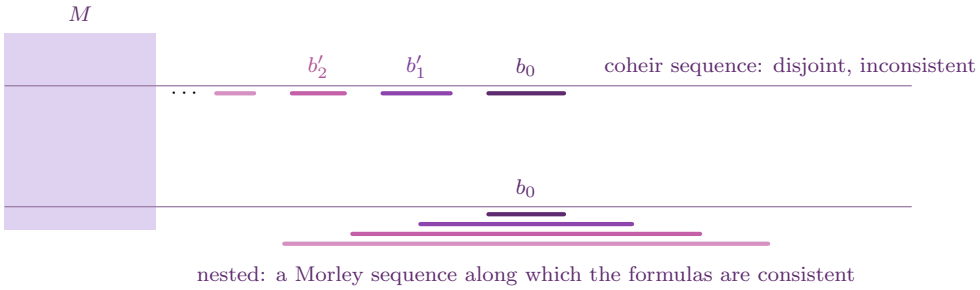


FIGURE 7.3. Example 7.2.3, in the theory of $(\mathbb{Q}, <)$: two Morley sequences of intervals over the shaded model M beginning with the same interval $b_0 = b$, one pairwise disjoint (top) and one nested (bottom).

Kim's lemma is the form in which simplicity enters the rest of the chapter. In the next section we use it to show that, over a model, a formula forks exactly when it divides; nondividing types then have nondividing extensions to every larger set. Extension and Kim's lemma together give symmetry, and symmetry gives transitivity.

7.3 Forking, symmetry and transitivity

Dividing is easy to test and hard to extend; forking is easy to extend and hard to test. A formula divides when one sequence of its conjugates is inconsistent, and such a sequence can be exhibited. A type that does not divide over a base set, however, need not have an extension to a larger set that still does not divide over the base. Indeed, a disjunction of formulas that divide need not divide (see Remark 7.3.2). Shelah's forking closes dividing under disjunctions and implications, and so makes extension automatic, with one more quantifier in the definition. In Theorem 7.3.3 below we show, with Kim's lemma, that in a simple theory the two notions coincide over a model, which gives us the advantages of both. With this

theorem and Kim's lemma, symmetry takes only a few lines (see Theorem 7.3.5); transitivity then comes from symmetry and transitivity on the left (see Theorem 7.3.6).

Throughout the section, the letters M and N denote models.

7.3.1 Forking

Definition 7.3.1. Let C be a small set, let $\varphi(x, y) \in L(C)$ be a formula, and let b be a tuple of the length of y.

(a) The formula $\varphi(x, b)$ *forks over* C if there are $r \geq 1$ and formulas $\psi_1(x, c_1), \ldots, \psi_r(x, c_r)$, with $\psi_l(x, z_l) \in L(C)$, each of which divides over C (in the sense of the parameter convention following Definition 7.1.1), such that

$$\mathbb{M} \models \forall x \Big(\varphi(x, b) \to \bigvee_{l=1}^{r} \psi_l(x, c_l) \Big).$$

(b) Let $B \supseteq C$ be small. A type p over B *forks over* C if $p \vdash \bigvee_{l=1}^{r} \psi_l(x, c_l)$ for some formulas $\psi_l(x, c_l)$ as in (a).

The definition agrees with the description of forking in Remark 6.3.2. A formula that divides over C forks over C (take $r = 1$ and $\psi_1 = \varphi$). The same holds for every formula that implies a formula forking over C, and for every finite disjunction of formulas forking over C. By compactness, a type that is closed under finite conjunctions, in particular a complete type, forks over C if and only if one of its formulas forks over C. Hence, as soon as forking and dividing agree for formulas, they agree for complete types as well.

Remark 7.3.2. Forking and dividing differ in general. The standard example is the theory of the dense circular order, with a ternary relation $R(x, y, z)$. This relation is interpreted as "going around the circle in the positive direction from x, one meets y before z" (see Tent and Ziegler [**TZ12**, Chapter 7]). Choose three distinct elements c_1, c_2, c_3 in this order around the circle. Every element is equal to some c_l or lies on one of the three open arcs between them. Consequently, the formula $x = x$ implies the disjunction of the six formulas $x = c_l$ and $R(c_l, x, c_{l+1})$ (indices modulo 3). Each of the six divides over $\emptyset$. Indeed, by quantifier elimination for this theory, all pairs of distinct points have the same type over $\emptyset$. For the same reason, a sequence of points $e_0, e_1, \ldots$ in positive order around the circle is indiscernible over $\emptyset$, and so is a sequence of pairs (e_i, f_i) with $e_0, f_0, e_1, f_1, \ldots$ in positive order. An automorphism carries e_0 to c_l, or (e_0, f_0) to (c_l, c_{l+1}). The formulas $x = e_i$ are 2-inconsistent, as are the formulas $R(e_i, x, f_i)$, whose arcs are pairwise disjoint. Hence the formula $x = x$ forks over $\emptyset$ (see Figure 7.4). The formula $x = x$ does not divide over $\emptyset$, since the formula has no parameters and is consistent. Over models of a simple theory, forking and dividing agree, as we now show.

7.3.2 Forking equals dividing

Theorem 7.3.3. *Assume that T is simple. Let b be a finite tuple, and let $\varphi(x, y) \in L(M)$ be a formula. Then the formula $\varphi(x, b)$ forks over M if and only if it divides over M. Consequently, for a small tuple a and a small set B, the type* $\mathrm{tp}(a/MB)$ *forks over M if and only if it divides over M.*

Proof. The idea is to realize $\varphi(x, b)$ along a Morley sequence and to let the pigeonhole principle choose a disjunct that the realization satisfies infinitely often. Kim's lemma then shows that this disjunct does not divide. A formula that divides over M forks over M, as observed after Definition 7.3.1. For the converse, suppose that $\varphi(x, b)$ does not divide over

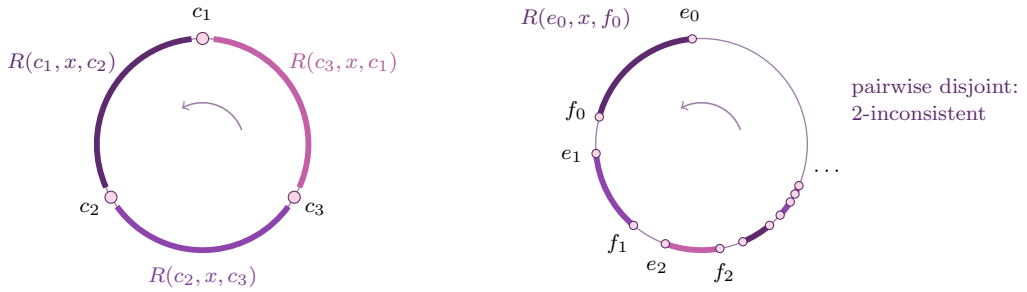


FIGURE 7.4. The dense circular order of Remark 7.3.2, with the positive direction counterclockwise: three points and three open arcs covering the circle (left), and the pairwise disjoint arcs $R(e_i, x, f_i)$ of an indiscernible sequence of pairs (right).

M, and let formulas $\psi_1(x, c_1), \dots, \psi_r(x, c_r)$ with $\psi_l(x, z_l) \in L(M)$ be given such that $\varphi(x, b)$ implies their disjunction. We show that some $\psi_l(x, c_l)$ does not divide over M.

Let $c = (b, c_1, \dots, c_r)$, and let $(c^i)_{i<\omega}$ be a Morley sequence over M with $c^0 = c$, which exists by Lemma 7.1.6. Each c^i splits as $(b^i, c^i_1, \dots, c^i_r)$, with the blocks of the lengths of $b, c_1, \dots, c_r$. The sequence $(b^i)_{i<\omega}$ is indiscernible over M, and monotonicity on both sides gives $b^i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b^{<i}$ for every i; hence this sequence is a Morley sequence over M with $b^0 = b$. The formula $\varphi(x, b)$ does not divide over M, so the set $\{\varphi(x, b^i) : i < \omega\}$ is consistent; let a realize it.

The $L(M)$-formula $\forall x(\varphi(x, y) \to \bigvee_l \psi_l(x, z_l))$ is satisfied by c, and also by every c^i, since $c^i \equiv_M c$; therefore the formula $\varphi(x, b^i)$ implies $\bigvee_l \psi_l(x, c^i_l)$ for every i. Hence for every i there is l with $\mathbb{M} \models \psi_l(a, c^i_l)$, and by the pigeonhole principle there are a single l and an infinite set $I \subseteq \omega$ such that $\mathbb{M} \models \psi_l(a, c^i_l)$ for every $i \in I$. We enumerate I as $i_0 < i_1 < \cdots$ and write d_k for $c^{i_k}_l$. As before, monotonicity shows that the sequence $(d_k)_{k<\omega}$ is a Morley sequence over M, and the set $\{\psi_l(x, d_k) : k < \omega\}$ is realized by a. By Kim's lemma (see Theorem 7.2.2(b)), the formula $\psi_l(x, d_0)$ does not divide over M. Since $d_0 \equiv_M c_l$, an automorphism of $\mathbb{M}$ over M carries d_0 to c_l, and so the formula $\psi_l(x, c_l)$ does not divide over M either.

Finally, the statement about types follows from the observations after Definition 7.3.1. ☺

Forking was designed to make extension possible, and extension is its first use.

Corollary 7.3.4 (Extension). *Assume that T is simple. Let a be a small tuple, and let $B \subseteq B'$ be small sets such that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$. Then there is $a' \equiv_{MB} a$ with $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B'$.*

Proof. As in the proof of Proposition 7.1.4, we realize $\mathrm{tp}(a/MB)$ while avoiding a set of formulas, here those over MB' that divide over M; the consistency of this requirement rests on Theorem 7.3.3. Let $\Sigma(x)$ be the union of $\mathrm{tp}(a/MB)$ and the set of formulas $\neg\psi(x, c)$ such that $\psi(x, z) \in L(M)$, the tuple c is from B', and the formula $\psi(x, c)$ divides over M. Suppose, for a contradiction, that Σ is inconsistent. Since $\mathrm{tp}(a/MB)$ is closed under conjunctions, compactness gives a formula $\varphi(x, b)$ of $\mathrm{tp}(a/MB)$ that implies a finite disjunction of formulas dividing over M. This formula forks over M, and by Theorem 7.3.3 it divides over M, contradicting $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$. Hence the set Σ is consistent, and by saturation it is realized by some tuple a'. This tuple satisfies $a' \equiv_{MB} a$, and no formula of $\mathrm{tp}(a'/MB')$ divides over M, since the negation of every formula over MB' that divides over M belongs to Σ; therefore the relation $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B'$ holds. ☺

7.3.3 Symmetry

Symmetry says that if one tuple tells us nothing new about another, then the second tells us nothing new about the first either. Arguing by contradiction, the proof uses extension to turn an asymmetric pair into a sequence of copies of the second tuple. Each of these copies is independent from the earlier ones, and all of them stand in the same relation to the first tuple. Kim's lemma forbids such a sequence (see Figure 7.5).

Theorem 7.3.5 (Symmetry). *Assume that T is simple, and let A and B be small sets. If $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$, then also $B \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} A$.*

Proof. By finite character, we may assume that A and B are finite tuples a and b. Suppose, for a contradiction, that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b$ and that $\mathrm{tp}(b/Ma)$ divides over M. Let $\psi(y, a)$ be a formula of $\mathrm{tp}(b/Ma)$, with $\psi(y, z) \in L(M)$, that divides over M.

We choose, by recursion on $i < \omega$, tuples a_i such that $a_i \equiv_{Mb} a$ and $a_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} ba_{<i}$. For $i = 0$ we take $a_0 = a$, which satisfies both conditions because $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b$. Given $a_{<i}$ with $i \geq 1$, Corollary 7.3.4, applied to a with $\{b\}$ and $\{b\} \cup a_{<i}$ in place of B and B', provides $a_i \equiv_{Mb} a$ with $a_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} ba_{<i}$.

Every a_i has the type of a over M and satisfies $a_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} a_{<i}$ by monotonicity. Since $\psi(y, a)$ divides over M, Kim's lemma (see Theorem 7.2.2(a)) says that $\{\psi(y, a_i) : i < \omega\}$ is inconsistent. On the other hand, the tuple b satisfies $\psi(y, a)$; as $a_i \equiv_{Mb} a$, this tuple satisfies $\psi(y, a_i)$ for every i and realizes the set. This is a contradiction. ☺

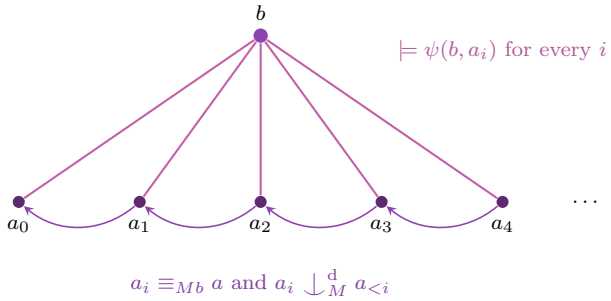


FIGURE 7.5. The proof of symmetry (see Theorem 7.3.5): copies a_i of a over $M \cup b$, each independent from the earlier ones, with b realizing every $\psi(y, a_i)$.

7.3.4 Transitivity, and the axioms that hold in simple theories

With symmetry in hand, transitivity reduces to transitivity on the left (see Lemma 7.1.7): one turns the relations around, applies that lemma, and turns the result around again. The theorem below collects the axioms that nondividing independence satisfies in a simple theory.

Theorem 7.3.6. *Assume that T is simple. Then the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$, with models as bases, satisfies* (INV), (MON), (BMON), (TR), (FC), (SYM), (EX) *and* (LOC).

Proof. All but transitivity and symmetry are in Proposition 7.1.8, and symmetry is Theorem 7.3.5. *Transitivity*: let $M \subseteq N$ be models, and suppose that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} N$ and $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} B$. By symmetry, we have $N \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} A$ and $B \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} A$; since $M \cup N = N$, the second relation says that $B \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{MN} A$. We apply transitivity on the left (see Lemma 7.1.7) with M and N in the

roles of C and A, and with A and B in those of B and D. The lemma yields $NB \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} A$, and a last application of symmetry gives $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} NB$. ☺

Over models, nondividing independence is nonforking independence (see Theorem 7.3.3), and we keep the notation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ for it. Of the properties of Definition 5.4.7, uniqueness is the only one not on the list, and in an unstable simple theory uniqueness does not hold. Indeed, uniqueness together with local character implies stability (see Theorem 6.2.2), while the random graph is simple and unstable (see Theorem 5.4.6, Example 4.1.4, and Proposition 4.1.2). In place of uniqueness, simple theories have the independence theorem, which we prove in the next section.

7.4 The independence theorem

In an unstable simple theory a type over a model may have several independent extensions to a larger set (see Theorem 7.5.2 below), and nothing singles one of them out. In place of uniqueness stands a weaker property which, as we show in Theorem 7.6.5, is the characteristic property of simplicity. Suppose that a type over a model has one independent extension to the model enlarged by a tuple, and another to the model enlarged by a second tuple. Provided only that the two tuples are themselves independent over the model, the two extensions can always be realized together, independently. This statement is the independence theorem of Kim and Pillay [**KP97**]; in the random graph it is the amalgamation of Theorem 5.4.9(c). In a stable theory it is a consequence of uniqueness; in a simple theory it has to be earned.

Throughout the section, the theory T is simple and the letter M denotes a model.

The proof of the independence theorem moves a parameter step by step. Suppose that φ and ψ are $L(M)$-formulas, and that a, b and b' are finite tuples with $b' \equiv_M b$ and $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} bb'$. Assuming that the formula $\varphi(x,a) \wedge \psi(x,b')$ does not divide over M, we wish to conclude the same of $\varphi(x,a) \wedge \psi(x,b)$. If b' and b begin an indiscernible sequence, the parameter can be moved from b' to b (see Proposition 7.4.5). In general, a third tuple, a coheir over everything in sight, serves as a stepping stone between b' and b. The step itself rests on a fact about conjunctions along indiscernible sequences (see Proposition 7.4.4), whose proof brings in a second idea. An arbitrary indiscernible sequence over M can be replaced by a Morley sequence over a larger model, and Kim's lemma applies to Morley sequences. The Morley sequence is built as a coheir sequence from a reversed copy of the original indiscernible sequence.

7.4.1 Two tools

The first tool extends the corollary on extension (see Corollary 7.3.4) from complete types to partial ones. A set of formulas is *closed under conjunctions* if it contains the conjunction of any two of its members. For such a set, the hypothesis of the next lemma says simply that no member of the set divides over M. It is in this form that we apply the lemma in the proof of Theorem 7.4.7.

Lemma 7.4.1. *Let $B \subseteq B'$ be small sets, and let $\pi(x)$ be a set of $L(MB)$-formulas such that no finite conjunction of members of π divides over M. Then some a realizes π with $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B'$.*

Proof. As in the proof of Corollary 7.3.4, let $\Sigma(x)$ be the union of π and the set of formulas $\neg\chi(x,c)$ with $\chi(x,z) \in L(M)$, the tuple c from B', and $\chi(x,c)$ dividing over M. If Σ were

inconsistent, some finite conjunction of members of π would imply a finite disjunction of formulas dividing over M (or be inconsistent, and then divide over M). The conjunction would then fork over M and, by Theorem 7.3.3, divide over M, contrary to the hypothesis on π. Hence the set Σ is consistent, and any realization a of Σ satisfies $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B'$. ☺

We also need a consequence of Theorem 7.3.3 that does not depend on the lemma. If a formula does not divide over M, then neither does any formula that it implies: forking is closed under implication (see the remarks after Definition 7.3.1), and over models forking equals dividing. The same holds over any model in place of M.

The second tool turns indiscernible sequences around and attaches companions to their terms. We say that tuples d_0, d_1 *begin an indiscernible sequence over* M if there are d_i for $2 \leq i < \omega$ such that $(d_i)_{i<\omega}$ is indiscernible over M.

Lemma 7.4.2. *If d_0, d_1 begin an indiscernible sequence over M, then so do d_1, d_0.*

Proof. We stretch the sequence to one indexed by $\mathbb{Z}$ and read it backwards. Let $(d_i)_{i<\omega}$ be a sequence that begins with d_0, d_1 and is indiscernible over M. By compactness and saturation there is a sequence $(d'_i)_{i\in\mathbb{Z}}$ such that $(d'_{i_1}, \ldots, d'_{i_n}) \equiv_M (d_0, \ldots, d_{n-1})$ for every n and all $i_1 < \cdots < i_n$ in $\mathbb{Z}$. Indeed, a finite part of the required type mentions finitely many indices, and an increasing map from those indices into ω realizes that finite part. We reverse the sequence, setting $e_i = d'_{1-i}$ for $i < \omega$. An increasing tuple of (e_i) is a decreasing tuple of (d'_i), and every decreasing n-tuple of (d'_i) has the type of $(d_{n-1}, \ldots, d_0)$ over M. Hence the sequence $(e_i)_{i<\omega}$ is indiscernible over M, and its first two terms satisfy $(e_0, e_1) \equiv_M (d_1, d_0)$. An automorphism of $\mathbb{M}$ over M carrying (e_0, e_1) to (d_1, d_0) carries (e_i) to the required sequence. ☺

Lemma 7.4.3. *Suppose that finite tuples b_0, b_1 begin an indiscernible sequence over M, and let a be a tuple with $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{Mb_0} b_1$. Then there is a^* such that the pairs (b_0, a) and (b_1, a^*) begin an indiscernible sequence over M.*

Proof. We first arrange that the sequence, from its second term on, is indiscernible over $M \cup b_0 \cup a$. Shifts of the sequence then attach a companion to every term, and extraction makes the pairs indiscernible. Let $(b_i)_{i<\omega}$ be a sequence that begins with b_0, b_1 and is indiscernible over M. The sequence $(b_i)_{1\leq i<\omega}$ is indiscernible over $C = M \cup b_0$ and begins with b_1, and $\mathrm{tp}(a/Cb_1)$ does not divide over C. By Lemma 7.1.2 there is $a' \equiv_{Cb_1} a$ such that $(b_i)_{i\geq 1}$ is indiscernible over $C \cup a'$. Applying an automorphism of $\mathbb{M}$ over $C \cup b_1$ that carries a' to a, and replacing b_i by its image for $i \geq 2$, we may assume that $(b_i)_{i\geq 1}$ is indiscernible over $M \cup b_0 \cup a$. Under this replacement the whole sequence remains indiscernible over M.

For $n < \omega$ the sequences $(b_i)_{i<\omega}$ and $(b_{n+i})_{i<\omega}$ have the same type over M, so there is $\sigma_n \in \mathrm{Aut}(\mathbb{M}/M)$ with $\sigma_n(b_i) = b_{n+i}$ for every i. Each σ_n carries a to a tuple $a_n = \sigma_n(a)$. For $n < m$ we have

$$(b_n, a_n, b_m) = \sigma_n(b_0, a, b_{m-n}) \equiv_M (b_0, a, b_{m-n}) \equiv_M (b_0, a, b_1),$$

the last step because $(b_i)_{i\geq 1}$ is indiscernible over $M \cup b_0 \cup a$. We now apply Lemma 5.2.2 over M to the sequence of pairs $((b_n, a_n))_{n<\omega}$. We regard $\mathrm{tp}(b_0, a, b_1/M)$ as a type of the first coordinate of one pair, the second coordinate of that pair, and the first coordinate of a later pair. Under this identification, the type $\mathrm{tp}(b_0, a, b_1/M)$ is contained in the Ehrenfeucht–Mostowski type of the sequence of pairs over M. The extraction lemma gives an indiscernible sequence $((b^*_n, a^*_n))_{n<\omega}$ over M with $(b^*_0, a^*_0, b^*_1) \equiv_M (b_0, a, b_1)$. An automorphism of $\mathbb{M}$ over M that carries (b^*_0, a^*_0, b^*_1) to (b_0, a, b_1) carries a^*_1 to the required a^*. ☺

7.4.2 Conjunctions along indiscernible sequences

Suppose that a formula does not divide over M and that an indiscernible sequence over M begins with the parameter of the formula. The instances of the formula along the sequence are then consistent, by the definition of dividing. The next proposition says more: finitely many of these instances together still do not divide. For a Morley sequence this stronger assertion is an application of Kim's lemma to blocks of the sequence (see Step 3 of the proof below). An arbitrary indiscernible sequence is first replaced by a Morley sequence over a larger model with the same type over M (see Steps 1 and 2). At the end, transitivity brings us back to M (see Step 4).

Proposition 7.4.4. *Let $\psi(x,y) \in L(M)$ be a formula, let b be a finite tuple such that $\psi(x,b)$ does not divide over M, and let $(b_i)_{i<\omega}$ be indiscernible over M with $b_0 = b$. Then for every $n \geq 1$ the formula $\bigwedge_{i<n} \psi(x,b_i)$ does not divide over M.*

Proof. ☺ **Step 1: a reversed copy.** By compactness and saturation, as in the proof of Lemma 7.4.2, there is a sequence $(h_j)_{j<\omega}$ such that $(h_{j_0},\dots,h_{j_{n-1}}) \equiv_M (b_0,\dots,b_{n-1})$ for every n and all $j_0 > j_1 > \cdots > j_{n-1}$. In other words, the decreasing tuples of (h_j) have the types of the increasing tuples of (b_i). Let N be a small model containing $M \cup \{h_j : j < \omega\}$.

☺ **Step 2: a Morley sequence over N.** Let $\mathcal{D}$ be a nonprincipal ultrafilter on ω, and let $\mathcal{V}$ be the ultrafilter on $N^{|y|}$ consisting of the sets X with $\{j : h_j \in X\} \in \mathcal{D}$. Let t_0 realize $q_{\mathcal{V}}|\emptyset$, the type over N determined by $\mathcal{V}$, and for $k \geq 1$ let t_k realize $q_{\mathcal{V}}|\{t_l : l < k\}$. By Lemma 7.1.6, applied over the model N, the sequence $(t_k)_{k<\omega}$ is a Morley sequence over N.

We claim that $(t_0,\dots,t_{n-1}) \equiv_M (b_0,\dots,b_{n-1})$ for every n. By induction on n, the definition of the coheir gives, for every $\theta(y_0,\dots,y_{n-1}) \in L(N)$,

$$\mathbb{M} \models \theta(t_0,\dots,t_{n-1}) \quad \text{if and only if} \quad \Big\{ j_{n-1} : \Big\{ j_{n-2} : \cdots \{ j_0 : \mathbb{M} \models \theta(h_{j_0},\dots,h_{j_{n-1}}) \} \in \mathcal{D} \cdots \Big\} \in \mathcal{D} \Big\} \in \mathcal{D}.$$

Indeed, the tuple t_{n-1} satisfies $\theta(t_0,\dots,t_{n-2},y)$ exactly when the set of j with $\mathbb{M} \models \theta(t_0,\dots,t_{n-2},h_j)$ belongs to $\mathcal{D}$, and for each j the formula $\theta(y_0,\dots,y_{n-2},h_j)$ has parameters in N. Let $\theta \in L(M)$ be given, and suppose that $\mathbb{M} \models \theta(b_0,\dots,b_{n-1})$; then the formula θ holds at every tuple $(h_{j_0},\dots,h_{j_{n-1}})$ with $j_0 > j_1 > \cdots > j_{n-1}$. The set $\{j : j > j'\}$ belongs to $\mathcal{D}$ for every j', because $\mathcal{D}$ is nonprincipal. Hence, by induction from the innermost set outward, the set of j_r at the level of nesting r contains $\{j_r : j_r > j_{r+1}\}$ for all $j_{r+1} > \cdots > j_{n-1}$, and therefore belongs to $\mathcal{D}$. At the outermost level we obtain $\mathbb{M} \models \theta(t_0,\dots,t_{n-1})$. The same argument, applied to $\neg\theta$, gives the converse and thereby proves the claim. In particular, the tuple t_0 has the type of b over M, and therefore the formula $\psi(x,t_0)$ does not divide over M.

☺ **Step 3: over N.** From now on, fix $n \geq 1$. By Lemma 7.4.1 there is a_0 with $\mathbb{M} \models \psi(a_0,t_0)$ and $a_0 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} N \cup \{t_0\}$. Let $p(x,y)$ be the set of $L(N)$-formulas $\theta(x,y)$ with $\mathbb{M} \models \theta(a_0,t_0)$; thus $p(x,t_0) = \mathrm{tp}(a_0/Nt_0)$. For $k < \omega$, we write $p(x,t_k)$ for the set of formulas $\theta(x,t_k)$ with $\theta \in p$. By Remark 5.2.5, the type $p(x,t_0)$ does not divide over N. Let $\theta \in p$ be given; since the formula $\theta(x,t_0)$ does not divide over N and the sequence (t_k) is indiscernible over N with first term t_0, the set $\{\theta(x,t_k) : k < \omega\}$ is consistent. The blocks $e_m = (t_{mn},\dots,t_{mn+n-1})$, for $m < \omega$, form an indiscernible sequence over N, and we claim that $e_m \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} e_{<m}$ for every m. We add the terms of the block one at a time, starting from $t_{mn} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} t_{<mn}$. Suppose that $(t_{mn},\dots,t_{mn+r}) \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} t_{<mn}$ for some $r < n-1$. Since $t_{mn+r+1} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} t_{<mn+r+1}$ (the Morley condition), monotonicity and Remark 5.2.5 give $t_{mn+r+1} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N\cup\{t_{mn},\dots,t_{mn+r}\}} t_{<mn}$,

and Lemma 7.1.7 yields $(t_{mn}, \dots, t_{mn+r+1}) \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} t_{<mn}$. Once all n terms of the block have been added, this is the claim, and the sequence (e_m) is therefore a Morley sequence over N. The formulas $\bigwedge_{k<n} \theta(x, t_{mn+k})$, for $m < \omega$, are consistent together, since all of them belong to the consistent set $\{\theta(x, t_k) : k < \omega\}$. Hence Kim's lemma, in the form of Theorem 7.2.2(b) applied over N, shows that $\bigwedge_{k<n} \theta(x, t_k)$ does not divide over N.

Each finite conjunction of members of $\bigcup_{k<n} p(x, t_k)$ is implied by a formula $\bigwedge_{k<n} \theta(x, t_k)$ with $\theta \in p$, because p is closed under conjunctions. Therefore, since a formula implied by a formula that does not divide over N does not divide over N either (see the remarks after Lemma 7.4.1), no such conjunction divides over N. By Lemma 7.4.1, applied over N, some a realizes this union with $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{N} \{t_k : k < n\}$.

☺ **Step 4: back to M.** The type $p(x, t_0)$ contains $\mathrm{tp}(a_0/N)$, which a therefore realizes. Since $a_0 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} N$ by monotonicity, and an automorphism of $\mathbb{M}$ over N carries a_0 to a, we obtain $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} N$, and transitivity (see Theorem 7.3.6) gives $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} N \cup \{t_k : k < n\}$. The tuple a satisfies $\psi(x, t_k)$ for every $k < n$, because $\psi(x, y) \in p$. Hence the formula $\bigwedge_{k<n} \psi(x, t_k)$ belongs to the type $\mathrm{tp}(a/M \cup \{t_k : k < n\})$, which does not divide over M. By the claim of Step 2, the tuples $(t_0, \dots, t_{n-1})$ and $(b_0, \dots, b_{n-1})$ have the same type over M, and therefore the formula $\bigwedge_{k<n} \psi(x, b_k)$ does not divide over M either. ☺

7.4.3 Moving a parameter

The tools are now in place for the step announced at the start of the section: the passage of a parameter from the first term of an indiscernible sequence to the second. This passage is effected by giving the second term a companion (see Lemma 7.4.3) and applying Proposition 7.4.4 to the resulting sequence of pairs.

Proposition 7.4.5. *Let $\varphi(x, z), \psi(x, y) \in L(M)$ be formulas, and let a, b_0, b_1 be finite tuples such that b_0, b_1 begin an indiscernible sequence over M and $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{Mb_0} b_1$. If $\varphi(x, a) \wedge \psi(x, b_0)$ does not divide over M, then the formula $\varphi(x, a) \wedge \psi(x, b_1)$ does not divide over M.*

Proof. By Lemma 7.4.3 there is a^* such that (b_0, a) and (b_1, a^*) begin an indiscernible sequence over M. We apply Proposition 7.4.4, with $n = 2$, to the formula $\chi(x; y, z) = \psi(x, y) \wedge \varphi(x, z)$ and this sequence of pairs. The hypothesis of Proposition 7.4.4 holds, since $\chi(x; b_0, a)$ is the formula $\varphi(x, a) \wedge \psi(x, b_0)$, and the conclusion is that the formula $\chi(x; b_0, a) \wedge \chi(x; b_1, a^*)$ does not divide over M. This conjunction implies $\varphi(x, a) \wedge \psi(x, b_1)$, which therefore does not divide over M. ☺

Two tuples with the same type over a model need not begin an indiscernible sequence; a coheir over both of them joins them (see Figure 7.6).

Proposition 7.4.6. *Let $\varphi(x, z), \psi(x, y) \in L(M)$ be formulas, and let a, b, b' be finite tuples with $b \equiv_M b'$ and $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} bb'$. If $\varphi(x, a) \wedge \psi(x, b')$ does not divide over M, then the formula $\varphi(x, a) \wedge \psi(x, b)$ does not divide over M.*

Proof. We pass from b' to b through a coheir e, in two steps, each of which is an application of Proposition 7.4.5. Let $\mathcal{U}$ be an ultrafilter on $M^{|b|}$ with $q_{\mathcal{U}}|\emptyset = \mathrm{tp}(b/M) = \mathrm{tp}(b'/M)$ (see Lemma 6.1.5(b)), and let e realize $q_{\mathcal{U}}|\{a, b, b'\}$. Then the tuple e realizes $q_{\mathcal{U}}|\{b'\}$ and $q_{\mathcal{U}}|\{b\}$, so by Lemma 7.1.6 each of the pairs b', e and b, e begins an indiscernible sequence over M, and by Lemma 7.4.2 so does the pair e, b. The type $\mathrm{tp}(e/Mabb')$ is finitely satisfiable in M, and Lemma 7.1.3 gives $e \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} abb'$. By symmetry we have $bb' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} a$, and monotonicity with Remark 5.2.5 gives $e \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{Mbb'} a$. Transitivity on the left (see Lemma 7.1.7) and symmetry

then yield $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} bb'e$. Monotonicity and Remark 5.2.5 now provide $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{Mb'} e$ and $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{Me} b$, the two relations that Proposition 7.4.5 requires.

Applied to the pair b', e, Proposition 7.4.5 shows that $\varphi(x,a) \wedge \psi(x,e)$ does not divide over M; applied to the pair e, b, it shows that $\varphi(x,a) \wedge \psi(x,b)$ does not divide over M. ☺

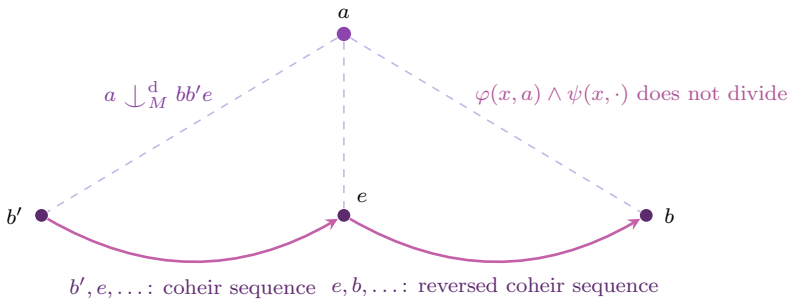


FIGURE 7.6. Proposition 7.4.6: a coheir e over $M \cup \{a, b, b'\}$ joining b' to b in two steps, each the start of an indiscernible sequence.

7.4.4 Amalgamation of independent extensions

The weight of the independence theorem rests on Proposition 7.4.6. What remains, for given formulas over the two independent sets of the theorem below, is to find a copy of the parameter of the second formula. The copy must have two properties, both of which Proposition 7.4.6 requires. First, the conjunction of the two formulas, with the copy in place of the original, must be known not to divide. Second, the parameter of the first formula must be independent over M from the original and the copy together.

Theorem 7.4.7 (Independence theorem). *Assume that T is simple. Let M be a model, let A and B be small sets with $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$, and let c_1, c_2 be tuples with $c_1 \equiv_M c_2$, $c_1 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} A$ and $c_2 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$. Then there is c with $c \equiv_{MA} c_1$, $c \equiv_{MB} c_2$ and $c \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} AB$.*

Proof. Let $\Sigma(x)$ be the closure under conjunctions of the union of the types $\mathrm{tp}(c_1/MA)$ and $\mathrm{tp}(c_2/MB)$. By Lemma 7.4.1 it suffices to show that no formula of Σ divides over M. Such a formula is implied by a formula $\varphi(x,a) \wedge \psi(x,b)$ with $\varphi(x,a) \in \mathrm{tp}(c_1/Ma)$, $\psi(x,b) \in \mathrm{tp}(c_2/Mb)$, $\varphi, \psi \in L(M)$, and a and b finite tuples from A and B. Fix such formulas and tuples, and recall from the remarks after Lemma 7.4.1 that a formula implied by a formula that does not divide over M does not divide over M either. It therefore suffices to show that $\varphi(x,a) \wedge \psi(x,b)$ does not divide over M. By monotonicity, the hypotheses give $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b$, $c_1 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} a$ and $c_2 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b$, and we look for a copy b' of b to which Proposition 7.4.6 applies.

Let $\sigma \in \mathrm{Aut}(\mathbb{M}/M)$ carry c_2 to c_1, and let $b_0 = \sigma(b)$. Then the pairs $c_1 b_0$ and $c_2 b$ have the same type over M; moreover, $b \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} c_2$ by symmetry, and invariance gives $b_0 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} c_1$. By Corollary 7.3.4 there is $b' \equiv_{Mc_1} b_0$ with $b' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} c_1 ab$. Thus the pairs $c_1 b'$ and $c_2 b$ have the same type over M, and in particular $b' \equiv_M b$.

We have $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} c_1$ by symmetry and $b' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{Ma} c_1$ by monotonicity and Remark 5.2.5, so Lemma 7.1.7 and symmetry give $c_1 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} ab'$. Hence the formula $\varphi(x,a) \wedge \psi(x,b')$, which belongs to $\mathrm{tp}(c_1/Mab')$ because $c_1 b' \equiv_M c_2 b$, does not divide over M. The relation $b \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} a$

follows from symmetry, and the relation $b' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{Mb} a$ from monotonicity together with Remark 5.2.5. As before, Lemma 7.1.7 turns these two relations into $bb' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} a$, which symmetry turns into $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} bb'$. By Proposition 7.4.6, the formula $\varphi(x,a) \wedge \psi(x,b)$ does not divide over M, and the proof is complete. ☺

In a simple theory, then, nondividing independence over models has every property of Definition 5.4.7 except possibly uniqueness (see Theorem 7.3.6), and satisfies the independence theorem as well. An analogue of the theorem was proved by hand in Chapter 5 for trivial independence, under which two sets are independent over a third when they meet only inside it (see Definition 5.4.8). The analogue holds over arbitrary small sets, in the Rado graph (see Theorem 5.4.9(c)) and, more generally, in Fraïssé limits in a finite relational language with disjoint 2- and 3-amalgamation (see Theorem 5.5.9). In Section 7.5 we show that uniqueness, added to the properties of nondividing independence, singles out exactly the stable theories. In Section 7.6 we prove that these properties, suitably strengthened, characterize the simple theories.

7.5 Stability through forking

Over a model, a simple theory has a notion of independence with every property of Definition 5.4.7 except possibly uniqueness. In Theorem 7.5.2 below we show that uniqueness adds exactly stability. One direction comes from Chapter 6, where we showed that uniqueness together with local character already forces stability. The other direction is a short argument with the independence theorem (see Lemma 7.5.1). If a type over a model had two different nondividing extensions, the independence theorem would let us realize them in any prescribed pattern along an independent sequence. A formula that distinguishes the two extensions would then shatter the sequence. In a simple NIP theory, therefore, nondividing extensions of types over models are unique.

Throughout the section, M is a model.

Lemma 7.5.1. *Assume that T is simple. Let a, a' be tuples with $a \equiv_M a'$, and let B be a small set with $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$ and $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} B$. If $a \not\equiv_{MB} a'$, then some formula has the independence property.*

Proof. Since $a \not\equiv_{MB} a'$, there are $\varphi(x,y) \in L(M)$, with x a finite tuple of variables, and a finite tuple b from B with $\mathbb{M} \models \varphi(a,b) \wedge \neg\varphi(a',b)$. By monotonicity, the tuples a and a' satisfy $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b$ and $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b$. Let $(b_i)_{i<\omega}$ be a Morley sequence over M with $b_0 = b$ (see Lemma 7.1.6); then every b_i satisfies $b_i \equiv_M b$ and $b_i \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<i}$.

Let $n \geq 1$ and $S \subseteq \{0, \ldots, n-1\}$ be given. We show by induction on $k \leq n$ that there is c_k with $c_k \equiv_M a$ and $c_k \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<k}$ such that for every $i < k$ we have $\mathbb{M} \models \varphi(c_k, b_i)$ if and only if $i \in S$. The case $k = 0$ is witnessed by $c_0 = a$. Suppose that c_k has been found, with $k < n$. Let $\sigma \in \mathrm{Aut}(\mathbb{M}/M)$ carry b to b_k, and let d be $\sigma(a)$ if $k \in S$ and $\sigma(a')$ if $k \notin S$. This element satisfies $d \equiv_M a \equiv_M c_k$, and invariance gives $d \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_k$; moreover, the formula $\varphi(d, b_k)$ holds if and only if $k \in S$. Since $b_{<k} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_k$ by symmetry, the independence theorem (see Theorem 7.4.7) applies to the sets $b_{<k}$ and b_k and to the tuples c_k and d. The theorem yields c_{k+1} with $c_{k+1} \equiv_{Mb_{<k}} c_k$, $c_{k+1} \equiv_{Mb_k} d$ and $c_{k+1} \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_{M} b_{<k+1}$, as required.

Thus the tuple c_n satisfies $\varphi(x, b_i)$ exactly for $i \in S$ (see Figure 7.7). The formula $\varphi(x,y)$ is $\varphi_0(x,y,m)$ for some $\varphi_0(x,y,z) \in L$ and some tuple m from M of the length of z. Since n

and S were arbitrary, for every n the tuples $(b_0, m), \ldots, (b_{n-1}, m)$ are shattered by $\varphi_0(x; y, z)$ (see Definition 4.1.1), and the formula φ_0 has the independence property. ☺

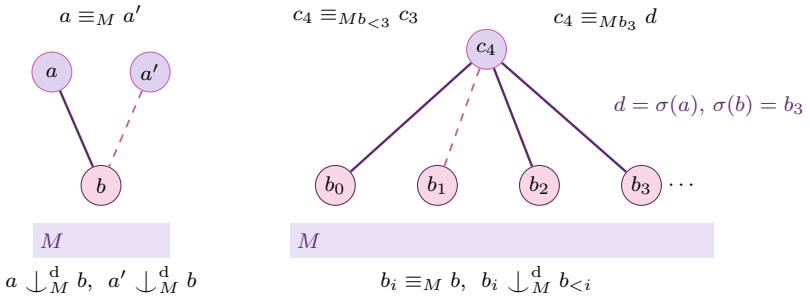


FIGURE 7.7. The construction in the proof of Lemma 7.5.1, for $n = 4$ terms of the Morley sequence and the pattern $S = \{0, 2, 3\}$, with $\varphi(x, y) \in L(M)$ the formula of the proof, solid lines where φ holds and dashed lines where it does not.

Theorem 7.5.2 (Stability through forking). *For a complete theory T, the following are equivalent:*

(1) *The theory T is stable.*

(2) *The theory T is simple, and the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$, with models as bases, satisfies* (UN).

(3) *The relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$, with models as bases, satisfies* (UN) *and* (LOC).

Moreover, if T is stable, then the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ coincides, over models, with the heir independence $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}$ of Definition 6.2.1.

Proof. (1) ⇒ (2). By Proposition 5.3.1, the theory T is simple. If (UN) failed for $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$, Lemma 7.5.1 would give a formula with the independence property, and by Proposition 4.1.2 that formula would be unstable, contrary to (1).

(2) ⇒ (3). In a simple theory, $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ satisfies (LOC) (see Proposition 7.1.8).

(3) ⇒ (1). This is the implication (3) ⇒ (1) of Theorem 6.2.2.

For the last statement, assume that T is stable, and let A and B be small sets, with $p = \mathrm{tp}(A/M)$. We first show that heirs do not divide. Let $\varphi(x, b)$ belong to the heir $p|B$, with $\varphi(x, y) \in L(M)$ and x a finite subtuple of the variables of A, and let $(b_i)_{i<\omega}$ be indiscernible over M with $b_0 = b$. The formula $d_p\varphi$ holds of b and has parameters in M, so it holds of every b_i; hence every $\varphi(x, b_i)$ belongs to the heir $p|\{b_i : i < \omega\}$, which is consistent by Lemma 6.1.3. Thus $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_M B$ implies $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$. Conversely, suppose that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$, and let A' realize $p|B$. This realization satisfies $A' \equiv_M A$ and $A' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_M B$, and hence $A' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$ by the first part. Since (UN) holds for $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ by (2), we obtain $A' \equiv_{MB} A$, and therefore $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{h}}_M B$. ☺

The theorem also gives a second proof of the harder half of Theorem 5.3.4, one that does not pass through Shelah's dichotomy (see Fact 5.3.3). If T is simple and every formula is NIP, then Lemma 7.5.1 shows that $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ satisfies (UN), and the theory T is stable by the implication (2) ⇒ (1). The two proofs divide the work differently. Shelah's dichotomy finds, in an unstable theory, either an independent configuration or a strict order. In the present

proof, simplicity excludes the strict order at the outset (see Proposition 5.3.2). The independent configuration is built by hand, with the independence theorem, from two incompatible extensions of one type. In the random graph the construction is directly visible: two new vertices with the same type over M, one joined to a vertex $b \notin M$ and one not, are both independent from b. The independence theorem, applied along a sequence of vertices, produces the shattering of Example 4.1.4.

7.6 The Kim–Pillay theorem

In Chapter 6 we characterized the stable theories by the existence of an independence relation with uniqueness. Kim and Pillay [**KP97**] did the same for the simple theories, with the independence theorem in place of uniqueness. Their theorem says that simplicity is not one property among several that a good notion of independence might have: it is what the existence of such a notion amounts to. In the absence of uniqueness, existence is taken in the stronger form of extension, in place of the weak form of Definition 5.4.7.

Definition 7.6.1. Let $\mathop{\smile\!\!\!\!\!\!|}$ be a relation $A \mathop{\smile\!\!\!\!\!\!|}_M B$ between small sets A, B and models M. We consider the following two axioms, for all small sets A, B, B', all models M and all tuples c_1, c_2.

(EXT) *Extension.* If $A \mathop{\smile\!\!\!\!\!\!|}_M B$ and $B \subseteq B'$, then there is $A' \equiv_{MB} A$ with $A' \mathop{\smile\!\!\!\!\!\!|}_M B'$.

(IT) *Independence theorem.* If $A \mathop{\smile\!\!\!\!\!\!|}_M B$, $c_1 \equiv_M c_2$, $c_1 \mathop{\smile\!\!\!\!\!\!|}_M A$ and $c_2 \mathop{\smile\!\!\!\!\!\!|}_M B$, then there is c with $c \equiv_{MA} c_1$, $c \equiv_{MB} c_2$ and $c \mathop{\smile\!\!\!\!\!\!|}_M AB$.

A *Kim–Pillay relation* is a relation satisfying (INV), (MON), (BMON), (TR), (FC), (SYM), (EX) and (LOC) of Definition 5.4.7, together with (EXT) and (IT).

In a simple theory, nondividing independence is a Kim–Pillay relation: it has the axioms of Definition 5.4.7 other than uniqueness by Theorem 7.3.6, the axiom (EXT) by Corollary 7.3.4, and the axiom (IT) by Theorem 7.4.7. The converse needs two tools (see Figure 7.8). The first, Proposition 7.6.2, produces from an indiscernible sequence an independent one with the same type. The second, Proposition 7.6.3, turns the tree property into a type that divides over every small submodel of a model.

7.6.1 Independent copies of indiscernible sequences

The first tool turns an indiscernible sequence into an independent one with the same type. We build a long sequence with this type as a coheir of a reversed copy, as in the proof of Proposition 7.4.4, and stop it at a stage of cofinality κ^+. Local character, applied to the term at that stage, gives a small base, and cofinality places this base inside an earlier model of the chain. Over that earlier model the rest of the sequence is indiscernible, so the independence of one term from its predecessors passes to every term, and a single stage suffices.

Proposition 7.6.2. *Let $\mathop{\smile\!\!\!\!\!\!|}$ be a relation over models satisfying* (INV), (MON), (BMON) *and* (LOC). *Let M be a model, and let $(b_i)_{i<\omega}$ be a sequence of finite tuples, indiscernible over M. Then there are a model $N \supseteq M$ and a sequence $(e_i)_{i<\omega}$, indiscernible over N, such that $(e_0, \dots, e_{n-1}) \equiv_M (b_0, \dots, b_{n-1})$ for every n and $e_i \mathop{\smile\!\!\!\!\!\!|}_N e_{<i}$ for every i.*

Proof. ☺ **Step 1: a coheir sequence of length $\kappa^+ + 1$.** We proceed as in Steps 1 and 2 of the proof of Proposition 7.4.4. Let $(h_j)_{j<\omega}$ be a sequence whose decreasing tuples have the types over M of the increasing tuples of (b_i), and let N_0 be a small model containing $M \cup \{h_j : j < \omega\}$. We fix a nonprincipal ultrafilter $\mathcal{D}$ on ω and take for $\mathcal{V}$ the ultrafilter on

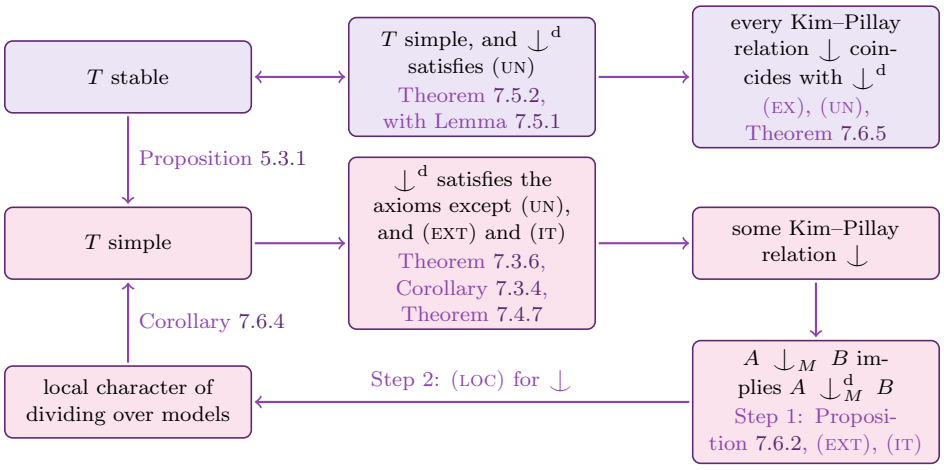


FIGURE 7.8. The proof of Theorem 7.6.5 (lower cycle) and its stable counterpart (upper lane), with the results behind each arrow in small type.

$N_0^{|b_0|}$ induced by $j \mapsto h_j$. Let $\delta = \kappa^+$, a regular cardinal by Lemma B.2.2. We choose by recursion on $\alpha \leq \delta$ small models N_α and tuples t_α as follows. The models form an increasing chain, continuous at limits, with $N_\alpha \supseteq N_0 \cup \{t_\gamma : \gamma < \alpha\}$ and $|N_\alpha| \leq |N_0| + |\alpha| + \kappa$, and the tuple t_α realizes $q_{\mathcal{V}}|N_\alpha$.

Let $\beta \leq \delta$ be arbitrary, and fix indices $\beta \leq \alpha_0 < \cdots < \alpha_{n-1} \leq \delta$ and a formula $\theta \in L(N_\beta)$. The computation of Step 2 of the proof of Proposition 7.4.4 gives

$$\mathbb{M} \models \theta(t_{\alpha_0}, \ldots, t_{\alpha_{n-1}}) \quad \text{if and only if} \quad \Big\{ j_{n-1} : \cdots \big\{ j_0 : \mathbb{M} \models \theta(h_{j_0}, \ldots, h_{j_{n-1}}) \big\} \in \mathcal{D} \cdots \Big\} \in \mathcal{D},$$

since t_{α_r} realizes the coheir of $\mathcal{V}$ over a model containing N_β and $t_{\alpha_0}, \ldots, t_{\alpha_{r-1}}$. The right side does not depend on $\alpha_0 < \cdots < \alpha_{n-1}$. Hence the sequence $(t_\alpha)_{\beta \leq \alpha \leq \delta}$ is indiscernible over N_β, and, as in the proof of Proposition 7.4.4, we have $(t_{\alpha_0}, \ldots, t_{\alpha_{n-1}}) \equiv_M (b_0, \ldots, b_{n-1})$ for all $\alpha_0 < \cdots < \alpha_{n-1}$.

☺ **Step 2: local character at the last stage.** By (LOC) there is a model $M' \subseteq N_\delta$ with $|M'| \leq \kappa$ and $t_\delta \downarrow_{M'} N_\delta$. The chain is continuous, so $N_\delta = \bigcup_{\alpha<\delta} N_\alpha$, and $|M'| \leq \kappa < \operatorname{cf} \delta$; hence Lemma B.2.3 gives $\beta < \delta$ with $M' \subseteq N_\beta$. Since $M' \subseteq N_\beta \subseteq N_\delta$, the axiom (BMON) gives $t_\delta \downarrow_{N_\beta} N_\delta$, and since N_δ contains every t_γ with $\gamma < \delta$, the axiom (MON) yields $t_\delta \downarrow_{N_\beta} \{t_\gamma : \beta \leq \gamma < \delta\}$.

☺ **Step 3: the sequence (e_i).** Let $N = N_\beta$ and $e_i = t_{\beta+i}$ for $i < \omega$; these indices lie below δ, because δ is a limit ordinal greater than β. By Step 1 the sequence (e_i) is indiscernible over N and has the required type over M. Fix $i < \omega$. The tuples $(t_\beta, \ldots, t_{\beta+i-1}, t_\delta)$ and $(t_\beta, \ldots, t_{\beta+i})$ are increasing tuples of the same length from the sequence $(t_\alpha)_{\beta \leq \alpha \leq \delta}$, which is indiscernible over N; hence some $\sigma \in \operatorname{Aut}(\mathbb{M}/N)$ fixes $e_0, \ldots, e_{i-1}$ and sends t_δ to e_i. By Step 2 and (MON) we have $t_\delta \downarrow_N e_{<i}$, and (INV) turns this into $e_i \downarrow_N e_{<i}$. ☺

A single stage suffices because the whole tail of the chain is indiscernible over each model N_β. Without indiscernibility, local character would supply a different small base at each stage, and a pressing-down argument would be needed to find one base serving infinitely many terms.

7.6.2 Trees give dividing over every small submodel

The second tool is a converse of the local character of dividing (see Theorem 5.3.6): if some formula has the tree property, then some type over a model divides over every small submodel. We grow a branch of length κ^+ through the tree property, one formula at a time, and make each new formula divide over everything chosen before it. To let the construction continue, we preserve above the branch a whole tree with which the branch is consistent.

Proposition 7.6.3. *Let $k \geq 2$, and suppose that some formula $\varphi(x, y)$ has the k-tree property. Then there are a model N and a finite tuple a such that $\mathrm{tp}(a/N)$ divides over every model $M \subseteq N$ with $|M| \leq \kappa$.*

Proof. By recursion on $\alpha \leq \kappa^+$ we choose models N_α with $|N_\alpha| \leq \kappa^+$, forming an increasing chain continuous at limits, and tuples $b_\gamma \in N_{\gamma+1}$ for $\gamma < \alpha$. With $\Sigma_\alpha = \{\varphi(x, b_\gamma) : \gamma < \alpha\}$, we require the following:

(i) for $\gamma < \alpha$, the formula $\varphi(x, b_\gamma)$ k-divides over N_γ;

(ii) there are tuples c_σ, for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, satisfying condition (b) of Definition 5.2.7, such that for every $\eta \in \omega^\omega$ the set $\Sigma_\alpha \cup \{\varphi(x, c_{\eta|n}) : n \geq 1\}$ is consistent.

Condition (ii) implies that Σ_α is consistent. For $\alpha = 0$, let N_0 be any model of size κ; then (i) is vacuous, and (ii) is the k-tree property.

Successor step. Let (c_σ) witness (ii) for α. For $i < \omega$ let s_i be the tuple $(c_{\langle i\rangle^\frown\tau})_{\tau \in \omega^{<\omega}}$, listing the subtree above the node $\langle i\rangle$, with $c_{\langle i\rangle}$ as its first entry. Here $\langle i\rangle^\frown\tau$ is the sequence obtained by prefixing i to τ. Applying Lemma 5.2.2 over N_α to the sequence $(s_i)_{i<\omega}$, we obtain a sequence $(s'_i)_{i<\omega}$, indiscernible over N_α, realizing the Ehrenfeucht–Mostowski type of (s_i) over N_α. The Ehrenfeucht–Mostowski type contains the following formulas, because they hold of all the tuples, or all the increasing k-tuples, of (s_i):

- $\neg\exists x \bigwedge_{l<k} \varphi(x, z_l)$, where z_l is the first entry of the l-th tuple;
- for each finite $\Sigma_0 \subseteq \Sigma_\alpha$, each n and each path ρ of length n in the subtree, the formula saying that Σ_0, $\varphi(x, z)$ and the formulas at the nodes of ρ are consistent, where z is the first entry;
- the formulas saying that siblings within one subtree are k-inconsistent.

Let b_α be the first entry of s'_0, and let $N_{\alpha+1}$ be a model of size $|N_\alpha|$ containing $N_\alpha \cup b_\alpha$. The first entries of the s'_i form a sequence indiscernible over N_α, beginning with b_α, along which φ is k-inconsistent, so the formula $\varphi(x, b_\alpha)$ k-divides over N_α. The remaining entries of s'_0, indexed by $\tau \neq \emptyset$, satisfy condition (b) of Definition 5.2.7, and by compactness every branch through them is consistent with $\Sigma_\alpha \cup \{\varphi(x, b_\alpha)\}$. These remaining entries therefore witness (ii) for $\alpha + 1$.

Limit step. For a limit $\delta \leq \kappa^+$, we take $N_\delta = \bigcup_{\alpha<\delta} N_\alpha$. Condition (ii) for δ is expressed by a set of formulas in variables z_σ, with parameters among the b_γ. A finite part of this set mentions only Σ_α for some $\alpha < \delta$, and is satisfied by the tree given by (ii) for α. By compactness and saturation, (ii) holds for δ, and (i) holds by the induction hypothesis.

Finally, with $N = N_{\kappa^+}$, let a realize Σ_{κ^+}, which is consistent by (ii). Let $M \subseteq N$ be a model with $|M| \leq \kappa$. Since κ^+ is regular, we have $M \subseteq N_\alpha$ for some $\alpha < \kappa^+$. The formula $\varphi(x, b_\alpha)$ belongs to $\mathrm{tp}(a/N)$ and k-divides over N_α, hence over M (see Remark 5.2.5). ☺

With Proposition 7.6.3 we obtain the converse of Theorem 5.3.6 promised in Remark 5.3.7, in the form with models.

Corollary 7.6.4. *A complete theory T is simple if and only if for every finite tuple a and every model N there is a model $M \subseteq N$ with $|M| \leq \kappa$ such that $\mathrm{tp}(a/N)$ does not divide over M.*

Proof. If T is simple, this is (LOC) in Proposition 7.1.8. If T is not simple, some formula has the k-tree property for some k, and Proposition 7.6.3 gives a and N for which no such M exists. ☺

7.6.3 The characterization of simple theories

With both tools in hand, the proof takes two steps. First, every Kim–Pillay relation is contained in nondividing independence; then the local character of the relation passes to dividing, and Corollary 7.6.4 yields simplicity.

Theorem 7.6.5 (Kim–Pillay). *A complete theory T is simple if and only if it carries a Kim–Pillay relation. Moreover, nondividing independence is then a Kim–Pillay relation, and every Kim–Pillay relation $\mathop{\smile\!\!\!\!\!\!|}$ is contained in nondividing independence: if $A \mathop{\smile\!\!\!\!\!\!|}_M B$, then $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$.*

Proof. If T is simple, nondividing independence is a Kim–Pillay relation, as observed after Definition 7.6.1. Conversely, let $\mathop{\smile\!\!\!\!\!\!|}$ be a Kim–Pillay relation.

☺ **Step 1: the relation $\mathop{\smile\!\!\!\!\!\!|}$ implies nondividing.** Let a and b be finite tuples with $a \mathop{\smile\!\!\!\!\!\!|}_M b$, and let $p(x, y)$ be the set of $L(M)$-formulas $\psi(x, y)$ with $\mathbb{M} \models \psi(a, b)$. Fix a sequence $(b_i)_{i<\omega}$, indiscernible over M, with $b_0 = b$. We show that $\bigcup_{i<\omega} p(x, b_i)$ is consistent; then no formula of $\mathrm{tp}(a/Mb)$ divides over M. The claim for arbitrary A and B follows, because $A \mathop{\smile\!\!\!\!\!\!|}_M B$ gives $a \mathop{\smile\!\!\!\!\!\!|}_M b$ for all finite subtuples by (MON), and dividing is witnessed by formulas, which involve only finite subtuples. By compactness it suffices to realize $\bigcup_{i<n} p(x, b_i)$ for each n.

The idea is to replace (b_i) by a sequence with the same type over M that is indiscernible over a larger model, over which each term is independent from the earlier ones. Along such a sequence, the axiom (IT) amalgamates the copies of p one term at a time. Let N and $(e_i)_{i<\omega}$ be given by Proposition 7.6.2, applied to M and $(b_i)_{i<\omega}$. Since $e_0 \equiv_M b$, there is $\sigma \in \mathrm{Aut}(\mathbb{M}/M)$ with $\sigma(b) = e_0$, and (INV) gives $\sigma(a) \mathop{\smile\!\!\!\!\!\!|}_M e_0$. By (EXT) there is $a^* \equiv_{Me_0} \sigma(a)$ with $a^* \mathop{\smile\!\!\!\!\!\!|}_M N \cup e_0$, and (BMON) and (MON) give $a^* \mathop{\smile\!\!\!\!\!\!|}_N e_0$. Let $q(x, e_0) = \mathrm{tp}(a^*/Ne_0)$; this type contains $p(x, e_0)$, because $a^* e_0 \equiv_M ab$. Since the sequence (e_i) is indiscernible over N, for each i an automorphism of $\mathbb{M}$ over N carries e_0 to e_i and a^* to some tuple d_i. The tuple d_i satisfies $d_i \equiv_N a^*$ and $d_i \mathop{\smile\!\!\!\!\!\!|}_N e_i$, and it realizes $q(x, e_i)$.

We show by induction on $m \leq n$ that some c_m realizes $\bigcup_{i<m} q(x, e_i)$ with $c_m \equiv_N a^*$ and $c_m \mathop{\smile\!\!\!\!\!\!|}_N e_{<m}$. For $m = 0$ we take $c_0 = a^*$; given c_m, we have $e_{<m} \mathop{\smile\!\!\!\!\!\!|}_N e_m$ by (SYM). The axiom (IT), applied over N to the sets $e_{<m}$ and e_m and to the tuples c_m and d_m, gives c_{m+1} as required. Thus the type $\bigcup_{i<n} p(x, e_i)$ is realized, and since $(e_0, \ldots, e_{n-1}) \equiv_M (b_0, \ldots, b_{n-1})$, so is $\bigcup_{i<n} p(x, b_i)$.

☺ **Step 2: the theory T is simple.** Let a finite tuple a and a model N be given. By (LOC) there is a model $M \subseteq N$ with $|M| \leq \kappa$ and $a \mathop{\smile\!\!\!\!\!\!|}_M N$, and by Step 1 the type $\mathrm{tp}(a/N)$ does not divide over M. By Corollary 7.6.4, the theory T is simple. ☺

The theorem identifies nondividing independence as the largest Kim–Pillay relation: every other is contained in it. For the analogous relations over arbitrary base sets, Kim and Pillay [KP97] show that there is only one: every such relation coincides with nonforking independence. In a stable theory, nondividing independence is likewise the only Kim–Pillay

relation, and this follows at once from what we have proved. Let $\mathop{\smile\!\!\!\!\!\!|}$ be a Kim–Pillay relation, and suppose that $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$. By (EX) there is $A' \equiv_M A$ with $A' \mathop{\smile\!\!\!\!\!\!|}_M B$, and Theorem 7.6.5 gives $A' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$. Uniqueness for $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ (see Theorem 7.5.2) now yields $A' \equiv_{MB} A$, whence $A \mathop{\smile\!\!\!\!\!\!|}_M B$ by (INV).

7.7 Exercises

The exercises follow the sections of the chapter, and the mark $(*)$ indicates those that ask for more than a direct application of the text.

Throughout, M is a model, and we write T_{RG} for the theory of the random graph, with edge relation E (see Example 5.4.1). As in the text, $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ denotes nondividing independence (see Definition 7.1.1), and the axioms keep their labels from Definitions 5.4.7 and 7.6.1. In the theory of $(\mathbb{Q}, <)$ we write $M < c$, as in Section 7.2, when the element c is greater than every element of M.

1. *(Section 7.1.)* Let b be an element with $b \notin M$. Show that the formula $x = b$ divides over M. Deduce that, in every complete theory, the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ satisfies anti-reflexivity: if $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M a$ for a finite tuple a, then every coordinate of a lies in M. Deduce also that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$ implies $a \cap B \subseteq M$. (Hint: a model is algebraically closed, so the type $\mathrm{tp}(b/M)$ has infinitely many realizations; use Lemma 5.2.2.)

2. *(Section 7.1.)* Work in T_{RG}. Let a and b be vertices outside M such that a is adjacent to every vertex of M and to b, while b is adjacent to no vertex of M. Show that such vertices exist, that $\mathrm{tp}(b/Ma)$ is finitely satisfiable in M, and that $\mathrm{tp}(a/Mb)$ is not. Conclude that *coheir independence*, the relation "$\mathrm{tp}(A/MB)$ is finitely satisfiable in M", is not symmetric, even in a simple theory. Show also that $\mathrm{tp}(a/Mb)$ does not divide over M (see Theorem 5.4.5), although it is not finitely satisfiable in M; thus the converse of Lemma 7.1.3 does not hold.

3. *(Section 7.2.)* Assume that T is simple, let b be a finite tuple, and let $\varphi(x, y), \psi(x, y) \in L(M)$ be given. Show that $\varphi(x, b) \vee \psi(x, b)$ divides over M if and only if both $\varphi(x, b)$ and $\psi(x, b)$ divide over M. (Hint: for the nontrivial direction, fix one Morley sequence $(b_i)_{i<\omega}$ over M with $b_0 = b$. By Kim's lemma (see Theorem 7.2.2), the instances of φ and of ψ along this sequence are inconsistent, hence k-inconsistent and l-inconsistent for some k and l (see Lemma 5.2.4). Show that the instances of the disjunction are $(k + l - 1)$-inconsistent.) Identify where the argument uses simplicity.

4. *(Section 7.2.)* Work in the theory of $(\mathbb{Q}, <)$, let $\varphi(x; y_1, y_2)$ be the formula $y_1 < x < y_2$, and let $b = (d_1, d_2)$ be a pair with $M < d_1 < d_2$, as in Example 7.2.3. By Example 7.2.3, the formula $\varphi(x, b)$ divides over M along some Morley sequence over M and not along another. Show that the formula $x = d_1$ divides along every indiscernible sequence over M that begins with d_1 and is not constant. Deduce from Exercise 1 that no Morley sequence over M beginning with d_1 is constant. Thus, in contrast with $\varphi(x, b)$, the formula $x = d_1$ divides along every Morley sequence over M beginning with d_1.

5. *(Section 7.3.)* Let b be a finite tuple, and let $\varphi(x, b)$ be a formula over $M \cup b$ that is satisfied by some tuple from M. Show that, in every complete theory, the formula $\varphi(x, b)$

does not fork over M. Deduce that, in the dense circular order of Remark 7.3.2, the formula $x = x$ forks over $\emptyset$ but over no model.

6. *(Section 7.3.)* Symmetry does not follow from invariance, monotonicity, finite character and existence alone. Using Exercise 2, show that the relation of coheir independence over models satisfies (INV), (MON), (FC) and (EX) in every complete theory (see Proposition 7.1.4) and is not symmetric in T_{RG}.

7. *(Section 7.4.)* Work in the theory of $(\mathbb{Q}, <)$. Let $a < b$ be elements greater than every element of M, and let c_1 and c_2 be elements with $M < c_1 < a$ and $b < c_2$. Show that $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M b$, $c_1 \equiv_M c_2$, $c_1 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M a$ and $c_2 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M b$, and that no c satisfies both $c \equiv_{Ma} c_1$ and $c \equiv_{Mb} c_2$. (Hint: a finite conjunction of formulas from $\mathrm{tp}(a/Mb)$ remains consistent when b is replaced by finitely many terms of an indiscernible sequence over M.) Thus $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ does not satisfy the independence theorem in this theory. Compare Theorem 7.6.5: since the theory is not simple, the relation $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ is not a Kim–Pillay relation, although the theorem does not say which axiom is missing. In this theory neither (IT) nor (LOC) holds (see Exercise 13).

8. *(Section 7.4.)* Assume that T is stable. Show that the independence theorem holds for $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}$ over models without the hypothesis $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$: if $c_1 \equiv_M c_2$, $c_1 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M A$ and $c_2 \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$, there is c with $c \equiv_{MA} c_1$, $c \equiv_{MB} c_2$ and $c \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M AB$. (Hint: realize a nondividing extension of $\mathrm{tp}(c_1/M)$ to $M \cup A \cup B$, and use (MON) and (UN); see Theorem 7.5.2.) Show by an example in T_{RG} that, in a simple theory, the hypothesis $A \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M B$ is essential.

9. (∗) *(Section 7.5.)* Assume that T is stable, and let C be a small set. Show that every sequence $(b_i)_{i<\omega}$ indiscernible over C is an indiscernible set over C. That is, for all n, all distinct $i_1, \ldots, i_n$ and all distinct $j_1, \ldots, j_n$, the tuples $(b_{i_1}, \ldots, b_{i_n})$ and $(b_{j_1}, \ldots, b_{j_n})$ have the same type over C. (Hint: by compactness, index the sequence by $\mathbb{Q}$. If a transposition of two adjacent positions changes the truth value of a formula, fix the other terms and obtain a formula that orders an infinite subsequence; see Definition 1.1.4.) Deduce that every Morley sequence over a model M of a stable theory is an indiscernible set over M.

10. *(Section 7.5.)* Work in T_{RG}. Find vertices a, a' and b outside M with $a \equiv_M a'$, $a \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M b$, $a' \mathop{\smile\!\!\!\!\!\!|}^{\mathrm{d}}_M b$ and $a \not\equiv_{Mb} a'$. Follow the proof of Lemma 7.5.1 for the formula xEy, and compare the tuples c_n that it produces with those given directly by the extension axioms.

11. (∗) *(Section 7.6.)* Work in T_{RG}, with trivial independence $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ (see Definition 5.4.8) and models as bases. Show that $\mathop{\smile\!\!\!\!\!\!|}^{\mathrm{t}}$ satisfies (EXT), and that Theorem 5.4.9(c), together with compactness, gives (IT) for small sets. Obtain from Theorem 7.6.5 a second proof that T_{RG} is simple (see Theorem 5.4.6), and a second proof of one direction of Theorem 5.4.5 over models.

12. *(Section 7.6.)* Show that every Kim–Pillay relation $\mathop{\smile\!\!\!\!\!\!|}$ satisfies anti-reflexivity: if $a \mathop{\smile\!\!\!\!\!\!|}_M a$ for a finite tuple a, then every coordinate of a lies in M. (Hint: Theorem 7.6.5 and Exercise 1.)

13. (∗) *(Section 7.6.)* Work in the theory of $(\mathbb{Q}, <)$, where $\kappa = |T| = \aleph_0$ in Corollary 7.6.4. Let N be a model, and let a realize a cut (L, R) of N such that no subset of L of size at

most κ is cofinal in L. Assume also that no subset of R of size at most κ is coinitial in R. Show that such N and a exist, and that $\mathrm{tp}(a/N)$ divides over every model $M \subseteq N$ with $|M| \leq \kappa$. (Hint: find $l \in L$ and $r \in R$ with no element of M between them, and argue as in Example 7.2.3.) Conclude from Corollary 7.6.4 that the theory is not simple.

7.8 Historical remarks

Section 7.1. Dividing and forking were introduced by Shelah in the first edition of *Classification Theory* (1978; see the second edition [She90]). In that book they are the principal instruments in the study of stable theories. Dividing is the combinatorial notion; forking is its closure under finite disjunctions, designed to guarantee that nonforking extensions always exist. Indiscernible sequences go back to Ehrenfeucht and Mostowski [EM56], who extracted them with Ramsey's theorem, as in Lemma 5.2.2. Morley [Mor65], in his proof of the categoricity theorem, made them a central tool of the subject, and the sequences whose terms are independent from the earlier ones now bear his name. Two ingredients of Section 7.1 come from the work of Lascar and Poizat [LP79] on forking in stable theories. The first is the fact that finitely satisfiable types do not divide; the second is the use of coheirs to produce Morley sequences. The characterization of nondividing by indiscernible sequences (see Lemma 7.1.2) is standard; we follow the presentation of Grossberg, Iovino and Lessmann [GIL02].

Section 7.2. Shelah named and studied the simple theories in his paper of 1980 [She80]. The local character of dividing in theories without the tree property, and its converse, already appear in Chapter III of *Classification Theory* [She90]. Kim [Kim98] proved that in a simple theory a formula divides exactly when it divides along some, equivalently every, Morley sequence. From this characterization he deduced that forking and dividing agree and that independence is symmetric and transitive. In a later paper [Kim01] he showed that each of symmetry and transitivity of forking characterizes the simple theories.

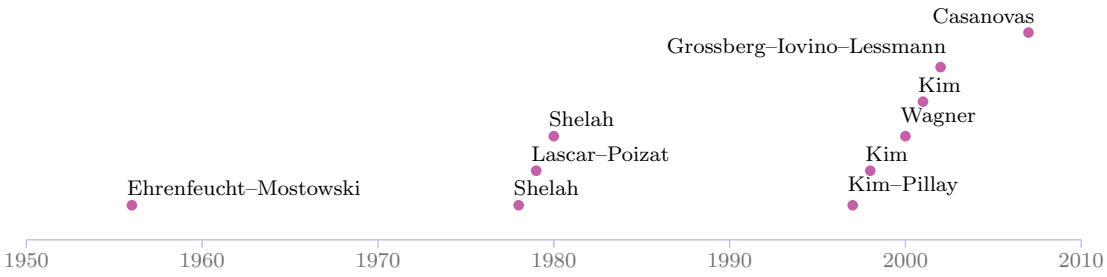
Section 7.3. Forking and dividing differ in general, and the dense circular order is the usual example (see Tent and Ziegler [TZ12, Chapter 7]). The derivation of symmetry from Kim's lemma, by a sequence of copies of one tuple that a single other tuple cannot follow, goes back to Kim [Kim98].

Section 7.4. The independence theorem is due to Kim and Pillay [KP97], who abstracted it from Hrushovski's work on pseudofinite fields [Hru02]. They proved it for Lascar strong types over arbitrary sets. The theorem for Lascar strong types appears in the monographs of Wagner [Wag00] and Casanovas [Cas11], in Casanovas's notes [Cas07], and in the primer of Grossberg, Iovino and Lessmann [GIL02]. Over a model, Lascar strong types are ordinary types, and the theorem takes the form of Theorem 7.4.7. Our proof follows the route of Casanovas, in which a parameter is moved along indiscernible sequences (see Propositions 7.4.5 and 7.4.6). Each move rests on the fact that conjunctions along an indiscernible sequence do not divide (see Proposition 7.4.4). Casanovas proves the nondividing of these conjunctions over arbitrary sets, using symmetry and transitivity of forking over arbitrary sets. To stay with models, we prove it by an argument specific to this book, in which the indiscernible sequence is traded for a Morley sequence over a larger model, obtained as a coheir of a reversed copy of the indiscernible sequence. Kim's lemma applies directly to this Morley sequence.

Section 7.5. That a theory is stable exactly when nonforking extensions of types over models are unique goes back to Shelah [She90]. Two arguments belong to the folklore of simple

theories (see, for instance, Casanovas [Cas11]). The first is the derivation of the independence property from two distinct nonforking extensions through the independence theorem; the second is the resulting proof that simple NIP theories are stable.

Section 7.6. The characterization of simple theories by an abstract independence relation is due to Kim and Pillay [KP97], who also proved that such a relation must be nonforking independence. As mentioned above, the converse of local character over arbitrary sets goes back to Chapter III of *Classification Theory* [She90], and Corollary 7.6.4 is its form with models. The construction of a long chain of dividing formulas from the tree property, keeping a tree above the chain, is the classical one. Our proof of Step 1 of Theorem 7.6.5 uses the reversed copy of Section 7.4, so as to stay with models as bases. Local character is applied at a single stage of cofinality κ^+, and indiscernibility over an earlier model of the chain does the rest, so that no pressing-down argument is needed. Over arbitrary sets, where coheirs are not available, Morley sequences are built instead by Morley's method, the extraction of indiscernible sequences through the Erdős–Rado theorem that Morley introduced to compute the Hanf number for omitting types; see Grossberg, Iovino and Lessmann [GIL02].



1956 **Ehrenfeucht–Mostowski.** Ehrenfeucht and Mostowski [EM56] use Ramsey's theorem to build models generated by indiscernible sequences.

1978 **Shelah.** Shelah introduces forking, dividing and the tree property in the first edition of *Classification Theory* (see the second edition [She90]), and shows that dividing has local character exactly when no formula has the tree property.

1979 **Lascar–Poizat.** Lascar and Poizat [LP79] present forking in stable theories through heirs and coheirs.

1980 **Shelah.** Shelah [She80] names the simple theories, those without the tree property, and studies independence in them.

1997 **Kim–Pillay.** Kim and Pillay [KP97] prove the independence theorem for Lascar strong types over arbitrary sets and characterize the simple theories by their independence relation.

1998 **Kim.** Kim [Kim98] proves that forking equals dividing and that forking is symmetric and transitive in simple theories.

2000 **Wagner.** In his monograph [Wag00], Wagner develops the theory of simple theories in full.

2001 **Kim.** Kim [Kim01] shows that symmetry of forking characterizes the simple theories, and that transitivity does as well.

2002 **Grossberg–Iovino–Lessmann.** Grossberg, Iovino and Lessmann [GIL02] give a self-contained introduction to simple theories.

2007 **Casanovas.** Casanovas [Cas07] streamlines the basic theory, including the independence theorem for Lascar strong types.

Appendix

A

Grothendieck's double limit criterion

Roadmap

Grothendieck's criterion of 1952 concerns bounded sets of continuous functions on a compact Hausdorff space. The criterion states that such a set is relatively weakly compact exactly when the two iterated limits agree along all sequences of functions and of points for which both exist. The criterion is the analytic form of the absence of arbitrarily long ladders that runs through Sections 1.2 and 4.2, and it is used in Subsection 1.2.4 and in Remarks 1.2.14 and 4.2.6. In Appendix A.1 we show that the ultrafilter form of the double limit condition used in this book agrees with Grothendieck's form (see Lemma A.1.2), and we prove the criterion (see Theorem A.1.3). We derive convergent subsequences with the Eberlein–Šmulian theorem (see Corollary A.1.6), and we deduce that every type of a stable function is the limit of a sequence of realized types (see Corollary A.1.8).

Throughout, we fix a compact Hausdorff space K, and we denote by $C(K)$ the Banach space of continuous real-valued functions on K with the supremum norm. The *weak topology* on $C(K)$ is the coarsest topology making every bounded linear functional continuous, and the *pointwise topology* is the one induced by the product topology of $\mathbb{R}^K$. The point evaluations $f \mapsto f(x)$, for $x \in K$, are bounded linear functionals on $C(K)$, so the weak topology is finer than the pointwise topology.

A.1 The criterion

We begin with the condition on iterated limits, in the ultrafilter form used throughout the book.

Definition A.1.1. A bounded set $H \subseteq C(K)$ satisfies the *double limit condition* if for all sequences $(x_m)_{m\in\mathbb{N}}$ in K and $(f_n)_{n\in\mathbb{N}}$ in H and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} f_n(x_m) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} f_n(x_m).$$

In the language of Definition 1.2.5, the double limit condition says that the evaluation function $K \times H \to \mathbb{R}$, $(x, f) \mapsto f(x)$, is stable. Grothendieck stated the condition with ordinary limits; in the next lemma we show that the two formulations are equivalent.

Lemma A.1.2. *A bounded set $H \subseteq C(K)$ satisfies the double limit condition if and only if*

$$\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$$

for all sequences $(x_m)_{m\in\mathbb{N}}$ in K and $(f_n)_{n\in\mathbb{N}}$ in H for which both iterated limits exist.

Proof. If both ordinary iterated limits exist, they agree with the iterated limits along any nonprincipal ultrafilters; hence the double limit condition implies the second condition.

Conversely, suppose that the double limit condition fails for sequences (x_m) and (f_n) along ultrafilters $\mathcal{U}$ and $\mathcal{V}$, with iterated limits $\alpha \neq \beta$. The idea is to pass to subsequences along which the ordinary iterated limits exist and equal α and β. After an affine change of scale we may assume that all values lie in $[0, 1]$. Applied to the function $(x, f) \mapsto f(x)$, Proposition 1.2.3 gives indices $i_0, i_1, \dots$ and $j_0, j_1, \dots$ in $\mathbb{N}$; writing x_m for x_{i_m} and f_n for f_{j_n}, we obtain sequences with $|f_n(x_m) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|f_n(x_m) - \beta| \leq 2^{1-n}$ for $n < m$. By a diagonal argument there is an infinite set $S \subseteq \mathbb{N}$ such that, for every $m \in S$, the values $f_n(x_m)$ converge as n tends to infinity through S. The same diagonal argument ensures, in addition, that for every $n \in S$ the values $f_n(x_m)$ converge as m tends to infinity through S. We keep only the terms x_m and f_n with $m, n \in S$ and renumber them in increasing order. Since one set of indices serves both sequences, the renumbering preserves the order between m and n, and since it does not increase an index, the inequalities still hold. Then the limit $\lim_n f_n(x_m)$ is within 2^{1-m} of α, whence $\lim_m \lim_n f_n(x_m) = \alpha$. The symmetric argument gives $\lim_n \lim_m f_n(x_m) = \beta$, and hence the second condition fails. ☺

Theorem A.1.3 (Grothendieck). *Let $H \subseteq C(K)$ be bounded. The following are equivalent:*

(1) *The set H is relatively weakly compact in $C(K)$.*

(2) *Every function in the pointwise closure of H in $\mathbb{R}^K$ is continuous.*

(3) *The set H satisfies the double limit condition.*

The equivalence of (2) and (3) is an exchange of ultrafilter limits, together with Proposition 1.2.3; the measure-theoretic input and the angelicity of pointwise compact sets enter only in (2) ⇒ (1). For a compact Hausdorff space, angelicity means that every point in the closure of a subset is the limit of a sequence from that subset.

Proof. (2) ⇒ (3). Let sequences (x_m) in K and (f_n) in H be given, together with ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, and let $x = \lim_{m,\mathcal{U}} x_m \in K$ and $f = \lim_{n,\mathcal{V}} f_n \in \mathbb{R}^K$ be the corresponding limits. By (2), the function f is continuous. Using the continuity of f and of each f_n, we obtain

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} f_n(x_m) = \lim_{m,\mathcal{U}} f(x_m) = f(x) = \lim_{n,\mathcal{V}} f_n(x) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} f_n(x_m).$$

(3) ⇒ (2). Suppose that f is in the pointwise closure of H but is not continuous at some $x \in K$. Then the function f is the limit $\lim_{i,\mathcal{U}} f_i$ of a family $(f_i)_{i\in I}$ in H along an ultrafilter $\mathcal{U}$ on I. There are also a family $(x_j)_{j\in J}$ in K and an ultrafilter $\mathcal{V}$ on J with $\lim_{j,\mathcal{V}} x_j = x$ and $\lim_{j,\mathcal{V}} f(x_j) = \beta \neq f(x)$. To find the points x_j and the ultrafilter $\mathcal{V}$, fix $\varepsilon > 0$ such that every neighborhood O of x contains a point x_O with $|f(x_O) - f(x)| \geq \varepsilon$. Let J be the set of neighborhoods of x, and let $\mathcal{V}$ be an ultrafilter on J containing, for each $O \in J$, the set of neighborhoods contained in O. Such an ultrafilter exists because these sets, one for each $O \in J$, have the finite intersection property. The family $(x_O)_{O\in J}$ converges to x along $\mathcal{V}$,

and the limit β of the bounded values $f(x_O)$ satisfies $|\beta - f(x)| \geq \varepsilon$. By continuity of each f_i, we have

$$\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} f_i(x_j) = \lim_{i,\mathcal{U}} f_i(x) = f(x), \qquad \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} f_i(x_j) = \lim_{j,\mathcal{V}} f(x_j) = \beta.$$

After an affine change of scale, Proposition 1.2.3 and the proof of Corollary 1.2.6 turn these families into sequences whose iterated limits along a nonprincipal ultrafilter are β and $f(x)$, in contradiction with (3).

(1) $\Rightarrow$ (2). Let W be the weak closure of H, which is weakly compact by (1). The identity map from $C(K)$ with the weak topology to $C(K)$ with the pointwise topology is continuous, so the set W is pointwise compact, hence closed in $\mathbb{R}^K$. Consequently, the pointwise closure of H is contained in $W \subseteq C(K)$.

(2) $\Rightarrow$ (1). Let $\bar{H}$ be the pointwise closure of H. This set is compact in $\mathbb{R}^K$, because H is bounded, and is contained in $C(K)$ by (2). It remains to be shown that the weak and pointwise topologies agree on $\bar{H}$, for then the set $\bar{H}$ is weakly compact. By the Riesz representation theorem, the weak topology is generated by the maps $h \mapsto \int h\, d\mu$ with μ a signed Radon measure on K. On the uniformly bounded set $\bar{H}$ each such map is sequentially continuous for the pointwise topology, by Lebesgue's dominated convergence theorem. Grothendieck proved that a pointwise compact subset of $C(K)$ is angelic [**Gro52**]; see also Pryce [**Pry71**]. On such a compact space, sequentially continuous functions are continuous. Each map $h \mapsto \int h\, d\mu$ is therefore continuous on $\bar{H}$ for the pointwise topology, and the two topologies agree on $\bar{H}$, which completes the proof. ☺

The simplest family of continuous functions with a discontinuous pointwise limit already shows the two iterated limits parting company.

Example A.1.4. Let $K = [0,1]$ and $H = \{f_n : n \geq 1\}$ with $f_n(x) = x^n$. The pointwise limit of (f_n) is 0 on $[0,1)$ and 1 at 1, which is not continuous, so (2) of Theorem A.1.3 fails. Condition (3) fails along $x_m = 1 - 1/m$: for each m we have $\lim_n f_n(x_m) = 0$, whereas for each n the sequence $(f_n(x_m))_m$ tends to 1; the two iterated limits are therefore 0 and 1 (see Figure A.1). Hence the set H is not relatively weakly compact in $C[0,1]$.

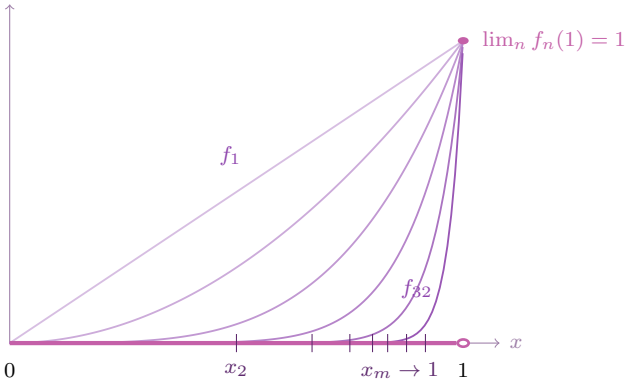


FIGURE A.1. The functions $f_n(x) = x^n$ of Example A.1.4 (purple, darker for larger n) and their pointwise limit (pink), which is 0 on $[0,1)$ and jumps to 1 at $x = 1$.

Weak compactness becomes a statement about sequences through the classical Eberlein–Šmulian theorem, and the criterion then yields convergent subsequences.

Fact A.1.5 (Eberlein–Šmulian). *In a Banach space, every sequence in a relatively weakly compact set has a weakly convergent subsequence* [Š40, Ebe47].

Corollary A.1.6. *If a bounded set $H \subseteq C(K)$ satisfies the double limit condition, then every sequence in H has a subsequence that converges pointwise on K, to a continuous function.*

Proof. By Theorem A.1.3, the set H is relatively weakly compact. The Eberlein–Šmulian theorem (see Fact A.1.5) then provides, for every sequence in H, a subsequence converging weakly to some $f \in C(K)$. Point evaluations are bounded linear functionals, so the subsequence converges pointwise to f. ☺

The criterion meets stability through the spaces of types of Section 1.2. Recall that, for a function $\varphi\colon A \times B \to [0,1]$, the space $\mathrm{LS}(\varphi)$ of left φ-types is the closure in $[0,1]^B$ of the realized types $\mathrm{ltp}_\varphi(a)\colon y \mapsto \varphi(a,y)$, for $a \in A$. Likewise, the space $\mathrm{RS}(\varphi)$ of right φ-types is the closure in $[0,1]^A$ of the realized types $\mathrm{rtp}_\varphi(b)\colon x \mapsto \varphi(x,b)$, for $b \in B$ (see Definition 1.2.7). From Subsection 1.2.4 we also use the evaluation maps attached to φ: the map $e_b\colon \mathrm{LS}(\varphi) \to [0,1]$, given by $e_b(p) = p(b)$, and the map $e_a\colon \mathrm{RS}(\varphi) \to [0,1]$, given by $e_a(q) = q(a)$.

Proposition A.1.7. *Let $\varphi\colon A \times B \to [0,1]$ be stable. Then the family $\{e_b : b \in B\}$ is relatively weakly compact in $C(\mathrm{LS}(\varphi))$, and the family $\{e_a : a \in A\}$ is relatively weakly compact in $C(\mathrm{RS}(\varphi))$. More precisely, the pointwise closure of $\{e_b : b \in B\}$ consists of the functions $\hat{\varphi}(\cdot, q)$ with $q \in \mathrm{RS}(\varphi)$, where $\hat{\varphi}$ is the function of Proposition 1.2.8.*

Proof. By (1.2) with a constant family, we have $e_b = \hat{\varphi}(\cdot, \mathrm{rtp}_\varphi(b))$. If $q = \lim_{j,\mathcal{V}} \mathrm{rtp}_\varphi(b_j)$, then by continuity of $\hat{\varphi}(p,\cdot)$ we get $\lim_{j,\mathcal{V}} e_{b_j}(p) = \hat{\varphi}(p,q)$ for every p. Thus every pointwise limit of functions e_b is of the form $\hat{\varphi}(\cdot,q)$, and conversely every such function is a pointwise limit of this kind. These functions are continuous on $\mathrm{LS}(\varphi)$ by the separate continuity of $\hat{\varphi}$, which is where the stability of φ enters. Hence condition (2) of Theorem A.1.3 holds, and the family $\{e_b : b \in B\}$ is relatively weakly compact. The statement for $\{e_a : a \in A\}$ is proved symmetrically. ☺

With Proposition A.1.7, the type spaces of a stable function enter one of the best-behaved classes of compact spaces beyond the metrizable ones, the Eberlein compacta. Recall that a compact space is an *Eberlein compactum* if it is homeomorphic to a weakly compact subset of a Banach space with the weak topology.

Corollary A.1.8. *Let $\varphi\colon A \times B \to [0,1]$ be stable. Then the spaces $\mathrm{RS}(\varphi)$ and $\mathrm{LS}(\varphi)$ are Eberlein compacta. In particular, every right φ-type is the limit of a sequence of realized right types, and likewise for left types.*

Proof. Let H be the pointwise closure of $\{e_b : b \in B\}$ in $\mathbb{R}^{\mathrm{LS}(\varphi)}$. By Proposition A.1.7, the weak closure W of $\{e_b : b \in B\}$ in $C(\mathrm{LS}(\varphi))$ is weakly compact. The identity map from W with the weak topology to W with the coarser pointwise topology is a continuous bijection from a compact space to a Hausdorff space. Consequently, the two topologies agree on W. In particular, the set W is pointwise compact, hence pointwise closed, and therefore $H \subseteq W$. The reverse inclusion $W \subseteq H$ holds because the weak topology is finer than the pointwise topology. Thus the sets H and W coincide, and on H the weak and pointwise topologies agree. The map $q \mapsto \hat{\varphi}(\cdot,q)$ from $\mathrm{RS}(\varphi)$ onto H is continuous for the pointwise topology, by the separate continuity of $\hat{\varphi}$, and injective, because $\hat{\varphi}(\mathrm{ltp}_\varphi(a), q) = q(a)$ for every $a \in A$. As before, a continuous bijection from a compact space to a Hausdorff space

is a homeomorphism. Hence the space $\mathrm{RS}(\varphi)$ is homeomorphic to H with the pointwise topology, which on H is the weak topology. The same argument, with the roles of A and B exchanged, applies to $\mathrm{LS}(\varphi)$. Finally, the set H is a pointwise compact subset of $C(\mathrm{LS}(\varphi))$, hence angelic (see the proof of Theorem A.1.3), so each point of H is the limit of a sequence from $\{e_b : b \in B\}$. Through the homeomorphism $q \mapsto \hat{\varphi}(\cdot, q)$, each $q \in \mathrm{RS}(\varphi)$ is the limit of a sequence of realized types $\mathrm{rtp}_\varphi(b_n)$. ☺

Stability is essential for the last statement of Corollary A.1.8: in an unstable structure a type may be a limit of realized types along an ultrafilter and along no sequence. The structure $(\omega_1, <)$, with $\varphi(x, y)$ the formula $x < y$, is an example. The left type p given by $p(\beta) = 0$ for every $\beta < \omega_1$ (an element above every countable ordinal) lies in $\mathrm{LS}(\varphi)$, because it agrees at any $\beta_1, \ldots, \beta_k < \omega_1$ with the realized type of $\max_i \beta_i$. Hence the type p is a limit of realized types along an ultrafilter, but not along any sequence. Indeed, if $\alpha_n < \omega_1$ for every n, then the supremum $\alpha = \sup_n \alpha_n$ is still below ω_1, and the realized types $\mathrm{ltp}_\varphi(\alpha_n)$ all take the value 1 at $\alpha + 1$. The sequence $(\mathrm{ltp}_\varphi(\alpha_n))$ therefore does not converge to p (see Figure A.2).

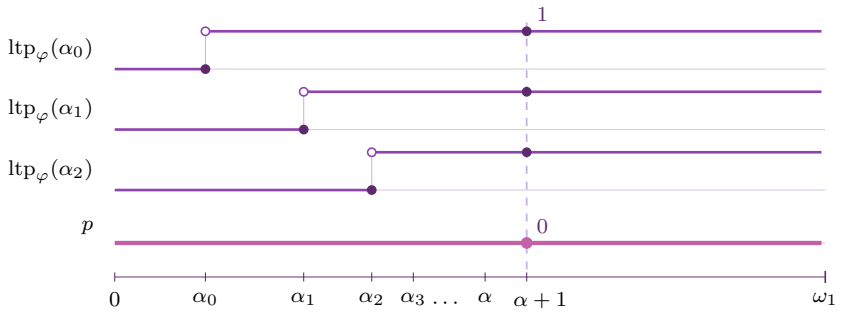


FIGURE A.2. The left types in $(\omega_1, <)$ with $\varphi(x, y)$ the formula $x < y$: the realized types $\mathrm{ltp}_\varphi(\alpha_n)$ and the type p, which is 0 everywhere.

A.2 HISTORICAL REMARKS

⧖ **Appendix A.1.** Grothendieck proved the criterion of Theorem A.1.3 in a paper published in 1952 [**Gro52**]. The paper had been received at the end of 1950, shortly after a note in the *Comptes rendus* in which he had announced his general compactness criteria [**Gro50**]. In the same paper he showed, in substance, that pointwise compact subsets of $C(K)$ are angelic; the word came later and is Fremlin's. The Eberlein–Šmulian theorem is due to Šmulian [**Š40**] and Eberlein [**Ebe47**]. Later proofs and extensions of the criterion, by Namioka, Pták, Young, Simons, Pryce and König, are listed in the timeline in the historical remarks on Subsection 1.2.1. The criterion can be read as a statement about stable functions, and this is the point of view of Definition A.1.1 and Proposition A.1.7. For formulas, this reading is due to Ben Yaacov [**BY14**] and Pillay [**Pil17**], following Iovino [**Iov99**] and Krivine and Maurey [**KM81**].

Appendix

B

Cardinals and cofinality

Roadmap

Two kinds of set-theoretic argument occur in this book. Counting arguments compare the number of types with the number of possible descriptions of them; capturing arguments place a small set inside one stage of a long chain of models. We develop the tools for both, assuming the axiom of choice throughout. The cardinal calculations needed for the tree arguments are proved in Appendices B.1 and B.3, and the capture of small sets, used in Proposition 7.6.2, in Appendix B.2.

The proofs are included so that each application can be checked without a separate course in set theory. For a fuller treatment, see Jech [**Jec03**].

B.1 Ordinals and cardinal calculations

An ordinal records a well-ordered sequence of stages, while a cardinal records only how many objects there are. We identify an ordinal with the set of its predecessors, so that $\alpha < \beta$ means $\alpha \in \beta$. The successor of an ordinal α is $\alpha + 1 = \alpha \cup \{\alpha\}$. A nonzero ordinal that is not a successor is a *limit ordinal*; for such an ordinal δ we have $\delta = \bigcup_{\alpha<\delta} \alpha$. More generally, the supremum of a set of ordinals is their union. The ordinal ω is the first infinite ordinal, the order type of $\mathbb{N}$.

By the axiom of choice, every set can be well ordered. The cardinality of a set is the least ordinal in bijection with the set, and cardinals are identified with these least ordinals. The notation κ^+ means the least cardinal greater than the cardinal κ. For infinite κ this is different from the ordinal successor $\kappa + 1$, which still has cardinality κ. The cardinals $\aleph_0$ and $\aleph_1$ are the cardinality of $\mathbb{N}$ and its successor cardinal. We also write ω_1 for the ordinal $\aleph_1$. More generally, the alephs enumerate the infinite cardinals: $\aleph_{\alpha+1} = (\aleph_\alpha)^+$, and $\aleph_\delta = \sup_{\alpha<\delta} \aleph_\alpha$ at every nonzero limit ordinal δ.

For cardinals κ, λ, their sum $\kappa + \lambda$ is the size of a disjoint union of sets of these sizes, their product $\kappa \cdot \lambda$ is the size of their Cartesian product, and λ^κ is the number of functions from κ to λ. In particular, 2^κ is the number of subsets of κ. Cantor's diagonal argument

gives $\kappa < 2^\kappa$: if $(A_\alpha)_{\alpha<\kappa}$ listed every subset, then $\{\alpha < \kappa : \alpha \notin A_\alpha\}$ would differ from every member of the list.

Lemma B.1.1. *For every infinite cardinal κ, we have $\kappa \cdot \kappa = \kappa$. Consequently, if κ, λ are infinite cardinals, then*

$$\kappa + \lambda = \kappa \cdot \lambda = \max(\kappa, \lambda).$$

The set of finite sequences from a set of infinite cardinality κ also has cardinality κ.

Proof. We well order pairs by their largest coordinate and argue by induction on the infinite cardinal κ. Order $\kappa \times \kappa$ first by $\max(\alpha, \beta)$, then by α, and then by β. This is a well ordering. Every predecessor of a pair with largest coordinate γ lies in $(\gamma+1)\times(\gamma+1)$. If γ is finite, the product $(\gamma + 1) \times (\gamma + 1)$ is finite; otherwise, with $\mu = |\gamma + 1| < \kappa$, the induction hypothesis gives the product cardinality $\mu < \kappa$. Thus every proper initial segment of the well ordering has cardinality less than κ. The order type of $\kappa \times \kappa$ is therefore at most κ, since a larger order type would have a proper initial segment of order type κ. This proves $|\kappa \times \kappa| \leq \kappa$; the reverse inequality follows from $\alpha \mapsto (\alpha, 0)$.

If $\kappa \leq \lambda$, both a disjoint union and a Cartesian product of sets of these sizes embed in $\lambda \times \lambda$, while each has at least λ elements. This proves the second assertion. By induction, $\kappa^n = \kappa$ for every positive integer n, and a countable union of sets of size at most κ has size at most $\aleph_0 \cdot \kappa = \kappa$. Taking the union over all finite lengths proves the last assertion. ☺

Exponentiation obeys a useful bookkeeping identity:

$$(\lambda^\kappa)^\mu = \lambda^{\kappa \cdot \mu}.$$

Indeed, a function from μ to the functions $\kappa \to \lambda$ is the same thing as a function $\mu \times \kappa \to \lambda$. For every infinite cardinal κ this gives

$$(2^\kappa)^\kappa = 2^{\kappa \cdot \kappa} = 2^\kappa. \tag{B.1}$$

If $\lambda \geq \kappa$ are infinite, a nonempty subset of λ of size at most κ is the range of a function $\kappa \to \lambda$. Including the empty set does not change the bound, so there are at most λ^κ subsets of λ of size at most κ. This is the calculation used in the proof of Theorem 6.2.2: with $\lambda = 2^\kappa$, there are at most λ choices of a small submodel, and at most $2^\kappa = \lambda$ types over each choice.

The next observation explains how a tree can have few nodes but many branches, without any assumption about the continuum function.

Lemma B.1.2. *Let λ be an infinite cardinal, and let μ be the least cardinal with $2^\mu > \lambda$. Then μ is infinite, $\mu \leq \lambda$, and the set*

$$2^{<\mu} = \bigcup_{\alpha<\mu} \{s : s\colon \alpha \to \{0, 1\}\}$$

has cardinality at most λ, whereas the set 2^μ of its branches has cardinality greater than λ.

Proof. Cantor's theorem shows that a suitable $\mu \leq \lambda$ exists, and no finite cardinal works. For every ordinal $\alpha < \mu$, the minimality of μ gives $2^{|\alpha|} \leq \lambda$. There are μ possible lengths, so

$$|2^{<\mu}| \leq \mu \cdot \lambda = \lambda.$$

The assertion about branches is the choice of μ. ☺

In Proposition 2.4.2, the nodes of this tree supply the parameters of a small model, while its branches supply distinct types over that model. The distinction between the number of nodes and the number of branches is the whole counting argument.

B.2 Cofinality and regular cardinals

A long construction may have many stages but still be approached through comparatively few of them. Cofinality measures the number of stages needed to approach its end.

Definition B.2.1. A subset A of a nonzero limit ordinal δ is *cofinal*, or *unbounded*, in δ if $\sup A = \delta$. The *cofinality* $\operatorname{cf}\delta$ is the least cardinality of such a subset. We define $\operatorname{cf}(\alpha+1) = 1$ and $\operatorname{cf} 0 = 0$. An infinite cardinal λ is *regular* if $\operatorname{cf}\lambda = \lambda$, and *singular* otherwise.

Every nonzero limit ordinal has infinite cofinality, since a finite set of smaller ordinals has a largest member. For example, $\operatorname{cf}\omega = \omega$ and $\operatorname{cf}(\omega+\omega) = \omega$, where the addition in the second expression is ordinal addition. Although the second ordinal is larger, its finite tail stages $\omega + n$, $n < \omega$, approach it. If $\operatorname{cf}\delta = \mu$, there is a strictly increasing cofinal sequence $(\delta_\xi)_{\xi<\mu}$ in δ. To construct one, enumerate a cofinal subset of size μ and choose each new term above both the next enumerated element and the preceding terms. At a stage $\xi < \mu$, the preceding terms form a set of size less than μ, which is therefore bounded in δ.

Lemma B.2.2. *The cofinality of a nonzero limit ordinal is a regular infinite cardinal. Every successor cardinal κ^+ of an infinite cardinal κ is regular.*

Proof. Let $\mu = \operatorname{cf}\delta$, and choose a strictly increasing cofinal sequence $(\delta_\xi)_{\xi<\mu}$. If μ had a cofinal subset D of cardinality less than μ, the set $\{\delta_\xi : \xi \in D\}$ would be cofinal in δ and would contradict the definition of μ. Thus μ is regular.

Suppose that $\operatorname{cf}(\kappa^+) \le \kappa$. A cofinal family of at most κ ordinals below κ^+ would have union κ^+. Each member of the family has cardinality at most κ, and therefore the union has cardinality at most $\kappa \cdot \kappa = \kappa$, a contradiction. ☺

One use of regularity is that fewer than λ objects, each of size less than λ, still have total size less than λ. More precisely, if λ is regular, $|I| < \lambda$, and $|A_i| < \lambda$ for $i \in I$, then $|\bigcup_{i\in I} A_i| < \lambda$. For $\lambda = \aleph_0$ this is the finite-union assertion. For uncountable λ, regularity bounds all the cardinals $|A_i|$ by one cardinal $\nu < \lambda$; replacing ν by $\max(\nu, |I|, \aleph_0)$ if necessary, Lemma B.1.1 bounds the union by $\nu \cdot \nu = \nu < \lambda$.

Here is the form of the same idea used for chains of models.

Lemma B.2.3. *Let δ be a nonzero limit ordinal, and let $(A_\xi)_{\xi<\delta}$ be an increasing family of sets. If $X \subseteq \bigcup_{\xi<\delta} A_\xi$ and $|X| < \operatorname{cf}\delta$, then $X \subseteq A_\beta$ for some $\beta < \delta$.*

Proof. For each $x \in X$, choose $\xi_x < \delta$ with $x \in A_{\xi_x}$. The set of indices chosen has size less than $\operatorname{cf}\delta$, so it is bounded in δ. Choose $\beta < \delta$ at least as large as all of them. Since the family is increasing, $X \subseteq A_\beta$. ☺

For instance, a countable set lying in the union of an increasing ω_1-chain is already contained in one stage. For an ω-chain the cofinality is ω, and a countable set need not lie in a single stage: each stage might contribute one new element of the set. In Proposition 7.6.2, the set to be captured has size at most κ, and the relevant stage has cofinality κ^+.

B.3 König's diagonal argument

Cantor's theorem defeats a proposed list by changing its entry on the diagonal. König's theorem uses the same idea when the entries on the diagonal are chosen from sets of different sizes.

Theorem B.3.1 (König). *Let $(\kappa_i)_{i\in I}$ and $(\lambda_i)_{i\in I}$ be families of cardinals with $\kappa_i < \lambda_i$ for every $i \in I$. Then*

$$\sum_{i\in I} \kappa_i < \prod_{i\in I} \lambda_i.$$

Proof. Choose pairwise disjoint sets A_i of cardinality κ_i and sets B_i of cardinality λ_i. Suppose that a map $F\colon \bigcup_{i\in I} A_i \to \prod_{i\in I} B_i$ were onto. For each i, the set of coordinates $\{F(a)(i) : a \in A_i\}$ has size at most $\kappa_i < \lambda_i$. Choose $b_i \in B_i$ outside it. The function $b = (b_i)_{i\in I}$ differs from $F(a)$ in coordinate i whenever $a \in A_i$, and therefore lies outside the range of F. Thus no such surjection exists, which proves the strict inequality. ☺

Corollary B.3.2. *Every infinite cardinal λ satisfies $\lambda^{\operatorname{cf}\lambda} > \lambda$.*

Proof. Let $\mu = \operatorname{cf}\lambda$, and choose a cofinal sequence $(\alpha_i)_{i<\mu}$ in λ. The sets $D_i = \alpha_i \setminus \bigcup_{j<i} \alpha_j$ partition λ, and each has cardinality $\kappa_i < \lambda$. Since $\kappa_i^+ \le \lambda$, König's theorem gives

$$\lambda = \sum_{i<\mu} \kappa_i < \prod_{i<\mu} \kappa_i^+ \le \lambda^\mu.$$

☺

Corollary B.3.3. *Let κ be an infinite cardinal, define $\lambda_0 = 2^\kappa$ and $\lambda_{n+1} = 2^{\lambda_n}$ for $n < \omega$, and write $\lambda = \sup_{n<\omega} \lambda_n$. Then $\operatorname{cf}\lambda = \omega$ and $\lambda^{\aleph_0} > \lambda$.*

Proof. The sequence (λ_n) is strictly increasing by Cantor's theorem and is cofinal in λ. Thus $\operatorname{cf}\lambda \le \omega$; since λ is a limit ordinal, its cofinality is infinite and must equal ω. The last inequality follows from Corollary B.3.2. ☺

This is the cardinal used in Proposition 5.3.1. A tree with λ successors at each node and height ω has only $\lambda^{<\omega} = \lambda$ nodes but $\lambda^{\aleph_0} > \lambda$ branches. Even if one type follows as many as $2^{\aleph_0}$ branches, at most λ types cannot cover them all, since $\lambda \cdot 2^{\aleph_0} = \lambda$.

B.4 Exercises

The first exercises practice the counts; the last ones practice cofinality and the capture of small sets.

1. Let κ be an infinite cardinal and let X have cardinality κ. Show that the set of finite subsets of X has cardinality κ, and that a language of cardinality at most κ has at most κ formulas with parameters in X. Deduce that, in a fixed finite tuple of variables, there are at most 2^κ complete types over X.

2. Let λ be an infinite cardinal, and let μ be the least cardinal with $2^\mu > \lambda$. Explain why the estimate $|2^{<\mu}| \le \lambda$ does not require μ to be regular. When $\lambda = \aleph_0$, identify μ and count the nodes and branches of the resulting tree.

3. Show that $\operatorname{cf}\aleph_\omega = \omega$, where $\aleph_\omega = \sup_{n<\omega} \aleph_n$. Use König's theorem to prove $\aleph_\omega^{\aleph_0} > \aleph_\omega$. Contrast this with $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$, and explain why no assumption about the size of the continuum is being made.

4. Let $(M_\alpha)_{\alpha<\lambda}$ be a continuous elementary chain, where λ is regular and uncountable. Show that every subset of $\bigcup_{\alpha<\lambda} M_\alpha$ of size less than λ is contained in one M_β. Give an ω-chain for which this fails for a countable subset of its union.

5. Show that the supremum of a strictly increasing sequence of ordinals of length θ has cofinality θ whenever θ is an infinite regular cardinal. Identify the use of regularity in the proof, and give an example showing that it cannot be omitted.

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